



BIOLOGY AND HEALTH SCIENCE

Senior Two



Student's Book

Experimental version

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FOREWORD

Dear Student's,

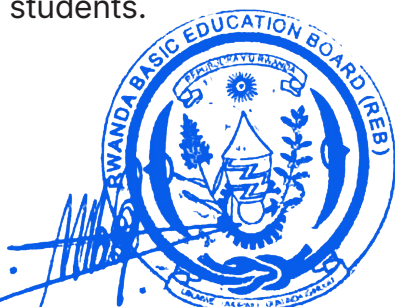
The Rwanda Basic Education Board (REB) is honoured to present the Biology for Lower Secondary student's book for Senior two. This Textbook is designed to support the effective implementation of the Competence-Based Curriculum (CBC), ensuring Student-cantered teaching practices that align with Rwanda's vision of empowering Students to reach their full potential, actively participate in society, and pursue meaningful careers.

As Rwanda advances toward building a knowledge-based economy, education plays a pivotal role. Teachers are central to this transformation, guiding Students to connect classroom knowledge with real-life experiences, thus fostering individual growth and national development.

This Student's book serves as a comprehensive resource for teaching Biology lower secondary for Senior two. The subject is an integral part of the CBC, aimed at equipping pupils with essential knowledge, skills, values, and attitudes necessary for becoming responsible and competent citizens. Through this subject, Students are encouraged to develop problem-solving abilities and contribute positively to their communities.

In pursuit of quality education, the Government of Rwanda emphasizes the importance of using teaching and learning materials that are aligned with the curriculum.

I sincerely thank everyone who contributed to the refinement of this textbook. Special recognition goes to the REB team who coordinated the entire process. We thank all partners, teachers, and individuals who were involved in the process from start to end. All feedback was highly valued and helped to improve learning outcomes of our students.



Dr. MBARUSHIMANA Nelson

Director General

ACKNOWLEDGEMENT

I would like to extend my sincere appreciation to everyone who contributed to the refinement of the Biology Lower secondary Student's Book for Senior Two. This achievement would not have been possible without the active involvement of various stakeholders.

Special thanks go to Rwanda Basic Education Board (REB) Staff, Subject experts, University lecturers and Teachers for their valuable work in the adaptation of this textbook. I am also grateful to all our education development partners whose support was instrumental throughout the process.



Joan Murungi,

Head of Curriculum, Teaching and Learning Resources Department/REB

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UNIT 1

CLASSIFICATION OF KINGDOM ANIMALIA



Key Unit Competence:

Classify animals into their main groups based on external features.

Learning objectives

By the end of this unit, you should be able to:

1 Knowledge and understanding

- Describe the general characteristics of animals
- Identify and explain the features and adaptations of chordates
- Classify arthropods and evaluate their importance to humans
- Recognize and outline the key traits of other animal phyla

2 Skills

- Distinguish different groups of animals using observable features.

3 Attitudes and values

- Appreciate the existence of animal diversity and increasing complexity from lower organisms to higher animal groups.
- Acknowledge economic importance of animals.
- Value the need for classification of animals.

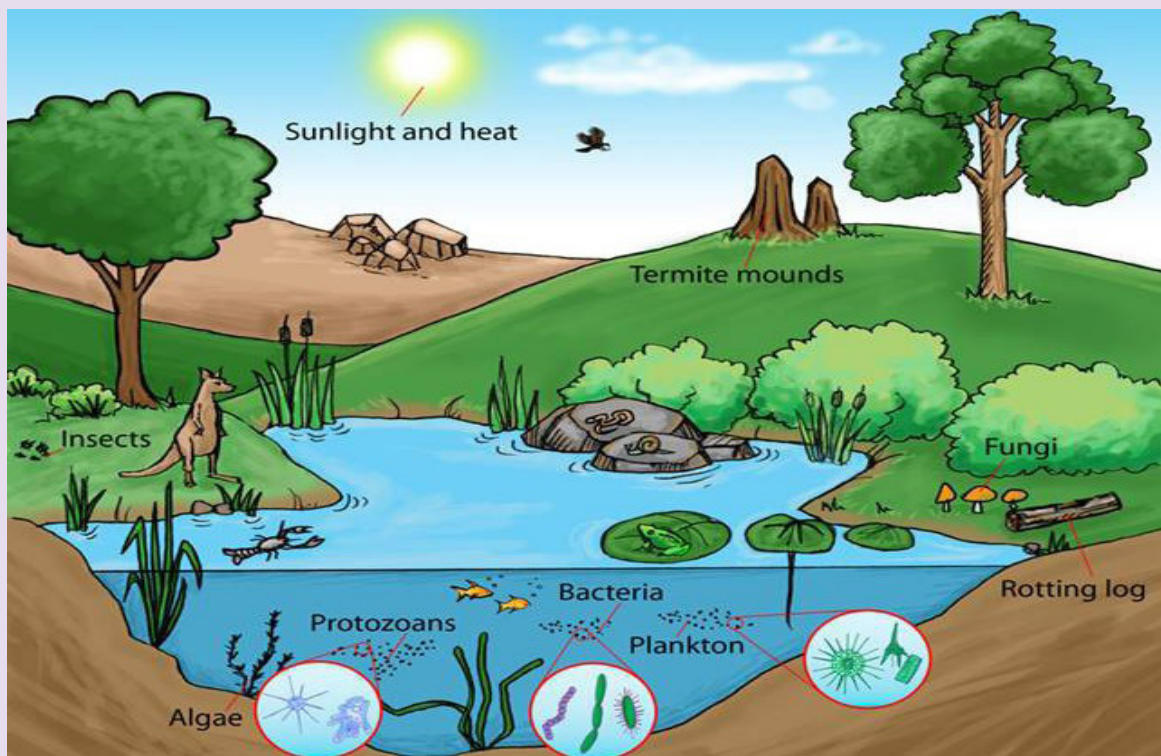
1.0. Introduction

The Kingdom Animalia comprises a wide variety of multicellular organisms that play important roles in ecosystems. Classification of animals is the scientific process of grouping animals according to their shared characteristics and evolutionary relationships. This helps scientists identify organisms, understand biodiversity, and study the relationships among different animal groups.

Animals are broadly classified into two major groups: **non-chordates**, which lack a notochord, and **chordates**, which possess a notochord at some stage of their development. Animal classification follows a hierarchical system developed by Carl Linnaeus and uses scientific (binomial) names to ensure accurate identification and communication worldwide.

Key Concepts: Kingdom Animalia, classification, taxonomy, biodiversity, binomial nomenclature, hierarchical classification, vertebrates, invertebrates, chordates, non-chordates, notochord, evolutionary relationships.

Introductory activity



In Senior 1, you learnt about biodiversity and classification of organisms in the environment. Now look at the picture and answer questions below.

1. Can you recall the five-kingdom system of classification and the key features of each kingdom?
2. Which animals can you see? Write down their external features.
3. Think about how the features can be used to classify the animals, then try grouping these animals into different categories.
4. Do you now have an idea of what you will learn in this unit?

1.1. General characteristics of animals.

Learning activity 1.1



Observe the collected specimens of animals

Requirements

- Specimen bottles
- Pair of forceps
- Gloves
- Sweep nets
- Pooter

Procedure

Go to the school field and collect animal specimen of different types.

1. Carefully observe the animals collected and note their external features.
2. Observe pictures or photographs of different animals and try to identify them.
3. Count the number of legs and wings (if present) for each animal.
4. Touch the outer covering of each animal and describe its texture.
5. Discuss with your classmates the external features you have observed.

Caution: Some animals can be harmful. Always wear gloves and use forceps when collecting them, then transfer them safely into specimen bottles and take them to the biology laboratory.

Task:

- a) Do all animals have the same number of legs and wings?
- b) What features are used to classify animals?
- c) What is the meaning of the following terms?

- Eukaryotic
- Multicellular
- Heterotrophic

There are many species of animals living on land (about 6.5 million), in water (about 2.2 million) and approximately 10,500 species in the air. However, only a few of these animals are clearly understood. It is therefore important to study and describe the characteristics of the main groups of the animal kingdom. This study enables us to be familiar with our environment as we interact with these animals in our day to day lives.

How are animals different from other organisms and from each other?

Animals have the following general characteristics:

- They are multicellular organisms.
- They have eukaryotic cells.
- Their cells are differentiated into tissues and organs.
- They are all heterotrophic, meaning they depend on other organisms for food.
- Their cells lack cell walls, cell sap and chloroplasts. They only have cell membranes, and this makes their cell to be irregular in shape.

- Most animals are able to move their whole bodies from one place to another (locomotion).
- Reproduction in most animals takes place through fusion of gametes.
- They respond to external stimuli.

Application activity 1.1



Choose the best answer for each of the following questions:

1. Which of the following is a characteristic shared by all animals?

- A. They produce their own food
- B. They have cell walls
- C. They are multicellular
- D. They perform photosynthesis

2. What type of nutrition do animals exhibit?

- A. Autotrophic
- B. Heterotrophic
- C. Chemosynthetic
- D. Saprophytic

3. Which of the following structures is absent in animal cells?

- A. Nucleus
- B. Mitochondria
- C. Cell wall
- D. Cytoplasm

4. Which kingdom do animals belong to?

- A. Plantae
- B. Fungi
- C. Protista
- D. Animalia

5. Which of the following best describes animal reproduction?

- A. Mostly asexual
- B. Mostly sexual
- C. Only external fertilization
- D. Only internal fertilization

1.2. Phylum Chordata

Learning activity 1.2



Compare general characteristics of Chordata

Observe and compare the external features of a fish, an amphibian, a reptile, a bird, and a mammal.

Task:

Examine the external features of each animal group carefully (body covering, limbs, shape, breathing structures, etc.).

1. Record your observations in a table or short notes.
2. Discuss similarities and differences
3. Prepare a short presentation to share your findings with the class.

The term **“Chordata”** comes from the Latin word chorda which means chord. Organisms in this phylum, at same stages in their lives have a chord-like structure called notochord.

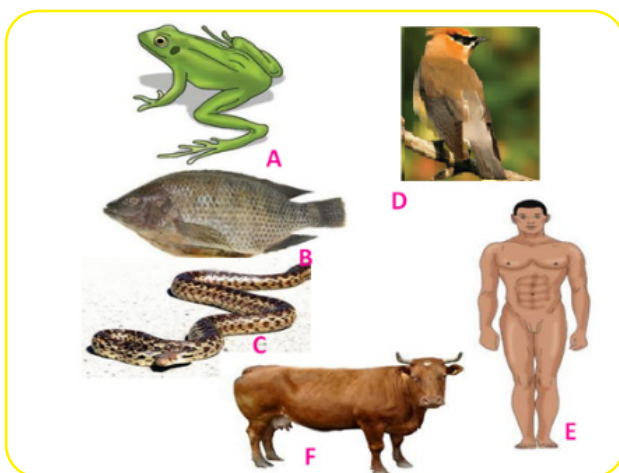


Figure 1.1 Examples of animals in the phylum Chordata

We differ from other organisms because we have a backbone that enables us to stand upright. The main characteristics of organisms in phylum Chordata are:

1. Presence of a chord-like structure called notochord. The notochord supports the body in lower chordates while in higher chordates (vertebrates) it is present only during embryonic stages. It is replaced by a vertebral column.
2. Presence of a vertebral column which forms part of an internal skeleton. Vertebral column is a bony structure made up of vertebrae. It protects the spinal cord.
3. Presence of a nervous system with a brain which is connected to a hollow nerve tube or a single tubular nerve cord. The nerve tube runs along the back and forms a brain anteriorly.
4. Bilateral symmetry: this means that the body can be divided along one plane into two equal halves that are roughly mirror images of each other.

All the animals in the picture belong to phylum Chordata. Based on this, what can you say about phylum Chordata?

Organisms in the phylum Chordata can further be subdivided into 5 different classes. These are:

Pisces (fishes) – Amphibia – Reptilia – Aves (birds) and Mammalia.

Application activity 1.2



1. Which of the following is a defining feature of all chordates at some stage of their development?
 - A. Jointed appendages
 - B. Notochord
 - C. Segmented exoskeleton
 - D. Compound eyes
2. The dorsal nerve cord in chordates is unique because it is:
 - A. Solid and ventral
 - B. Hollow and dorsal
 - C. Hollow and ventral
 - D. Solid and dorsal
3. Pharyngeal gill slits in chordates are primarily associated with:
 - A. Locomotion
 - B. Respiration or filter feeding
 - C. Reproduction
 - D. Excretion
4. Which structure in chordates develops into the vertebral column in vertebrates?
 - A. post-anal tail
 - B. Gill arches
 - C. Notochord
 - D. Dorsal nerve cord
5. The presence of a post-anal tail in chordates mainly helps in:
 - A. Circulation
 - B. Excretion
 - C. Movement and balance
 - D. Temperature regulation

1.3. Class of Pisces or Fish

Activity 1.3



Examine external features of fish

Requirements

- Tilapia or any other type of fish (freshly killed)
- Hand lenses
- White tiles
- Gloves
- A pair of forceps

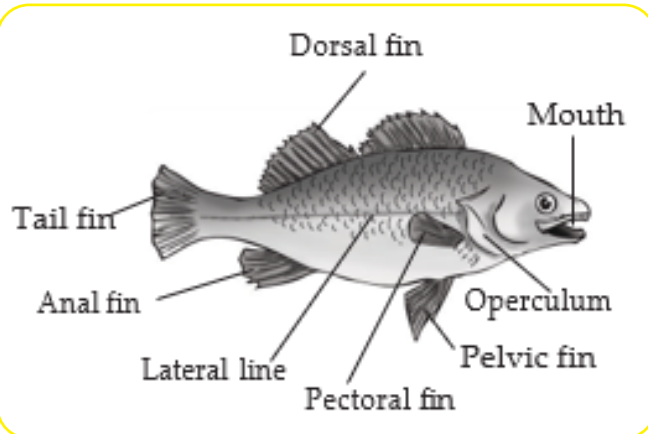
Procedure

1. Observe the body surface of the fish and note the arrangement of the scales.
2. Place the hand on the anterior part; slowly move your hand over the body surface of the fish towards the tail region. How do you feel?
3. Gently lift the flap-like structure (operculum) covering the gill chamber to expose the gills.
4. Discuss the following questions and present your findings to the rest of the class.
 - a) Note the shape of the fish's body.
 - b) State the observable external features which all fish possess.
 - c) Draw the arrangement of scales on fish.
 - d) Draw the structure of a gill.
 - e) What is the function of the gills you observed in the gill chamber?

1.3.1. Main characteristics of animals in class Pisces

- a) All fish are aquatic. They live in places such as seas, lakes, oceans, rivers and dams.
- b) The skin of the fish is covered with scales which overlap backwards.
- c) Fish have gills which are used for gaseous exchange. The gills are located in a space called the gill cavity on the side of the head. The number of gills in a fish ranges from 4 - 7 pairs.
- d) Fish have fins that aid in movement. The fins on the sides of the body are usually found in pairs, for example, the pelvic and pectoral fins. The dorsal, ventral and tail fins are usually unpaired.

- e) Fish are poikilothermic: Their body temperature is dependent on environmental temperature because they cannot regulate their own body temperature.
- f) Fish exhibit external fertilisation where eggs are first laid by the female then the male sheds sperms over them.
- g) Fish have a lateral line on their body for sensitivity.
- h) Fish have a single circulatory system with a two chambered heart.



Note: Fish use gills for gaseous exchange except lung fish that live in oxygen deficient swamps and use lungs for breathing.

Fig 1.2: External features of Tilapia fish.

1.3.2. Adaptation of fish to aquatic environment

The following features enable fish to survive in water:

a. Gills

Fish use gills to breathe under water. Most fish have to swim constantly. This enables water to pass through the gills to allow for gaseous exchange.

b. Streamlined body

The body shape of a fish is well suited to its particular habitat. Most fish have a streamlined body to allow water to easily pass over them, reducing friction (resistance) as they swim.

c. Fins and tails

Fins and tails allow fish to move through water. The tail propels the fish while fins guide their movement in water by controlling their direction and balance.

d. Lateral lines

The lateral line allows fish to detect vibrations in water, alerting them of predators.

e. Huge number of eggs

A single fish can lay more than a million eggs, which can all be fertilised. However, a lot of eggs are eaten by predators while others are washed away by water currents. A large number of eggs ensures that at least some will survive to maturity.

Application activity 1.3



1. What are two characteristics that help a fish live in water, and how do these characteristics help it to survive?
2. a) Draw a fish and label these parts: gills, fins, scales, and tail.
b) What is the function of each part?

1.4. Class Amphibia

Learning activity 1.4



Observe an amphibian

Requirements

- Preserved specimen of toads and frogs or freshly killed.
- Photos and illustration of newts and salamanders.

Procedure

1. Examine the external features of the specimen provided (do not touch the warty skin of toads, it contains glands which produce poison). Note the following:
 - a) The nature of the skin.
 - b) Presence or absence of scales.
 - c) Number of legs.
2. Draw a labelled diagram of a toad, a newt and a frog. Then answer the following questions:
 - a) What is the difference between a toad and a frog?
 - b) Do these amphibians have scales on their skins?
 - c) How many legs do these amphibians have?
 - d) Share your findings with the rest of the class.

The word **amphibia** comes from the word '**amphi**' which means '**dual**' or '**two**'. This class of chordates can live both on land and in water. Most adult amphibians live on land. However, they go back to the water to breed. Examples are:



Newt



Salamander



Frog

Toad

Fig 1.3. Examples of amphibians.

1.4.1. Main characteristics of animals in class Amphibia

- They have mucus glands under the skin to keep it moist.
- The skin has no scales unlike that of fish and reptiles. (Reptiles will be discussed later in this unit).
- Adult amphibians use the lungs, moist skin and mouth cavity for gaseous exchange. However, their young ones use external gills for gaseous exchange while in water.
- The adult female amphibians always lay their eggs in water.
- They exhibit external fertilisation.
- They have two pairs of limbs.

Application 1.4



What does the word "Amphibian" mean, and why are frogs called amphibians?

Describe two characteristics or adaptations that help amphibians live both in water and on land.

1.5. Class Reptilia

Learning activity 1.5



Observe the Characteristics of reptiles

Requirements

- Live specimens of non-poisonous reptiles such as lizards
- Preserved specimens
- Photos and pictures
- Films and videos on lives of reptiles

Procedure

Observe the specimens, photos and illustrations provided carefully. Then do the following:

- a) Note the nature of the skin.
- b) Note the presence and number of limbs.
- c) Count the number of limbs

Caution: Do not attempt to catch live animals like snakes or even go close to them because they are poisonous and dangerous.

Study questions

- a) How are the scales attached to the skin?
- b) Is the skin of reptiles moist or is it dry?
- c) Compare the scales of reptiles to those of fish.
- d) Which reptile does not have legs?
- e) How many legs do the other reptiles have?
- f) What is the habitat of each of the reptiles studied?

The term 'Reptilia' comes from a Latin word, 'Reptilis' which means 'crawl'. The animals in this class move by creeping or crawling. Unlike amphibians, reptiles do not breed in water because they lay eggs with leathery shells; hence cannot dry out. Examples of reptiles are snakes, turtles, tortoises, crocodiles and lizards



Fig 1.4. Main groups of reptiles

1.5.1. Main characteristics of animals in Class Reptilia

They have a dry scaly skin. Some like the tortoise have scales which have hardened to form a shell.

They are mostly terrestrial with a few being partially aquatic.

They undergo internal fertilisation, where the male introduces sperms into the female body. The eggs laid thereafter are covered with a shell.

Most of them have two pairs of legs except the snake.

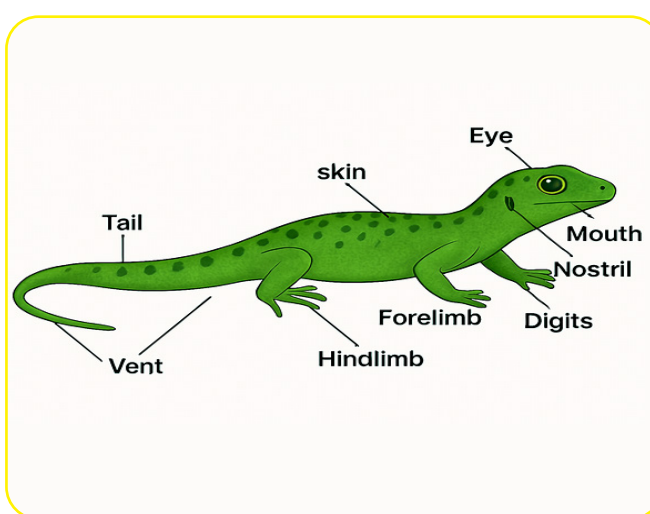


Fig 1:5 External Parts of a lizard.

Application activity 1.5



Fill in the blanks with appropriate terms.

Reptiles have _____ skin, they lay their eggs on _____, they breathe using their _____, their body temperature depends on the _____, and most reptiles move using their strong _____.

1.6. Class of Aves (Birds)

Learning activity 1.6



Observe and examine the Characteristics and adaptations of birds

Requirements

- Photos and illustrations of different kinds of birds.
- Live specimen of caged or domestic birds.

Procedure

Examine the birds' photographs and illustrations carefully. Then note the following:

- a) The nature of the skin covering on their body.
- b) The skin on their legs.
- c) Number of limbs.
- d) Adaptation of the limbs to their function.
- e) The nature of the feathers.
- f) Draw and label different parts of the bird.

Task

1. What covers the body of the bird? How does this cover help the bird?
2. How is the skin on the bird's legs different from the rest of its body?
3. How many limbs does the bird have? What are they modified into?
4. How are the forelimbs (wings) adapted for their function?
5. What types of feathers can you observe? What roles do they play (e.g., flight, warmth, protection)?
6. How are hind limbs (legs and feet) adapted for activities such as walking, perching, or swimming?

The term '**Aves**' comes from a Latin word '**Avis**' which means bird. Animals in this class consist of birds such as the hummingbird, ostrich, fowl, sparrow, hawk, eagle, sea gull, parrot, crow and ibis.

Look at the following pictures. Have you ever seen them?

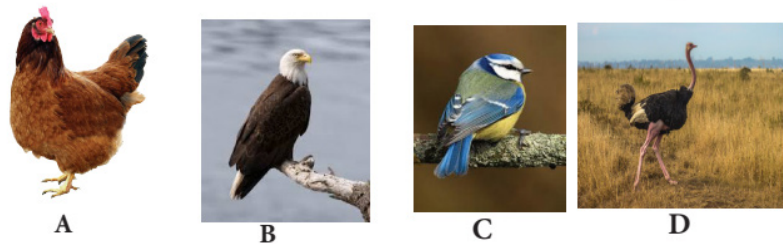


Figure 1.6: Examples of birds

1.6.1. Main characteristics of Class Aves

- Their bodies are covered with feathers.
- Their legs are covered with scales.
- Their front limbs are modified to form wings. Hind legs are used for walking, running, swimming among other uses.
- They have hollow bones that make them light for flight.
- They have toothless jaws covered by a horny beak or bill.
- They lay eggs with a hard shell made of calcium.
- They carry out internal fertilisation.

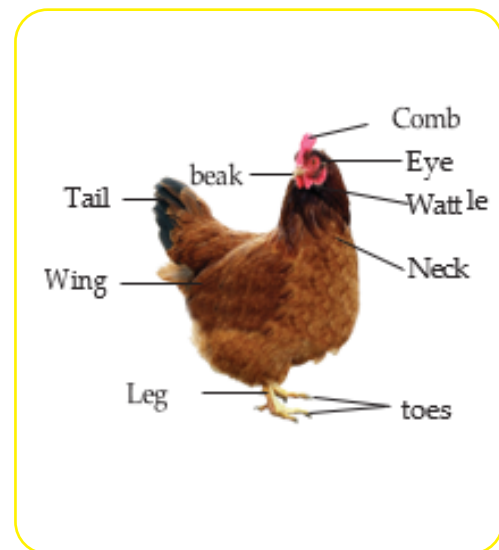


Fig 1.7: External Parts of a hen.

1.6.2. Adaptations of birds to their environment

The adaptations for their flight are the following:

a) Light weight skeleton

Birds have hollow (air-filled) bones, which make their body light without losing strength.

b) Feathers for lift and control

Wing feathers are shaped to help push air downward, giving lift. Tail feathers help with steering and balance.

c) Strong flight muscles

Large chest (pectoral) muscles provide the powerful wing strokes needed for take-off and sustained flight.

d) Streamlined body shape

Their smooth, pointed body reduces air resistance, allowing them to move easily through the air.

e) Efficient respiratory system

Birds have air sacs and lungs that supply a continuous flow of oxygen, giving them energy for long or fast flights

f) Adapted beaks and feet

Birds have different forms of beaks and feet to help birds survive in various environments. These include:

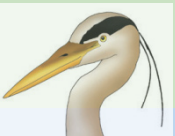
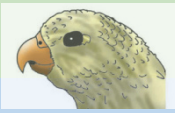
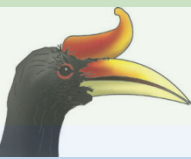
- Seed-eaters (e.g., finches: short, thick conical beak and perching feet)
- Insect-eaters (e.g., warblers): slender pointed beak and perching feet)
- Nectar-feeders (e.g., sunbirds): long curved beak and perching feet)
- Fish-eaters (e.g., kingfishers): long sharp beak and strong gripping feet)
- Meat-eaters (e.g., eagles): hooked beak and powerful talons)
- Filter-feeders (e.g., ducks): broad flat beak and webbed feet)
- Wading feeders (e.g., herons): long spear-like beak and long stilt-like legs)

Application activity 1.6

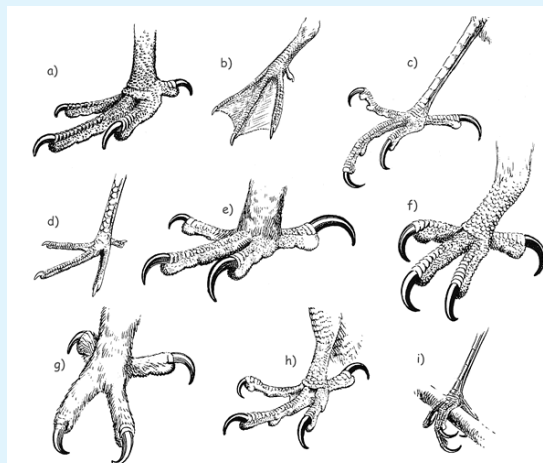


1. Identify the mode of feeding of the birds whose beaks are shown below.

Table 1.1: Birds and their mode of feeding

Bird	Mode of feeding
A. 	
B. 	
C. 	
D. 	
E. 	
F. 	

2. Identify the feeding system of birds based on their feet structure below.



1.7. Class of Mammalia

Learning activity 1.7



Observation and examination of external features of mammals

Requirements

- Live specimen of animals in class Mammalia in the school compound such as rabbits or rats.
- Photos of aquatic mammals/Illustrations and videos of other members of class Mammalia.

Procedure

1. Examine the animals of class Mammalia found in your school compound or in the school surroundings.
2. Recall a goat or a cow in your home or at your neighbour.
3. Note the following:
 - Presence of mammary glands
 - Presence of hair or fur on the skin.
 - The nature of the ears.
4. Locate a mammal with young ones. For instance, a rabbit with its litter.

What is the behaviour of the mother towards its young ones?

Study questions

- a) What is the importance of mammary glands in a mammal?
- b) What is the covering on the skin of mammals?
- c) Describe the ears of mammals.

The term of Mammalia is derived from the Latin word “mammalis” which means “mammal” or “mamma” which means milk-secreting organ of female mammals. Most mammals are terrestrial except a few like a dolphins and whales which are aquatic.

1.7.1. Main characteristics of animals in Class Mammalia

- a) They have mammary glands. They suckle and take care of their young ones.
- b) Their bodies are covered with fur, hair or wool.
- c) They give birth to young ones except the duck-billed platypus and spiny anteater which lay eggs and feed mainly on ants and termites using its long sticky tongue.
- d) They have external ears. These are the only class of chordates that have external ears.
- e) They exhibit internal fertilization.
- f) They are homeothermic. Their body temperatures do not depend on the environment. It is maintained constantly.
- g) They have differentiated teeth (i.e incisors, canines, pre-molars and molars) each with different functions. They are therefore referred to as heterodonts. Others with uniform teeth differing only in size are homodonts

1.7.2. Classification of Class Mammalia

The class **Mammalia** is commonly divided into three main subclasses: These subclasses are based mainly on differences in reproduction and development.

- a) **Prototheria** – Egg-laying mammals (monotremes), such as the platypus and echidnas.



Fig. 1.8: *Echidna* (Spiny Anteater)

- They are egg laying mammals of Australia.
- They lay amniotic eggs that hatch outside the body and attach to the mother
- They suckle the attached mammalian
- They are unable to regulate their body temperature like other mammals
- Milk oozes from the mammary glands and young sucks the milk.

b) Metatheria – Marsupials, which give birth to underdeveloped young that usually continue developing in a pouch (e.g., kangaroos, koalas).

- They are pouched mammals
- The young develops in a pouch where it is attached to mammary glands.
- They are non-placental mammals.

c) Eutheria – Placental mammals, where the young develop inside the uterus with the help of a placenta (e.g., humans, cows, whales, goat).

- They are placental mammals.
- Young develops in the uterus and nourished by the placenta.
- They have mammary glands to nourish the newborn.

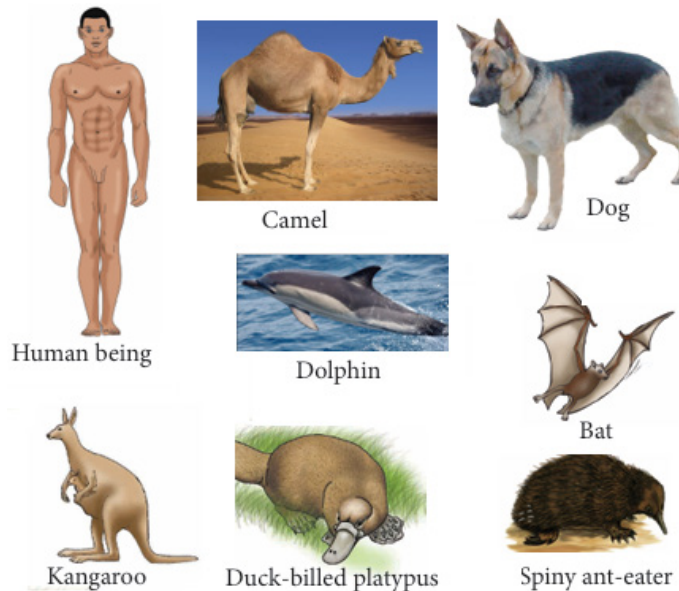


Fig. 1.9: Examples of mammals.

Application activity 1.7



1. Study the two rats below.



Which features seen in the picture show that rats are mammals?

2. Using your knowledge of the external features of the five classes of chordates, complete the table below, using a tick (✓) where the feature is present and (X) if absent.

Class of phylum chordate	Scales	External ear	Mammary glands	Feathers
Birds				
Reptilia				
Amphibia				
Mammalia				
Pisces				

1.8. Phylum Arthropoda

Learning activity 1.8



Observe external features of arthropods

Consider the following pictures or photographs of animals in phylum Arthropoda.



1. Observe the external features.
2. Discuss the features observed.
3. Write down the common features possessed by the different groups of organisms.
4. Note the number of body parts, legs, antennae and eyes.
5. Summarise the features of the different groups of organisms in a table.

The word arthropod comes from two words '**Arthros**' meaning 'jointed' and 'poda' meaning 'leg' or 'foot'. Therefore, arthropods are animals with jointed appendages. It is the largest phylum in the kingdom Animalia.

Animals in this group live on land, in water, and in soil. Some arthropods are useful in many ways, for example:

- Butterflies and bees act as pollinators of flowering plants.
- Bees make honey.

Some arthropods are harmful to other living organisms, for instance:

- Ticks transmit diseases to other animals.
- Mosquitos transmit malaria.
- Tsetse flies transmit trypanosomiasis.
- Aphids destroy crops such as maize, coffee and cassava.

1.8.1. Main characteristics of organisms in Phylum Arthropoda

- a) They have jointed legs (appendages).
- b) They have a tough coat or covering made of chitin. This coat forms an outer skeleton known as the exoskeleton. The tough coat protects the internal organs against damage. The exoskeleton does not grow. It is usually shed in a process known as **moulting** to allow the organism to grow.

- c) They have **bilateral symmetry**. This means that they can be cut into two similar halves in only one way. Each half is a mirror image of the other.

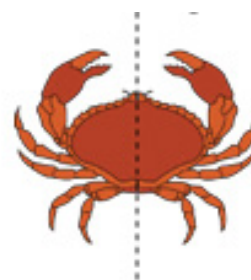


Fig 1.10: Bilateral symmetry in arthropods.

- d) Muscles for movement are attached to the cuticle or exoskeleton, for example in insects.
- e) They have a fluid-filled body cavity called haemocoel for example in earthworms.
- f) Their bodies are segmented.

Organisms in phylum Arthropoda can further be grouped into various classes. The organisms are grouped based on:

- Number of legs
- Presence or absence of antennae
- Number of antennae
- Number of body parts
- Type of their eyes.

The five classes of phylum Arthropoda are: Insect, Arachnida Crustacea, Diplopoda and Chilopoda.

Application activity 1.8



Collect a housefly, a spider, a crab and a millipede. Observe their external features and then fill the table below.

Specimen	Number of body parts	Number of legs	Wings (presence and number)	Presence and number of antennae	Type of eyes and number
Housefly					
Spider					
Crab					
Millipede					

1.9. Class of Insects

Learning activity 1.9



Observe and examine the external features of insects

Requirements

Sweep nets - Hand lenses - Glass jars - Gloves

Procedure

Collect a large variety of insects from the school compound.

Caution: Avoid catching dangerous insects like wasps and bees as they can sting you.

Use sweep nets to catch flying or jumping insects like houseflies and grasshoppers.

Put them in glass jars and take them back to the laboratory for observation. Note: To make observation easier, some insects can be immobilised using chloroform.

Precaution: Do not inhale chloroform. It is toxic.

Examine the specimen carefully and note the following features.

- The number of body parts.
- The types of eyes: simple or compound.
- Presence or absence of antennae.
- Presence or absence of wings
- (Number of legs.

Draw and label clear diagrams of the specimen.

This is the largest class in the phylum Arthropoda. The term 'insecta' comes from the word 'incised' which means 'cut.' The bodies of organisms in this class are divided into three distinct parts; that is head, thorax and abdomen.

Look at the pictures below.
Can you identify the different insects?

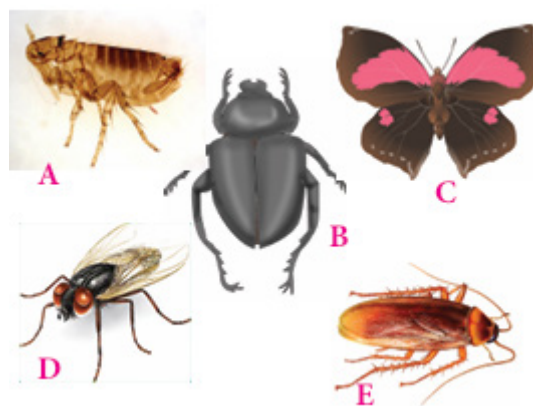


Fig 1.11: Examples of insects.

1.9.1. Main characteristics of organisms in class Insecta

- a) They have three distinctive body parts: head, thorax and abdomen.
- b) They have a pair of long antennae.
- c) They have three pairs of jointed legs, which are attached to their thorax.
- d) They have a pair of large compound eyes.
- e) Some have one or two pairs of wings that are attached to the thorax.
- f) They breathe by means of **spiracles**, which are found on the sides of the abdomen and thorax.

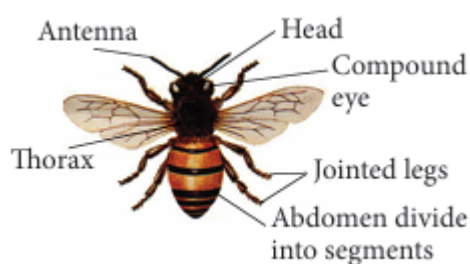


Fig 1.12: External parts of an Insect.

1.9.2. Classification of insects

The class **Insecta** is divided into two main groups/ subclass: **Apterygota**, which includes wingless insects (e.g. silverfish), and **Pterygota**, which includes winged insects (e.g. bees and mosquitoes).

Table1.1. Main orders of Pterygota

Order	Examples
1. Dictyoptera Dorsally-ventrally flattened body Leathery fore wings Membranous hind wings	e.g. Cockroach (Blatta) 
2. Diptera They have a pair of wings. They have halteres for balancing body during flight Their mouth parts are modified for sucking, piercing and biting. They have complete metamorphosis	e.g. House fly (Musca domestica); Mosquito 

3. Coleoptera - They have two pairs of wings. The fore wing is horny. The hind wing is membranous. Mouths parts are adapted for biting and chewing.	e.g. Bean weevil 
4. Hymenoptera They are social insects, They have four membranous wings. Their mouth parts are adapted for chewing or sucking and lapping Abdomen is constricted - Ovipositor always present and modified for sawing, piercing or stinging.	e.g. Bees, Termites 
5. Lepidoptera - They have four wings covered with tiny scales. They have a coiled proboscis for sucking nectar	E.g. Butterflies 

Application activity 1.9



1. Which feature is found in all insects?

- A. Four pairs of legs
- B. Two pairs of antennae
- C. Three-part body plan (head, thorax, abdomen)
- D. Two pairs of wings

2. How many legs does a typical adult insect possess?

- A. Four
- B. Six
- C. Eight
- D. Ten

3. Which of the following is NOT a characteristic of insects?

- A. Exoskeleton made of chitin
- B. Compound eyes
- C. Dorsal nerve cord
- D. One pair of antennae

4. What type of respiratory system is found in insects?

- A. Lungs
- B. Gills
- C. Tracheal system
- D. Book lungs

5. Butterflies and moths belong to which order?

- A. Diptera
- B. Lepidoptera
- C. Coleoptera
- D. Hymenoptera

6. Which order includes insects with two wings such as mosquitoes and houseflies?

- A. Diptera
- B. Hemiptera
- C. Orthoptera
- D. Odonata

7. Beetles belong to which order?

- A. Coleoptera
- B. Orthoptera
- C. Isoptera
- D. Hemiptera

8. Which of the following insects belongs to the order Hymenoptera?

- A. Dragonfly
- B. Cockroach
- C. Bee
- D. Grasshopper

1.10. Other Classes of Phylum Arthropoda

Learning activity 1.10



Observe and examine the external features of Arachnida, Crustaceans, of Diplopoda and Chilopoda

Requirements:

- Hand lens
- Live or preserved specimen or photographs of any of the following organisms: spiders, crabs, lobster, millipedes and centipedes, placed on paper or in Petri dishes.

Caution: Some animals in this class like spiders can be poisonous.

Procedure

Examine the provided specimen and record your observations on the following:

- Number of body parts - Number of body parts
- Number of legs
- Presence or absence of antennae
- Presence or absence of wings
- Number and type of eyes.

Study Questions

- a) How many body parts can be observed in the organisms?
- b) How many legs do they have?
- c) Do they have any antennae?
- d) Describe the type and number of eyes present in the organisms.
- e) Do the organisms have wings?
- f) Make a search on characteristics and examples of other classes of Phylum Arthropoda
- g) Discuss economic importance of Arthropoda.

Class of Arachnida (e.g. Spider, Scorpion)

This class of arthropods includes spiders, mites, ticks and scorpions.

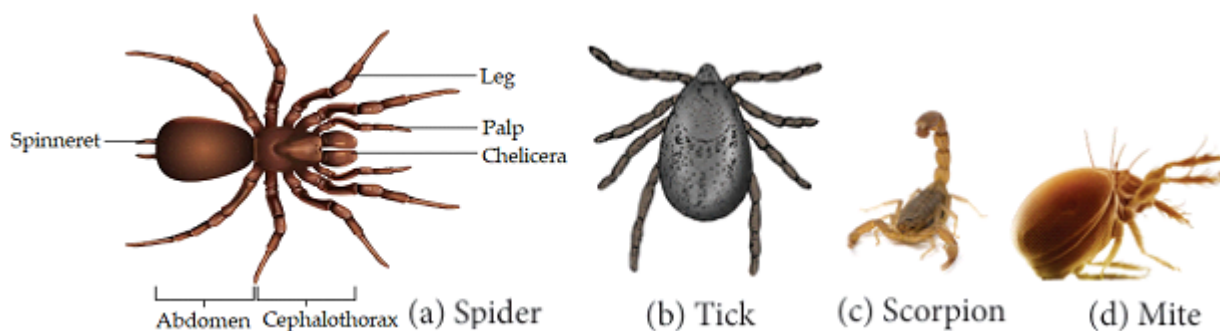


Fig. 1.13: Examples of Arachnida.

The main characteristics of organisms in class Arachnida are the following:

The body is divided into two parts: cephalothorax and abdomen. The head and thorax are joined to form a **cephalothorax**.

They have four pairs of jointed legs attached to the **cephalothorax**.

They have simple eyes about 8 in number.

They do not have antennae. Instead, they have a pair of pedipalps.

They possess a pair of poison glands called **chelicerae**

❖ Class of Crustaceans

Have you come across a crab, wood louse or water flea? All these are Crustaceans. Other members of this class include lobsters, crayfish, shrimps and barnacles.

The term crustacea is derived from "crusta" which means a 'hard shiny coat'. They are aquatic arthropods except for wood lice which are the only fully terrestrial crustaceans.

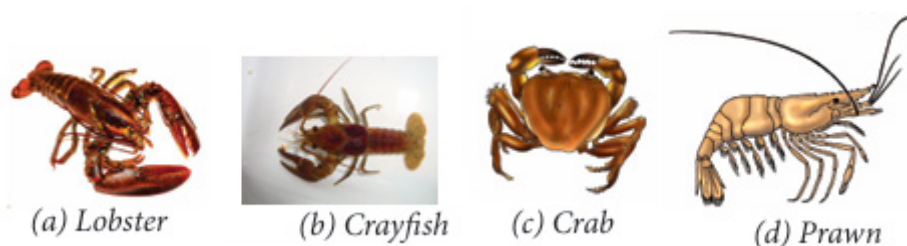


Fig: 1.14: Examples of crustaceans

❖ Main characteristics of organisms in class Crustacea are the following:

1. Their body is divided into two parts: the cephalothorax and the abdomen. The head and the thorax are joined to form the cephalothorax. The cephalothorax is covered by a shiny coat known as carapace.
2. They have different types of appendages. Their appendages are modified to form legs for walking, feeding, protection, and for swimming.
3. They have two pairs of antennae.
4. They have a pair of compound eyes at the end of stalks.

5. They have ten or more legs. The walking legs are usually located on the cephalothorax while the swimming legs are located in the abdomen.
6. They breathe by use of external gills on the underside of the carapace.

❖ Class of Diplopoda

Diplopoda refers to arthropods that have two pairs of legs per body segment. This class consists of millipedes, which occur in a variety of sizes.

They have elongated cylindrical bodies.

Diplopods are herbivores and mainly feed on vegetables and dead organic matter.

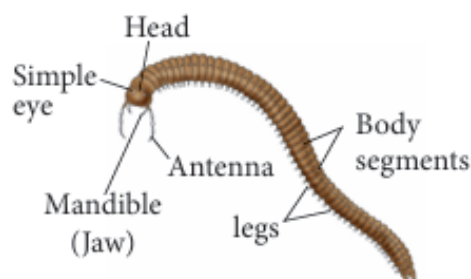


Fig 1.15: External features of a millipede.

The main characteristics of organisms in class Diplopoda are the following:

- a) They have elongated and cylindrical bodies.
- b) Their bodies are divided into two main parts: the head and a segmented trunk. The number of body segments ranges from 25 – 100.
- c) Each body segment has two pairs of legs except the first thoracic segments that have one pair of leg each.
- d) They have a pair of antennae.
- e) They may have simple or compound eyes, or in some cases no eyes are present.
- f) They breathe through spiracles found on the sides of the body segments.
- g) Millipedes roll their bodies when disturbed. After coiling, they produce a stinking substance from their stink glands. This is self-protective behaviour against enemies and predators.

❖ Class of Chilopoda

This class is composed of centipedes. They are found in virtually all habitats but unlike the millipedes, they have less number of legs. Centipedes have only one pair of legs per segment.

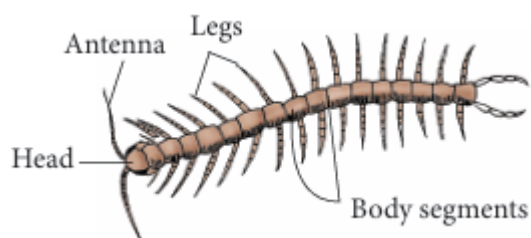


Fig 1.16: External feature of a centipede.

The main characteristics of organisms in class Chilopoda are the following:

- a) They have flat bodies with legs positioned on either side of the body.
- b) They have a pair of legs in each segment.

- c) Their body is divided into a head and a segmented trunk. However, they have 15 -21 segments, which are fewer than those of animals in the class Diplopoda.
- d) They are carnivorous. They use their poison claws to kill their prey.
- e) They may have compound eyes, simple eyes or no eyes.
- f) They breathe through spiracles found on each side of the body segments.

Economic importance of arthropoda

Arthropods have both advantages and disadvantages.

1. Some arthropods are useful to us in many ways. They include:
 - Butterflies and bees act as pollinators of flowering plants.
 - Bees make honey.
 - Lobsters and prawns are used as food.
2. However, most arthropods are harmful to other living organisms. For instance:
 - Ticks transmit diseases in animals, for example East Coast Fever
 - Mosquitos transmit malaria.
 - Tsetse flies transmit trypanosomiasis in human beings and Nagana in cattle.
 - Aphids destroy crops such as maize, coffee and cassava in the fields while weevils destroy stored grains.
3. Some arthropods cause harm and injury to human beings as well. Some are poisonous if they bite, for example, spiders, wasps, centipedes and crabs

Application activity 1.10



1. For each of the following sentences say whether it is true (T) or false (F):
 - a. Arachnids have three pairs of legs and no antennae. _____
 - b. Crustaceans usually possess two pairs of antennae. _____
 - c. Chilopods have two pairs of legs on each body segment. _____
 - d. Diplopods have two pairs of legs on most body segments. _____
 - e. The body of an arachnid is divided into cephalothorax and abdomen. _____
 - f. Chilopods are slow-moving herbivores while diplopods are fast-moving carnivores. _____
2. Match the animals in Column A with the correct arthropod class in Column B.

Column A

- a. Crab
- b. Spider
- c. Centipede
- d. Millipede
- e. Butterfly

Column B

1. Insecta
2. Arachnida
3. Crustacea
4. Chilopoda
5. Diplopoda

1.11. Other Phyla in the Kingdom Animalia.

Learning activity 1.11



Examine external features of specimens collected

Requirements

- Specimen bottles or jars
- Forceps
- Preserved specimen of roundworm (Ascaris) and tapeworm in different petri dishes.
- Petri dishes
- Labels
- Gloves

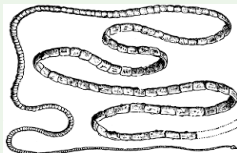
Procedure

1. Visit a damp place around the school compound. Carry with you specimen bottles and petri dishes.
2. With your gloves on and using a pair of forceps, collect the following specimen: snail and earthworm.
3. Put the snail in a specimen bottle and the earthworm in a petri dish. Label them and take them to the laboratory for examination.
4. Examine the specimens carefully with a hand lens.
5. Note the external features for each organism.
6. Construct a dichotomous key using the features you have noted in (5) to place each organism in the correct phylum.
7. Discuss the steps you followed to identify each organism and present your work to the rest of the class.

Apart from the two phyla, Chordata and Arthropoda, Kingdom Animalia comprises other lower-level organisms that belong to different phyla. These phyla include Platyhelminthes, Nematoda, Annelida, Mollusca, Coelenterata, Cnidaria, Porifera/ sponges, Echinodermata. These animals are complex and therefore are not studied at this level in details.

The main features that are used to group animals further into other phyla include: Body symmetry type of skeleton and the body form

Table 1.2 The different phyla, their main characteristic and examples of organisms in each.

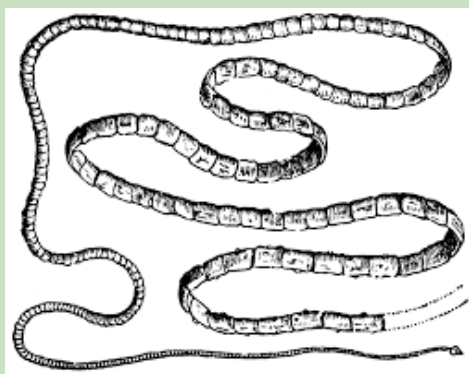
Phyla	Main characteristics	Examples	
Platyhelminthes	Have a mouth but no anus. Dorso-ventrally flattened bodies with bilateral symmetry. Mainly hermaphroditic.	Tapeworm, liver flukes, flat worms and planaria 	
Nematoda	Rounded unsegmented body with tapering ends. Body covered with thick elastic cuticle. Have a simple alimentary canal with mouth anteriorly and anus posteriorly.	Round worms, hook worms, ascaris spp 	
Annelida	The bodies have segments (metameric). Body segments are separated by sheet-like septa. They lack legs; instead, each segment has bristle- like chaetae to aid in locomotion. Have an alimentary canal with oral and anal opening.	Earthworms, lugworms, leeches 	
Mollusca	The dorsal side of the body is covered by a hard external shell, although some Molluscs do not have a shell. The ventral side of the body has one soft muscular foot adapted for movement, attachment to surfaces and food capture. Have sensitivity structures called tentacles on which eyes are located.	Octopus, slugs, squids, snails and oysters 	

Coelenterate (cnidarian)	Have a sac-like body cavity that also acts as a gut. The body has radial symmetry i.e. any section passing through the body gives two equal halves. Possess tentacles.	The sea anemones, hydra and jellyfish. 
Porifera	They are sessile. They attach themselves on rocks or dead corals. They have a simple body made of a cavity or interconnected cavities opening to the outside through pores. The body cavities are lined with flagellated cells that create water currents. They lack a nervous system.	Sponges 
Echinodermata	The skin has calcareous exoskeleton and spines. Body of the larva has bilateral symmetry while adult body has a five-way radial symmetry (pentaradial). The mouth is located on the lower side (oral) while the anus is located on the upper (aboral) side.	Sea cucumbers, sea urchins and the starfish 

Application activity 1.11



1. Some of the important characteristics of some phyla belonging to the Kingdom Animalia are listed below. Against each characteristic, write the name of the phylum.
 - (i) Animals with a two layered body wall enclosing a single cavity.
 - (ii) Animal whose gut has only one opening.
 - (iii) Simplest multicellular animals with a tube-like body and pores in the body wall.
 - (iv) Unsegmented sea animals with an exoskeleton of spines.
 - (v) Animals with unsegmented bodies with tapering ends.
2. Match the animals in column A with the phylum they each belong in column B.

A	B
	Platyhelminthes
	Coelenterata
	Mollusca



1.12.1. Project Activity: Bees, rabbits, goats and chickens rearing,

Plan this project at school level for rearing bees, rabbits, chicken or goat. Work on a single project. Below is an example of detailed project description

Project activity: Establishing a School Rabbit-Rearing Project

1. Context/Background

Rabbit rearing is an important agricultural activity that provides meat, manure, and income. Rabbits grow quickly, require relatively small space, and can be raised using locally available resources. School-based rabbit rearing projects help students develop practical agricultural skills, entrepreneurship, responsibility, and teamwork.

2. Challenge/Problem Statement

Many schools and communities face challenges related to inadequate sources of protein, limited income-generating activities, and lack of practical agricultural skills among learners.

Key Question

How can rabbit rearing contribute to improved nutrition, income generation, and practical agricultural skills at school?

3. Purpose of the Project

The purpose of this project is to enable students to acquire practical skills in rabbit rearing, understand the importance of animal husbandry, and appreciate how livestock farming can contribute to food security, income generation, and environmental sustainability.

4. Instructions

- Plan and establish a small rabbit-rearing unit at school.
- Identify suitable rabbit breeds and housing requirements.
- Investigate feeding, health care, and management practices.
- Keep records of feeding, growth, and general management.
- Present findings and recommendations at the end of the project.

5. Your Tasks

- Design and construct a simple rabbit hutch using locally available materials.
- Identify appropriate feeds and feeding schedules.
- Observe and record rabbit growth and behaviour.
- Maintain cleanliness and ensure proper health management.
- Calculate estimated costs and potential benefits of rabbit rearing.
- Prepare a report on the project.

6. Expected Student Outputs

- A functional rabbit hutch.
- Records of feeding, growth, and management practices.
- A cost-benefit analysis of rabbit rearing.
- A project report with conclusions and recommendations.
- A presentation on lessons learned from the project.

7. Constraints and Limitations

Scope: The project should focus on managing a small rabbit-rearing unit within the school environment.

Materials and Methodology: Use locally available materials for housing, feeding equipment, and record keeping. Data should be collected through observation, measurement, and record keeping.

Time Frame: 8 weeks.

8. Guiding Questions

1. What housing conditions are necessary for successful rabbit rearing?
2. What types of feed are suitable for rabbits?
3. How can diseases and parasites be prevented in rabbits?
4. What benefits can a school obtain from rabbit rearing?
5. How can rabbit rearing contribute to food security and income generation?
6. What challenges may be encountered during rabbit rearing, and how can they be addressed?

1.12.2. Assessment questions

A. Knowledge and understanding

1. The kingdom Animalia is divided into taxa such as order, family, class, species, phylum and genus. Rewrite these categories in the descending order starting with kingdom.
2. Identify an echinoderm from the following.
A. Starfish B. Roundworm C. Sponge D. Elephant
3. Which characteristics distinguish humans from other mammals.
Man has sense organs, which are more efficient.
Man is born with the ability to speak.
Man has a well-developed brain.
Man possesses hands, which are more skilled than forelimbs of other mammals.

4. When trying to determine whether two birds (male and female) belong to the same species, the most certain method would be.
- A. Mate the two birds.
 - B. Compare their feathers.
 - C. Mate the two birds and see if they produce fertile offspring.
 - D. Compare the colour of their feathers.
5. Which of the following statements is correct?
- A. Grasshoppers, mosquitoes, cockroaches and spiders are all insects.
 - B. Crab, scorpion, millipede and centipede are all arachnids.
 - C. Centipedes, crabs, spiders and mosquitoes are all crustaceans.
 - D. Crab, cockroaches, mosquitoes and spiders are all arthropods.
6. (a) Name the two largest groups into which all animals are classified.
(b) What is the main difference between these two groups?
7. Name the phylum to which all vertebrates belong.
8. Below is a list of important characteristics of different phyla belonging to the animal kingdom.
Write the name of the phylum corresponding to each characteristic.
- a. Moves by means of a muscular foot _____.
 - b. Possess a notochord and a tubular dorsal nerve chord _____.
 - c. Segmented animals with jointed limbs and an exoskeleton of chitin _____.
 - d. Unsegmented soft bodied animals with a hard calcareous protective shell _____.
9. Insects are both useful and harmful. Explain.
10. Name the characteristic that is common to:
- a. Fish, amphibians, and reptiles.
 - b. Amphibians and reptiles but not fish.
 - c. Birds and mammals, but not reptiles.

B. Skill

11. Study the following diagrams representing organisms in a given phylum.



- To which phylum do the above organisms belong? Give reasons for your answer.
 - How does animal Z protect itself from enemies?
 - Which animal above belongs to a class that lacks antennae? Name the class.
12. You are provided with specimens below. They represent animals in different phyla under kingdom Animalia.



- Choose one of the animals and describe its external characteristics.
- Draw and label the diagram of the animal described above.
- Which animal above belongs to a class that lacks antennae? Name the class.

C. Attitudes and values

13. Organize a discussion in your class on the topic "are insects important to our environment?" Explain
14. Some animals under class Arachnida are dangerous. They can both bite and sting. According to your experience in Activity 1.11, suggest ways in which such organisms can be handled.

UNIT 2

INTRODUCTION TO ENVIRONMENTAL BIOLOGY



Key Unit Competence:

Apply environmental concepts in everyday life.

Learning objectives

By the end of this unit you should be able to:

Knowledge and understanding

1

- Explain key terms such as Ecology, Biosphere, biome, ecosystem, population, community, habitat, ecological niche, biotic and abiotic factors.
- Explain the role of abiotic factors on organisms living in an ecosystem.
- Describe how energy is lost between trophic levels.

Skills

2

- Interpret simple food chains and food webs.
- Construct pyramids of biomass and numbers.
- Carry out analysis of the diagram showing a food chain.
- Differentiate between biotic and abiotic factors of an ecosystem.
- Classify biotic components of an ecosystem into Producers – Consumers and – Decomposers

Attitudes and values

3

- Appreciate the role of green plants in terms of conversion and supply of energy to all living organisms.
- Appreciate the interdependence of living organisms.

2.0. Introduction

Environmental biology is the study of the interactions between living organisms and their physical surroundings. It helps us understand how plants, animals, microorganisms, and humans depend on one another and on the environment for survival. Through the study of ecosystems, students explore how energy flows, how nutrients are recycled, and how organisms adapt to different habitats. Environmental biology also highlights the importance of conserving natural resources, protecting biodiversity, and promoting sustainable use of the environment to maintain ecological balance and support life on Earth.

Key Concepts: environment, ecology, ecosystem, habitat, population, community, biodiversity, biotic factors, abiotic factors, food chain, food web, energy flow, nutrient cycling, conservation, sustainability.

Introductory activity



Look at the picture below. Can you explain what is happening? What is the importance of the action in the picture? Supposing one of the animals (Lion and Buffalo) in the picture becomes extinct, what would happen?



Talk to your friend about how such relationships in your area are regulated. What does this tell you about what you will learn in this unit?

2.1. Concepts of ecology

Learning activity 2.1



Reading and understanding the important concepts of ecology

Read the text below and then answer questions that follow.

" Ecology is the branch of biology that studies how living organisms interact with each other and with their environment. All life on Earth exists within the **biosphere**, which includes many **biomes** such as tropical forests, deserts, and tundra—each with its own climate and species. Within a biome, an **ecosystem** is made up of a **biotope** (the physical environment) and a **biocenosis** (the community of living organisms).

A **population** refers to all individuals of one species in an area, while a community includes all the different populations living together. Each organism lives in a specific **habitat** and plays a role called its **ecological niche**, which includes how it gets food, reproduces, and interacts with others. Ecosystems are shaped by **biotic factors** like plants and animals, and **abiotic factors** such as sunlight, temperature, water, and soil”.

1. Define each term that is bolded in the text above
2. Arrange the following terms in an increasing order: Ecosystem – Biosphere – Population – Biocenosis – A bird – Biome – Cell

An old saying goes '*no man is an island*'. This seems to apply to all other organisms on earth. Organisms do not exist in isolation. They depend on each other for survival.

Plants obtain their energy from the sun. Herbivores like cows eat the plants for survival. On the other hand, carnivores like lions feed on herbivores if they are to survive. Through this interdependence, organisms have developed feeding relationships. The scientific study of these relationships between organisms and their environment is called **ecology**.

Therefore, ecology seeks to explain the connection between plants and animals and their surroundings. It also provides information about the benefits of the ecosystem and how we can use the Earth's resources efficiently. The study of ecology enables us to appreciate the environment and leave it healthy for future generations.

During the study of ecology, the following terms are widely used.

- **Ecosystem:** This is a stable unit of nature consisting of all communities interacting with each other and their surrounding physical environment. Examples of ecosystem include a pond, a grassland and desert.
- **Biosphere:** This is the part of the Earth and its atmosphere capable of supporting life. It is an area where organisms live, including the ground and the air.
- **Producer:** This refers to all green plants (for example, beans, mango tree and pine) which manufacture their own food by the process of photosynthesis.
- **Habitat:** An area in which an organism lives. The habitat for a leopard should have the right amount of food (gazelle, rabbits and impala) water (a lake, river or spring), and shelter (trees or dens on the forest floor).
- **Population:** This is the total number of organisms of the same species living together in a specific area at a certain time.



Fig. 2.1: Flamingos in their habitat

- **Community:** A combination of different species of organisms living together in an area. An example is a forest of trees and undergrowth plants, inhabited by animals and rooted in soil containing bacteria and fungi.
- **Biotic factors:** These are the living components of an ecosystem. These may be plants, animals, fungi, and any other living things.
- **Abiotic factors:** These are the non-living components of an ecosystem. They include pH, sunlight intensity and temperature.
- **Niche:** This is the position that an organism occupies in a habitat. It includes where it lives, its role and feeding habits, for example, a garden spider is a predator that hunts for prey among plants, while an oak tree grows to dominate a forest canopy, turning sunlight into food.
- **Carrying capacity:** The maximum size of a population that a given area can support without straining.
- **Edaphic factors:** These are all non-living components of an ecosystem both chemical and physical. They are related to abiotic factors.
- **Environment:** This is a combination of all factors that affect the life of an organism.
- **Biodiversity:** This is the variety of life in the world or in a particular habitat or ecosystem.
- **Biome:** This is a large naturally occurring community of animals (fauna) and plants (flora) occupying a major habitat. Examples include forest, grassland, freshwater, marine and desert.
- **Biomass:** This is the total mass of organisms in a given area or volume.

Application activity 2.1



Fill in the blanks with the correct ecological terms.

"Ecology studies how living things interact with each other and their environment. The place where a plant or an animal normally lives is called its _____. A group of organisms of the same species living in the same area at the same time is called a _____. The community of different species living together in a particular area is called _____. The non-living environment, such as soil, water, and climate, is referred to as the _____. When living organisms and their non-living environment interact together, they form an _____. All the ecosystems on Earth, including land, water, and air, make up the _____."

2.2. Ecosystems

Learning activity 2.2



Observation and analysis/examination of pictures of Ecosystems



Observe pictures or photographs of different ecosystems. Study them carefully to identify the ecosystems shown.

1. (2). identify both the biotic and abiotic factors present in each picture.
2. Suggest how important wetlands are to the community.
3. What problems will the community face when wetlands are interfered with?

Ecosystems are also known as **biomes**. They are made up of both the living component (biotic factors) and the non-living component (abiotic factors). There is a close interaction between the two components whereby each affects the other. Ecosystems are broadly subdivided into two: terrestrial and aquatic ecosystems.

Aquatic ecosystem

Aquatic biomes or ecosystems form the largest part of the biosphere. Since water bodies contain different amounts of salt in them, they are classified into two: marine and freshwater ecosystem.

a. Marine ecosystems

These are aquatic ecosystems made up of water with high salt concentration. Oceans, seas and all organisms in them form part of such ecosystems. This covers the largest part of the earth's surface.



d) A mangrove from forest forms part of the marine ecosystem



b) Fresh water ecosystem



c) Akagera is a terrestrial ecosystem

Fig 2.2: Different types of ecosystems

b. Freshwater ecosystems

These are made up of water with a less salt concentration. Lakes, wetlands and rivers with all organisms in them fall under this category of ecosystems.

Terrestrial ecosystems

This is the land ecosystem. Terrestrial ecosystems are always named according to the climate and physical features that define them. They include savanna grasslands, deserts, temperate grasslands and forests. Terrestrial biomes usually crosscut into each other without clear boundaries.

Communities and their physical surroundings constitute an **ecosystem**. The environmental conditions that affect a community are temperature, light, pH, wind, salinity and atmospheric pressure. These physical factors that affect the type of organisms and their distribution are called abiotic factors. Communities include living organisms that is, plants and animals. These are known as **biotic factors**.

1. Abiotic factors

These are the environmental factors in an ecosystem. They describe the physical conditions in the ecosystem.

These factors influence the type of organisms living in an ecosystem. They also influence the distribution of the organisms in that ecosystem. The abiotic factors in an ecosystem include the following:

- a) **Light** - The sun is the main source of light energy in all ecosystems on earth. Light is important in any ecosystem because plants use it to make their own food.
- b) **Temperature** – Temperature variation in an ecosystem affects the type and distribution of organisms found in it. Some organisms prefer high temperatures while others prefer low temperatures. Therefore, they will inhabit different parts of the ecosystem.
- c) **Atmospheric pressure** – This is the pressure exerted by air in the atmosphere. Atmospheric pressure affects the amount of oxygen in the air. At the sea level, air pressure is high. This reduces at higher altitudes. Therefore, different organisms will occupy different altitudes based on their endurance.
- d) **Salinity** – This is the degree of salt concentration in water. This is a factor that is mainly found in aquatic habitats. Some organisms live in salty environments while others live in freshwater environments.
- e) **Humidity** – This is the amount of water vapour in the atmosphere. It influences evaporation and transpiration rates.
- f) **pH**– This is the degree of acidity or alkalinity. The pH of an ecosystem affects the type of distribution of organisms found in it.
- g) **Wind** – This is moving air. Air moves in different directions and at different speeds. Wind influences environmental factors such as temperature and humidity.

2. Biotic factors

There are many different people who form the school community. There are teachers, parents, students, secretaries, cooks and watchmen. The school cannot be able to run efficiently if all these people do not perform the jobs or tasks they are supposed to perform. If the night guard slept while on duty, then robbers might get into the school and steal all the students' books. This would affect the learning process. With the interrelationship among different people, the school runs smoothly.

A similar form of interrelationship exists among organisms in any ecosystem. Biotic factors are the entire living components in an environment. They affect each other either positively or negatively. Autotrophic organisms such as plants use energy from the sun and simple elements to make food. Without them other species of organisms in the area cannot survive. Herbivores eat food from plants; they affect the population of the plants and at the same time support other organisms that depend on them. Organisms in an ecosystem interact in many ways.

Application activity 2.2



You are to observe a picture of an ecosystem.

Analyse it and answer to the questions that follow.



- Describe the nature of this ecosystem.
- State any abiotic factor and biotic factor of that ecosystem.

2.3. Food chains and ecological pyramids

Learning activity 2.3



Think – Pair – share work

Madame Mwiza observes in her maize garden that 100 caterpillars attacked the leaves of her 500 plants, 20 small birds feed on caterpillars, 4 cats feed on birds and 2 wild foxes feed on cats.

But her son KALISA says that in addition to what her mother Mwiza reported, he also saw those birds feeding on both caterpillars and maize leaves, and 6 rats feeding on maize. The wild foxes were feeding on birds, rats and cats. Kalisa managed to kill one wild fox and after a week he found it decomposing with numerous bacteria and microscopic fungi (molds).

1. What is the role of each of the following in this scenario?
 - (i) Maize plants
 - (ii) Caterpillars, small birds, cats, wild foxes
 - (iii) Bacteria and microscopic fungi on the dead fox
2. Based on the report of Madame Mwiza, use arrows to show the feeding relationship
3. Based on the report of Mr. Kalisa, use arrows to show the feeding relationship
4. What difference is there between the feeding system of the report from Mwiza and that from Kalisa?
5. Use boxes of different sizes according to the number of individuals stated by Madame Mwiza to draw a pyramid starting from plants to wild foxes.

Organisms that share the same habitat are grouped into feeding levels or trophic levels: producers, consumers and decomposers.

Producers are green plants like maize and algae that make their own food using sunlight through photosynthesis, and they form the base of all food chains because they provide energy for other organisms. . They are also known as **autotrophs**.

Consumers are animals that cannot make their own food. **Primary consumers** such as caterpillars eat plants, **secondary consumers** like birds eat herbivores, and **tertiary consumers** like foxes feed on smaller animals.

It should be noted that sometimes in an ecosystem, some organisms can feed on both plants and animals. In this case, they are known as **omnivorous** organisms. Examples of such organisms are pigs and human beings.

Decomposers such as bacteria and fungi break down dead plants, dead animals, and waste materials, returning nutrients to the soil for plants to use again.

In ecology, a trophic level refers to the position an organism occupies in a food chain. Green plants form the first **trophic level** known as the producers.

A **food chain** shows the simple flow of energy in one direction, for example: maize → caterpillar → bird → fox.

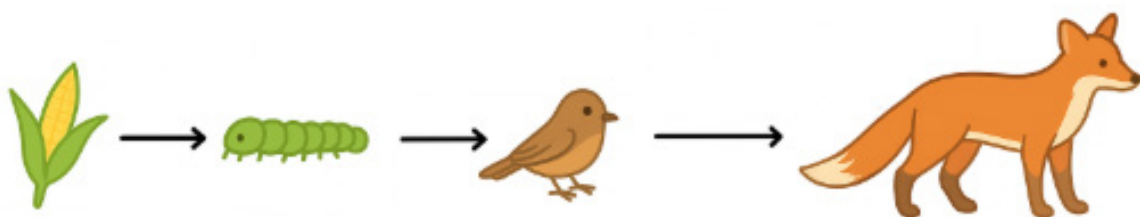
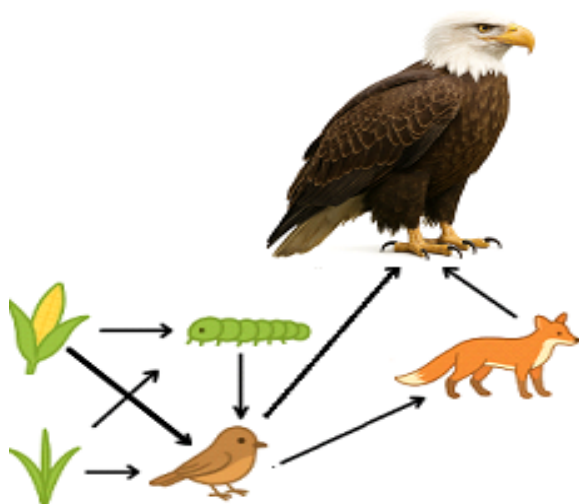


Fig2.3. Simple food chain



A **food web** is more realistic because it links many food chains together and shows that most animals feed on more than one type of organism in the garden.

Fig2.4. Simple food web

Pyramids of numbers show how the number of organisms becomes fewer at each level from producers to top predators, while **pyramids of biomass** show the total dry mass of living things, which is largest at the bottom and decreases at higher levels because energy is lost at each step through movement, heat, and life processes.

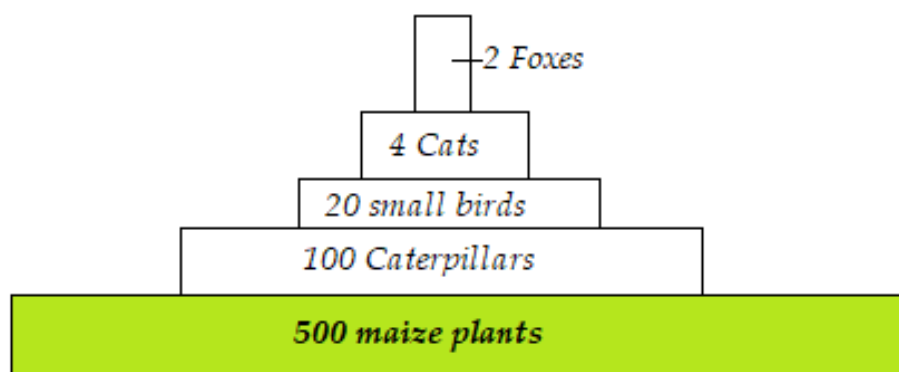


Fig2.5. Pyramid of numbers.

Application activity 2.3



Complete the following statement:

A food chain shows how _____ moves from one organism to another, while a food web includes many interconnected _____. In a pyramid of numbers, the number of organisms usually _____ from producers to consumers, because producers are the _____ group and top predators are the _____.

2.4. Energy flow

Learning activity 2.4



Observation and discussion of the flow of energy in an ecosystem

1. Form a group of 5–6 students.
2. Together, assign each member a role: **Sun, Producer, Herbivore, Carnivore, or Decomposer.**
3. Stand in a line showing the correct order of energy flow the Sun to Producers, then to Herbivores, Carnivores, and finally Decomposers.

Pass a small object (e.g. printed paper) along the line to represent the **flow of energy**

Task

1. Observe and discuss as a group how the energy decreases at each level.
2. Share with the class what you learned about how energy moves through an ecosystem

The primary source of energy is the sun. Energy enters ecosystems in the form of sunlight. Part of it is then converted into chemical energy by the green plants (**producers**) through the process of **photosynthesis**. Plants pass on this energy to the **grazers** who are **primary consumers**, also known as **herbivores**. Herbivores fix part of the energy into their bodies; the rest is lost into the surroundings. They are also in turn eaten by the **carnivores** who take in part of this energy losing the largest part in form of heat and waste excreta. The carnivores are grouped into first level, second level up to the topmost level. The topmost carnivores receive the least portion of energy as most of it has been lost. The last trophic level in an ecosystem is that of the **decomposers**. They are **saprophytic** organisms. They act on the dead remains of organisms in all the other levels.

First level carnivores like the lizards form the third trophic level, followed by snakes on the fourth level and then eagles come on the fifth level as top level carnivores. Some consumers called the detritivores derive their energy from non-living organic material such as faeces and fallen leaves.

How energy flows in an Ecosystem? Study the diagram below. What does it show?

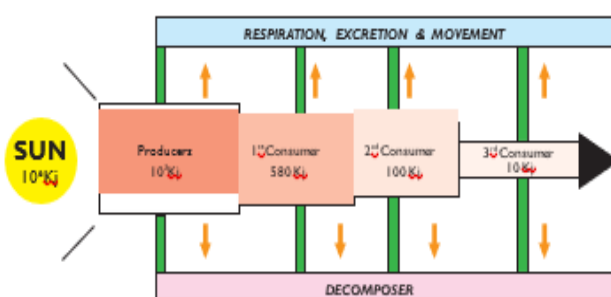


Fig. 2.6: Energy flow in an ecosystem

From this diagram, we can see that energy from the sun flows through producers to the consumers. It also shows that not all energy from the producers is transferred to the consumer.

Energy flows into a food chain from the sun to plants through the process of photosynthesis. It then moves up the food chain to higher trophic levels. Since the transfer of energy from one trophic level to the next is inefficient, less energy enters higher trophic levels.

What brings about the loss of energy at each trophic level?

Several factors account for the loss of energy as one moves from one trophic level to the next. They include the following:

- Respiration
- Part of the energy is lost as undigested food matter.

The rest is lost as excretory products to the surroundings.

Application activity 2.4



1. What is the main source of energy for all living things in an ecosystem?
2. In a food chain, why does the amount of energy decrease as it moves from producers to top predators?
3. What role do decomposers play in the flow of energy in an ecosystem?

2.5. End of Unit Assessment



2.5.1. Project activity: Investigating Trophic Levels and Food Relationships in the School Ecosystem

1. Context/Background

Living organisms in an ecosystem depend on one another for food and energy. Energy flows through ecosystems from producers to consumers and decomposers, forming food chains and food webs. Understanding trophic levels helps explain how organisms interact and how changes in one part of an ecosystem can affect the entire community.

2. Challenge/Problem Statement

Many people do not fully understand how organisms in an ecosystem depend on one another for survival. Disturbance or loss of one group of organisms may affect the balance of the entire ecosystem.

Key Question

How do trophic levels and feeding relationships contribute to the stability of an ecosystem?

3. Purpose of the Project

The purpose of this project is to help students identify organisms within a local ecosystem, classify them according to their trophic levels, and understand the feeding relationships that exist among them through food chains and food webs.

4. Instructions

- Visit the school compound or another nearby ecosystem.
- Observe and identify different organisms found in the area.
- Classify the organisms according to their trophic levels.
- Construct food chains and a food web using the observed organisms.
- Analyze the relationships among organisms and the importance of each trophic level.

5. Your Tasks

- List all plants and animals observed in the study area.
- Group the organisms into producers, primary consumers, secondary consumers, tertiary consumers, and decomposers.
- Record observations in a table form.
- Construct at least three food chains using the organisms identified.
- Develop a simple food web showing feeding relationships.
- Explain the role of each trophic level in the ecosystem.
- Investigate the possible effects of removing one trophic level from the ecosystem.

6. Expected Student Outputs

- A table showing organisms and their trophic levels.
- At least three correctly constructed food chains.
- A food web illustrating feeding relationships.
- A report explaining the importance of trophic levels.
- Conclusions and recommendations on ecosystem conservation.

7. Constraints and Limitations

Scope: The study should focus on organisms found within the school compound or a nearby ecosystem.

Materials and Methodology: Use observation, field notes, sketches, cameras (where available), charts, and textbooks for identification and classification of organisms.

Time Frame: 2 weeks.

8. Guiding Questions

1. What organisms are found in the study area?
2. Which organisms are producers, consumers, and decomposers?
3. How is energy transferred from one trophic level to another?
4. Which organisms occupy more than one feeding relationship within the food web?
5. Why is each trophic level important in maintaining ecosystem balance?
6. What would happen if one trophic level were removed from the ecosystem?
7. How can humans help maintain balanced food chains and food webs?

Expected Learning Outcome

By the end of the project, students should be able to classify organisms according to trophic levels, construct food chains and food webs, explain energy flow in ecosystems, and appreciate the importance of conserving biodiversity and ecosystem balance.

2.5.2. Assessment questions

A. Knowledge and understanding

1. Explain in your own terms each of the following biological concepts:
 - a) Biosphere
 - b) Biotope
 - c) Biotic factors
 - d) Habitat
 - e) Ecosystem
 - f) Trophic level
2. Consider this food chain: Grass → Grasshopper → Birds.
 - a) The birds in this food chain are:
 - A. Producers
 - B. Primary consumers
 - C. Secondary consumers
 - D. Tertiary consumers
 - b) If a disease killed all the birds, what effect will it have in this food chain?
 - c) Explain why food chains are not endless; they are limited to four or five and rarely six organisms.
3. A vulture feeds on dead bodies of animals. The vulture is best described as:
 - A. An omnivore
 - B. A carnivore
 - C. A decomposer
 - D. A scavenger

B. Skills

4. An ecologist carried out a survey to estimate the number of organisms in a certain dam. The following table shows the record of the survey.

Organisms	Estimated number
Microscopic algae	450,000
Small fish	290
Mosquito larvae	5,500
Crocodiles	10
Large fish	160

- (a) Which of the above organisms are consumers of the last order?
- (b) Which organism is likely to get finished first in the dam? Give a reason to support your answer.
- (c) Draw a possible food chain that shows the energy flow in the dam.
5. Plot the following data by placing the producers at the bottom. Use the scale $1\text{cm} = 50\text{g/m}^2$ for length and 1 cm height for each level

Trophic level	Dry weight (gm/m ²)
Carnivores	10
Herbivores	60
Producers	500

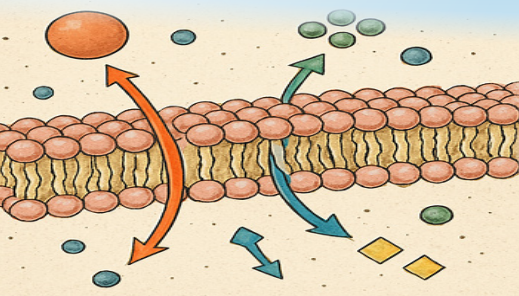
- (a) What does this pyramid indicate?
- (b) Explain why the dry weight of the carnivores must be low.

C. Attitude and value

6. Do you agree with the statement below?
- "All food chains begin with green plants". Explain your answer.

UNIT 3

TRANSPORT ACROSS THE CELL MEMBRANE



Key Unit Competence:

Demonstrate different movements of substances across the cell membrane.

Learning objectives

By the end of this unit you should be able to:

Knowledge and understanding

1

- Define key transport-related terms including semi-permeable membrane, passive transport, diffusion, osmosis, turgidity, active transport, exocytosis, and endocytosis.
- Describe mechanisms of substance movement across cell membranes, focusing on diffusion, osmosis, and active transport.
- Explain the biological significance of membrane transport, especially the role of osmosis in plant cells and the overall importance of transport for cellular function.
- Analyse active transport processes, including influencing factors, examples, and the role of carrier proteins in particle movement.

Skills

2

- Illustrate and label the structure of the cell membrane, highlighting its components and transport pathways.
- Design experiments to demonstrate diffusion in gases and liquids, and osmosis in biological contexts.
- Analyse transport mechanisms, comparing passive and active transport and identifying influencing factors.
- Use visual and digital tools (charts, animations, simulations) to demonstrate active transport, endocytosis, and exocytosis.

Attitudes and values

3

- Recognize why plants and animals need different transport processes, including the importance of turgidity in plant support and the role of active transport in living organisms.
- Appreciate that energy is required for active transport, especially in the absorption of mineral salts that help plants germinate and grow

3.0. Introduction

Transport across the cell membrane is an essential process that enables cells to exchange substances with their surrounding environment. Cells require nutrients and oxygen for their activities and must remove waste products to remain healthy and functional. The cell membrane acts as a selectively permeable barrier, controlling the movement of substances into and out of the cell. This unit introduces students to the structure and function of the cell membrane and the different mechanisms of transport, including diffusion, osmosis, and active transport. Understanding these processes helps students explain how cells maintain internal balance and support the life processes of organisms.

Key Concepts: cell membrane, selectively permeable membrane, transport, diffusion, osmosis, active transport, concentration gradient, passive transport, water potential, nutrients, waste removal, homeostasis.

Introductory activity



Rwanda tea is always strong and delicious. A few tea leaves are diluted in boiled water and thereafter filtered to make you fit along the day.



- Why hot colourless water turns brown when tea leaves are poured into it?
- Is there any similarity between the cell membrane and the filter gauze?

3.1. Structure and role of the cell membrane

Learning activity 3.1



Understanding the structure and the function of the cell membrane

In Senior one, you learnt about the cell organelles. One of them is the cell membrane. The structure of the cell membrane as seen under the light microscope is like a small simple layer. But under the electron microscope the cell membrane appears to have two soft layers phospholipids through which are floating proteins.

1. Draw a labelled diagram of the cell membrane structure.
2. Make research to find the properties of the cell membrane
3. Explain why the transport in and out of the cell is important

The cell membrane is a thin, flexible barrier that surrounds every cell and controls how the cell interacts with its environment.

It is mainly made of a **phospholipid bilayer**, where the **hydrophilic heads** face water and the **hydrophobic tails** face inward, creating a stable structure. Scattered within this bilayer are **proteins** that help the cell communicate, transport substances, and recognize signals from outside. Because of this structure, the membrane is **semipermeable**, meaning it allows some substances to pass through easily while blocking others. This selective control helps the cell maintain **homeostasis**, keeping internal conditions stable even when the outside environment changes.

The exchange through the cell membrane is vital because it ensures the cell receives nutrients, eliminates waste, and stays alive and functional.

The membrane allows essential substances like **oxygen**, **water**, and **glucose** to enter the cell, while removing waste products such as **carbon dioxide**.

Without the membrane's flexibility, **selective permeability**, and **transport mechanisms**, cells would not survive or carry out their roles in the body.

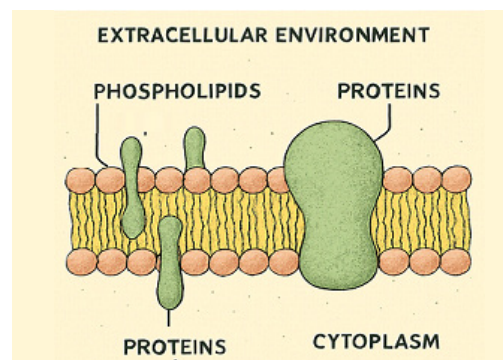


Fig3.1. Ultrastructure of the cell membrane

Application activity 3.1



Fill in the blanks with correct terms:

The cell membrane is made of a _____ with _____ embedded in it. It is _____, allowing only some substances to pass through. Materials like oxygen and water move by diffusion or osmosis, while others require _____. This exchange helps the cell maintain _____

3.2. Passive transport

Learning activity 3.2



Investigation of the diffusion using Parfum in air and KMnO_4 in water

a) Diffusion of parfum

- Spray perfume once into the air (away from faces).
- Stand still and wait to notice when they first smell it.
- Raise their hand the moment they detect the scent.
- Observe how long it takes for the smell to reach different students in the room.
- Move a few steps farther away and repeat to see if the smell takes longer to reach them.
- Write down what they noticed about how the scent spreads.

Explain in one sentence why the smell reaches some students earlier than others

b) Diffusion of Potassium permanganate in water

Requirements

- Crystals of potassium permanganate
- 250 ml beaker
- Water
- Glass rod (open ended)

Procedure

1. Insert the open-ended glass rod into the empty beaker.
2. Drop a crystal of potassium permanganate to the bottom of the beaker through the upper end of the glass rod.



3. With the glass rod still intact, pour water to fill the beaker.
4. Gently remove the glass rod so as not to tamper with the potassium permanganate crystal.
5. Make your observations.

6. Share your findings with the rest of the class.

Study questions

- a) What observation did you make when the glass rod is lifted from the beaker?
- b) Describe the movement of particles of the potassium permanganate crystals.
- c) Why does the movement occur?

3.2.1. Diffusion

Some materials move across the cell membrane naturally through **passive transport** which does not require any energy. There are two main types of passive transport: **diffusion** and **osmosis**. On the other hand, **active transport** require energy because substances are moving against a concentration gradient.

Diffusion involves movement of particles (ions or molecules) from a region of high concentration to a region of low concentration. This process continues until the particles are uniformly distributed throughout the system or until equilibrium is reached.

Diffusion is a product of constant random motion (**kinetic energy**) of all atoms, molecules, or ions in a solution. The area with higher concentration of the particles has more random motion resulting to the net movement of the particles to the area with lower concentration.

Net movement of particles will always take place whenever there is a difference in concentration of particles between two regions. This difference is known as concentration gradient. Diffusion is important since it enables useful molecules to enter the cell and waste products to be removed.

(i) Factors that affect the rate of diffusion

The rate of diffusion of particles refers to the time taken for the particles to move within an available space (fixed) until they are evenly distributed. Several factors affect the rate of diffusion. They include:

a. Temperature

When the temperature of particles is high, their kinetic energy increases, and the particles move faster. Therefore, the higher the temperature, the faster the particles will diffuse while the lower the temperature the lower the rate of diffusion.

b. Concentration gradient

This is the difference in the number of particles present in two regions. A greater difference in concentration of particles between two regions, results in a steeper concentration gradient which causes diffusion rate to be faster. When the concentration gradient is low, diffusion rate is also slower.

c. Size of molecules

Larger molecules are heavier and will diffuse at a slower rate compared to smaller molecules which are lighter.

d. Diffusion distance

The rate of diffusion depends on the distance that particles have to travel in order to be evenly distributed within the available space. An even distribution of particles is reached faster when the distance involved in diffusion is small compared to longer distances. It takes a longer time for molecules to diffuse across a thick membrane while it takes less time for molecules to diffuse across a thin membrane.

e. Surface area to volume ratio

When the surface area to volume ratio is large, more of the substance diffuses across it than when it is small. This takes place as long as the concentration and temperature of the diffusing molecules remain the same. Note: A larger surface area to volume ratio does not increase the speed of diffusion of the particles. It simply enables more particles to diffuse across it in a given time. The figures below show two cubes, calculate the surface area (S.A) of the two cubes).

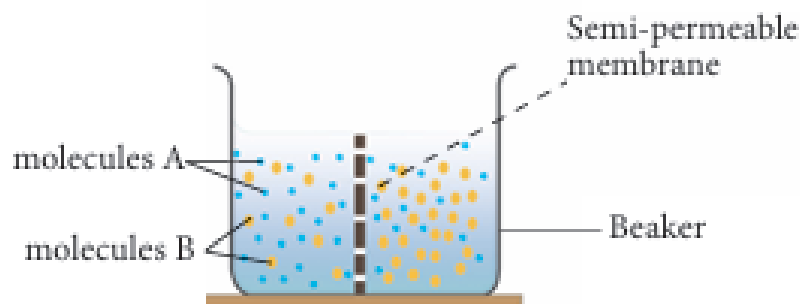


Fig 3.2: Difference in concentration gradient

3.2.2. Importance of diffusion in plants and animals

- Plants absorb water, mineral salts and oxygen from the soil through the root hairs by diffusion.
- Digested food such as glucose and amino acids moves from the small intestine into the blood of animals by diffusion. These substances move from the blood to the cells and tissues by diffusion as well.
- Cells and unicellular organisms such as Amoeba get rid of waste substances by diffusion.
- Diffusion is involved in exchange of gases in stomata, skin of frogs and in the lungs of animals.

Application activity 3.2



1. Diffusion is best described as:
 - A. Movement of particles from low concentration to high concentration
 - B. Movement of particles from high concentration to low concentration
 - C. Movement of particles using energy from the cell
 - D. Movement of water through a partially permeable membrane
2. Which of the following is a real-life example of diffusion?
 - A. Water moving into plant roots
 - B. Air being pumped into a bicycle tire
 - C. Smell of perfume spreading in a room
 - D. Food being chewed in the mouth
3. Diffusion happens faster when:
 - A. Temperature is low
 - B. Particles are large
 - C. Temperature is high
 - D. Concentration difference is small
4. Which statement about diffusion is true?
 - A. It requires energy from the cell
 - B. It only happens in liquids
 - C. It happens because particles move randomly
 - D. It stops working in gases

3.3. Osmosis

Learning activity 3.3



Demonstrating osmosis using Visking tube and tubers

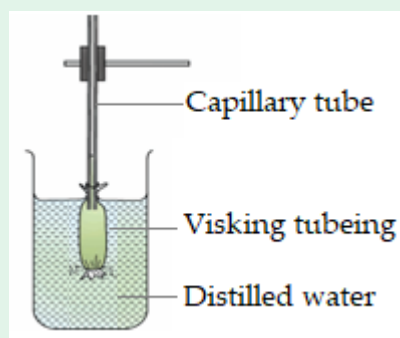
A. Osmosis using visking tubing

Requirements

- 10 cm long visking tubing
- Distilled water
- Concentrated salt solution
- Capillary tube
- Two pieces of strings each measuring 30 cm
- 250 ml beaker

Procedure

1. Tie one end of the visking tubing using a string.
2. Open the other end of the visking tubing and half-fill it with the salt solution.
3. Tightly tie the open end of the visking tubing and allow part of the string to hang.
4. Fill the beaker with distilled water.



5. Gently put the visking tubing containing the salt solution into the beaker with distilled water.
6. Tie the hanging end of the string on the visking tubing onto the capillary tube and inverse the visking tubing into the distilled water
7. Support the capillary tube using a retort stand as shown below
Note: The capillary tube level of the solution in the visking tubing and the beaker at the beginning of the experiment.
8. Leave the set-up undisturbed for at least 30 minutes.
9. Remove the visking tubing from the beaker and make your observations at the end of the experiment.

Study questions

- (a) State three observations you made at the end of the experiment.
- (b) Account for your observations in (a) above.
- (c) Discuss your findings with your class members.

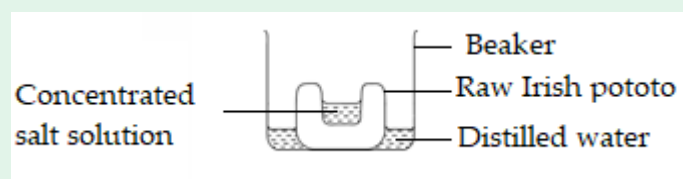
B. Osmosis in a tuber

Requirements

- Fresh arrow roots, cassava, sweet potatoes or Irish potatoes
- Strong salt solution
- Distilled water
- Scalpel
- Large beaker or small basin

Procedure

1. Peel a large Irish potato. You can also use an arrow root, cassava or sweet potato.
2. Cut off a piece so that it stands at least 6 cm high.
3. Cut and scoop out a deep hollow portion in the middle, then pour a strong salt solution halfway into the hollow
4. Mark the level of the salt solution using a scalpel.
5. Place the potato in a petri-dish containing distilled water. Let it stand for several hours then note the level of solution in potato.



6. Repeat the experiment with boiled pieces of Irish potato, sweet potato, cassava or arrow roots.

Study questions

- (a) Draw a diagram to represent the results of the experiment.
- (b) Is the level of the salt solution still the same at the end of the experiment?
- (c) Explain what brings about the change in level of the salt solution.
- (d) Compare these results with those when boiled Irish potato is used.

Osmosis is the movement of water molecules from a region of high-water potential (dilute solution) to a region of low water potential (concentrated solution) through a partially permeable membrane.

Since the concentration of solutions is defined in terms of solute concentration and not in terms of water content; water molecules diffuse from less concentrated solution (fewer solutes, more water) to a more concentrated solution (more solute, less water).

Types of solutions from your knowledge of chemistry; can you recall what the word solution means? What about solute and solvent? When a solid is dissolved in water, we get a solution. The solid that is dissolved in this solution is called the solute. The liquid that dissolves the solid is known as the solvent.

Solute + Solvent = Solution

Concentration of a solution depends on the amount of solute dissolved. A dilute solution has more water molecules compared to solute molecules whereas a concentrated solution has more solute molecules than water molecules.

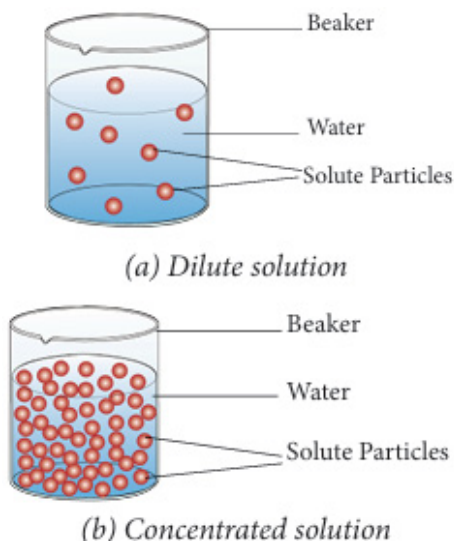


Fig 3.3: Dilute and concentrated solutions.

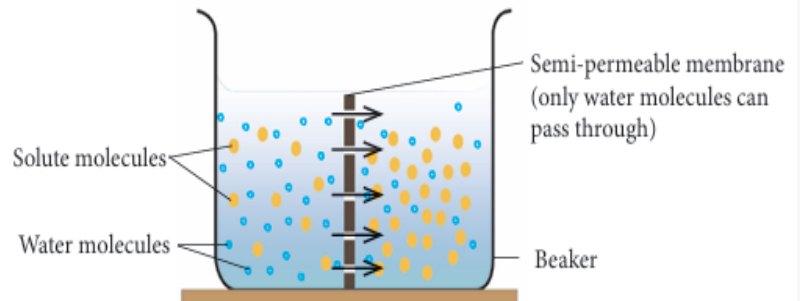


Fig 3.4: Movement of water molecules through a semi-permeable membrane

Suppose a dilute solution is separated from a concentrated solution by a semi-permeable membrane as shown in figure 3.5. Water molecules will move from the dilute solution to the concentrated solution. This is because the dilute solution has more water molecules than the concentrated one. Water molecules are very small and pass easily through the channels or pores of the cell membrane. On the other hand, solute molecules are too large to pass through the pores.

Note:

- Medium of diffusion also influences the rate of diffusion. Particles diffuse faster through gases than liquids.
- The concentration of water in a solution containing mixtures of different solute molecules depends only on the total solute concentration and not the types of solutes.
- If the total concentration of all ions and solutes on both sides of a membrane is the same, there will be no **osmosis**. It is important to first understand the terms used to describe the solutions of different concentration with respect to cells function.

Table 3.2 Terms and their definitions

Term	Meaning
Isotonic	'Iso' means 'equal' When two solutions of equal concentration are separated by a semi-permeable membrane, the net movement of water out of a cell balances water movement into the cell.
Hypotonic	'Hypo' means 'under' When two solutions that have different concentrations are separated by a semi-permeable membrane, the dilute solution is said to be hypotonic to the more concentrated solution. This results into net movement of water from the hypotonic solution to the concentrated solution. If a solution has a lower concentration of solutes than the cytoplasm of a cell, water molecules move into the cytoplasm. The cell expands as water moves in.
Hypertonic	'Hyper' means 'above' A hypertonic solution has a higher concentration of solutes than the cytoplasm of the cell. In this situation, water molecules move, from the cytoplasm, which is now the area of higher water concentration, to the area outside the cell. The cell shrinks as its cytoplasm loses water.
Water potential	This is a measure of the concentration of free water molecules in a solution. A solution with more free water molecules has a higher water potential. It has a greater tendency to lose water molecules to another solution separated by a partially permeable membrane.
Osmotic pressure	This is a force generated by a solution causing it to draw water molecules from another solution separated from it by a partially permeable membrane.

Water relations in plant cells

Living cells are surrounded by a fluid medium which may be isotonic, hypertonic or hypotonic to the cell contents. If the fluid is **isotonic**, there will be no net movement of water into or out of the cell. If the external fluid is **hypertonic** to the cell contents, then, water leaves the cell. If it is **hypotonic**, then water enters the cell.

The movement of water into and out of the cell and the effects that such movements have on the cell may be described as **water relations in the cell**.

In the next section, we find out how various solutions affect how plant cells behave.

a. Plant cells in hypotonic solutions

Plant cells have a large vacuole that contains a fluid called **cell sap**. The sap contains salt and sugar molecules. If a plant cell is surrounded by a hypotonic solution, water molecules move from the surrounding fluid through the cell wall and cell membrane

into the vacuole by **osmosis**. As it receives water, the vacuole swells and pushes the cytoplasm and nucleus outwards against the cell wall. This pressure exerted by the cell contents against the cell wall is called **turgor pressure**. As turgor pressure increases due to intake of more water by osmosis, the cell wall exerts a pressure that is equal to turgor pressure on the protoplasm called **wall pressure**. A point will be reached when no more water can enter the plant cell. At this point, the wall pressure is equal to turgor pressure but opposite in direction. Because the cell wall is made of rigid cellulose material, it does not stretch very much, and the cell does not burst.

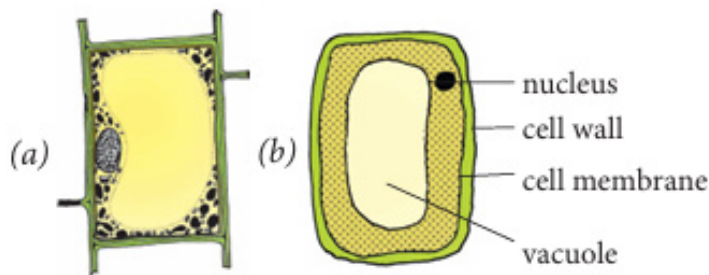


Fig. 3.5: (a) Normal and (b) turgid plant cell

However, turgor pressure causes the cell to become **stiff or firm**. Such a cell is described as being **turgid**. A plant in which all the cells are turgid always appears firm and erect.

Turgidity in plant cells is important because the stiff cells give support to the soft tissues such as petals and sepals. Turgor pressure enables soft, non woody plant stems to remain upright despite the downward force of gravity. If there is inadequate water in the environment, turgor pressure cannot be maintained in a plant.

b. Plant cells in hypertonic solutions

A plant cell that is surrounded by a hypertonic solution will lose water. Water is lost from the cytoplasm then from the vacuole. The turgor pressure in the cell begins to decrease. If this continues, the cell membrane and cytoplasm shrink away from the cell wall. The vacuole in turn reduces in size. The moving of the cell membrane and cytoplasm away from the cell wall is called **plasmolysis**. The cell is said to be **plasmolysed**. When a plasmolysed cell is placed in distilled water, it will become turgid again. The cell is said to undergo **deplasmolysis**. The point at which plasmolysis just begins is called **incipient plasmolysis**. At this point, the cell protoplasm that is, cell membrane and cytoplasm no longer exert any pressure against the cell wall. The turgor pressure is zero, and the cell therefore loses its turgidity. It is now a **flaccid cell**.

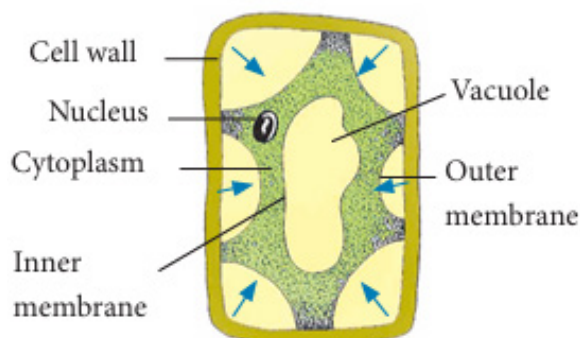


Fig 3.6: A plasmolysed plant cell

Plants that are deprived of water for several days have their leaf cells plasmolysed. The overall effect is a drooping plant which is said to be **wilting**. If this persists for long, the plant cells become **flaccid**. The plant then dies. However, if the plant is well watered before the cells become flaccid, its turgidity is restored.

Note:

Turgidity in plant cells is caused by **osmotic flow** of water from an area of low solute concentration outside the cell into the cell vacuole. The cell vacuole has a higher solute concentration. Healthy plant cells are turgid. Plants rely on turgidity to maintain rigidity and stand upright.

Application activity 3.3



1. What is osmosis?
2. Through what type of membrane does osmosis occur?
3. In which direction does water move during osmosis?
4. Name one living tissue that can be used to demonstrate osmosis

3.4. Role of osmosis in plants and animal cells

Learning activity 3.4



Discussion of the role of osmosis in plants

1. Discuss the following with a classmate.
 - a) How are plants supported by turgor pressure?
 - b) What is the importance of osmosis in plant and animal cells?
2. Share your findings with other members of your class.

(1) Uptake of water by roots

Plants absorb water from the soil by osmosis. The cell sap in root hair cells usually has a higher solute concentration than water in the soil. Therefore, root hair cells are able to take up water from the soil by osmosis.

(2) Movement of water from cell to cell in tissues

When a cell takes up water by osmosis, its solute concentration becomes lower (it becomes more dilute) compared to the cell adjacent to it. As a result, water moves from it to the adjacent cell whose solute concentration is higher (more concentrated). In this way, water moves from the root hair cells to the inner cells in the root.

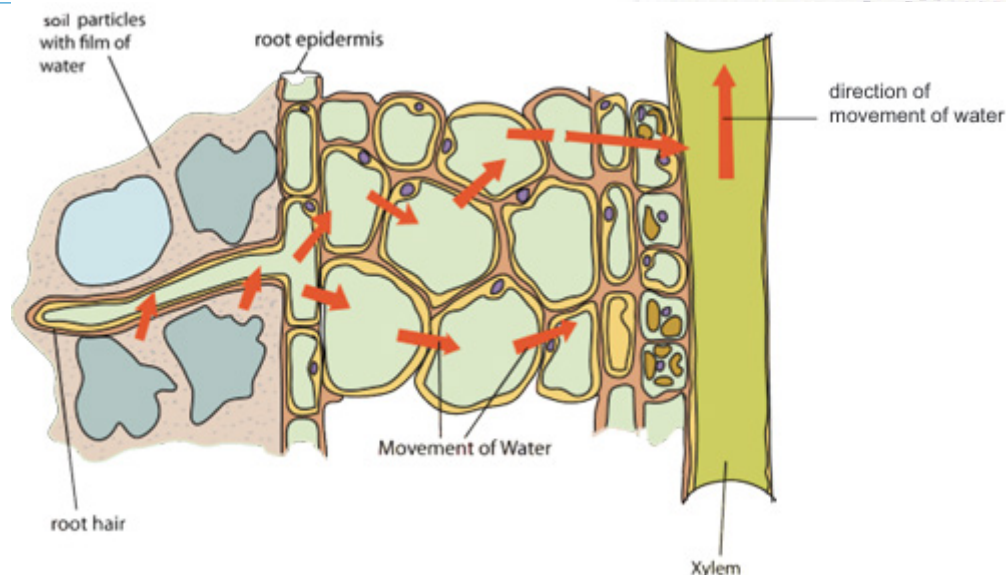


Fig 3.7: Movement of water from cell to cell in tissues

(3) Opening and closing of stomata

When guard cells gain water by osmosis from the surrounding cells, they become turgid. This changes their shape, making the stomata to open. As a result:

- Gaseous exchange takes place through the open stomata.
- Water vapour also exits through the open stomata.

When guard cells lose water through osmosis to surrounding cells, their size and shape changes. This makes the stomata to close.

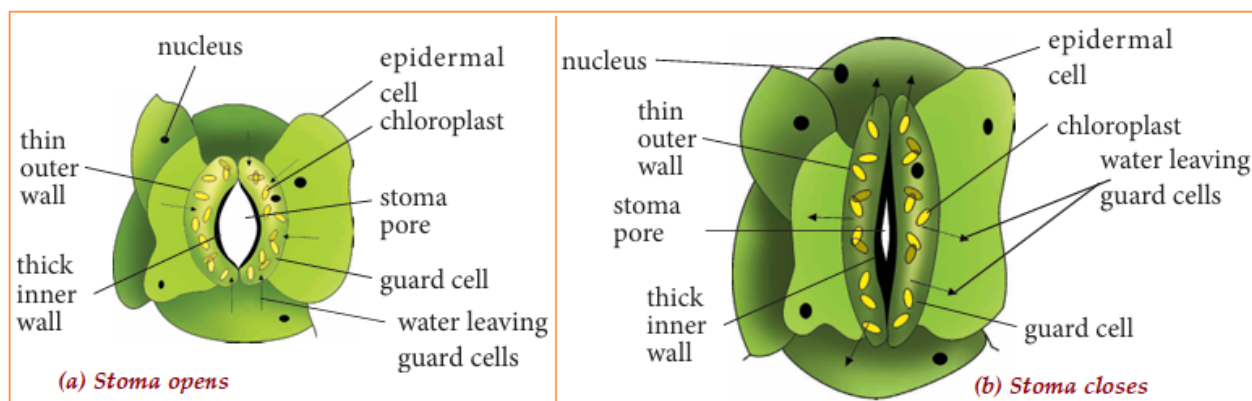


Fig 3. 8 Structure of open (a) stoma and closed stoma (b)

(4) Feeding in insectivorous plants

Insectivorous plants such as Pitcher plant prey on insects. They trap insects when there is a sudden change in their turgor pressure when disturbed by the insect.



Fig3. 9. Pitcher plant

Application activity 3.4



1. How does osmosis help plant roots absorb water from the soil?
2. Why is osmosis important in maintaining turgidity of plant cells?
3. How does osmosis help in opening and closing of stomata?
4. What happens to a plant when osmosis does not occur properly?

3.5. Active transport

Learning activity 3.5



1. Observe the charts and animations carefully.
 - Can you tell what is taking place in the pictures?
 - Which part of the body is the process taking place?
 - What is the role of proteins in active transport?
2. Record what you see in the pictures.
3. Discuss with your partner the importance of the process involved.

Study questions

- (a) Explain in your terms what is meant by active transport
- (b) What is the role of active transport in organisms?
- (c) What is the role of the membrane proteins in active transport?

Remember that passive transport does not require energy and substances move in gradient of concentration. Now active transport always requires energy in the form of ATP (Adenosine Triphosphate) and substances move against the gradient of concentration i.e. from lower concentration to high concentration.

(i) Meaning of active transport

Active transport is the movement of particles through the cell membrane from a region of low concentration to a region of high concentration using energy. This process requires the expenditure of energy since particles move against a concentration gradient.

In active transport, the cell must use its own energy to move the molecules against a concentration gradient. Such energy is supplied by the cell's **mitochondria**. Therefore, active transport takes place only in living cells.

(ii) Role of active transport in living organisms

1. Through active transport, root hair cells in plants absorb mineral salts and ions from the surrounding soil. This takes place even when the concentration of these mineral salts is already higher in the cells than in the soil.
2. The cells lining the human small intestines continue to absorb food molecules by active transport even when the concentration of these molecules is higher in the cells than in the intestinal lumen.
3. Nerve cells need sodium ions and potassium ions to function. The concentration of sodium ions outside a nerve cell is higher than the concentration on the inside. The concentration of potassium ions on the outside is lower than on the inside. The nerve cells maintain these concentrations in this way by active transport.
4. When urine is first formed in the kidney, it contains useful substances like glucose in addition to the waste substances. The useful substances are reabsorbed into the blood by active transport.
5. Active transport is involved in accumulation of substances in the body of some organisms to offset osmotic imbalances in arid and saline environments.

(iii) The role of proteins in active transport

The movement of substances across a membrane by active transport requires carrier protein molecules. They pick up molecules of a substance from one side of the cell membrane and transport them across. Carrier protein molecules have binding sites for the molecules. The molecules to be transported bind to the carrier protein using energy. The carrier protein changes shape and the particles are transported to the other side of the membrane; and they diffuse away from the carrier protein.

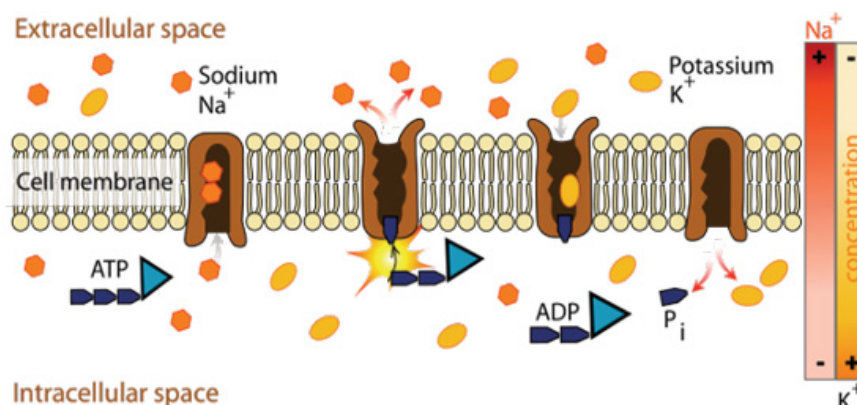


Fig 3.10: How a carrier protein transports a molecule

Application activity 3.5



1. Absorption of mineral salts from the soil to the root hair requires energy. Which of these describe the process used?
A. Diffusion B. Osmosis C. Active transport D. Respiration
2. Which one is a function of the proteins in the cell membrane?
A. Communication B. Cellular respiration C. Cell transport D. Cell repair

3.6. Factors that affect active transport

Learning activity 3.6



Demonstration of the factors affecting active transport

Learning Resources:

- Diagram/chart of a cell membrane showing protein pumps
- Small stones/beans
- Small cups or containers
- Cards labelled: *Energy (ATP)*, *No Energy*, *High Temperature*, *Low Temperature*
- Chalkboard / flip chart

Procedure

- Students place many beans in one cup (high concentration) and few beans in another (low concentration).
- Students move beans from low concentration to high concentration only when holding the Energy (ATP) card.
- Teacher removes the energy card; students attempt to move beans and stop.
- Students repeat the activity using high temperature and low temperature cards, observing speed differences.
- Students block the protein pump on the diagram and observe that movement stops.

Study questions

- Why did movement stop when energy was removed?
- What role do protein pumps play?
- How does temperature affect active transport?

Any factor that affects energy production also affects active transport.

a. Concentration of oxygen

At higher oxygen concentration, the cells produce more energy. This leads to rapid active transport by the protein carriers.

b. Concentration of glucose

The amount of energy produced by a cell depends on the amount of glucose available. When more glucose is available, the cells produce more energy. This leads to rapid active transport.

c. Temperature

High temperatures can destroy (denature) the protein carriers in the cell membrane. This stops active transport. Active transport works best under normal temperature, which in human beings is 37°C.

d. Enzyme inhibitors

An enzyme is a substance, which speeds up reactions taking place in the cell. Enzyme inhibitors are chemical substances, which 'poison' or make an enzyme inactive. If enzymes involved in the release of energy during respiration are affected, there will be lack of energy. That will prevent active transport from taking place.

e. pH

Abnormal changes in pH in the environment of the cell can alter the structure of the protein carriers. This will reduce their ability to transport molecules across the cell membrane.

Application activity 3.6



1. State any four factors that affect the active transport.
2. In their experiment two students from S2 discovered that roots of Oxalis absorb mineral salts by active transport when the plant is still alive. Once dead the roots lose mineral salts. Explain.

3.7. Endocytosis and exocytosis.

Learning activity 3.7



Conducting research and discussion on the process of endocytosis and exocytosis
Search and illustrate the process of endocytosis and exocytosis.

1. Discuss the process involved.
2. Share your findings with the rest of the class.

The movement of macromolecules such as proteins or polysaccharides into or out of the cell is called **bulk transport**. There are two types of bulk transport, **exocytosis** and **endocytosis**, and both require the expenditure of energy.

3.7.1. Endocytosis

This is the in folding of the cell surface membrane to form a vesicle or vacuole. A portion of the membrane wraps around the particle and fuses. Thereafter, the particle is surrounded by the plasma membrane. The portion of the membrane containing the particle then pinches off leaving the plasma. Endocytosis involves the bulk uptake of materials through the plasma membrane. It takes place in two ways: phagocytosis and pinocytosis.

(i) Phagocytosis

Phagocytosis is a type of endocytosis in which a cell engulfs large solid particles, such as food particles, bacteria, or dead cells, by surrounding them with its cell membrane. The engulfed material is enclosed in a vesicle called a **phagosome** inside the cell. Phagocytosis is an active process that requires energy (ATP).

Below is diagrammatic representation of phagocytosis

Food particle/Bacterium → Pseudopodia surround particle → Phagosome formed → Fusion with lysosome → Digestion

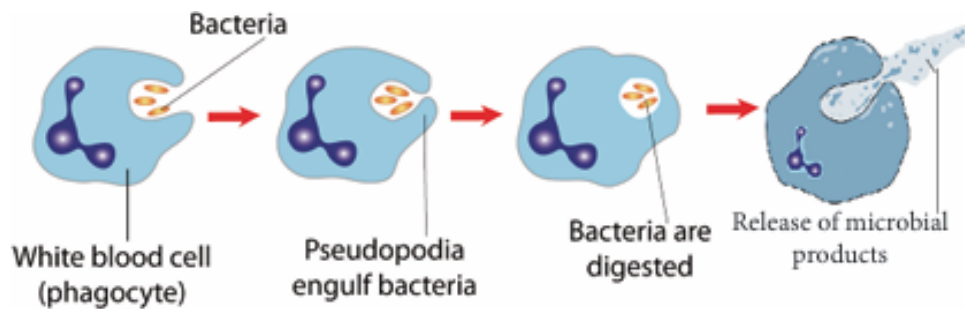


Fig 3.11: Blood cells engulfing bacteria by phagocytosis

Process of Phagocytosis

1. The cell detects a solid particle outside its membrane.
2. The cell membrane extends around the particle, forming pseudopodia (temporary projections).
3. The pseudopodia surround and engulf the particle.
4. A vesicle called a **phagosome** forms inside the cell.
5. The phagosome fuses with a lysosome containing digestive enzymes.
6. The particle is digested and useful substances are absorbed by the cell.
7. Undigested waste may be expelled from the cell by exocytosis.

Examples of Phagocytosis

- **Amoeba** engulfing food particles.
- **White blood cells (phagocytes)** engulf bacteria and other pathogens.
- Macrophages removing dead cells and cellular debris.

Characteristics of Phagocytosis

- It is a form of endocytosis.
- It involves the uptake of large solid particles.
- It requires energy (ATP).
- It involves the formation of pseudopodia and vesicles.
- It helps in feeding and protection against infection.

Importance of Phagocytosis

Phagocytosis:

- Enables some unicellular organisms to obtain food.
- Helps the body defend itself against disease-causing microorganisms.
- Removes dead or damaged cells from tissues.
- Contributes to the immune response.

(ii) Pinocytosis

Pinocytosis means “**cell drinking**.” It is a type of endocytosis in which a cell takes in extracellular fluid and the dissolved substances it contains. Unlike phagocytosis, which involves the uptake of large solid particles, pinocytosis involves the uptake of materials in liquid form.

During pinocytosis, the cell membrane folds inward to form small vesicles that enclose droplets of extracellular fluid and dissolved molecules. These vesicles are generally smaller than those formed during phagocytosis. Pinocytosis is an active process that requires energy (ATP) and enables cells to absorb nutrients, ions, and other dissolved substances from their surroundings.

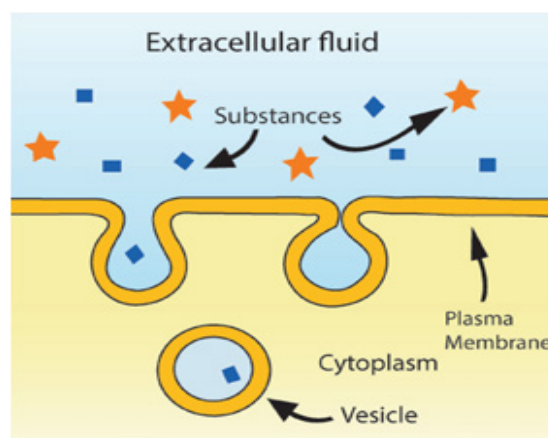


Fig 3.12: Pinocytosis

Examples of pinocytosis include:

- Absorption of nutrients by intestinal cells.
- Uptake of dissolved substances by kidney tubule cells.
- Intake of extracellular fluids by many animal cells.

Key characteristics of pinocytosis:

- It is a type of endocytosis.
- It involves the uptake of liquids and dissolved substances.
- Small vesicles are formed from the cell membrane.
- It requires energy (ATP).
- It occurs in many types of animal cells

3.7.2. Exocytosis

Exocytosis is the process by which substances are transported out of a cell. It occurs when membrane-bound vesicles move to the cell membrane, fuse with it, and release their contents outside the cell. Exocytosis is an **active transport process**, meaning it requires energy in the form of ATP.

Process of Exocytosis

1. Materials to be secreted are packaged into vehicles inside the cell.
2. The vesicles move towards the plasma membrane.
3. The vesicle membrane fuses with the plasma membrane.
4. The contents of the vesicle are released outside the cell.
5. The vesicle membrane becomes part of the plasma membrane.

Illustration of Exocytosis

Vesicle → Fusion with cell membrane → Release of substances outside the cell

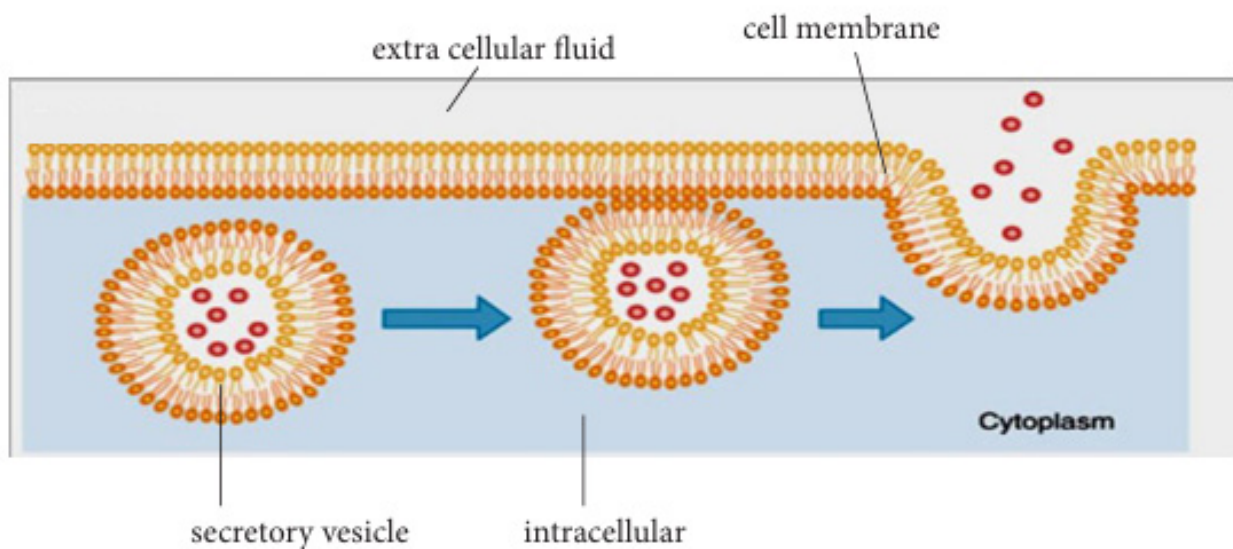


Fig 3.13: Exocytosis

Examples of Exocytosis

- Release of insulin by pancreatic cells.
- Secretion of digestive enzymes by gland cells.
- Release of neurotransmitters at synapses.
- Removal of cellular waste products.

Characteristics of Exocytosis

- It is an active process requiring ATP.
- It involves vesicles.
- Substances move from inside the cell to the outside.
- The plasma membrane participates in the process by fusing with vesicles.

Importance of Exocytosis

Exocytosis is important because it:

- Removes waste products from cells.
- Secretes hormones, enzymes, and other substances.

- Releases neurotransmitters from nerve cells.
- Helps in cell communication.
- Contributes to the growth and repair of the cell membrane.

Application activity 3.7



1. State one difference and one similarity between endocytosis and active transport.
2. A white blood cell ingests solid particles by forming vesicles. This describes which process?
3. Given these characteristics:
 - (i) Requires carrier proteins
 - (ii) Requires energy
 - (iii) Requires membrane channels
 - (iv) Requires vesicles
4. Choose the characteristics that define exocytosis.

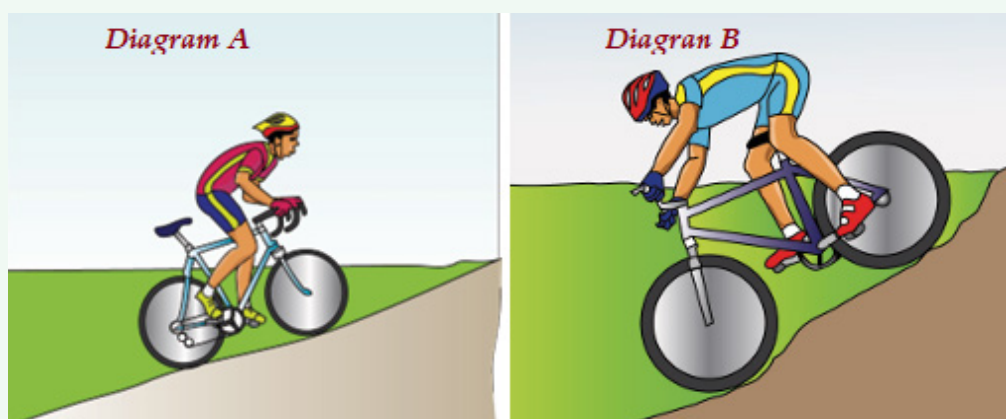
3.8. Difference between passive and active transport

Learning activity 3.8



Illustration of active and passive transports

Study the picture below and then answer to the questions that follow.



- a) What is going on in the pictures?
- b) Which is easier, climbing uphill or going downhill?
- c) Why is that the case?
- d) How do actions in the pictures relate to passive and active transport?

Based on knowledge gained about passive and active transport give four similarities and four differences.

Similarities between Passive and Active Transport

1. Both are processes by which substances move across the cell membrane.
2. Both help the cell take in useful substances and remove waste.
3. Both involve movement of molecules such as ions, gases, or nutrients.
4. Both are important for maintaining a stable internal environment in cells.

Differences between Passive and Active Transport

The table below shows the difference between passive and active transport.

Table 3.3: Difference between passive and active transport

Passive transport	Active transport
Transport occurs from a high concentration of molecules and ions to low concentration, in order to maintain equilibrium in the cells.	Transport occurs from a low concentration of solute to high concentration of solute.
Does not require cellular energy.	Requires cellular energy.
Involves osmosis and diffusion processes.	Involved endocytosis and exocytosis processes.
Transports anything- soluble water, oxygen and carbon dioxide.	Transports proteins, ions, large cells and complex sugars.

Application activity 3.8



1. You are provided with the following processes: Pinocytosis, simple diffusion, Plasmolysis, exocytosis, turgidity and endocytosis. Divide them into active and passive transport.
2. Compare passive and active transport.

3.9 End of unit Assessment



3.9.1. End of unit 3 Assessment

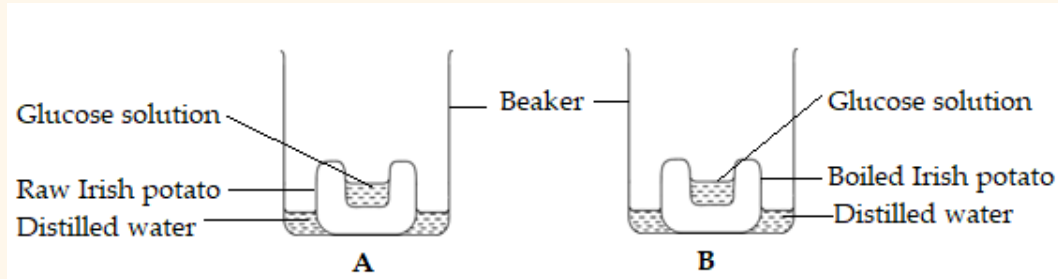
A. Knowledge and understanding

1. Why are diffusion and osmosis referred to as passive processes?
2. Complete the following statements:
 - (a) _____ is the movement of the molecules of gas, liquid or solute from a region of their _____ to a region of their _____ and which results in _____ distribution.
 - (b) When two solutions are separated from one another by a special selectively _____ or semi _____ membrane, there is net flow of water molecules from a solution in which water molecules are _____ into a solution in which water molecules are _____ through the special membrane. This process is known as _____. It is described as a special type of diffusion in which molecules move from a _____ solution into a _____ solution.
3. Which of the statements below about osmosis is true?
 - a. The greater the solute concentration, the smaller the osmotic pressure.
 - b. Osmosis occurs due to hydrostatic pressure outside the cell
 - c. Osmosis moves water molecules from a greater solute solution to a lesser solute concentration.
 - d. The higher the osmotic pressure of a solution, the greater the tendency of water to move into the solution.
4. Active transport is affected by the following factors except?
 - a. Oxygen concentration
 - b. glucose concentration
 - c. thickness of the membrane
 - d. enzyme inhibitors

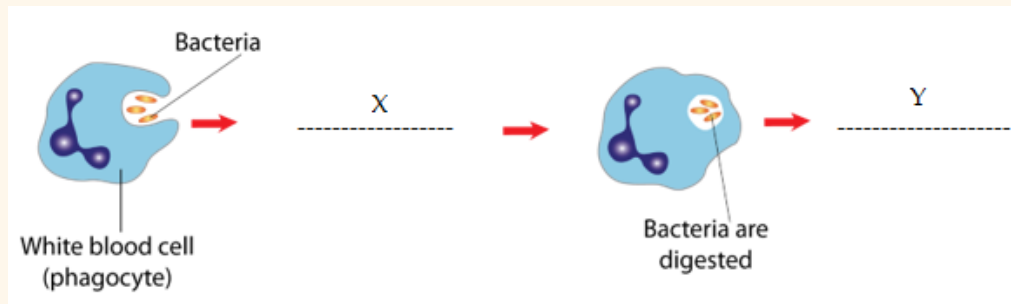
B. Skills

5. (a) Draw a plant cell as it would appear when placed in a hypertonic solution for several minutes.
 - (b) What name is given to such a cell?
 - (c) Explain the appearance of this cell.
 - (d) Compare (c) above with what would happen with mammalian red blood cells in the same solution.

6. Two potatoes were peeled. One of them was boiled. A cavity was made in each of the potatoes and the experiment was set up as shown below. The experiment was left undisturbed for 24 hours.



- (a) Draw a diagram to illustrate the result in each figure.
(b) Give reasons to support your answer in (a) above.
7. Complete the process of phagocytosis, by drawing the missing diagram



C. Attitudes and values

8. What would happen if osmosis was not available in plants?
9. Concentration of oxygen affects active transport. Explain
10. Why does salting fresh fish make it dry faster than non-salted fish?

UNIT 4

IDENTIFICATION OF FOOD COMPONENTS



Key Unit Competence:

Identify the composition of food nutrients in various food samples through chemical testing experiments.

Learning objectives

By the end of this unit you should be able to:

Knowledge and understanding

1

- State the chemical reagents used in the identification of each class of foods.
- List the suitable apparatuses required for the test of a given food sample
- Recall the colours observed from the chemical test of a given food sample.

Skills

2

- Design a tabular reporting procedure, showing observation and deduction for given food sample tests.
- Carry out tests to identify food nutrients in each sample.
- Carry out an investigation to determine the composition of an unknown food substance.
- Observe changes in colour during food tests and relate them to the chemical reactions.

Attitudes and values

3

- Appreciate the importance of different classes of nutrients in one food sample.
- Show perseverance when observing changes in colour during practical lessons.
- Acknowledge the need to carry out practical activities with less supervision to develop independence and skills in handling apparatus.

4.0. Introduction

Food provides the nutrients that the body needs for energy, growth, repair, and protection against diseases. In Rwanda, common foods such as maize, rice, potatoes, cassava, beans, milk, fruits, and vegetables contain different food components that support healthy living.

Identifying these food components helps students understand the nutritional value of foods and the importance of making healthy dietary choices.

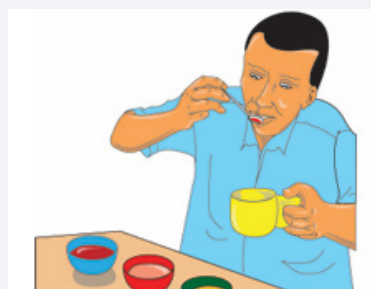
In this unit, students explore the major food components, their functions in the body, and simple tests used to identify them in different food samples. This knowledge will promote awareness of balanced diets and good nutrition for maintaining health and wellbeing.

Key Concepts: food components, nutrients, carbohydrates, proteins, fats and oils, vitamins, minerals, water, balanced diet, nutrition, food tests, energy, growth, health.

Introductory activity



Have you ever visited a tea coffee-processing factory? The picture below shows what happens during coffee or tea grading. What are the good qualities of coffee? How the person on the picture is determining the quality of good coffee?



Coffee or tea is tasted to determine its quality during processing.

The fragrance of the brewed tea or coffee, the deep flavours when drinking and the pleasant after-taste are used to tell the quality. Based on this, do you think it is possible to know the components of the various foods that we eat by tasting? Why? What can we use to do this?

4.1. Components of food substances

Learning activity 4.1



Exploring your daily diet

- Share with classmates the list of the foods you usually eat at home, for breakfast, lunchtime and evening supper.
- Make another list of the foods you eat at school and compare it with the foods eaten by students in a boarding school.
- From both lists let everyone choose his or her favorite food.
- Explain why you prefer such food.
- Read the notes below and there after deduce which food stuff is more important.

A **balanced diet** means eating the various food types in the correct proportions to keep your body healthy. There are seven main components of a balanced diet. These are carbohydrates, lipids, proteins, vitamins, minerals, roughage and water. If your body lacks any one of these different component types for any length of time, you could become weak and ill.

The foods that we eat contain different types of nutrients. These nutrients serve different functions in the body. It is, therefore, essential that we know the components of the food that we eat in order to live healthy lives. Drinks and foods that are manufactured in factories need to be tested as well.

This is done to confirm the type and percentage of nutrients they contain. That enables the consumer to be informed on the product contents. Food contains mainly two classes of nutrients: inorganic and organic nutrients. The **inorganic nutrients** include mineral salts like calcium, phosphorous and others like water. The **organic nutrients** include proteins, carbohydrates, lipids and vitamins.

The process by which foods and drinks are taken to the laboratory to confirm the type of food components that they contain is called food testing. **Food testing** is done to confirm the presence of starch, reducing sugars, proteins, fats and vitamin C in a food sample.

Application activity 4.1



Fill-in-the-blank

A balanced diet means eating different food types in the correct proportions to keep the body _____. (1). Foods contain different _____ (2) that perform various functions in the body. Manufactured foods and drinks are tested to confirm the type and _____ (3) of nutrients they contain. Nutrients are grouped into two main classes: inorganic and _____ (4) nutrients. The process of confirming the presence of substances such as starch, proteins and vitamin C in food samples is called _____ (5).

4.2. Testing for starch and reducing sugars

Learning activity 4.2



Test for starch and reducing sugars

(a) Materials

- Test tubes and test-tube rack
- Dropper/pipette
- Beaker with hot water (water bath)
- Iodine solution
- Benedict's solution
- Food samples: Starch solution (or cassava, rice, maize juice) and Glucose solution (Glucose biscuit)
- Heat source (teacher supervised)

(b) Procedure

1. Work in your assigned group and label two test tubes: Starch and Glucose.
2. Put a small amount of starch solution into the first test tube.
3. Add a few drops of iodine solution to the starch solution.
4. Observe and record the colour change.
5. Put a small amount of glucose solution into the second test tube.
6. Add an equal amount of Benedict's solution to the glucose solution.
7. Place the test tube in a hot water bath for 2–3 minutes.
8. Observe and record the colour change.
9. Record all observations in a table with columns: Food sample, Reagent used, observation, and Food *component present*.
10. Clean the apparatus and wash your hands.

▲ Safety rule: Do not taste any food sample and handle hot water carefully.

Study Questions

1. Which reagent is used to test for starch?
2. What colour change shows the presence of starch?
3. Which reagent is used to test for reducing sugars?
4. Why is it heating needed when testing for reducing sugars?
5. Which of the two food components mainly provides energy to the body?

4.2.1. Starch

Starch is a carbohydrate found mainly in plants. It occurs in foods such as maize, rice, cassava, potatoes, bananas, and bread, which are commonly eaten in Rwanda. In plants, starch is used as a storage food, and in humans, it is broken down into glucose to provide energy for body activities.

Starch is a **polysaccharide** comprising of glucose monomers linked together. It is insoluble in cold water and does not have a sweet taste.

Starch can be identified using a simple food test by adding **iodine solution** to the food sample. A **blue-black** colour shows the presence of starch. Examples of starchy foods include maize flour, rice, potatoes, and cassava.



Fig4.1. Test for starch.

4.2.2. Reducing Sugars

Reducing sugars are simple carbohydrates found in fruits, honey, milk, and germinating seeds. Their main biological role is to provide quick energy for living cells during respiration. Reducing sugars are tested by adding blue **Benedict's solution** to the food sample and heating it or putting it in a water bath; a colour change from blue to green, yellow, orange, or brick-red indicates their presence. They are called **reducing sugars** because they can reduce the ions of **copper (II)** in the Benedict's solution into **Copper (I) oxide** due to the presence of a free sugar group. Examples of reducing sugars include glucose, fructose, maltose, and lactose.

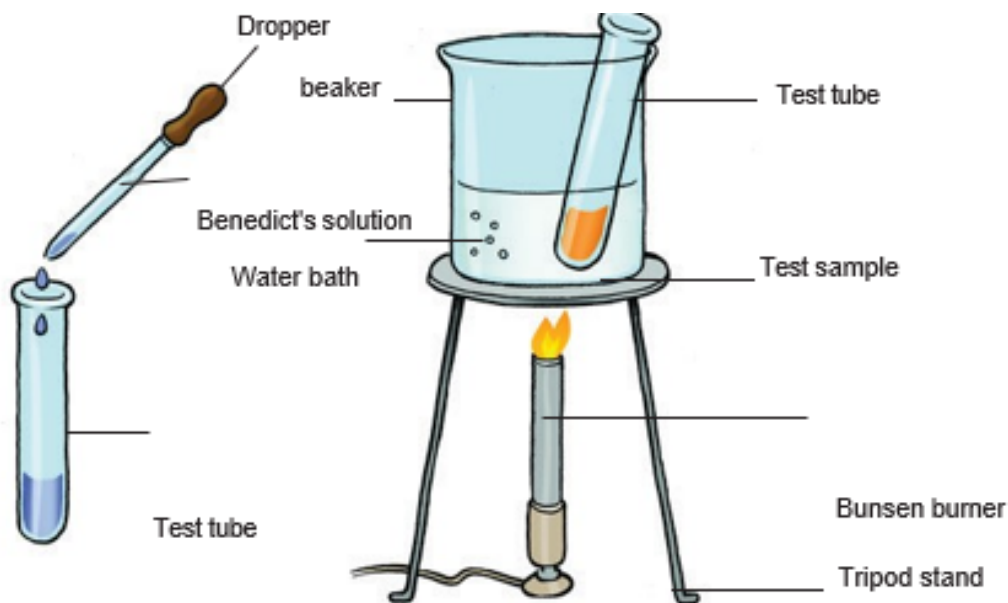


Fig 4.2: Testing for reducing sugars

Application activity 4.2



1. What is starch, and why is it important in the diet of humans?
2. Describe how you would test a food sample for the presence of starch in the laboratory.
3. What are reducing sugars and give two examples of foods that contain them.
4. Explain how Benedict's test shows the presence of reducing sugars and what colour change means.

4.3. Tests for proteins, lipids and vitamin c

Learning activity 4.3



Identification of protein, lipids and vitamin C

Materials

- Test tubes and test-tube rack
- Droppers / pipettes
- Beaker and hot water bath
- Food samples (e.g. egg white or milk, cooking oil or groundnuts, orange or lemon juice)
- Sodium hydroxide solution
- Copper (II) sulfate solution
- Ethanol
- Distilled water
- DCPIP solution

Procedure

(a) Test for Proteins (Biuret test)

- Put a small amount of the food sample into a test tube.
- Add sodium hydroxide solution and mix well.
- Add a few drops of copper (II) sulfate solution.
- Observe and record the colour change.

(b) Test for Lipids (Emulsion test)

- Put the food sample into a clean test tube.
- Add ethanol and shake well.
- Add distilled water and shake gently.
- Observe and record the appearance of the mixture.

(c) Test for Vitamin C (DCPIP test)

- Put DCPIP solution into a test tube.
- Add the food extract drop by drop.
- Shake gently and observe the colour change.

Study Questions

1. Which colour change shows the presence of proteins?
2. What observation confirms the presence of lipids?
3. Which nutrient causes DCPIP to change from blue to colourless?
4. Why should clean test tubes be used for each test?
5. Name one food source rich in proteins, lipids and vitamin C.

Food contains different nutrients that are important for good health. Among these nutrients are proteins, lipids and vitamins. Scientists use simple food tests to identify them.

Proteins

Proteins are found in foods such as beans, peas, eggs, meat, fish and milk. They are needed for the growth, the repair of body tissues, and for making enzymes and hormones. They are tested using **Biuret test**. A small amount of food is put in a test tube, then **Sodium hydroxide** solution is added, followed by a few drops of **copper sulfate** solution. A **purple** or **violet** colour shows that protein is present.

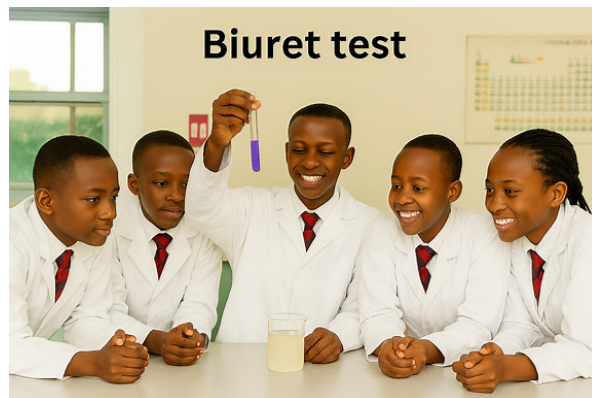


Fig.4.3. *Biuret test of protein*

Lipids (Fats and Oils)

Lipids are found in cooking oil, butter, margarine, groundnuts, avocado and fatty meat. Lipids provide energy, help to keep the body warm, and protect internal organs. Lipids are tested using **Emulsion test**. The food sample is shaken with ethanol (alcohol) and then water is added. A **milky white emulsion** shows that lipids are present.



Fig 4.4. *Emulsion test for lipids*

Vitamin C

Vitamin C is found in oranges, lemons, guava, tomatoes and green vegetables. Vitamin C helps in wound healing, keeps gums healthy, and helps the body fight diseases. Vitamin C is tested using **DCPIP** (Dichloro-Phenol -Indophenol). The food sample is added drop by drop to blue DCPIP solution. If vitamin C is present, the **blue colour** disappears and becomes **colourless**.



Fig. 4.6. *Steps in testing for vitamin C*

Table 4.1. Summary Table of Food Tests

Food Substance	Test Used	Reagent(s)	Procedure (Simple)	Positive Result	Conclusion
Starch	Iodine test	Iodine solution	Add a few drops of iodine solution to the food sample.	Blueblack colour appears	Starch is present
Reducing sugar (e.g. glucose)	Benedict's test	Benedict's solution	Add Benedict's solution to the food sample and heat in a hot water bath for 3–5 minutes.	Green, yellow, orange or brickred precipitate	Reducing sugar is present
Protein	Biuret test	Sodium hydroxide + copper (II) sulfate (Biuret reagent)	Add sodium hydroxide to the food sample, then add a few drops of copper (II) sulfate.	Purple / violet colour	Protein is present
Lipid (fat/oil)	Emulsion test	Ethanol + water	Shake the food sample with ethanol, then add water.	White milky emulsion	Lipid is present
Vitamin C	DCPIP test	DCPIP solution	Add food extract drop by drop to DCPIP solution.	Blue DCPIP turns colourless	Vitamin C is present

Application activity 4.3

Complete the table below by filling in the missing parts.

Food test (nutrient)	Procedure	Observation	Deduction
Proteins (Biuret test)	Add sodium hydroxide to the food sample, then add a few drops of _____	The mixture turns _____ in colour	Proteins are _____
Lipids (Emulsion test)	Shake the food sample with _____, then add water	A _____ emulsion is formed	Lipids are _____
Vitamin C (DCPIP test)	Add the food extract drop by drop to _____ solution	The blue colour becomes _____	Vitamin C is _____

4.4. End of unit assessment



4.4.1. Assessment questions

A. Knowledge and Understanding

1. What is meant by a food test?
2. Name the reagent used to test for starch and state the positive result.
3. Which food substances are tested using Benedict's solution, Biuret test, emulsion test, and DCPIP?

B. Skills

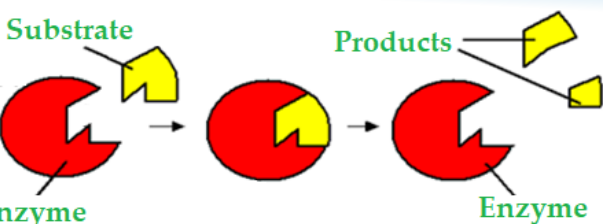
4. Describe the steps you would follow to test a food sample for reducing sugars.
5. Explain how you would test proteins in a food sample and record the result.
6. A student adds ethanol and water to a food sample and observes a white milky mixture.
 - (a) Which nutrient is being tested?
 - (b) What does the result show?

C. Attitudes and Values

7. Why is it important to handle chemicals and food samples carefully during food testing experiments?
8. Explain how knowing the results of food tests can help people make healthy food choices.

UNIT 5

ENZYMES



Key Unit Competence:

Investigate the role of enzymes as catalysts of biological processes

Learning objectives

By the end of this unit you should be able to:

1 Knowledge and understanding

- Explain the term catalyst and enzyme.
- Outline the characteristics of enzymes.
- Describe the enzyme mode of action through the lock and key hypothesis scheme.
- Explain the factors affecting enzyme activity limited to temperature and pH.
- Describe the importance of enzymes in everyday life. Recognize and outline the key traits of other animal phyla

2 Skills

- Conduct an experiment to show the effect of temperature and pH on enzyme activity.
- Interpret graphs of the rate of enzyme activity on temperature and pH.
- Illustrate enzyme activity as Lock and key hypothesis model.

3 Attitudes and values

- Appreciate the importance of enzymes in speeding up reactions to sustain life.
- Appreciate the specificity of enzymes.

5.0. Introduction

Enzymes are special proteins that act as biological catalysts, speeding up metabolic reactions without being used up in the process. They play an essential role in life by helping organisms digest food, release energy, build new cells and tissues, and carry out many other vital functions.

For example, the enzyme amylase found in saliva breaks starch into simpler sugars that can be absorbed and used by the body. This unit introduces students to nature and functions of enzymes, how they work, and the factors that affect their activity. Understanding enzymes helps explain how living organisms efficiently maintain their life processes.

Key Concepts: enzyme, biological catalyst, protein, substrate, product, active site, enzyme specificity, metabolism, digestion, enzyme activity, temperature, pH, denaturation.

Introductory activity



Look at the photographs below. Can you identify what is in the photographs? What is the relationship between them?



5.1. Enzymes as catalyst

Learning activity 5.1



(a) Demonstration of the role of amylase as catalyst

Materials:

- Test tubes
- Starch solution
- Commercial amylase solution
- Benedict's solution
- Droppers/pipettes
- Stopwatch or timer
- Water bath (optional)

Procedure:

1. Label one test tube A and the other B.
2. Pour the same amount of starch solution into both test tubes.

3. Add saliva to test tube A only. Do not add saliva to test tube B.
4. Place both test tubes in warm water for about 10–15 minutes.
5. After waiting, add 1ml of Benedict's solution to each test tube and boil them in water bathe.
6. Observe and record the color change in both test tubes.
7. Compare the results of both test tubes.

(b) Investigating the presence of enzymes in living tissues

Requirements:

- Samples of liver, potato or yeast
- 8 cm³ volume hydrogen peroxide
- Forceps, pestle and mortar
- Wooden splint
- Matchbox
- Fine sand
- Test tubes
- Labels

Procedure

1. Label 4 test tubes as A, B, C and D.
2. Measure 2cm³ of hydrogen peroxide solution and pour into each of the labelled test tubes.
3. Into test tube A, do not add anything.

Note any changes or observations.

4. Into test tube B, add some sand. Shake the test tube and note down any changes.
5. Into test tube C add some pieces of potato that have been ground into a paste (using a pestle and mortar).
 - Shake and observe what happens.
 - Test for oxygen with a glowing splint at the mouth of the test tube.
6. Repeat procedure (5) in test tube D using some liver which has been cut up then, pounded in a mortar using a pestle.
7. Record your observations in a table like the one below.

Test tube	Contents	Observations	Conclusion
A	Hydrogen peroxide only		
B	Hydrogen peroxide + sand		
C	Hydrogen peroxide + potato tissue		
D	Hydrogen peroxide + liver		

Try washing off the stain in a piece of cloth with water only, then using the detergent. Where was it easy to remove the stain? Detergents which are used to remove stains, act as enzymes.

During the washing process, water and dirt are quickly combined when the detergent is available to form a new product. This new product is easily removed from the cloth as we wash. On the other hand, water without detergent or soap can take a long time to remove the stains.

Most metabolic reactions in the body are catalysed by enzymes; hence increasing efficiency of these processes in living organisms. It is important therefore to understand the nature, the properties and the function of enzymes.

Enzymes are special proteins found in all living cells that act as catalysts.

A **catalyst** is a substance that speeds up a chemical reaction without being used up.

Enzymes help important reactions in the body happening very fast. Without enzymes, digestion, breathing, and growth would be too slow.

Each enzyme works on a specific substance called a **substrate**. For example, the enzyme amylase breaks starch into simple sugars.

Some enzymes are used within the cells that produce them. These are called **intracellular enzymes**, for example, respiratory enzymes and catalase enzymes in the liver cells. Other enzymes are transported from the cells that produce them to be used in other cells. These are called extracellular enzymes. Some examples of extracellular enzymes are those secreted into the alimentary canal to help in digestion. They are also different types of enzymes based on the foods they act on, for example:

When scientists name enzymes, they follow a simple rule called **nomenclature**. Most enzyme names end with the letters "**-ase**".

The name usually shows the substance the enzyme acts on. For example, **amylase** acts on starch called amylose, **Lipase** breaks down fats (lipids), Protease breaks down proteins, carbohydrases break down carbohydrates.

Some enzymes are named after what they do, **like catalase**, which breaks down hydrogen peroxide.

This naming system helps us knowing an enzyme's job easily. It also helps scientists to communicate clearly.

So, enzyme names tell us both *what they are* and *what they do*.

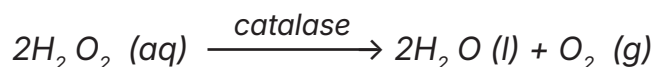
However, some enzymes have **special or specific names**. These names were given before the naming rules were made. An example is pepsin; a protease found in the stomach.

Living tissues contain an enzyme called **catalase** which breaks down hydrogen peroxide to water and oxygen. Catalase is found in many cells, especially in liver cells. The liver produces harmful substances during metabolism. One of these substances is hydrogen peroxide.

Catalase breaks down hydrogen peroxide very quickly. This reaction produces water and oxygen.

Water is harmless to the body. The oxygen is released as bubbles.

Hydrogen peroxide $\xrightarrow{\text{catalase}}$ water + oxygen



The gas produced relights a glowing splint as seen in procedure 5 and 6. No gas is produced in Test tube 1 and 2 which do not contain living tissue.

By breaking down harmful chemicals, catalase helps in **detoxification**. This protects liver cells and keeps the body healthy

Application activity 5.1



1. What is an enzyme? Explain your answer with an example.
2. Describe the nature of enzymes and state what they are made of.
3. Explain why enzymes are called biological catalysts.
4. Describe how enzymes are named and give one example of an enzyme with a special name.

5.2. Mode of action of enzymes

Learning activity 5.2



Investigating the fitting of substrate to enzyme active site

Requirements

- A set of 15 keys
- 10 different padlocks

Procedure

- Working in groups, each member of the group should try to find the right key for the two padlocks.

Study questions

- (a) Were you able to find out the right key?
- (b) What conclusion can you make from the activity?
- (c) What do the following correspond to regarding enzymes and substrates?

Enzymes work by using a special part of their structure called the **active site**, which is the region where chemical reactions take place.

The **active site** has a particular shape that allows only one type of substance to attach to it.

When the substrate comes into contact with the enzyme, it fits exactly into the active site.

This fitting is very specific and accurate.

The **lock and key hypothesis** is used to explain this process. According to this hypothesis, the enzyme acts like a lock. The substrate acts like a key that fits into the lock.

Only the correct substrate can fit into a particular enzyme.

After the reaction, the products are released, and the enzyme remains unchanged and ready to work again. When the substrate fits into the active site, an **enzyme-substrate complex is formed**. This reaction complex can be explained by a key and lock mechanism as illustrated below.

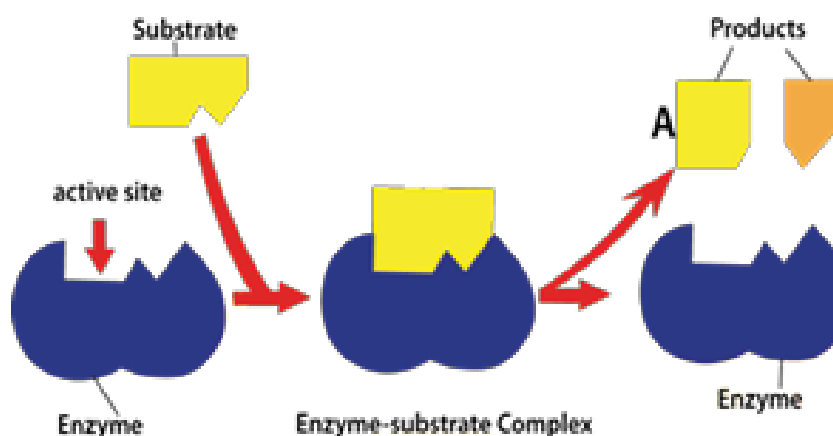


Fig 5.1: Key and lock mechanism

Application activity 5.2



Fill in the blanks

Enzymes work by using a special region called the _____ where chemical reactions take place.

The substance that an enzyme acts on is called the _____.

The _____ hypothesis explains how enzymes and substrates fit together.

In this hypothesis, the enzyme is compared to a _____.

The substrate fits into the enzyme like a _____.

5.3. Factors affecting enzyme activity

Learning activity 5.3



Investigation of the effect of pH on action of salivary amylase on starch

Requirements

- 1% amylase solution
- Starch solution
- Iodine solution
- Benedict's solution
- 0.1M Hydrochloric acid (HCl)
- Distilled water
- 0.1% sodium hydroxide (NaOH)
- Metal test tube rack
- Boiling tubes
- Labels
- Source of heat
- Water bath at 37° C
- Thermometer

Procedure

1. Take a sample of starch solution and test for the presence of starch and reducing sugars. Divide the remaining solution into three test tubes and label them as A, B and C.
2. To test tube A, add 1cm³ of dilute hydrochloric acid and shake. Add 1cm³ of the enzyme amylase and shake.
3. To test tube B, add 1cm³ of dilute sodium hydroxide solution then 1cm³ of the enzyme amylase and shake.
4. To test tube C, add 1cm³ of distilled water into the test tube then add 1cm³ of the enzyme amylase and shake.
5. Place the labelled test tubes into a water bath maintained at 37°C for 20 minutes.
6. Test the contents of the test tubes for the presence of starch and reducing sugars.
7. Record your results and conclusions in a table below

Test tube	Test for starch		Test for reducing sugars	
	Observation	Conclusion	Observation	Conclusion
A				
B				
C				

8. Discuss your findings with a teammate.
9. Share your findings with the rest of the class.
10. Account for your results in the experiment above.

Learning activity 5.3.2



Investigation of the effect of temperature on action of ptyalin

Requirements

- 1% Amylase solution
- Starch solution
- Iodine and Benedict's solution
- Metal rack
- Source of heat
- Pair of tongs
- 3 boiling tubes
- Water bath at 37° C
- Labels
- 2 beakers
- Ice cubes and cold water

Procedure

1. Place 2 cm³ each of starch solution into three different test tubes labelled A, B and C.
2. To each test tube, add 1cm³ of amylase enzyme.
3. Immerse test tube A into a beaker of cold water (preferably with ice cubes).
4. Put test tube B in a water bath maintained at 37°C.
5. Boil the contents of test tube C.
6. Test the contents of each test tube for the presence of starch and reducing sugars.
7. Record your results in a table below

Test tube	Test for starch		Test for reducing sugars	
	Observation	Conclusion	Observation	Conclusion
A				
B				
C				

Any change in the structure of an enzyme interferes with its ability to catalyse a reaction. We therefore need to investigate the factors that affect enzyme activity. They include temperature, enzyme concentration, substrate concentration and presence of inhibitor.

a. pH

Different enzymes have different pH at which they work best. Most enzymes in the human body however, work best at pH 7 (neutral pH). Some digestive enzymes have different optimum pH though. For instance, pepsin digests proteins in the stomach at a pH of 2 while trypsin digests proteins in the duodenum at a pH of 9. Extreme pH causes enzymes to denature and permanently lose their function.

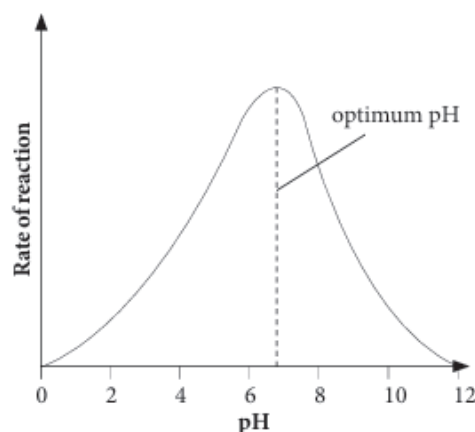


Fig. 5.2. Graph showing optimum pH (7) of an enzyme

b. Temperature

Enzyme activity is higher at high temperature since molecules gain more kinetic energy. There is therefore a higher chance of substrates bumping into enzyme's active site. The temperature at which enzymes work best is called optimum temperature. Temperatures lower than the optimum cause enzymes to be inactive and make them to work at a slow rate.

Higher temperatures than the optimum temperature destroy enzymes, because such temperatures alter or change the structure of the enzyme. They destroy the binding site of enzymes and denature them. A denatured enzyme cannot function. Most enzymes begin to denature at 40°C. However, enzymes of bacteria found in hot springs may begin to denature at temperatures higher than 80° C.

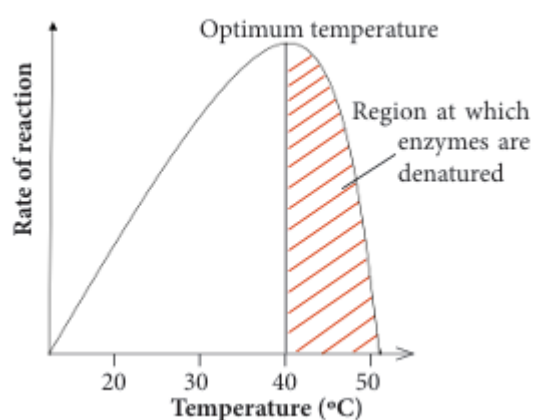


Fig. 5.3: Effect of temperature on enzyme action

Note: Other factors of that affect enzyme activity are substrate concentration, enzyme concentration, presence of inhibitors and co-factors.

Application activity 5.3



Complete each of the following statement by True or False

1. Enzymes work best at a specific pH called the optimum pH. _____
2. High temperatures always increase the rate of enzyme activity indefinitely. _____
3. Extreme pH levels can denature enzymes, making them inactive. _____
4. Enzyme activity decreases if the temperature is much lower than the optimum. _____
5. Each enzyme has a temperature range within which it functions best _____

5.4. Importance of enzymes

Learning activity 5.4



Conducting research of on the importance of Enzymes (Research and Gallery Walk)

1. Work individually. Each student will research the importance of enzymes.
2. Use textbooks or a search engine to find information about what enzymes are and why they are important in living organisms.
3. Write short notes in your notebook, focusing on at least three importance of enzymes (for example: digestion, respiration, or growth).
4. Summarize your findings on a sheet of paper using simple words and clear points.
5. Add drawings or examples to make your work easy to understand.
6. Display your work on the classroom walls or desks as instructed by the teacher.
7. Walk around the class quietly to read other students' work (gallery walk).
8. Take notes of new ideas you learn from your classmates.
9. Return to your seat and be ready to discuss what you learned

Enzymes are very important because they help chemical reactions in living organisms to happen faster without being used up, and without enzymes many life processes would be too slow to keep us alive. In our bodies, enzymes help in **digestion**, where enzymes such as amylases, protease, and lipase break down food into small molecules so that the body can absorb and use them for energy, growth, and repair.

Enzymes are also important in **respiration**, where they help release energy from food so that cells can carry out activities such as movement, thinking, and keeping the body warm. Without enzymes, the energy needed by cells would not be released fast enough, and we would feel weak and tired.

In everyday life, enzymes are used in **washing powders** and **detergents**, where they help remove stains like blood, fat, and food at low temperatures, saving energy and making clothes cleaner. Enzymes are also used in the **food industry**, for example in baking where yeast enzymes help dough rise, and in making yoghurt and cheese where enzymes help change milk into other products.

Enzymes are used in **medicine**, where some enzymes help in diagnosing diseases and others help in treating conditions such as poor digestion. In agriculture, enzymes help plants grow by controlling important chemical reactions in plant cells, which leads to better crop production.

In summary, enzymes are essential for life because they control and speed up chemical reactions in living organisms, and they are very useful in everyday activities such as eating, washing, making food, and staying healthy.

5.5. End of Unit Assessment



5.5.1. Project activity: Using Amylase to Make Maize Porridge Sweet Without Adding Sugar

1. Context/Background

Many foods contain starch, a complex carbohydrate that is not sweet. During digestion, enzymes break down starch into simpler sugars that have a sweeter taste. Amylase is an enzyme found in saliva that helps convert starch into sugars. Understanding how enzymes work helps explain important biological processes such as digestion and nutrient absorption.

2. Challenge/Problem Statement

Can maize porridge be made sweet without adding table sugar? Investigate how the enzyme amylase can change starch in maize porridge into sugar and affect its taste.

Key Question

How can amylase be used to make maize porridge sweet without adding table sugar?

3. Purpose of the Project

The purpose of this project is to help students understand the action of enzymes by investigating how amylase breaks down starch into simpler sugars. Students will observe changes in porridge treated with amylase and relate these changes to the role of enzymes in digestion.

4. Instructions

- Work in groups.
- Prepare maize porridge using the materials provided.
- Divide the porridge into two samples.
- Add commercial salivary amylase to one sample and leave the other sample untreated as a control.
- Keep both samples under the same conditions and observe any changes over time.
- Record observations carefully and compare the two samples.
- Use scientific explanations to account for the observed results.

5. Your Tasks

- Prepare and label the two porridge samples.
- Observe and record changes in taste, texture, or consistency.
- Compare the treated and untreated samples.
- Explain the role of amylase in the observed changes.
- Present your findings to the class.

6. Expected Student Outputs

- Observation records and data tables.
- A comparison of treated and untreated porridge samples.
- An explanation of how amylase acts on starch.
- A group report containing results, conclusions, and recommendations.
- A brief presentation of findings.

7. Constraints and Limitations

Scope

The project should focus only on investigating the effect of commercial salivary amylase on maize porridge.

Materials and Methodology

Use only the provided maize flour, commercial salivary amylase, water, containers, and simple observation tools. Observations should be recorded systematically in tables and notes.

Timeframe

The project should be completed within one week, including preparation, observation, analysis, and presentation of results.

8. Guiding Questions

1. What differences do you observe between the porridge treated with amylase and the untreated porridge?
2. Why does the porridge treated with amylase become sweeter?
3. What role does amylase play in the digestion of starch?
4. Why is a control sample important in this investigation?
5. How do the results demonstrate that enzymes are biological catalysts?

5.5.2. Assessment questions

A. Knowledge and understanding

1. What is an enzyme?

- A. A type of vitamin
- B. A biological catalyst that speeds up chemical reactions
- C. A waste product in cells
- D. A source of energy

2. Most enzymes are made of which substance?

- A. Carbohydrates
- B. Lipids
- C. Proteins
- D. Minerals

3. The part of an enzyme where the substrate fits in is called the:

- A. Cell membrane
- B. Active site
- C. Cytoplasm
- D. Nucleus

4. Which name best shows how enzymes are usually named?

- A. According to colour
- B. According to the food they digest and ending with “-ase”
- C. According to where they are found
- D. According to their size

5. Which enzyme helps in the digestion of starch in the mouth?

- A. Pepsin
- B. Lipase
- C. Amylase
- D. Protease

6. Enzymes work best at a particular pH and temperature called:

- A. Maximum point
- B. Average point
- C. Optimum condition
- D. Normal level

B. Skills

7. A student heats saliva with starch to a very high temperature and no sugar is formed. What is the most likely reason?

- A. The enzyme was too slow
- B. The enzyme was denatured by heat
- C. There was too much starch
- D. The saliva contained no enzyme

8. Which diagram best explains how an enzyme works?

- A. Key fitting into a lock
- B. Water flowing in a pipe
- C. Air filling a balloon
- D. Light reflecting from a mirror

9. Why does changing pH affect enzyme activity?

- A. It changes the colour of the enzyme
- B. It changes the shape of the active site
- C. It increases the amount of substrate
- D. It adds more energy to the reaction

10. Which action would increase the rate of an enzyme-controlled reaction?

- A. Lowering temperature below freezing
- B. Using the correct pH for the enzyme
- C. Removing the substrate
- D. Boiling the enzyme

C. Attitudes and Values

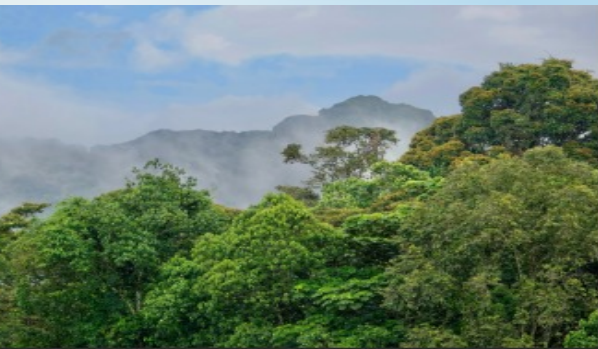
11. Why is it important to use enzymes in industries like baking and brewing?

- A. They make products more expensive
- B. They slow down reactions
- C. They make processes faster and save energy
- D. They replace all machines

12. Which behaviour shows responsible use of enzymes in daily life?

- A. Storing enzymes at very high temperatures
- B. Using detergents with enzymes to save water and energy
- C. Mixing all chemicals without instructions
- D. Heating enzymes before use

UNIT 6 PHOTOSYNTHESIS



Key Unit Competence:

Assess the process of photosynthesis and how various environmental factors affect its rate.

Learning objectives

By the end of this unit you should be able to:

Knowledge and understanding

1

- Explain photosynthesis by describing its chemical equation, required conditions, products, and limiting factors.
- Describe how leaf structure is adapted for photosynthesis, including both internal and external features.
- Explain the importance of photosynthesis in the maintenance of life and ecological balance.
- Describe the role of mineral nutrients and fertilisers, including the importance of nitrate and magnesium in chlorophyll synthesis, as well as the uses and dangers of nitrogen and other fertilisers.

Skills

2

- Demonstrate practical skills in investigating photosynthesis, including testing for starch in green leaves and experimentally showing the production of oxygen by plant leaves.
- Analyse factors influencing the rate of photosynthesis by conducting investigations on the necessity of light, chlorophyll, and carbon dioxide, and by interpreting graphical data related to light intensity and carbon dioxide concentration.
- Demonstrate experimental competence in plant mineral nutrition by investigating and explaining the effects of nitrate and magnesium ion deficiencies on plant growth

Attitudes and values

3

- Appreciate the importance of photosynthesis in reducing atmospheric carbon dioxide and producing oxygen essential for life.
- Demonstrate awareness of plant adaptation and laboratory safety, including how plants inhabit different geographical regions and safe handling of flammable substances such as ethanol.

6.0. Introduction

Photosynthesis is one of the most important biological processes on Earth and is fundamental to the survival of living organisms. It is the process by which green plants use light energy, carbon dioxide, and water to produce food, releasing oxygen as a by-product. This process provides the energy and organic matter that support nearly all food chains and ecosystems.

Photosynthesis also plays a vital role in maintaining the balance of oxygen and carbon dioxide in the atmosphere, contributing to climate stability and environmental sustainability. Without photosynthesis, plants could not produce food, and animals, including humans, would lack both a primary source of energy and the oxygen needed for respiration.

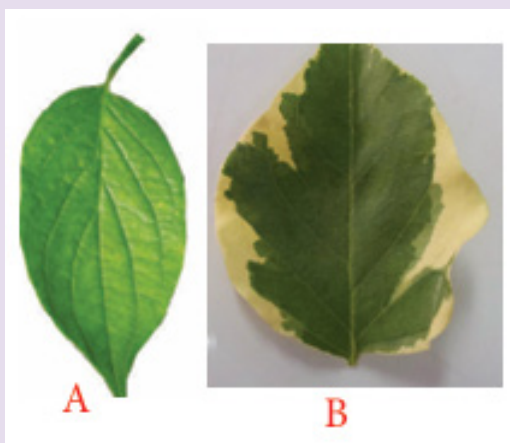
Understanding photosynthesis helps students appreciate the interdependence of living organisms, the flow of energy through ecosystems, and the critical role of plants in sustaining life on Earth.

Key Concepts: photosynthesis, chlorophyll, chloroplast, light energy, glucose, starch, carbon dioxide, water, oxygen, food production, producers, food chain, ecosystem, energy flow, respiration, limiting factors, climate stability, environmental sustainability.

Introductory activity



By now, you have likely heard about photosynthesis. Look at the plant leaves below



1. What do you understand by the term *photosynthesis*?
2. How does it occur?
3. What are the raw materials needed for photosynthesis to occur?
4. Which leaf shown in the picture do you think is best suited for photosynthesis? Why?

6.1. Photosynthesis Process

Learning activity 6.1



Investigating the photosynthesis

- Leaves
- Beaker with water
- Boiling tube
- Tripod stand and wire gauze
- Methylated spirit or alcohol
- Iodine solution
- Pair of forceps
- Source of heat

Requirements:

Step 1: Exploring the idea of photosynthesis

1. Observe a green plant or a labelled diagram provided by the teacher.
2. Discuss in groups how plants obtain their food.
3. Suggest and write the meaning of photosynthesis in your own words.

Step 2: Understanding the chemical process

4. Examine a chart or equation of photosynthesis.
5. Identify and list the raw materials used in photosynthesis.
6. State and explain the conditions required for the process to occur.
7. Name and describe the products formed during photosynthesis.

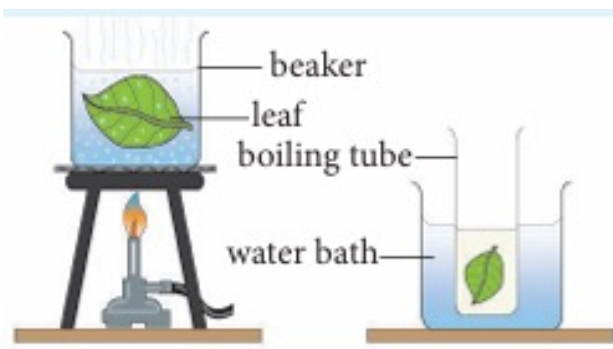
Step 3: Demonstrating evidence of photosynthesis (starch test)

8. Pluck a green leaf from a plant exposed to sunlight.
9. Boil the leaf in water to kill it.
10. Heat the leaf in ethanol using a water bath to remove chlorophyll (observe safety precautions).
11. Rinse the leaf in warm water to soften it.
12. Add iodine solution to the leaf.
13. Observe and record the colour change.

Step 4: Drawing conclusions

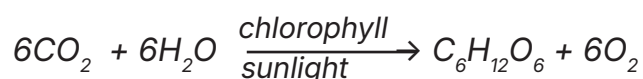
14. Interpret the results of the starch test.
15. Conclude how the presence of starch provides evidence that photosynthesis has taken place

Photosynthesis is a vital biological process in which green plants manufacture their own food using light energy absorbed by chlorophyll. During this process, carbon dioxide from the atmosphere and water absorbed by the roots are converted into organic food substances. Photosynthesis is the foundation of all food chains and supports life on Earth by supplying energy and oxygen.



Chemical equation, conditions, and products

Plants, algae and certain types of bacteria are **autotrophs**. This means that they can synthesise their own food. Plants use energy from the sun, carbon dioxide and water as raw materials to make food in a process known as photosynthesis. They have chlorophyll which traps light energy from the sun. In the process, oxygen is given off as a by-product. The overall chemical equation of photosynthesis is:



The essential conditions for photosynthesis include sunlight as the energy source, chlorophyll to trap light energy, carbon dioxide from the air, water from the soil, and a suitable temperature for enzyme activity.

The main product of photosynthesis is glucose, which is used by the plant for respiration, growth, and synthesis of other substances. Excess glucose is converted into starch for storage. Oxygen is released into the atmosphere as a by-product.

Evidence of photosynthesis (starch)

Starch formation in green leaves provides clear evidence that photosynthesis has taken place. Since starch is an insoluble storage form of glucose, it accumulates in leaves after exposure to light.

To test for starch, a leaf is boiled in water to kill it, then heated in ethanol to remove chlorophyll, and finally treated with iodine solution. A blue-black colour indicates the presence of starch, confirming that photosynthesis has occurred.

Application activity 1.1



1. Why is photosynthesis considered as the foundation of all food chains and life on Earth?
2. How do the conditions required for photosynthesis influence the rate at which the process occurs in plants?
3. Explain how glucose produced during photosynthesis is used and stored by plants.
4. Why is chlorophyll essential in the process of photosynthesis, and what would happen if it were absent?
5. How does the test for starch provide evidence that photosynthesis has taken place in green leaves?

6.2. Conditions for photosynthesis

Learning activity 6.2



Investigating the necessity of chlorophyll, Light, carbon dioxide for photosynthesis

a) Investigating the necessity of chlorophyll for photosynthesis

Requirements:

- Potted plant with variegated leaves
- Ethanol or methylated spirit
- 2 beakers (250 ml)
- Iodine solution
- White tile
- Source of heat
- Water in a wash bottle

Procedure:

1. Expose the variegated potted plant (that had previously been put in darkness for 48 hours - a process called destarching) to sunlight for about 3 hours.
2. Remove a variegated leaf from the plant and draw it.
3. Label the green part and the white part.
4. Test the leaf for starch. (Refer to activity 6.1)
5. After the test, draw the leaf after the test and label the brown parts and the blue-black parts.

Study questions

- a) In the test for starch, what is the colour of the part of the leaf that was: (i) Green? (ii) Yellow?
- b) Explain the results that you get on testing the variegated leaf for starch.
- c) Which part of the leaf acts as a control?

Caution:

- Put on eye protection because ethanol is highly flammable.
- Remember to turn out the flame before putting the boiling tube with the ethanol into the hot water.

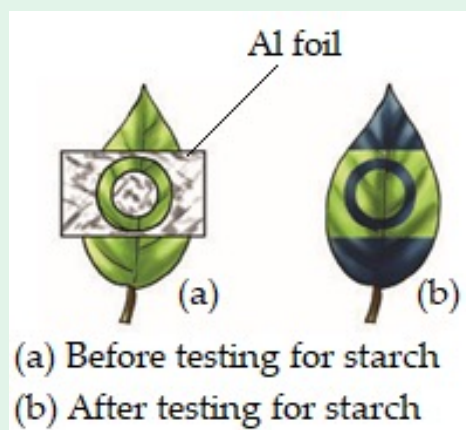
b) Investigating necessity of light in photosynthesis

Requirements

- | | |
|------------------|--------------------------|
| • A potted plant | • Water in a wash bottle |
| • Aluminium foil | • Forceps |
| • Paper clips | • White tile |
| • 250ml beakers | • Iodine solution |
| • Water bath | • Ethanol |

Procedure

1. Cut out a star shape or circle in the middle of an aluminium foil or black carbon paper that is big enough to cover a big portion of a leaf.
2. Use a paper clip to attach this paper securely round one of the green leaves of a potted plant as shown in figure below. Cover the leaf with aluminium foil only leaving out a star shaped space that exposes part of the leaf.
3. Expose the plant to sufficient light for about 3 hours.
4. Pick the leaf that had been covered and one other uncovered leaf and carry out a starch test on the leaves.
5. Record your results.
6. Account for the observations.
7. Discuss with a classmate your findings.



Study questions

- a) What do you observe when you test the leaf for starch?
- b) Draw the different parts of the leaf as observed in the experiment.
- c) Which part of the leaf acts as a control?
- d) Explain your answer in (c) above

c) Investigating the necessity of carbon dioxide in photosynthesis

Requirements

- Two potted plants growing in small plastic containers kept in the dark for 24 hours.
- Transparent plastic bags.
- Sodium hydroxide (soda lime) or potassium hydroxide (potash) solution.
- Rubber bands to fit around plastic containers for the solutions.
- Methylated spirit.
- Beaker with water.
- Boiling tube.
- Sodium hydrogen carbonate (sodium bicarbonate).

Procedure

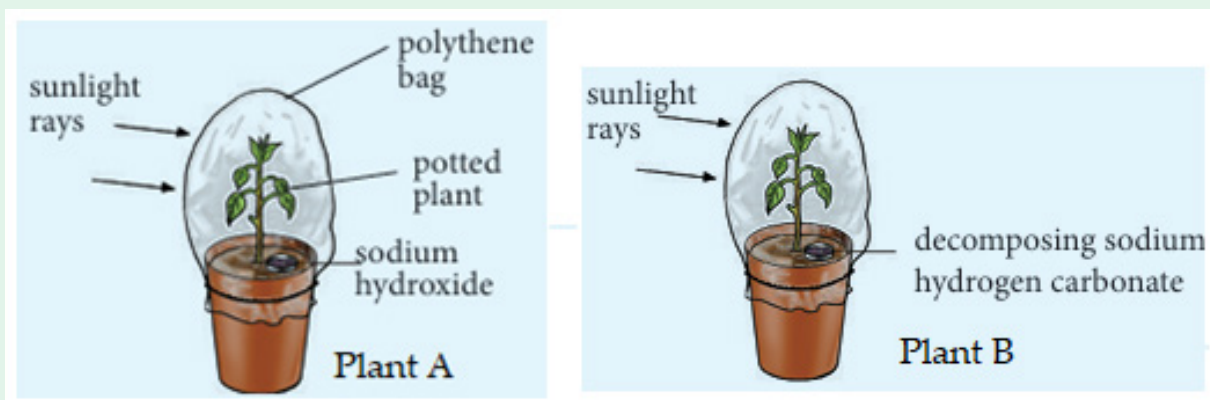
Set up the two potted plants as follows:

Plant A

1. Put some sodium hydroxide or potassium hydroxide into the small plastic container.
Place the container carefully on the soil holding the plant.
2. Take the transparent polythene bag and cover the whole plant with it.
Secure the bottom by tying it with one or two elastic bands.

Plant B

3. Repeat the procedure but place sodium hydrogen carbonate (sodium bicarbonate) in the plastic container.



4. Leave the set-up in a well-lit part of the laboratory or in sunlight for several hours or 1 to 2 days.
5. Detach leaves from each setup and test for the presence of starch.

Study questions

1. What is the role of the sodium hydroxide (potassium hydroxide) in set-up A?
2. What did you find out when you tested for starch in leaves from set-up A and B?
3. What conclusion can you make from the results of set-up A and B?
4. Between set-up A and B which one acts as the control experiment?
5. What is the role of sodium bicarbonate in set-up B?

A variegated leaf is one whose surface shows two colours, for example, green on some parts and white on others.

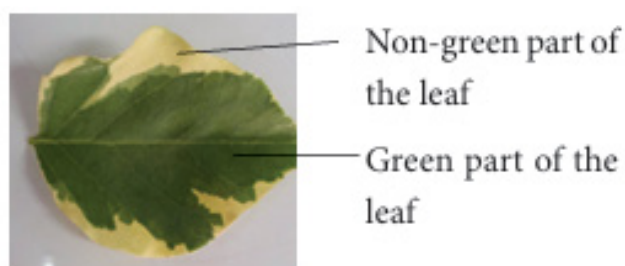


Fig. 6.1: A variegated leaf

The green part of the leaf has cells with chlorophyll. Therefore, photosynthesis will take place and form starch. Starch present in this part will turn iodine from brown to blue-black colour. The yellow part has cells that do not contain chlorophyll. These cells will not carry out photosynthesis; therefore, no starch will be formed. The starch test will be negative. The green part of the leaf acts as a control experiment because it has all the conditions required for photosynthesis.

The part of the plant not covered with the foil had all conditions needed for photosynthesis. Therefore, on testing with iodine, it appeared blue-black. This is because starch was present. This part of the leaf acts as a control set-up. When a test for starch was carried on the part covered with aluminium foil, the colour of iodine persisted. This shows that starch was absent. The foil prevented sunlight from reaching this part of the leaf and hence photosynthesis did not take place.

The Photosynthesis can only take place when certain conditions are present; if any one of them is missing, the process stops, and the plant cannot make its own food. Among the most important conditions are chlorophyll, light, and carbon dioxide.

a) Chlorophyll

Chlorophyll is the green pigment found mainly in the leaves of plants. Its main role is to absorb light energy from the sun. This absorbed energy is used to drive the process of making food.

Only green part of a plant can photosynthesize because they contain chlorophyll. Without chlorophyll, plants would be unable to trap light energy and photosynthesis would not occur.

b) Light

The light provides energy needed for photosynthesis. Sunlight supplies the energy that allows plants to change carbon dioxide and water into glucose. When plants are placed in the dark, they cannot photosynthesize because there is no light energy available. This explains why plants grow better in well-lit areas than in shaded or dark places.

c) Carbon Dioxide (CO₂)

Carbon dioxide is an important raw material for photosynthesis. Plants take in carbon dioxide from the air through small openings in the leaves called stomata. The carbon from carbon dioxide is used to build glucose, which is the plant's food. If carbon dioxide is absent, photosynthesis cannot take place even when light and chlorophyll are present.

Application activity 6.2



Complete the sentences by filling in the correct words.

Photosynthesis can only occur when certain conditions are present. _____ (1) is the green pigment that absorbs light energy in plant leaves. _____ (2) provides the energy needed for the process to take place. Plants obtain _____ (3) from the air through tiny openings in the leaves called _____ (4). If one of these conditions is absent, _____ (5) cannot occur.

6.3. Limiting factors for photosynthesis

Learning activity 6.3



A science teacher planted two bean plants of the same age in separate pots using the same type of soil and the same amount of water.

- **Plant A** was placed outside where it received bright sunlight, fresh air, and warm daytime temperatures.
- **Plant B** was placed inside a classroom near a window where the light was weak, the air was limited, and the room was cooler.

After three weeks, Plant A grew taller, had dark green leaves, and looked healthy. Plant B grew slowly, had pale leaves, and was smaller in size.

Why did the two bean plants grow differently?

The teacher explained that although both plants had water and soil nutrients, some conditions were not enough for Plant B. These conditions slowed down the rate of photosynthesis. Such conditions are called limiting factors because they limit how fast photosynthesis can take place.

Questions for students

1. What do you mean by “limiting factor” in photosynthesis?
2. What conditions were different between the Plant A and - B, and how did they influence the growth of the plants?
3. How can low light intensity reduce photosynthetic rates in plants?
4. Why are temperature and carbon dioxide concentrations also considered photosynthetic limiting factors?

The limiting factor is the variable that limits the rate of a biological process such as photosynthesis. The photosynthetic process is not always effective. Sometimes there are factors that prevent it from doing so. Photosynthesis may be inhibited by the lack of the availability of one or more raw materials or other factors required for photosynthesis.

Some of these factors are listed below.

a) Light intensity

The intensity of the sun varies depending on the time of day, the season and the location of the plants on the surface of the earth. Light is essential for photosynthesis. Plants can't photosynthesise in the dark. With low light intensity, photosynthesis rates are low. But as the intensity of light increases, so does the rate of photosynthesis. The point come when the plant can no longer photosynthesise at faster rate even with a further increase in light intensity. At this point, any other factor that affect the rate of photosynthesis are going to be holding back the rate of photosynthesis.

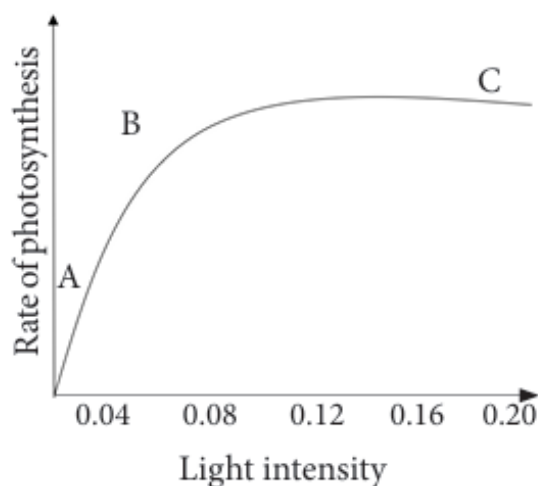


Fig. 6.2: A graph showing effect of light intensity on rate of photosynthesis

- Between points A and B; the rate of photosynthesis is influenced by sunlight. The increased light intensity cause the photosynthesis rate to increase. Light is therefore the limiting factor.
- However, between B and C, the plant cannot photosynthesise any more rapidly, even if the intensity of the light is increased. At that point, light no longer is an obstacle. Instead, other factors limit it.

b) Carbon dioxide concentration

Carbon dioxide levels in the atmosphere are relatively low (0.03%). Therefore, it may also be a limiting factor in photosynthesis. The increase in carbon dioxide increases photosynthesis rate. But this is only last until a certain point when the rate of photosynthesis no longer with increasing carbon dioxide, because other factors that influence photosynthesis become more restrictive.

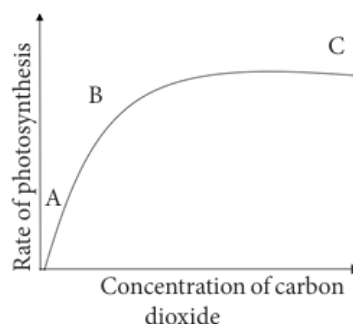


Fig. 6.3: Graph of effect of carbon dioxide concentration on the rate of photosynthesis

A-B: The increase in carbon dioxide concentration of causes the photosynthetic rate to increase.

B-C: The limiting factors have set in and further increases in the concentration of carbon dioxide do not result in corresponding increases in the rate of photosynthesis.

c) Temperature

Photosynthesis is an enzymatic reaction- and is therefore temperature dependent. When temperatures are low, plants photosynthesise slowly. but when temperature rise, the rate of photosynthesis increases. The rate of photosynthesis is highest at optimal temperature. Further temperature increases a above optimum results in reduced photosynthetic rate because enzymes are denatured.

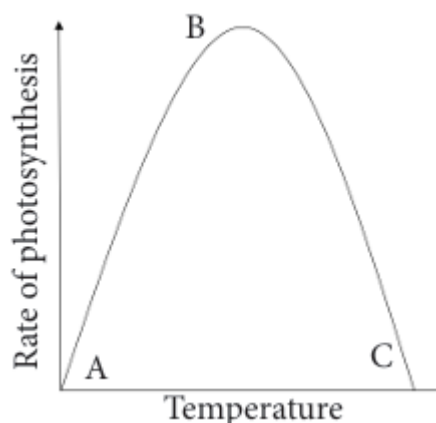


Fig. 6.4: Effect of temperature on the rate of photosynthesis

A-B: The increase in temperature causes the corresponding increase in photosynthesis rate p. photosynthesis rate is highest at B.

B-C: Any further temperature increase will denature the enzyme. So, the rate of photosynthesis is going down.

d) Water

Only about one percent of the water that plants take is used for photosynthesis. Water scarcity affects – photosynthesis indirectly. For example, when water is scarce, stomata close. This reduces the gas exchange between the leaf and the atmosphere. As a result, less carbon dioxide is transferred to the leaf through the stomata. This slow down photosynthesis.

e) Minerals

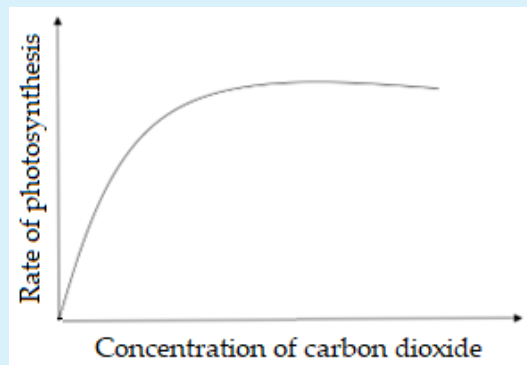
Minerals such as manganese, magnesium, nitrogen and phosphorus are all components of photosynthetic substances.

Application activity 6.3



1. Which of the following is not a limiting factor for photosynthesis?
 - a) Amount of oxygen
 - b) Concentration of chlorophyll
 - c) Light intensity
 - d) Temperature
 - e) Carbon dioxide concentration

2. The graph below illustrates the effect of increased carbon dioxide on photosynthesis rate.



Explain why the rate of photosynthesis does not continue to increase with increase in carbon dioxide concentration.

3. The temperature is – one factor limiting the rate of photosynthesis. Explain why the photosynthesis rate begins to decrease as temperature rise above 40°C and finally stops at 60°C.
4. (a) Discuss with your classmates the effects of the following on the process of photosynthesis under the following conditions:
 - i. Very low temperatures
 - ii. Cloudy conditions
 - iii. Very high light intensity
 - iv. Low carbon dioxide(b) What does this tell you about the process of photosynthesis

6.4. Adaptation of leaf for photosynthesis

Learning activity 6.4



Observing the internal structure of the leaf

Materials Needed

- Fresh green leaf (e.g. bean, cassava, mango, or hibiscus)
- Razor blade or sharp knife (teacher supervises)
- Microscope or hand lens
- Microscope slide and cover slip
- Dropper
- Water
- Tissue paper
- Diagram/chart of a leaf cross-section
- Pencil and notebook

Procedure

1. Collect a fresh green leaf and observe its colour, shape, flat surface, and thinness, then record your observations.
2. With the help of the teacher, carefully cut a very thin cross-section of the leaf using a razor blade.
3. Place the thin leaf section on a clean microscope slide and add one drop of water.
4. Gently place a cover slip over the section and remove excess water with tissue paper.
5. Put the slide on the microscope stage and observe it under low power, then adjust the focus until the structures are clear.
6. Observe and identify the different internal parts of the leaf such as the upper epidermis, palisade mesophyll, spongy mesophyll, air spaces, veins, and lower epidermis with stomata.
7. Draw a neat diagram of the leaf cross-section in your notebook and label all the observed parts.
8. Discuss how each structure is adapted for photosynthesis, for example trapping light, allowing gas exchange, and transporting water and food.
9. Write a short conclusion explaining how the internal structure of the leaf helps it to carry out photosynthesis.

Safety Rules

- Do not play with the razor blade.
- Cut slowly and carefully.
- Wash hands after the activity.

a) Internal structure of a leaf

The leaf consists of several internal layers that work together to carry out **photosynthesis**. The outermost layer on the upper side is called the **upper epidermis**. It protects leaves and is **transparent**, allowing **sunlight** to pass through. Beneath the upper epidermis is the **palisade mesophyll**, which consists of long, closely packed cells containing many **chloroplasts**. This layer is the main site of **photosynthesis**. Beneath the palisade mesophyll lies the **spongy mesophyll**, composed of loosely arranged cells with large **air spaces** between them. These air spaces allow **gases** to move easily within the leaf. Running through the leaf are **vascular bundles** or **veins**, which contain **xylem** for transporting **water and minerals** and **phloem** for carrying away **food** made during photosynthesis. The **lower epidermis** forms the bottom layer of the leaf and contains **stomata**, which are small openings that allow **gas exchange**.

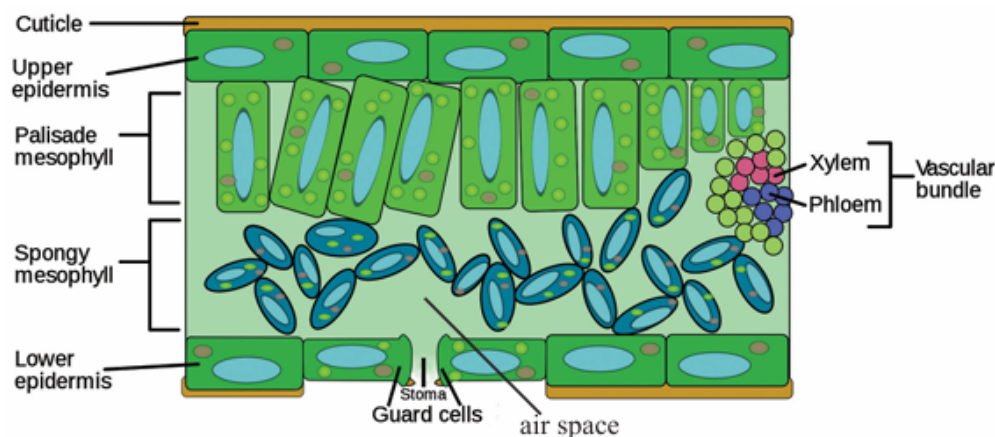


Fig. 6.5. Internal structure of the leaf

b) Adaptation of a leaf for photosynthesis

The leaf has several features that make it well adapted for **photosynthesis**. Its **broad and flat shape** provides a large **surface area** for absorbing **sunlight**. The leaf is **thin**, which reduces the distance **gases** must travel during **diffusion**. The **palisade mesophyll** contains many **chloroplasts**, enabling the leaf to trap maximum **light energy**. The **upper epidermis** is **transparent** so that light can easily reach the palisade cells. Large **air spaces** in the **spongy mesophyll** allow **carbon dioxide** and **oxygen** to move freely within the leaf. A network of **veins** ensures that **water and minerals** are supplied to the leaf and that the **food** produced is transported to other parts of the plant. **Stomata** in the **lower epidermis** allows **carbon dioxide** to enter the leaf and **oxygen** to leave, allowing photosynthesis to take place efficiently.

Application activity 6.4



1. (i) Name the pigment responsible for photosynthesis in green plants?
- (ii) Which cell organelle contains chlorophyll?
- (iii) What is the name of the opening through which gases enter and leave the leaf?
- (iv) Which cells control the opening and closing of the stoma?
- (vi) Name the gas required by plants for photosynthesis.

6.5. Extraction of Chlorophyll and Its Importance

Learning activity 6.5



Extraction of chlorophyll

Materials Needed:

- Fresh green leaf (e.g. bean, cassava, mango, or hibiscus)
- Scissors.
- Mortar
- Alcohol (methylated spirit or ethanol)
- Piece of clean cloth or filter paper
- Test tube or small container.

Procedure:

1. Collect fresh green leaves and wash them with clean water to remove dust.
2. Cut the leaves into small pieces using scissors.
3. Put the cut leaf pieces into a mortar.
4. Add a small amount of **alcohol (methylated spirit or ethanol)** to the leaves.
5. Grind the leaves gently with a pestle until the alcohol turns green.
6. Filter the mixture using a piece of clean cloth or filter paper into a test tube or small container.
7. Observe the green solution obtained; this solution contains **chlorophyll**.

Study questions

1. Why are fresh green leaves washed with clean water before starting the experiment?
2. Why is alcohol (ethanol or methylated spirit) added to the crushed leaves during the experiment?
3. What change is observed in the alcohol after grinding the leaves, and what does this change show?
4. Why is the mixture filtered at the end of the experiment?

Extracting chlorophyll allows students to see that **green leaves contain a special green pigment called chlorophyll**. This pigment is essential because it **absorbs sunlight**, the main source of energy that is needed for **photosynthesis**. During photosynthesis, plants use this absorbed light energy to **convert carbon dioxide and water into glucose**, which is their food, and **release oxygen** into the atmosphere, which is vital to life in animals and humans. By extracting chlorophyll, students understand that **chlorophyll is essential to photosynthesis**. Without it, plants cannot capture light, energy, produce food and release oxygen, and ultimately, they cannot survive. This practical demonstrates that chlorophyll is essential for plant life on Earth.

Application activity 6.5



1. If a student forgets to add alcohol while grinding the leaves, what result would be expected at the end of the experiment? Explain why.
2. Why is alcohol used instead of water to extract chlorophyll from the leaves
3. Suppose the leaves used are yellow instead of green. What difference would you expect in the colour of the final solution? Give a reason.
4. How would the result of the experiment be affected if the mixture is not filtered before observation?

6.6. Mineral requirements for chlorophyll synthesis

Learning activity 6.6



Investigating the role of minerals in chlorophyll production

Aim: To understand the **role of magnesium and nitrate ions** in the production of chlorophyll and to study their influence on colour and content of chlorophyll in leaves.

Materials :

- Fresh green leaves (spinach or bean leaves)
- Petri dishes or small containers
- Alcohol (ethanol/methylated spirit)
- Mortar and pestle
- Filter paper or cloth
- Test tubes
- Water
- Solutions of **magnesium sulfate (MgSO_4)** and **potassium nitrate (KNO_3)**
- Forceps, scissors, and safety gloves



Procedure:

- Collect fresh green leaves and wash them thoroughly to remove dust.
- Observe the leaves carefully and note their natural color.
- Cut the leaves into small pieces using scissors and place them in a mortar.
- Add a small amount of alcohol (ethanol or methylated spirit) and gently grind the leaves until the alcohol turns green.
- Filter the mixture through clean cloth or filter paper into a test tube or small container to obtain a green solution containing chlorophyll.
- Take two sets of leaves from similar plants for mineral treatment and leave one set untreated as a control.

- Soak one set of leaves in magnesium sulfate solution and the other set in potassium nitrate solution for a few hours.
- After treatment, cut and grind the leaves in alcohol as described earlier, then filter to obtain the chlorophyll solutions.
- Observe and compare the intensity of the green color in treated and untreated leaves, recording the differences

Study questions

1. Discuss how the leaves treated with magnesium or nitrate solutions may show darker green color compared to untreated leaves, relating this to the **role of magnesium ions in forming chlorophyll** and **nitrate ions in chlorophyll synthesis**.
2. Reflect on how deficiencies in these minerals can affect chlorophyll production, leaf color, and the plant's ability to perform photosynthesis.

plants need a number of minerals to grow and to perform photosynthesis effectively. The two most important minerals for the production of chlorophyll are magnesium and nitrate **ions**.

c) The role of magnesium ions

Magnesium is a major element in **chlorophyll molecules**. Without magnesium, the chlorophyll pigment cannot form properly, and the leaf loses its green colour, which may impair the ability of the plant to absorb sunlight. The magnesium ions therefore play a **key role in capturing the light energy** required for photosynthesis to occur. Magnesium deficiency often leads to **leaf discolouration (chlorosis)** and poor plant growth as photosynthesis becomes less efficient.

d) Role of nitrate ions

Nitrate ions are the main source of nitrogen, which is an essential component and builds block of amino acids, proteins, and nucleic acids. Nitrogen is also important for synthesizing **chlorophyll molecules**. Without sufficient nitrate ions, plants cannot produce enough chlorophyll, leading to poor photosynthesis, stunted growth and pale or yellow leaves. By providing nitrate ions, plants are able to maintain healthy green leaves, which ensures **efficient energy capture for food production**.

Application activity 6.6



1. Explain why chlorophyll is important for photosynthesis and what would happen to the plant if it were to be deprived of it.
2. Describe the role of magnesium ions in the formation of chlorophyll and how deficiency might affect the plant.
3. Discuss how nitrate ions contribute to chlorophyll synthesis and overall growth of the plant.
4. How does extracting chlorophyll from leaves help students to understand the process of photosynthesis?
5. If a plant has pale or yellow leaves, what could be the possible reasons related to minerals, and how would this affect photosynthesis?

6.7. Use and dangers of nitrogen and other fertilisers

Learning activity 6.7



Case study on Use and dangers of nitrogen and other fertilizers.

In rural agricultural communities, most farmers rely on fertilizers to increase the yield of their crops. Nitrogen- fertilizers, such as urea and NPK (Nitrogen Phosphate and Potassium) are widely used because they provide basic nutrients such nitrogen, phosphorus and potassium, that help plants grow faster and produce more food. Farmers have noticed that crops such corn, tomatoes and vegetables grow more vigorously when applied with fertilizers.

More recently, some farmers have noticed unusual changes in their crops and in their surroundings. Some plants have developed yellowing leaves and stunted growth despite repeated fertilization. Others reported reduced decrease in soil fertility over time. In addition, the nearby river has become cloudy and stinking, and fish populations have begun to decline. Local health officials have also reported an increase in cases of stomach and skin problems among farmers who use unprotected fertilizers.

Discussion questions

1. How does nitrogen in fertilizers help plants to grow?
2. Why do some crops grow poorly despite fertilizers use?
3. What are the environmental problems caused by fertilizer use?
4. What precautions should be taken by farmers when handling fertilizers?
5. Suggest ways of reducing the harmful effects of fertilizers while maintaining the yield of crops.

The soils contain naturally a lot of nutrients such as nitrogen, phosphorus, calcium, and potassium. These nutrients allow the plants to grow healthy. When soil nutrients are absent or scarce, plants suffer nutrient deficiencies. Their growth and development are affected and therefore the plant cannot function properly to produce foodstuffs. Fertilisers are applied to soils lacking nutrients. Fertilisers add plant nutrients such as nitrogen, phosphorus, and potassium which are not present in the soil, and they also replenish the soil. Fertilisers are simply nutrients applied to the fields to in the missing elements in the soil. Nitrogen comes naturally from the atmosphere and enters soils by precipitation and lightning strikes. It is also found in beneficial microorganisms (some types of bacteria) living in the soil. It is the misuse and over-application of nitrogen fertilisers that has had a serious impact on the natural ecosystems on Earth. It also depleted the minerals from the soil.

Dangerous levels of nitrates in the environment. The excessive use of nitrogen fertilisers is linked to diseases such as diabetes, cancer, and Parkinsonian. Consumption of groundwater containing high levels of nitrates can cause health problems in young children.

Harmful effects of nitrogen fertilisers to the environment

- a) Nitrogen fertilisers help plants to grow. However, weeds and non-native plants tend to grow more readily with an increased supply of nitrogen.



Fig 6.6. Nitrogen fertiliser

- b) Excess nitrogen in soil cause nutrients imbalance. This causes depletion of other minerals that are important for the production of energy such as calcium, phosphorus and magnesium.
- c) High nitrate content in the soil lead to an increased acidity. This affects plants that need an alkaline environment in order to grow.
- d) Nitrogen-emitting air, caused by nitrates from cars and industrial installations, lead to soil acidification when acid rain falls.
- e) Surface runoff containing high quantities of nitrates enter water bodies as a result of eutrophication. When nitrogen levels rise in rivers and streams, it encourages algae growth. As the algae die and decompose, the organic matter in the water is increased. This process uses up the oxygen available and cause a drop in the level., Fish, crabs and other aquatic organisms die without oxygen.
- f) Nitrogen is a solubular substance. After a storm or irrigation, it sinks deep into the soil. It can get into ground water and nearby well contaminating them.

Briefly:

1. Benefits of Fertilizers:

- Provide basic **nutrients** (nitrogen, phosphorus, potassium) for plant growth.
- Increase **crop yield** and improving food security.
- Support for **rapid growth** in vegetables and cereals.

2. Dangers from Fertilizers:

- **Excessive nitrogen** may cause leaf burning, discolouration, or stunted growth.
- **Leaching** fertilizers into rivers can lead to **pollution** and **eutrophication**, which reduce oxygen content of the water and kills aquatic organisms.
- Excessive use may **damage the structure of the soil**, reduce its fertility over time, and make crops dependent on fertilizers.
- Direct contact with or inhalation of fertilizers may cause **health problems** such as skin irritation, breathing problems, and stomach problems.

Application activity 6.7



1. What is NPK fertilizer?
2. Would you advise Rwanda farmers to apply NPK fertilizers' or not? Support your opinion.

6.8. End of unit assessment



6.8.1. Investigating the Effect of Light Intensity on Oxygen Production by Aquatic Algae

1. Context/Background

Photosynthesis is the process by which green plants and algae use light energy to produce food and release oxygen. The rate of photosynthesis can be affected by environmental factors such as light intensity, carbon dioxide concentration, and temperature. Aquatic algae are important producers in aquatic ecosystems because they supply oxygen and form the base of many food chains. Understanding how light intensity affects oxygen production helps explain how environmental conditions influence photosynthesis and ecosystem productivity.

2. Challenge/Problem Statement

Light is essential for photosynthesis, but changes in light intensity may affect the amount of oxygen produced by aquatic algae. Investigate how different light intensities influence oxygen production in aquatic algae.

Key Question

How does light intensity affect the amount of oxygen produced by aquatic algae during photosynthesis?

3. Purpose of the Project

The purpose of this project is to help students investigate the relationship between light intensity and the rate of photosynthesis by measuring oxygen production in aquatic algae. Students will collect, analyze, and interpret data to determine how changes in light intensity affect photosynthetic activity.

4. Instructions

- Work in groups.
- Obtain a sample of aquatic algae and place it in a transparent container filled with water.
- Expose the algae to different light intensities by varying the distance between the algae and a light source or by placing the algae in areas with different light levels.

- Observe and measure oxygen production over a fixed period.
- Record observations and measurements systematically.
- Compare the results obtained under different light intensities.
- Draw conclusions based on the collected evidence.

5. Your Tasks

- Design a fair investigation to test the effect of light intensity on oxygen production.
- Identify and control variables such as temperature, number of algae, and observation time.
- Measure and record oxygen production at different light intensities.
- Present data using tables, charts, or graphs.
- Analyze patterns and explain the results using knowledge of photosynthesis.
- Draw conclusions and make recommendations for improving the investigation.

6. Expected Student Outputs

- An investigation plan.
- Data tables showing oxygen production at different light intensities.
- Graphs or charts illustrating the relationship between light intensity and oxygen production.
- Analysis of the results and identified trends.
- A project report including introduction, methodology, results, discussion, conclusion, and recommendations.
- A group presentation of findings.

7. Constraints and Limitations

Scope

The project should focus only on investigating the effect of light intensity on oxygen production by aquatic algae under controlled conditions.

Materials and Methodology

Use readily available materials such as aquatic algae, transparent containers, water, a light source, rulers, stopwatches, and observation sheets. Data should be recorded accurately using tables, graphs, and written observations.

Timeframe

The project should be completed within **2–3 weeks**, including planning, data collection, analysis, report writing, and presentation.

8. Guiding Questions

1. How will you vary light intensity while ensuring a fair test?
2. What method will you use to measure oxygen production by algae?
3. Which variables must be controlled to ensure reliable results?
4. What patterns do you observe between light intensity and oxygen production?
5. Why does oxygen production change as light intensity changes?
6. What conclusions can you draw about the relationship between light intensity and photosynthesis?

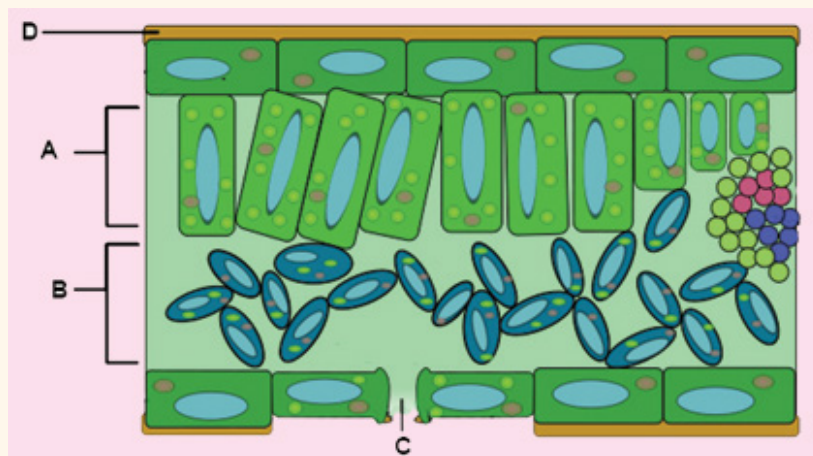
Success Criteria

- The investigation is planned and conducted systematically.
- Data is collected accurately and presented clearly.
- Appropriate graphs or charts are used to display results.
- Explanations correctly relate oxygen production to the rate of photosynthesis.
- Conclusions are supported by evidence from the investigation.

6.8.2. Assessment questions

a) Knowledge and understanding

1. Use the diagram of the cross-section through the leaf to answer the questions that follow.



- a) Indicate on the diagram the following: guard cells, spongy mesophyll layer, palisade layer and vascular bundles.
- b) Which cell type absorbs most carbon dioxide during the day?
- c) State the role of the part labelled D in photosynthesis.
- d) Describe ways in which cell type B are suited for photosynthesis.
- e) Of what importance is the shape of cell type C in photosynthesis?

2. Differentiate between the following.

- a) Epidermal cell and guard cell.
- b) Palisade layer and spongy layer.

3. Explain the role of vascular bundles in photosynthesis.

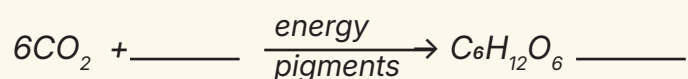
4. Describe how each of the following helps a leaf to photosynthesize.

- a) Chlorophyll molecules in the chloroplast.
- b) Lack of chloroplasts in the epidermal cells.
- c) Transparent cuticle

5. Which of the following statements is not true?

- a) Plants respire all the time.
- b) At night plants give out carbon dioxide.
- c) During the day, the rate of photosynthesis exceeds that of respiration.
- d) Plants only respire.

6. Study the equation given below.

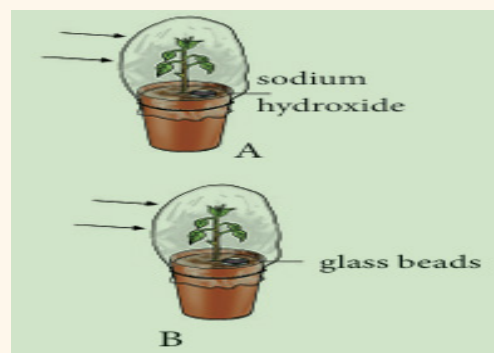
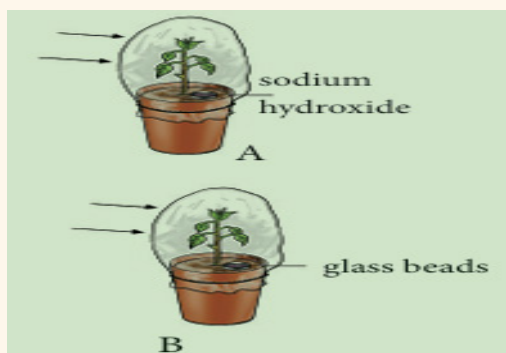


- a) Complete the equation.
- b) What is the source of energy for this reaction?
- c) Name the pigment needed for this reaction to occur.
- d) What is the role of this pigment in photosynthesis?
- e) State the mineral ion required for the synthesis of this pigment.

B. Skills

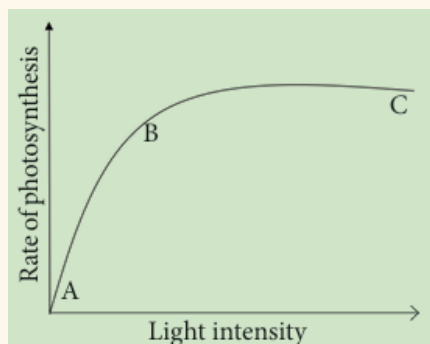
7. An experiment was carried out to find out if carbon dioxide is necessary for photosynthesis. Two plants of the same species were destarched.

The plants were then placed in sealed glass containers for 24 hours as shown in figure below and exposed to sunlight. Sodium hydroxide pellets were placed in glass container of plant A while glass beads were placed in the glass container of plant B. After 24 hours, starch test was carried out on plant A and B.

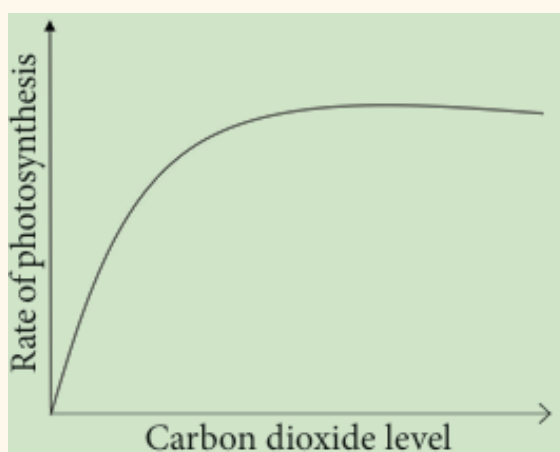


- Explain why the plants were destarched before beginning the experiment.
 - Why is chlorophyll removed from the leaves before carrying out a starch test?
 - What is the role of sodium hydroxide in the experiment?
 - State the expected results of the starch test.
 - Account for the results stated in (d).
 - Why was plant B included in the experiment?
8. Gahigi grew some wheat plants in a culture solution that contained all mineral ions except nitrate. The wheat did not grow as well as those that were given all mineral ions.
- What is the role of nitrate in plant growth?
 - What was the appearance of the plants grown without any nitrate?
 - If plants get very little magnesium ions, they have poor growth. Explain.

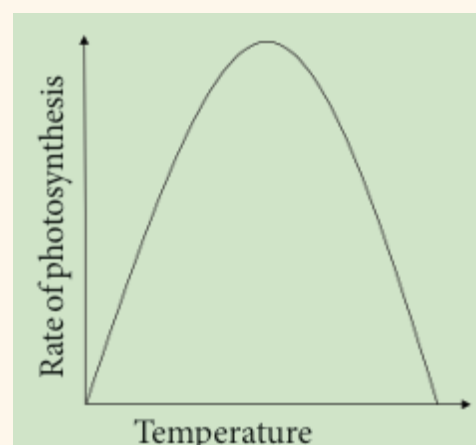
9. Study the graphs below.



- What happens to the rate of photosynthesis as light intensity is increased between A and B?
- What limiting factors are involved on further increase of light intensity from B to C? Explain how these factors affect the rate of photosynthesis



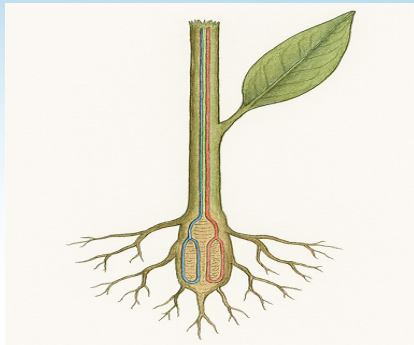
According to the above graph, what is the effect of increased carbon dioxide levels to photosynthesis? Explain your answer.



- Explain the changes that occur in the above graph to the rate of photosynthesis as the temperature rises.

C) Attitudes and values

10. The growth and yields of crops grown in open fields are limited by environmental factors. Describe how these factors are controlled in commercial greenhouses for crops to produce high yields.



Key Unit Competence:

Demonstrate the process of transport of mineral and organic substances in plants

Learning objectives

By the end of this unit you should be able to:

Knowledge and understanding

1

- Describe the structure and functions of xylem and phloem, including the mechanisms of water transport and translocation of food in plants.
- Explain transpiration in plants, its significance, and the factors affecting its rate, such as temperature and humidity.
- Describe plant adaptations for water regulation, including leaf adaptations to control water loss and adaptations of plants to different environments.
- Explain plant water balance problems, including the causes of wilting and how water movement from roots to leaves maintains turgidity

Skills

2

- Create a table for the comparison of transpiration and translocation.
- Use a potometer to measure the rate of water uptake of a given plant.
- Demonstrate practical evidence of the translocation process.
- Compare the role of transpiration and translocation in the transport of materials from sources to sinks, within a plant during different seasons.
- Demonstrate how some parts of a plant may act as a source or a sink for food stores at different times during their lifespan

Attitudes and values

3

- Appreciate the importance of absorption and transport of water in plants.
- Understand other perspectives related to transport such as absorption and transport of pesticides.
- Show concern for the control of aphid populations among plants since their feeding process through phloem content permits spread of viruses among plants.

7.0. Introduction

Plants, like all living organisms, require the movement of substances from one part of the body to another in order to survive and function effectively. Water and mineral salts absorbed by the roots must be transported to the leaves, while food produced in the leaves through photosynthesis must be distributed to other parts of the plant. Although plants do not move from place to place, they possess specialized transport tissues that enable roots, stems, and leaves to work together as an integrated system. Efficient transport is essential for growth, photosynthesis, respiration, reproduction, and maintaining the firmness of plant tissues.

Understanding transport in plants helps students appreciate how plants adapt to different environmental conditions and respond to factors such as drought, temperature, and humidity. It also explains common observations such as wilting, water uptake, and the movement of nutrients within plants. Knowledge of plant transport systems is important for understanding plant growth, crop production, and ecosystem functioning.

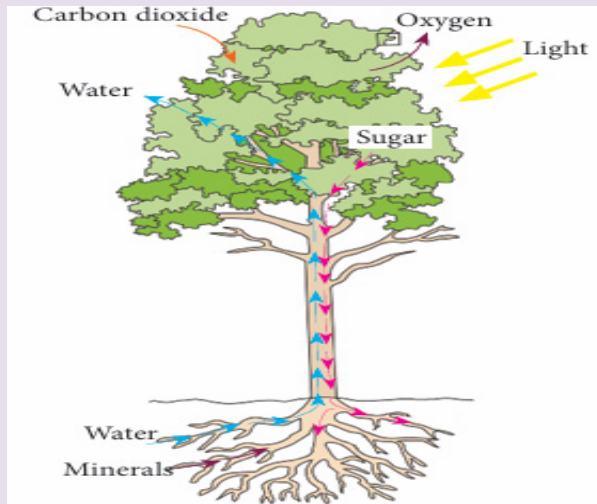
Key Concepts: transport in plants, xylem, phloem, water uptake, mineral salts, translocation, transpiration, root hair cells, stomata, transpiration stream, photosynthesis, turgidity, wilting, vascular tissues, adaptation.

Introductory activity



Plants produce food in their leaves. They absorb water and mineral salts from the soil and use them as raw material.

Observe the tree beside you and answer the questions below.



1. How are these essential substances transported to other parts of the plant?
2. What's happening in that tree?
3. The tree is large and intricate; how does the food produced in the leaves reach the stem and roots?
4. By what method does water absorbed from the soil reach the stem and leaves of the plant?
5. What does this suggest about what you will learn in this unit?

7.1. Transport system in vascular plants

Learning activity 7.1



Observing permanent slides of dicotyledonous and monocotyledonous roots

Requirements

- Microscopes
- Permanent slides of transverse sections of:
 - Dicotyledonous roots
 - Monocotyledonous roots

Procedure

1. Place a prepared slide of a dicotyledonous root under the microscope. Observe under low power and high-power objective lenses.
2. Note the different tissues present and their location.
3. Draw a plan diagram to show the position and layout of different layers of tissue. Do not draw any cells or shade.
4. Compare your diagram with that of the plan diagram of the dicotyledonous root section in Fig 7.1 (a) and use it to identify the tissues.
5. Repeat this procedure for the monocotyledon root and compare your drawing with that in Fig 7.1 (b).

Study questions:

- a) Describe the pattern in the arrangement of xylem in relation to phloem in the
 - Dicotyledonous root
 - Monocotyledonous root
- b) What is the difference in distribution of tissues between the dicot root and the monocot root?

In unicellular organisms, diffusion is sufficient for the transport of materials into and out of the body. However, plants are complex multicellular organisms, so diffusion alone is not sufficient to transport materials within them. This is because they have a lower surface area, to volume ratio, and many of their cells are far from the external environment where these materials are found. Therefore, plants require a transport system to move materials throughout the organism.

All living organisms consist of cells. To survive, these cells take up useful substances from the environment. They also generate and dispose of wastes. Water and mineral salts are transferred from the soil to the leaves and other parts of the plant by roots. The food produced is also transported from the leaves to other parts of the plant. The transport system of living organisms consists of specialised tissue and organ systems. In plants, the transport system consists of specialised tissue called vascular bundles. The vascular tissue comprises two types: **xylem** and **phloem**. Substances which must be transported in plants are:

- Water needed for photosynthesis.
- Mineral salts used in various plant processes.
- Organic substances produced primarily by photosynthesis in leaves.

Phloem and **xylem** tissues are present in root, stem and leaves.

Phloem tissues transport nutrients such as glucose and amino acids from leaves to other plant tissues, where they can be used or stored in the body.

The **xylem** tissues transfer water and mineral salts absorbed from the roots to various parts of the plant.

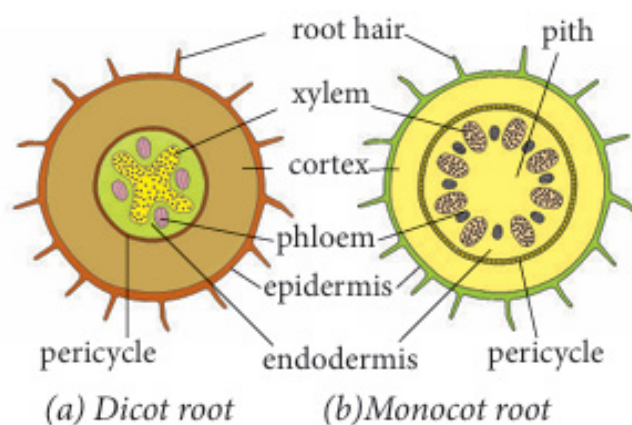


Fig 7.1: Transverse section of dicot and monocot roots

Application activity 7.1



1. What is the importance of the plant transport system?
2. What are the principal transport tissues of plants?
3. What substances are transported by each of the tissues?

7.2. Taking up Water and minerals

Learning activity 7.2



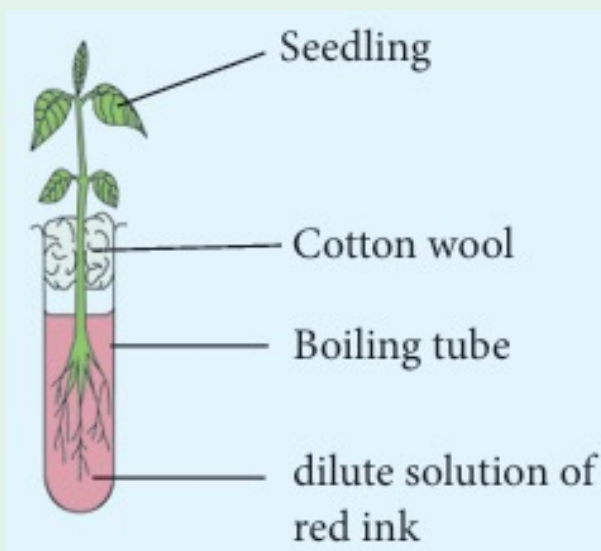
The investigation of the intake of water and mineral salts in plants

Requirement

- Boiling tube
- Dilute solution of red ink or dye
- Cotton wool
- Microscope or hand lens
- *Oxalis latifolia*

Procedure

1. Set up the apparatus as shown below.



2. Colour the solution with red ink or eosin stain.
3. Add a few drops of food colouring into the water.
4. Leave the set up for three or four hours then remove the plant and cut sections of the root, stem and leaf.
5. Examine the sections under low power of the microscope or with a hand lens.
6. What do you observe in the cut section?

Study questions

- a) Which tissues are stained red by the dye?
- b) Suggest a control for this experiment.

Water and mineral salts are taken up from the soil by the hair cells of the roots. To understand how this happens, we first look at root hairs and the root internal structure.

The distribution of tissues in a transverse section of the dicotyledonous root is not the same as that in the monocotyledonous root. The vascular bundle occupies a central position in both monocot and dicot roots. In a dicotyledonous root, the xylem occupies the centre and forms the shape of the star. The phloem is found between two stars. The vascular bundles are arranged in a ring. In a monocotyledonous root, the xylem and phloem are arranged in a ring with the xylem tissue alternating with the phloem tissue.

Root hair cells are found growing from the outer surface of the root to the tip of branch but not at the base of the branch. They absorb water and mineral salts. Adaptations of hair cells in the root:

- i. They are numerous to increase the surface area over which water and mineral salts are absorbed.
- ii. They are thin and fine, in order to penetrate the spaces between particles of the soil where water is present.

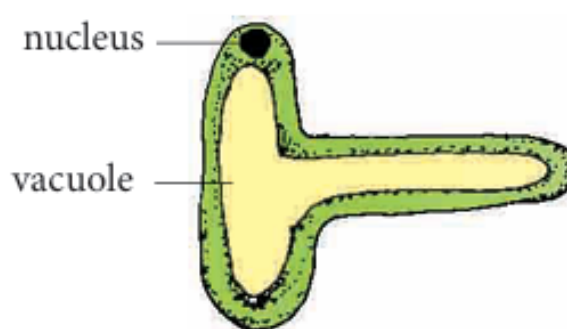


Fig. 7.2: Internal structure of a root hair cell

Soil particles are usually surrounded by water, except in drought condition. The hair cells of the root absorb water from the soil by osmosis.

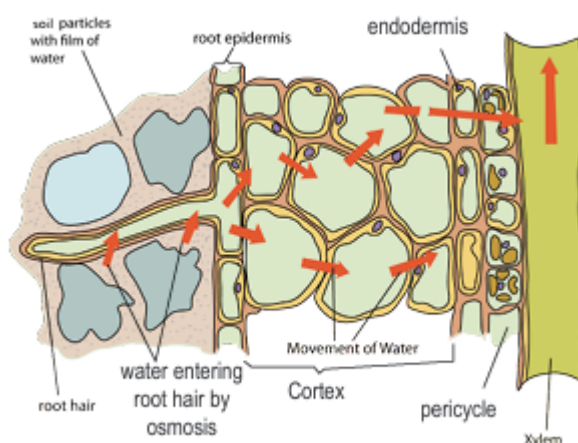


Fig 7.3: Absorption of water and how it moves from the root hair cell to the root xylem

The cell sap in the vacuole of the hair cell has high concentrations of salts and sugars. It is therefore hypertonic with water found in soil particles. because of this concentration gradient, water molecules are transported from soil by osmosis through the semi-permeable membrane of hair cell of the roots into the cell sap. Root hair cells absorb water if their salt concentration in the cell was higher than the soil concentration and if the pressure the cell wall is not high enough to prevent osmosis. Within the roots, water is transferred from cell to cell by osmosis. This is because

the neighbouring cells contain different amounts water. various theories have been proposed to explain water movement in plants. These include capillarity, root pressure, transpiration pull and the theory of cohesion tension...

a) Capillarity

In plants, xylems are narrow channels through which water flows. Water rises in a xylem due to strong forces of attraction between water molecules and the cell walls of the tubes or vessels of the xylem vessels, hence capillary action.

b) Root pressure is responsible for the rising of water up the stem.

When the water finally finds its way into the xylem vessels of the stem, the transport process is taken over by two other processes. The root pressure is shown when the tree trunk is cut and water is poured out, as shown in figure 7.4.

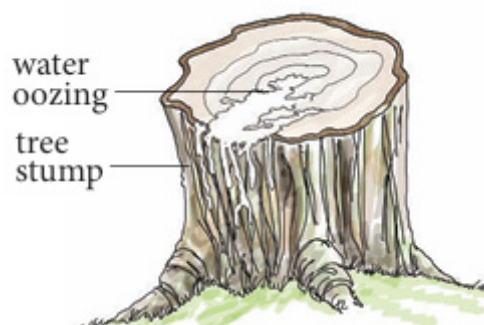
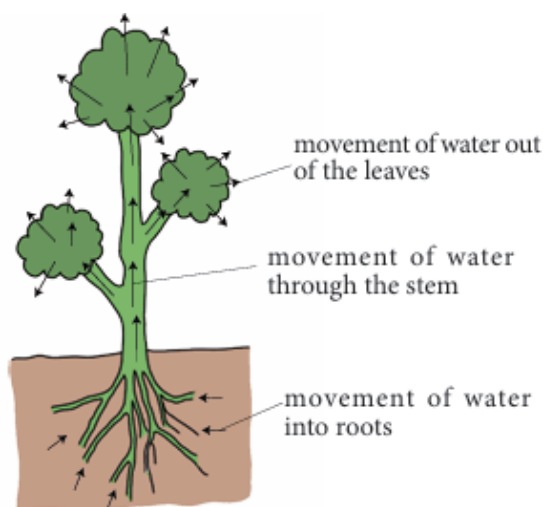


Fig 7.4: Water oozing from a tree stump

c) Transpiration pulls and cohesion tension theory

The transpiration pull is due to water being lost from the leaves. The leaves have small pores called stomata. Stomata are constantly losing water vapor. Because the water molecules stick together, a pull is formed, called the pull of the water. It replaces water lost through the dead, narrow xylem vessels. The adhesion forces between the water molecules and the walls of the xylem vessels allow water to rise. Cohesion forces allow the continuous water stream to rise up the plant. The constant movement of water in columns in the root xylem, the stem xylem and the leaf xylem towards the air spaces of the leaf is called transpiration stream.



Once absorbed into the hair cell of the root, the water and mineral salts must reach the xylem tissue that lies in the centre of the root. It is in the xylem that the water and mineral salts are transferred to other parts of the plant. Once in the root xylem, the water moves to the stem xylem where it finally reaches the leaf xylem.

Fig 7.5: Transpiration stream

Application activity 7.2



1. Why should plants have an elaborate transport system?
2. Differentiate between monocotyledonous and dicotyledonous roots.
3. With a diagram illustrate how mineral salts are transported from roots to leaves.
4. Describe the three differences between xylem and phloem.

7.3. Transpiration

Learning activity 7.3



Demonstration of the process of transpiration

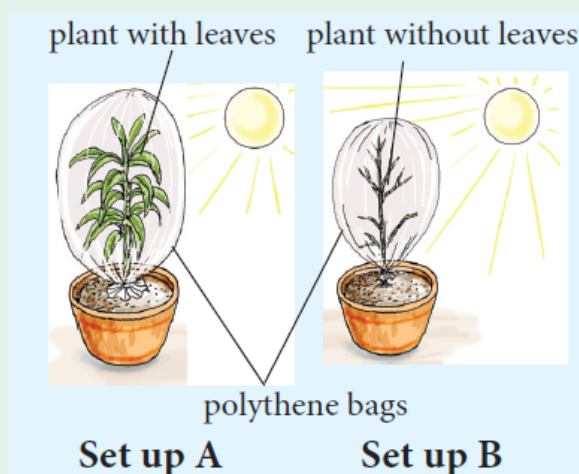
Requirements

- Potted plants (one with leaves, the other with its leaves removed)
- Two polythene bags
- Strings

2. Leave the two set-ups in a sunny place.
3. Collect and test the liquid collected in the plastic bag with anhydrous copper (II) sulphate or anhydrous cobalt chloride paper.

Procedure

1. Set up the potted plants and cover each with a polythene bag. Tie the polythene bag round the stem as shown below.



Study questions

- a) What observations have been made after several hours in set-ups A and B??
- b) What conclusion can be drawn from the observations?
- c) what is a control experiment and What is it used for
- d) What changes are noted in the anhydrous cobalt chloride paper?

Transpiration is the process where plants **lose water vapour from their aerial parts**, mainly via the **leaf stomata**. The stomata, the microscopic pores on the surface of the leaf, provides a network of internal air passage that allows the mesophyll cells to take up the carbon dioxide they need to photosynthesise. The air in these spaces is saturated with water vapour because it is in contact with the wet leaf walls.

On most days, the air outside the leaf is dried and therefore has a lower water potential than the air inside. Thus, gaseous water as it passes through its concentration gradient, leave the leaf through the stomata. It's this loss of water vapor from the leave that we call **transpiration**.

Water lost through the stomata is replaced by water drawn from xylem of the leaf xylem.

Importance of transpiration in plants

Water is important to a plant's survival. However, the transpiration causes a great loss of water from the plant. Too much transpiration may be detrimental to the plant because it may interfere with various biological processes. Also, excessive water loss by transpiration may lead result in a decrease of cell turgidity and consequently loss of mechanical support in the plant.

Though may seem like a waste of water, transpiration performs four critical functions:

- **Nutrient Transport:** It creates the suction needed to pull minerals and water from the roots up to the highest leaves.
- **Cooling:** Just like human sweat, the evaporation of water from leaves cools the plant down during hot weather.
- **Turgidity:** It keeps the plant cells "inflated" (turgid), which provides structural support and prevents the plant from wilting.
- **Photosynthesis:** It facilitates the opening of stomata, which allows the plant to take in carbon dioxide for making food.

Types of transpiration

Transpiration may occur from the following three sites:

- **Stomata (stomatal transpiration):** by evaporation of water from cells and diffusion of the water vapour through stomata, the pores found in the epidermis of leaves and green stems. (About 90 % of the water is lost this way.)
- **Cuticle (cuticular transpiration):** by evaporation of water from the outer walls of epidermal cells through the waxy cuticle covering the epidermis of leaves and stems. (About 10 % of the water lost, varying with thickness of cuticle.)
- **Lenticels (lenticular transpiration):** by evaporation of water through lenticels. These are small slits in the stems and bark of trees for gas exchange. (Minute proportions, although this is the main method of water loss from deciduous trees after leaf fall.)

Application activity 7.3



1. Explain why plants grown in a closed environment (greenhouse) have a lower rate of transpiration than plants grown in an open field?
2. Explain the following observation. The freshly cut tree trunk will continue to release water for a while.

7.4. Translocation of organic foods

Learning activity 7.4



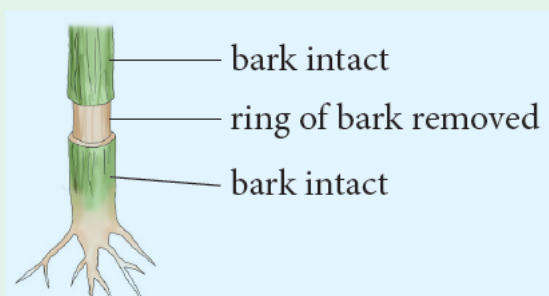
Ringing experiment

Requirements

- Tree or shrubs with many branches
- Sharp knife

Procedure

1. Remove completely a ring of bark with its phloem from two branches. The xylem tissue which makes up the bulk of the stem is left intact.



2. The setup is left undisturbed for four weeks.

Study question

- a) What observations have been made in the stem after four weeks of treatment of the stem? Explain why?
- b) Discuss with your friends how to harvest the bark of medicinal trees without killing these important trees.

Photosynthetic products (photosynthates) are the substances produced by plants during photosynthesis. They are usually simple sugars, such as sucrose. Photosynthates are formed in sources and are transferred to sinks. The points where sugar is supplied, such as roots, young shoots and developing seeds, are referred to as sink points. Seeds, tubers and bulbs may be source or a sink, depending on the stage of development of the plant and the season.

- The movement of organic products of photosynthesis from leaves to other parts of the plant is called **translocation**.
- Sugars are produced in leaves (source), but other non-photosynthetic parts of the plant such as roots and stems also require a certain amount of sugar to be fed. Therefore, organic food products are translocated from source to sink.
- Some important sinks are roots, flowers, fruits, stems and emerging leaves. Leaves are particularly interesting in this regard because they are a sink when they are young and a source later, when they are about half-grown.
- Organic products of photosynthesis (photosynthates) are translocated through the phloem tissue.

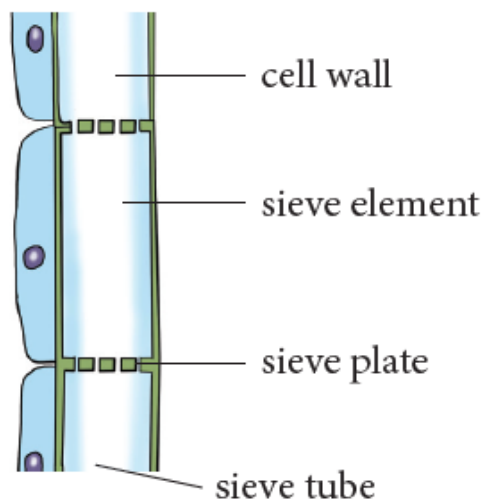
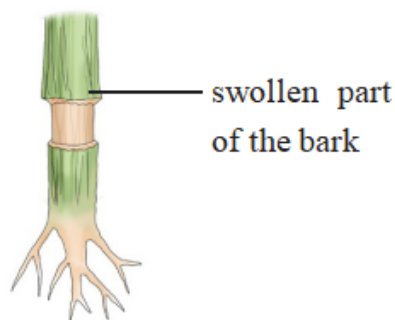


Fig 7.6: Phloem tissue

Photosynthetic products are directed primarily to the roots during early development, to shoots and leaves during vegetative growth, and to seeds and fruits during reproductive development. The products from the source are usually translocated to the nearest sink through the phloem.

The high percentage of sugar in phloem sap causes water to move from the xylem into the phloem. This increases water pressure inside the phloem, causing the sap to move from source to sink. When the ring of bark is removed, the phloem beneath it is also removed. After several weeks, swelling above the cut ring is noted.



The swelling is caused by the accumulation of substances in food. They were being transferred from the leaves, but they couldn't get through the debarked portion of the stem. That is why the lower part of the ring doesn't swell.

Fig 7.7: Swollen tissue above the cut part after four weeks.

Application activity 7.4



1. Differentiate between a source and a sink?
2. The movement of organic products of photosynthesis from to other parts of the plant is called.....

7.5. Adaptations of plants to different environmental conditions

Learning activity 7.5



Examination of the adaptive features of different plants

You will be provided with samples pictures and photographs of different plants.

Procedure

1. Study each of the plants provided and note the following for each plant.
 - a) Their leaves, stems and roots
 - b) Where they are found
 - c) Their sizes
2. Why do you think plants have the features above? Find out from textbooks or the internet.
3. Share your findings with other class members.

Plants are structurally modified in ways that reduce water loss, depending on the environment in which they live. The following are some examples of ways in which plants growing in different environments reduce their rate of transpiration

a) Xerophytes

Xerophytes are plants with properties suitable for areas with very little water and very high temperatures. These plants grow in arid regions, such as arid and semi-arid regions.

Adaptations of xerophytes to their habitat

1. They have high capacity to absorb soil water. Their roots are usually more developed and grow deep into the soil and extend over a large area.
2. They have a water retaining tissue. Many xerophytes have swollen stems and leaves that contain special tissues that store water. These plants are called succulent plants. Examples of succulent plants are *Bryophyllum*, sisal, cactus, Euphorbia, and *Aloe vera*. A baobab tree has a thick stem for water storage.



Fig 7.8: Cactus plant with leaves reduced to thorns in order to reduce area for transpiration

3. Xerophytes reduce water loss through transpiration in many ways:
- a) They have a dense waxy cuticle that prevents excessive water loss from the leaf by evaporation.
 - b) They have hairs which trap moist air near the leaf surface and cause saturation.
 - c) In certain xerophytes, such as *Nerium oleander*, stomata are found sunken in pits. The rate of transpiration is reduced because there is a space of moist air between the stomata and the air.
 - d) Many xerophytes have small leaves to reduce the area of their surface for transpiration.
 - e) Some xerophytes have very few stomata, which are located in the lower epidermis, which is away from direct sunlight. Others have reversed the stomata rhythm. Their stomata open at night and close during the day, unlike in normal plants.
4. Some xerophytes have life cycles that allow them to avoid drought conditions, for example some of them shed their leaves in the dry season.

b) Mesophytes

Mesophytes are plants that grow under average conditions of water and temperature. These plants grow well on the soil and develop into forests and grasslands.

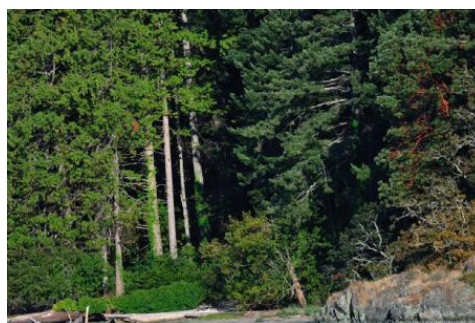


Fig 7.9: A forest

Adaptations of mesophytes to their habitats

1. They have thin leaves that allow carbon dioxide to diffuse rapidly from the stomata to the photosynthetic cells.
2. They have broad, flat leaves that provide a large surface area for absorbing light and carbon dioxide.
3. The mosaic arrangement of leaves on the plant to ensure that each leaf receives the maximum amount of sunlight.
4. Presence of stomata on the epidermis of the upper and lower epidermis of leaves to allow efficient exchange of gases as well as transpiration.
5. The internal structures of their leaves have an air passage that allows them to circulate freely.
6. Their leaves have chlorophyll cells, allowing photosynthesis to take place.
7. They have thick transparent cuticles to avoid water loss.
8. They have a well-developed root system with long tap or fibrous roots.

c) Hydrophytes

Hydrophytes are plants that live in moist places or water. Examples are *Nymphaea* and water hyacinth.

There are three types of water plants:

- i. **Emergent plants** : they have roots and part of stem under water, while their leaves are above water. They have a problem of taking in excess water, for example, reeds.
- ii. **Floating plants** : they float on the water surface with roots in water, for example, in water lilies.
- iii. **Submerged plants**: these are found completely under water, for example, spirogyra.

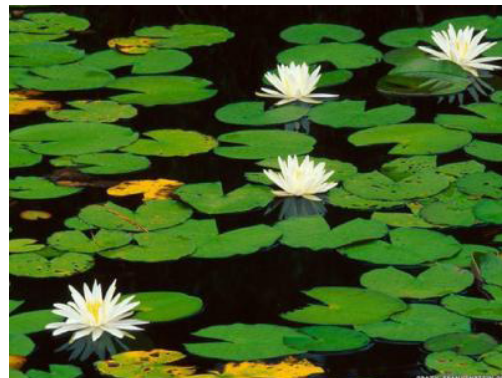


Fig 8.10: Water lilies

Adaptations of hydrophytes:

- a) Their cuticle is thin or absent. This allows the plant to absorb water, minerals and carbon dioxide throughout the entire surface of the plant.
- b) As some hydrophytes absorb water throughout their body, there they do not have well-developed roots. The roots can be used for anchorage or used for absorption of minerals, e.g in water lilies.
- c) The presence of a number of air passages in the stem and leaf tissue; a special tissue called **aerenchyma, which** makes the plants buoyant and allow for support and exchange of gas.
- d) They contain small amounts of xylem and supporting tissue. They are supported by aerenchyma and the water buoyancy.
- e) The submerged leaves have no stomata, and the floating types have many on the upper surface.

d) Halophytes

They are plants that grow in saline environment such as rocky coasts, seas and sand dunes along the coast. Some halophytes, such as *Atriplex* and mangroves, grow close to the surface of the sea.



Fig 7.11: Mangrove plants

So, they have a problem of extracting water from their salty environment. They have cells that soak up salt. As a result, they create higher osmotic pressure, which allows the plant to absorb water. Because they consume a lot of salt, they expel excess salt by means of salt glands.

The salt is washed away from the plant surface by rain. Some halophytes absorb salt from their environment and remove it by shedding leaves that have been saturated with salt.

Application activity 7.5



1. What is the significance of modified leaves in xerophytes?
2. Halophytes are found in salty environments. How do they avoid having too much salt in their tissues?
3. Mutuyimana found a plant without stomata. Where do you think he found that plant?

7.6. End of Unit Assessment



7.6.1. Project activity: Investigating the Process of Transpiration in Plants Using a Potometer

1. Context/Background

Plants continuously lose water vapour from their leaves through a process called transpiration. This process plays an important role in the transport of water and mineral salts from the roots to the rest of the plant, helps cool the plant, and contributes to the movement of water through the xylem. Environmental factors such as light, temperature, humidity, and wind can affect the rate of transpiration. A potometer is a scientific apparatus used to estimate the rate of water uptake by a plant, which is closely related to the rate of transpiration.

2. Challenge/Problem Statement

Plants lose water through transpiration, but the rate of water loss may vary under different environmental conditions. Investigate the process of transpiration and determine how environmental factors influence the rate of water uptake in a plant shoot using a potometer.

Key Question

How can a potometer be used to investigate transpiration, and how do environmental factors affect the rate of transpiration in plants?

3. Purpose of the Project

The purpose of this project is to help students understand the process of transpiration and its importance in plants. Students will use a potometer to measure water uptake, investigate factors affecting transpiration, and interpret results using scientific principles related to plant transport.

4. Instructions

- Work in groups.
- Assemble or observe a potometer set-up with a healthy leafy shoot.
- Introduce an air bubble into the capillary tube and mark its starting position.
- Measure the movement of the air bubble over a fixed period to estimate water uptake.
- Investigate one environmental factor (light intensity, wind, temperature, or humidity) while keeping other factors constant.
- Record all measurements and observations accurately.
- Present results using tables, graphs, and written explanations.

5. Your Tasks

- Set up and use the potometer correctly.
- Measure water uptake under different environmental conditions.
- Record and organize data systematically.
- Compare transpiration rates under the different conditions tested.
- Analyze the relationship between environmental factors and transpiration.
- Draw conclusions based on the evidence collected.

6. Expected Student Outputs

- A labelled diagram of the potometer.
- Observation and measurement tables.
- Graphs showing the effect of environmental factors on transpiration rates.
- Analysis and interpretation of results.
- A project report including introduction, methodology, results, discussion, conclusion, and recommendations.
- A group presentation of findings.

7. Constraints and Limitations

Scope

The investigation should focus on transpiration in a leafy plant shoot and the effect of one selected environmental factor on the rate of transpiration.

Materials and Methodology

Use a potometer, leafy shoot, water, stopwatch, ruler, petroleum jelly (where applicable), and other simple materials required to manipulate environmental conditions. Ensure that measurements are taken consistently and accurately.

Timeframe

The project should be completed within **2–3 weeks**, including planning, experimentation, data analysis, report writing, and presentation.

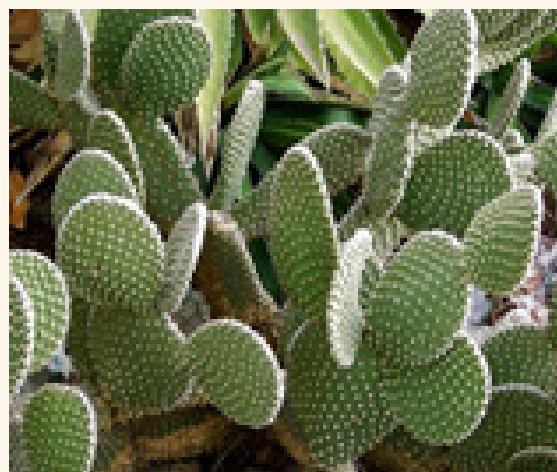
8. Guiding Questions

1. What is transpiration, and why is it important to plants?
2. How does a potometer help estimate the rate of transpiration?
3. Which environmental factor are you investigating, and how will you control the other variables?
4. What changes occur in the movement of the air bubble under different conditions?
5. How do your results explain the effect of the selected factor on transpiration?
6. What conclusions can you draw about the relationship between environmental conditions and transpiration rate?

7.6.2. Assessment questions

Knowledge and understanding

1. Removal of the ring of bark, from the tree trunk primarily kills the tree. Why is that?
 - a) Supply of food to the roots is cut off.
 - b) Supply of carbon dioxide to the leaves is cut off.
 - c) Supply of water to the leaves is cut off.
 - d) Trunk loses the required strength and protection
2. One of the following transports manufactured food away from the leaves. Which one is it?
 - a) Xylem
 - b) Phloem
 - c) Petiole
 - d) Epidermis
3. Explain why in a unicellular organism, an elaborate transport system is unnecessary.
4. What mechanism do plants use to absorb nutrients?
5. What is the adaptation of the plant below to its habitat?



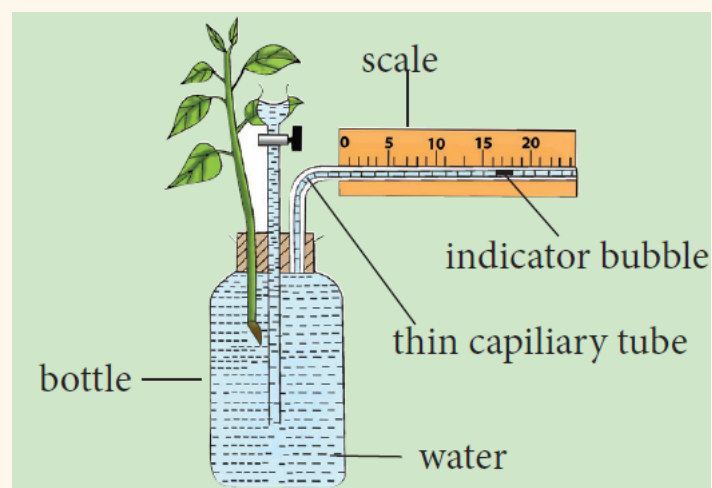
6. Describe the differences between xylem and phloem by filling the following table.

Xylem	Phloem

7. What is the purpose of companion cells in the phloem tissue?
8. (a) Transpiration is a necessary evil. Explain
(b) Identify three environmental conditions that increase the rate of transpiration.

B. Skills

9. The instrument shown below is used to measure the rate of transpiration. Explain how it works.



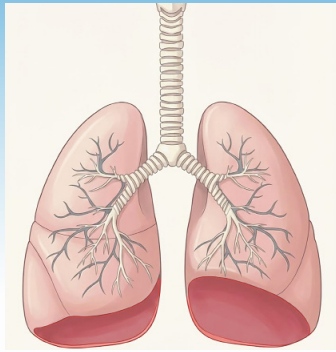
10. Give an explanation as to why, when a cut leafy shoot is placed in a beaker of water containing red dye, its leaves start to turn red thirty minutes later.
11. Amoeba uses simple diffusion for all its requirements while plants and animals use a complex transport system. Suggest reasons for this difference.

C. Attitudes and values

12. How does understanding transpiration encourage environmental conservation?
13. Why should farmers water crops early in the morning or late in the evening?

UNIT 8

GASEOUS EXCHANGE IN HUMANS AND PLANTS



Key Unit Competence:

Develop models demonstrating gaseous exchange in humans and plants.

Learning objectives

By the end of this unit you should be able to:

Knowledge and understanding

1

- Explain gaseous exchange process and its importance.
- Explain the features and adaptations of gaseous exchange surfaces.
- Observe the process of gaseous exchange at the alveolus.
- Discuss common respiratory diseases and suggest preventive measures and treatment.
- Explain the process of gaseous exchange in plants.

Skills

2

- Demonstrate the processes of inspiration/inhalation and expiration/exhalation.
- Conduct a dissection of a small mammal to display its respiratory system.
- Construct a bell-jar model of mammalian thoracic cavity from locally available materials.
- Demonstrate the movement of air into and out of the lungs during breathing.

Attitudes and values

3

- Appreciate learning enhanced by the use of available local materials.
- Value practices that enhance healthy gaseous exchange organs

8.0. Introduction

Gaseous exchange is an essential life process through which organisms obtain oxygen and remove carbon dioxide. In humans, oxygen is needed for respiration to release energy from food, while carbon dioxide produced during respiration must be removed from the body. In plants, gaseous exchange supports both photosynthesis and respiration, allowing carbon dioxide and oxygen to move between the plant and its environment. Although humans and plants have different structures for gaseous exchange, both rely on the diffusion of gases across specialized surfaces.

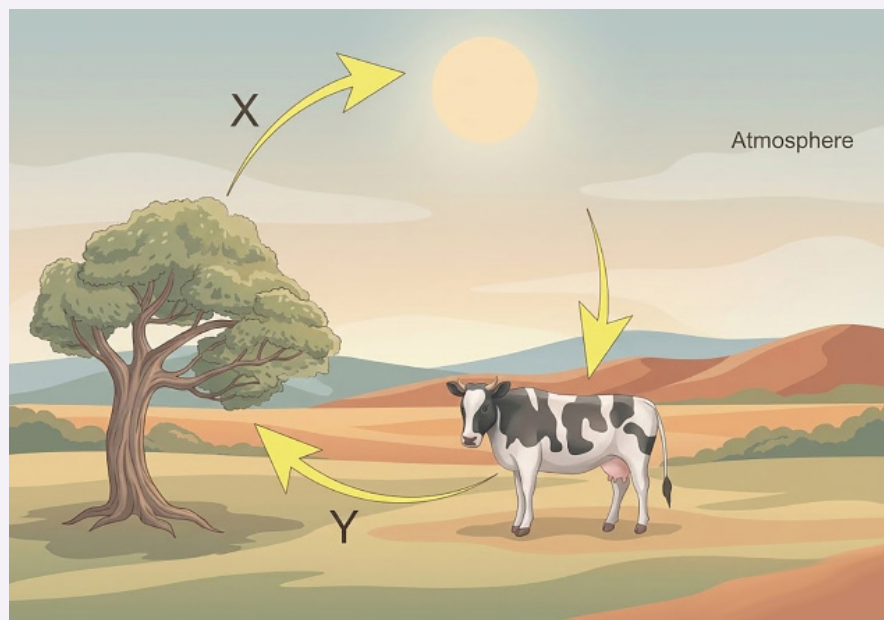
This unit introduces students to the structures involved in gaseous exchange in humans and plants, the mechanisms by which gases are exchanged, and the factors that influence the efficiency of this process. Understanding gaseous exchange helps explain how organisms obtain energy, maintain life processes, and interact with their environment.

Key Concepts: gaseous exchange, diffusion, respiration, photosynthesis, oxygen, carbon dioxide, lungs, alveoli, breathing, ventilation, stomata, lenticels, intercellular air spaces, concentration gradient, exchange surfaces.

Introductory activity



In Senior One, you learned about gas exchange in human beings. How did you define gas exchange? What did you learn about its importance? Look at the carbon/oxygen cycle below, then answer the related questions.



1. What is the importance of the cycle shown in the diagram?
2. Can you identify the gases X and Y?
3. What is their relevance to each organism represented in the cycle?
4. Based on the diagram above, you can predict what you will learn in this unit.

8.1. Concept of respiration and respiratory surfaces

Learning activity 8.1



Discussions on the concepts of respiration and respiratory surfaces

Find out the following with your classmates and write down in your notebooks

- What is a respiratory surface?
- What are the characteristics of respiratory surface?
- Identify the surface of gas exchange in humans.
- The difference between cellular respiration and gas exchange.

Respiration is a process by which food consumed by organisms is burnt to produce the energy they need to function. During respiration, oxygen is used, and carbon dioxide is eliminated.

Cellular respiration occurs continuously in all living cells. It creates a constant demand for oxygen and a need to get rid of the carbon dioxide gas. Gas exchange is a biological process whereby gases are transported through an organism's body through a specialized respiratory surface. Organisms like plants need oxygen and carbon dioxide to be taken up or released at some point during their respiration and photosynthesis. Animals, on the other hand, always take oxygen and release carbon dioxide during respiration. Gas exchange is therefore:

- Exchange of respiratory gases in the respiratory tract of animals.
- Exchange of gases involved in photosynthesis and respiration in plants.

An efficient system of gas exchange is therefore of great importance for living organisms. Gas exchange is necessary because organisms can extract useful gases from their environment and dispose of waste gases in the environment.

Environments with which organisms exchange gases include air for some organisms and water for others. The main sources of oxygen and carbon dioxide are in the air. Organisms living on land exchange these gases directly with air. Air-generated oxygen and other gases are released into lakes, rivers and oceans. Air dissolved in water is used to exchange gases by organisms living in water.

In large multicellular animals, the ratio of surface area to volume is small. Many cells are deep inside the animal's body, far from the surface of the organism. Gas diffusion alone is not powerful enough to move gases to and from all cells. Large multicellular animals, therefore, have special structures or organs with special surfaces on which gases are exchanged. These special surfaces are called **respiratory surfaces**. Examples of specialized structures for the exchange of gas in animals include:

- Cell membrane
- Tracheal system in insects
- Buccal cavity in frogs
- Skin in frogs
- Gills in fish
- Lungs in mammals, birds, reptiles and amphibians.

The gas exchange occurs through the respiration surface. The respiration surface has a number of characteristics that make it efficient for the exchange of gases. Some of these features include:

- Thin walls for faster diffusion of gases across it.
- It is moist to dissolve gases as they diffuse across them.
- It has a large surface area for maximum gas exchange.
- In animals with a transport system, the respiratory surface has a rich supply of blood capillaries (highly vascularized) to quickly transport gases to and from the cells.

Not all surfaces of the respiratory track are in direct contact with the medium through which the gas is exchanged, such as water or air surrounding organism. A procedure is therefore needed to ensure a constant supply of fresh water or air to and from the respiratory surface. This is achieved by the process of **ventilation** that continuously moves water or air oxygen content to the respiration surface. It also removes water and air that contains high amounts of carbon dioxide. Ventilation is therefore important in maintaining a high gradient of diffusion at the respiration surface. It also ensures a high exchange rate for gas. Breathing is one example of a ventilation system.

Application activity 8.1



1. What are the properties of respiration surfaces?
2. What is the difference between respiration and gas exchange?
3. What is a human respiratory surface?

8.2. Mechanism of breathing in humans

Learning activity 8.2



Observation of the movement of the chest during breathing

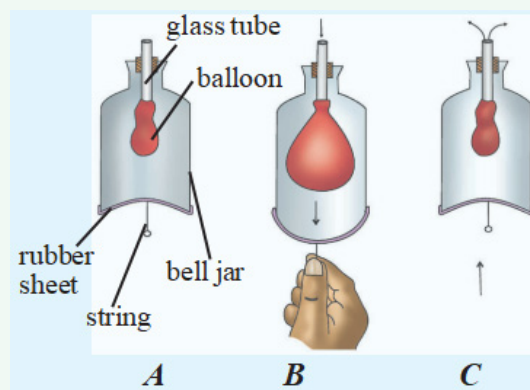
- Ask your partner to take a deep breath and then hold it.
 - Describe what you see happening to your partner's chest as he or she takes a deep breath in.
 - Does it remain in the same position? Let your partner now breathe out.
 - What do you see happening to the chest? Record the form of movements noticed. What is the difference between inhalation and exhalation?
- Constructing and using a model to demonstrate respiration mechanism

Requirements

- Bell jar
- Two balloons
- Rubber stopper with a hole
- Y-Shaped glass tube
- Rubber sheet and rubber band
- String

Procedure

1. Assemble the materials as shown below.



2. Study the model and show what the following sections represent in the human respiratory system
 - Bell jar
 - Balloon
 - Rubber sheet
3. Pull down the rubber sheet at the bottom of the bell jar.
 - What happens to balloons?
4. Now push up the rubber sheet.
 - What happens to balloons?

Study questions

- a) Draw sketches to show the appearance of the balloons in (2) and (3) above.
 - b) Explain your observation.
5. Explain how the model is similar to the working of the thorax in human beings during breathing.

In humans, the process of breathing (ventilation) is the first step in the process of gas exchange. The other step is the exchange of these gases between blood and lungs. Breathing provides a constant supply of fresh air to the surface of gas exchange. It also helps maintain high gradient of diffusion along the surface of the gas exchange. The volume of gases exchanged during respiration varies depending on physiological demands, such as exercise. The respiration rate is controlled by brain's respiratory center.

The ventilation(breathing) rate is affected by certain factors such as blood carbon dioxide concentration, physical activity, fear, and excitement.

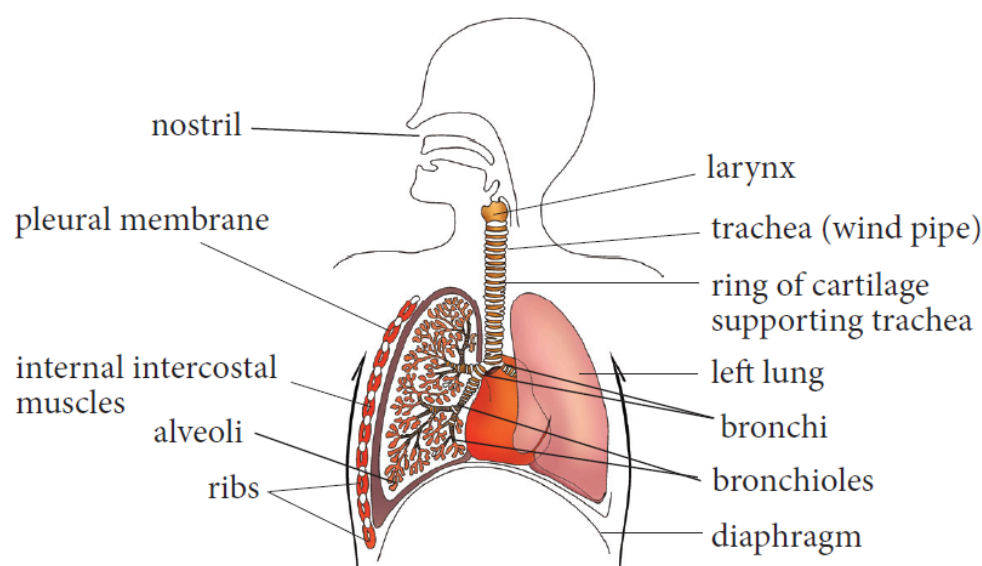


Fig 8.1: Human respiratory system

The process of bringing air into the lungs and then clearing it out again is called respiration or breathing. It consists of two phases called **inhalation** and **exhalation**. Because the lungs don't contain any muscle tissue, they can't move independently movement. However, they are elastic and are forced to expand or contract during respiration due to pressure variations around them. These pressure changes are due to the movement of the diaphragm muscle, ribs and intercostal muscles (rib muscle) and to the atmospheric pressure.

a) Inhalation (breathing in)

Inhalation is also known as inspiration. This is the active phase of respiration pulling air into your lungs. During inhalation, the diaphragm muscles contract, causing the airways to compress. Look at Fig. 8.2. In the area of the ribs the external intercostal muscles are contracting while the internal intercostal muscles are relaxing. This causes the rib cage to move inwards and outwards.

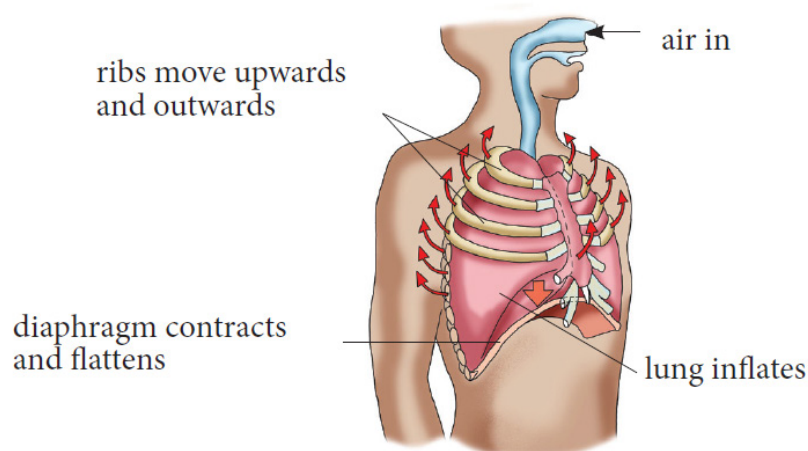


Fig 8.2: Inspiration (Inhalation)

The contraction of the diaphragm and external intercostal muscles increases the volume of the chest cavity. However, it causes the pressure of air inside to be reduced

relative to the air outside. The Air is pushed through the airways into the lungs, forcing them to expand.

b) Exhalation (breathing out)

Exhalation is also called expiration. This is the phase of respiration that pushes air out of the lungs. During exhalation, the diaphragm muscle relaxes causing the airway to move upward and regain their dome shape. The external intercostal muscles relax and the internal intercostal muscles contract. This causes the rib cage to move inwards and downward. The volume of the chest cavity is reduced, and the pressure is increased in comparison with air. The increased pressure forces air out of the lungs, causing them to be deflated.

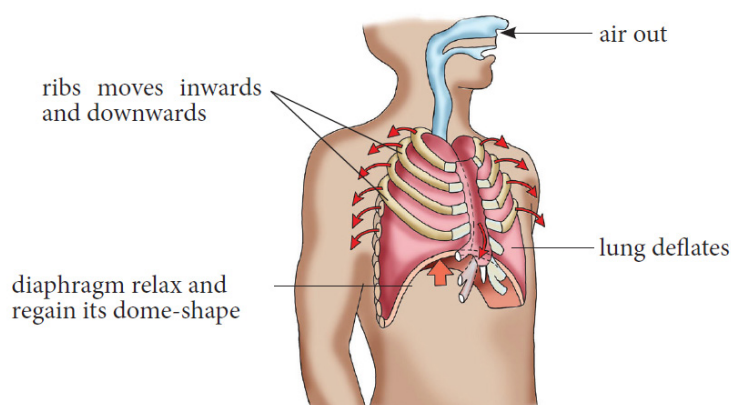


Fig 8.3: Expiration (Exhalation)

Table 8.1: Differences between inhalation and exhalation

Inhalation	Exhalation
1. External intercostal muscles contract. Internal intercostal muscles relax.	1. External intercostal muscles relax. Internal intercostal muscles contract.
2. Rib cage moves upwards and outwards.	2. Rib cage moves downwards and inwards.
3. Diaphragm muscles contract and diaphragm flattens.	3. Diaphragm muscles relax, and diaphragm forms a dome shape.
4. Volume of the thoracic cavity increases.	4. Volume of the thoracic cavity decreases.
5. Air pressure in the lungs and thoracic cavity decreases compared to external atmospheric pressure.	5. Air pressure in the lungs and thoracic cavity increases compared to external atmospheric pressure.
6. External air is driven into the lungs due to the pressure difference between the inside and the outside.	6. Air in the lungs is compressed and forced out.
7. Lungs inflate	7. Lungs deflate

Application activity 8.2



1. What role do the nasal passages and diaphragm play in the process of gas exchange?
2. Describe the mechanism and the characteristics of a respiratory surface.
3. Describe the mechanism by which carbon dioxide in the capillaries of the blood around the alveoli is transferred to the alveoli and finally released out.

8.3. Gas exchange in the alveoli

Learning activity 8.3



Dissection of a small mammal to display the structures of the respiratory system

Requirements

- Dissection board
- Pins
- Scissors
- Scalpels
- Cotton wool
- Rat or rabbit (freshly killed)
- Forceps
- String
- Rubber pipette
- Gloves

1. Pin the animal onto the dissection board with the ventral side up.
2. Open the thorax by cutting with scissors along the midsection from the neck to the hind legs.



3. Cut the skin further to expose the abdomen as shown in Figure 9.6 above. Note the following:
 - Muscles of the diaphragm.
 - Intercostal muscles between the ribs.
 - Lungs.
4. Remove the lungs and place on the board.
 - How does it feel to touch?
5. Cut through the lungs.
 - What do you see?
 - Identify the main airways.
 - Note your observations.
 - Share your findings with other class.

When air travels down the trachea, it passes through the nose or mouth. The trachea is divided into a left and a right bronchus. Each bronchus is divided into smaller bronchioles, each bronchiole in turn is connected to a number of alveolar sacs. The clusters of alveoli have the same shape as a bunch of grapes.

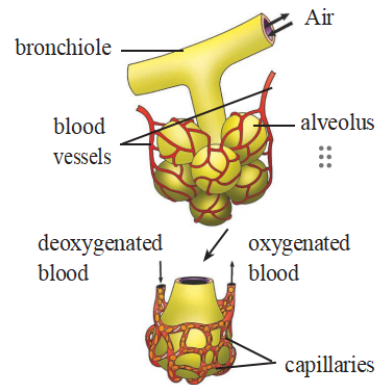


Fig 8.4: Alveoli

The exchange of gases in the alveoli occurs between the inhalation and exhalation phases. The alveoli are suitable for gaseous exchange because:

- They are supplied by blood, which carries the exchanged gases.
- They have a very thin wall through which gases are diffused
- They are lined with a thin film of water to dissolve the diffusing gases.
- The ventilation process draws in and out air containing the gases being exchanged.
- It has a large number of alveoli, which increases the surface area for gas exchange

Gas exchange between air in the alveoli and capillaries of the lungs occurs by diffusion. Oxygen in the air and in the alveoli is higher than in the blood capillaries. It is therefore first dissolved in the water in the alveoli lining then diffuses across the alveoli and then through capillary walls to the red blood cells. This is oxygenated blood which is transported to the heart by the pulmonary vein.

Carbon dioxide in the blood diffuses through the capillary and alveolar walls to the alveolar space where it is finally removed in exhalation.

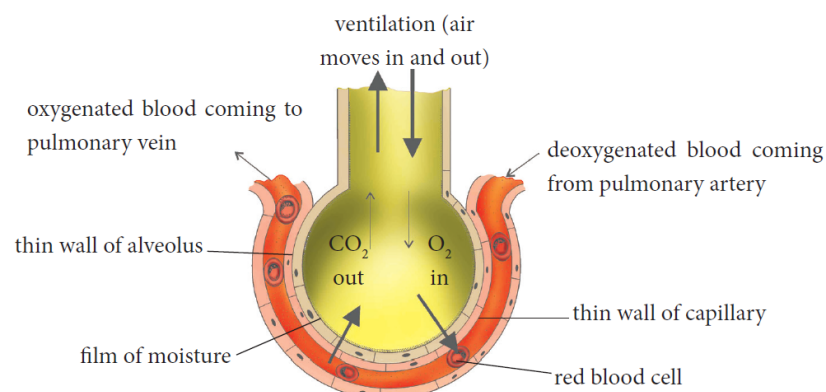


Fig 8.5: Gaseous exchange in the alveolus

The diffusion gradient is necessary for the rapid gas exchange in the alveoli. The following factors contribute to maintaining this diffusion gradient.

- Lung ventilation:** Breathing movements transport gases to and from the alveoli.
- Blood flow:** it continuously replaces the oxygenated blood with deoxygenated blood.

Pulmonary artery carries the blood with low oxygen and high carbon dioxide concentrations. Pulmonary vein drains blood with high concentrations of oxygen and low concentrations in carbon dioxide.

- c) **Haemoglobin:** rapidly combines with oxygen and prevents it from accumulating in the alveoli

Application activity 8.3



1. Explain the way gases cross the alveolus
2. Identify the characteristics of an alveolus that makes it a good respiratory surface
3. Why should an alveolus have a large surface area?
4. What is the importance of moisture on the surface of the alveolus?

8.4. Respiratory diseases and smoking

Learning activity 8.4



Research-based learning on respiratory diseases and smoking

1. Use your handbooks, textbooks, photographs, and the internet for research on the following topics:
 - a) Respiratory diseases.
 - b) Causes of respiratory diseases in humans.
 - c) Prevention and treatment of respiratory diseases.
2. Share your findings with the class. disorders. Some are caused by microorganisms, while others result from genetic factors. The most common respiratory diseases include Tuberculosis, asthma, pneumonia, bronchitis, whooping cough, common cold and influenza.

1. Asthma

This is a disease that occurs when air passages in the lungs become suddenly narrower due to the contraction of their smooth muscles. It is also characterized by the influx of mucus, which clogs the narrow passages even more. Some substances can cause asthma attacks. They are called **allergens**. They include pollen grains, certain types of proteins in milk, pet hair, dust and even flavors in food. It may also be caused by stress and anxiety. In some families, the disease is hereditary.

Symptoms of asthma

- Difficulty breathing. Breathing may be so difficult or rapid that the patient may faint.
- Wheezing sounds when breathing.

Prevention and treatment

Asthma is treated with two types of medicines: long-term control and quick-relief medicines that help to reduce the inflammation of the airway and prevent asthma symptoms.

- Quick-relief medicines relieve the symptoms of asthma that flare up.
- People with asthma should avoid an environment likely to cause asthma attacks, by avoiding contact with allergens.

Asthma sufferers are advised to carry inhalers containing a medicine that helps to relieve the condition.



Fig 8.6: Asthma patient using an inhaler

2. Bronchitis

It is an infection of the inner walls of the bronchi, caused by bacteria or air pollutants such as smoke in the air we breathe. The condition causes the excess mucus to build up in mucous membrane of the respiration airway. This causes the cells that line the bronchi and bronchioles in the lungs do not work properly. As a result, the passage of air to the alveoli is blocked by mucus, and severe coughing occurs as the patient tries to clear the excess mucus. It becomes difficult to breathe, and bronchitis may be acute or chronic.

Acute bronchitis usually starts very quickly and stops withing few days. Acute bronchitis symptoms are similar to a cold.

Chronic bronchitis starts slowly and lasts long It's a much more serious condition. This is usually due to smoking and air pollution.

Symptoms of bronchitis

- Secretion of excess mucus
- Coughing
- Difficulties in breathing.

Prevention and treatment

- Acute bronchitis can be treated with simple measures such as getting plenty of rest, drinking plenty of fluids, and using a cough suppressant.
- People with chronic bronchitis should use antibiotics when prescribed, especially if they have a fever.
- A doctor should be consulted at the early stages of bronchitis.
- Avoid smoking whether directly or passively.
- Avoid polluted air.

3. Emphysema

This is due to long-term untreated bronchitis, when the bronchioles in the lungs become blocked. This causes damage to the delicate alveoli walls as a result of the high pressure exerted by coughing. This results in a reduction of the surface area for the gas exchange. The lungs become so compressed and inelastic that the gases cannot be exchanged effectively. The patient becomes weak because of the lack of oxygen to the tissues. Running and walking can be difficult if one has this condition

Prevention and treatment

Emphysema is treated according to the severity of the symptoms. Bronchodilators are normally used to help with coughing, shortness of breath and breathing difficulties.

Early antibiotics treatment of bronchitis to prevent secondary infection may help prevent the development of emphysema

4. Pneumonia

Pneumonia is a lung infection. It is caused by a bacterium called *pneumococcus*, which is spread by air. It may also be caused by a virus or fungus. The infection spreads from the mouth to the lungs. As a result of the infection, a liquid is formed that collects in the alveoli. The lungs become solid and there is no air in them. This prevents the gases from circulating in the lungs.

Signs and symptoms

- Sudden chills and high fever.
- Rapid shallow breathing and sometimes wheezing.
- Cough with yellow, greenish colour or mucus with some blood.
- Chest pains.

Prevention and treatment

- Crowded places should be avoided, and good ventilation in the living room should be provided.
- Treatment of pneumonia includes treatment of the infection and prevention of complications. It also depends on the causative agent: bacterial, viral, or fungal.
- Bacterial and fungal pneumonia are treated with medicines while viral pneumonia is cleared on its own.

5. Tuberculosis

Tuberculosis (TB) is caused by a bacterium called *Mycobacterium tuberculosis*. The source of infection may be bacteria droplets that are blown out of airways when someone breathes or sneezes. It may also be caused by infected dry sputum in dust particles. Tuberculosis bacteria can attack any part of the body, but they usually attack the lungs, causing pulmonary tuberculosis. Drinking raw milk from cows suffering from **bovine** tuberculosis is another source of infection.

Signs and symptoms

- Lung tuberculosis starts with dry cough, followed by blood spatter, fever and night sweats as the infection progresses.
- If left untreated, the patient will lose weight and eventually die.



Fig 8.7: Severe coughing may be a sign of TB

In addition to tuberculosis of the lungs, there are other forms of disease where bacteria attack the lymphatic glands, bones and other parts of the body.

Prevention and treatment

- Patients should consult their doctors for appropriate treatment. Treatment of tuberculosis usually involves a long course of antibiotics for 6 to 9 months.
- Overcrowding increases the risk of spread of tuberculosis.
- Avoid drinking raw milk. Boil all milk or drink pasteurised milk.
- Immunization with B.C.G. vaccine in children.
- Isolating patients

6. Whooping cough

Whooping cough is caused by a bacterium called *Bordetella pertussis*. The mode of transmission is from person to person by inhalation of infected droplets.

Signs and Symptoms

- Whooping cough starts like a cold with fever, running nose and cough.
- Two weeks later, the whooping begins. The patient coughs rapidly many times without taking a breath, until the patient coughs out a mass of sticky mucus, and the air sucks back into the lungs with a loud whoop sound. After the "whoop," the patient may vomit. Between bouts of coughing, the patient appears to be healthy.

Prevention and treatment

1. Patients should consult their doctors for appropriate treatment. Treatment of whooping cough includes the use antibiotics early before coughing fits start. Patients should be isolated from contact with others.
2. Vaccination with whooping cough vaccines. In children, whooping cough vaccines are usually combined with those for diphtheria, tetanus and poliomyelitis.

7. Common cold

Common cold is a disease caused by a virus that infects the nose. Colds also affect the sinuses, ears, and bronchial tubes. Colds last an average of one week. Mild colds may last for only two or three days, while severe colds can last for up to two weeks. Colds are a milder illness than influenza.

Nausea-related secretions containing cold viruses infect the hands of people with cold when they blow their nose, cover their sneezes and touch their nose. Cold viruses can contaminate objects and surfaces in the patient environment.

Note: Children are particularly susceptible to colds.

The cold virus is released into the air by coughing and sneezing, can fall into someone else's hands and cause infection. Hence the transmission.

Signs and symptoms

- Sneezing
- Runny nose
- Nasal obstruction
- Sore or scratchy throat
- Cough
- Hoarseness
- Mild general symptoms like headache, feverishness, chilliness, and not feeling well in general.

Prevention and treatment

The cold is caused by the virus and will therefore disappear after a few days. Treatment is to reduce symptoms. If the cold persists, seek medical advice.

To avoid catching a cold, avoid contact with known cold sufferers, especially during the first three days of the disease. Take precautions to prevent cold viruses from entering your nose:

- Wash hands after contact with cold sufferers, objects and surfaces they may have contaminated.
- Keep fingers out of the eyes and nose.
- Avoid having cold patients cough and sneeze on you or in your direction.

8. Influenza

Influenza or 'the flu' is a highly contagious disease caused by infection with influenza A or B (or rarely C) viruses. These viruses affect the upper airways and lungs. flu is a highly contagious disease. The flu is not the same as the common cold and may be a serious disease. For some people, such as older people and people with underlying medical conditions, flu may cause serious complications requiring hospitalisation. It can sometimes cause death. The flu is usually spread by infected people who cough and sneeze, temporarily contaminating the air and surfaces with infected droplets.

Signs and Symptoms

Symptoms usually occur 1 to 3 days after infection. A person may spread flu to others 1 to 2 days before becoming sick and up to 5 days after the onset of symptoms. Symptoms of flu may include:

- fever
- dry cough
- muscle and joint pain
- tiredness or extreme exhaustion
- headache
- sore throat
- Stuffy nose

Prevention and treatment

1. In general, uncomplicated flu can be managed by simply resting in bed, drinking plenty of fluids (especially water) and taking medicines to help relieve the symptoms.
2. Antiviral medicines reduce the duration of symptoms and help people who have contracted the disease to return to their normal daily routines earlier.
3. Good hygiene is essential to protect yourself and others.
4. You can reduce the risk of infection by getting vaccinated.

9. Covid-19

This means Corona virus 19 disease, a pandemic virus disease that broke out in 2019 and became a worldwide outbreak in 2020.

The most common symptoms are fever, dry cough, tiredness.

Serious symptoms include difficulty breathing or shortness of breath, chest pain or pressure, and loss of speech or movement.

If you experience severe symptoms, seek urgent medical attention. Always call before visiting a doctor or hospital. It takes on average up to 14 days for symptoms to appear after someone is infected with the virus.

Effect of smoking on the respiratory system

The respiratory system isn't designed to deal with the smell of tobacco. Tobacco smoke also weakens the cilia in the respiratory tract and stops them from moving.

Furthermore, tobacco smoke increases the production of mucus in the airway. Smoking-related cough is an attempt to remove excess mucus from the respiratory system. In addition to irritating trachea and bronchi, smoke interferes with the uptake of oxygen from the airways.

When cigarette smoke is inhaled, about a third of the particles stay in the alveoli. Phagocytic cells called macrophages can slowly remove a lot of the particles from the blood stream. However, the accumulation of particles from smoking or other sources of air pollution breaks down the air sac walls and causes inelastic tissue to form. This reduces the functional surface area of the respiratory tract and can lead to a disease called **emphysema**. In some cases, **lung cancer** may also develop.

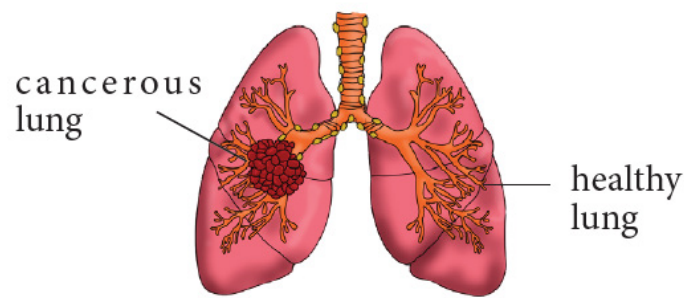


Fig 8.8: Normal lung and lung with cancer

Application activity 8.4



1. Suppose you are treating a patient suffering from respiratory illness. How can you avoid getting the disease?
2. Is it advisable to avoid friends who smoke?
3. Why is it not recommended to use antibiotics without a prescription of your doctor?

8.5. Gaseous exchange in plants

The main surface for the exchange of gases in plants is the leaf. Plant leaves have stomatal pores on their surface, through which gases are exchanged. Plants don't have specialized respiratory systems like animals. This is because they have less metabolic activity than animals.

Learning activity 8.5



The demonstration of the presence of stomata in leaves

Requirements

- Plant leaves (e.g. hibiscus plant leaves)
- A bottle of clear or colourless nail polish
- A clear cellophane tape (or clear packing tape)
- Microscope slides
- Scissors
- Microscope

Procedure

1. Paint the underside surface of the leaf under examination with 1 cm² (or more) of thick nail polish. Allow the nail polish to fully dry.
2. When the nail polish strip is dry, attach a piece of clear cellophane tape to the strip on the leaf.
3. Pull the tape tightly at the corner and gently remove the nail polish strip from the leaf. The layer of cells that remain on the nail polish is what will be examined under the microscope

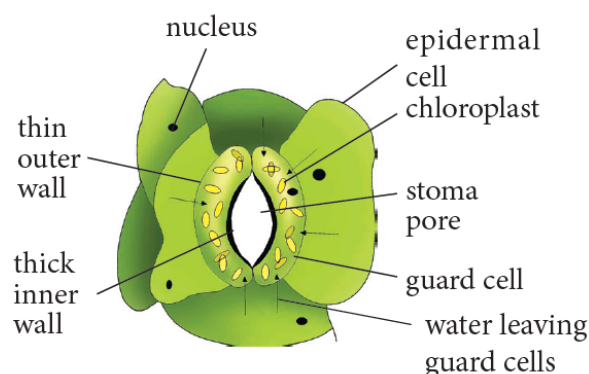
4. Attach the peeled nail polish and its cell layer to a clean microscope slide and cut any excess from the nail polish
5. Teacher will help you to predict and explain the different types of cells you see and how to locate the stomata.
6. Place the slide on the microscope stage. Using the low power objective lens, focus on the slide until the stomata are visible. Each stoma is bordered by two sausage-shaped cells, called guard cells, which are normally smaller than epidermal cells. Unlike other cells in the epidermis, guard cells contain chloroplasts.
7. Sketch what you observe under the microscope, labelling the stoma, guard cells, epidermal cells, and chloroplasts.
8. Count the number of stomata in your field of view, then estimate the number of stomata on the sample being examined.

Stomata are the pores between pairs of guard cells in the epidermis. They are present in the upper and lower epidermis, or both. The stomata allow:

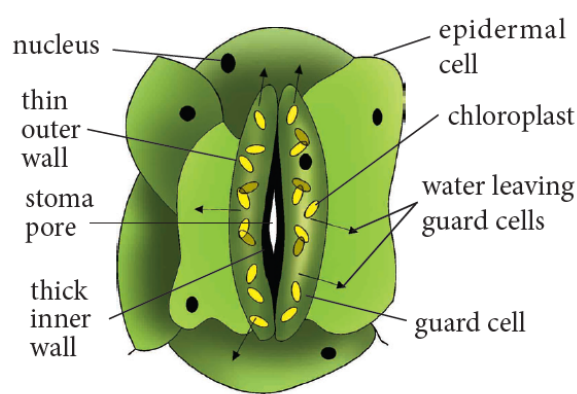
- The entry of carbon dioxide into the leaf for photosynthesis.
- The release of oxygen.
- Evaporation of water.

Stomata are the major structures for exchange of gases in plant leaves.

Let us examine the structure of stomata to understand how gaseous exchange takes place through them.



(a) Open stoma



(b) Closed stoma

Fig 8.9: Open and closed stomata

The guard cells control the opening and closing of each stoma.

Gas exchange in plants

When the stomata are opened, gases in the leaf while and other gases leave. The

oxygen produced by photosynthesis is diffused from the palisade cells into the air spaces. Eventually it gets out through the stomata into the atmosphere. On the other hand, carbon dioxide from the atmosphere enters the leaf through the stomata and is distributed through the airways. When the stomata are open, air from the surrounding air space is diffused into the palisade cells, where it is used for photosynthesis.

When the stomata are open, air from the surrounding enters the leaf and occupies the air spaces. Oxygen and carbon dioxide diffuse into or out of the leaf cells along their concentration gradient.

During photosynthesis, carbon dioxide is used up by the palisade cells, and oxygen is produced.

This means that the concentration of carbon dioxide in the palisade cells becomes lower than in the air spaces outside the cells. Therefore, during photosynthesis carbon dioxide diffuses into the palisade cell from the airspace. On the other hand, photosynthesis produces oxygen whose concentration becomes higher inside the palisade cells than in the air spaces surrounding the cells. Oxygen therefore diffuses out of the palisade cells into the air spaces. This is the process of gaseous exchange in leaves.

Gases first dissolve in the film of moisture surrounding the cell, before they diffuse into or out of the cell. Note that when air is entering or leaving the plant through stomata, the process is not gaseous exchange, but diffusion. Gaseous exchange refers to the movement of gases between the cells and their surroundings.

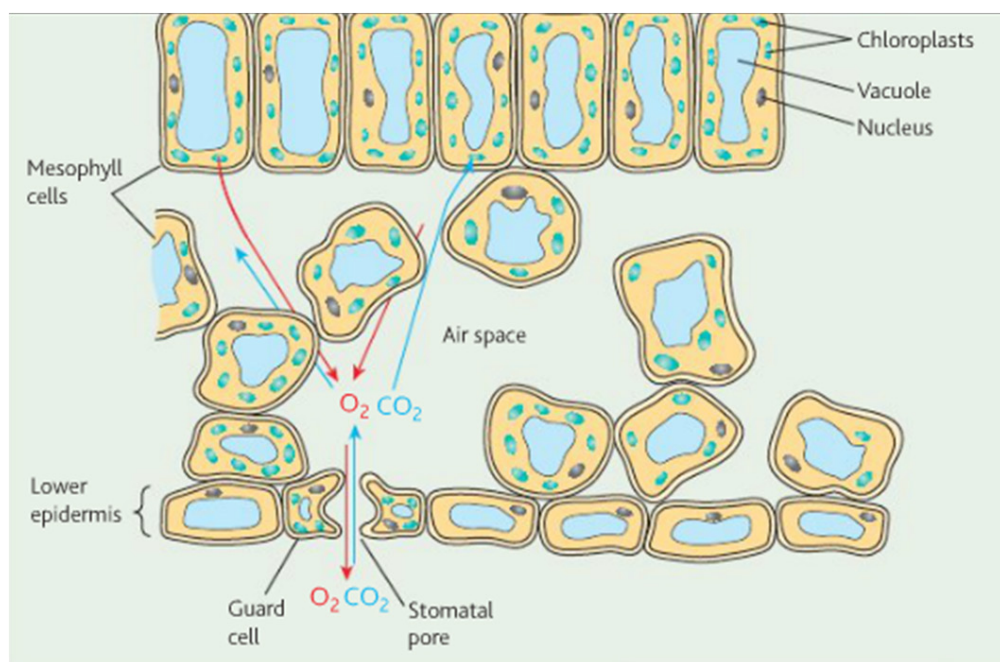


Fig. 8.10: Gaseous exchange in the leaf of a terrestrial plant

Stems of woody terrestrial trees and shrubs have areas of loosely arranged cells with large air spaces between them. These cells together form a structure called a **lenticel**. Lenticels are formed when the epidermis is replaced by the bark. Lenticels appear scattered on the surface of the stem as small, raised openings. They allow gaseous exchange of oxygen and carbon dioxide between the atmosphere and the internal tissues of the stem.

Application activity 8.5



1. What are the structural differences between guard cells and other epidermal cells?
2. Although the leaves are the main organs of gas exchange in plants, the roots also absorb oxygen in the soil. Why is this necessary?
3. What are stomata?
4. How important are stomata for plants?

8.5. End of unit assessment



8.5.1. Designing and Constructing a Model to Demonstrate Gas Exchange in Humans

1. Context/Background

Gas exchange is a vital process that enables the human body to obtain oxygen and remove carbon dioxide. This process takes place in the lungs, particularly in tiny air sacs called alveoli, where gases diffuse between the air and the blood. Understanding how the respiratory system functions help students appreciate the importance of breathing and maintaining respiratory health. Models can be used to simplify complex biological processes and make them easier to understand.

2. Challenge/Problem Statement

The process of gas exchange in the human respiratory system cannot be observed directly. How can a simple model made from locally available materials be used to demonstrate and explain gas exchange in humans?

Key Question

How can a simple model be designed and used to demonstrate the process of gas exchange in the human respiratory system?

3. Purpose of the Project

The purpose of this project is to help students understand the process of gas exchange in humans by designing and constructing a model using locally available materials. Through the project, students will demonstrate how the lungs function during breathing and explain how oxygen and carbon dioxide are exchanged in the respiratory system.

4. Instructions

- Work in groups.

- Research the structure and function of the human respiratory system.
- Identify locally available materials that can be used to construct a model.
- Design and build a model showing the major parts involved in breathing and gas exchange.
- Use the model to demonstrate inhalation, exhalation, and gas exchange.
- Record the construction process and explain how the model represents the real respiratory system.
- Present the model and findings to the class.

5. Your Tasks

- Investigate how gas exchange occurs in humans.
- Design a model using locally available materials.
- Construct and test the model.
- Label the major parts represented in the model.
- Demonstrate how the model shows breathing and gas exchange.
- Explain the strengths and limitations of the model.
- Present your findings and recommendations.

6. Expected Student Outputs

- A labelled model of the human respiratory system.
- Design sketches and descriptions of materials used.
- A demonstration showing inhalation and exhalation.
- An explanation of how gas exchange occurs in the lungs.
- A project report containing the design process, observations, conclusions, and recommendations.
- A group presentation of the model and findings.

7. Constraints and Limitations

Scope

The project should focus on demonstrating the process of breathing and gas exchange in humans using a simple model. The model does not need to represent all structures of the respiratory system in detail.

Materials and Methodology

Use safe and locally available materials such as plastic bottles, balloons, straws, rubber sheets, cardboard, paper, tape, glue, and string. The model should clearly represent the major structures involved in breathing and gas exchange.

Timeframe

The project should be completed within **2–3 weeks**, including planning, model construction, testing, report writing, and presentation.

8. Guiding Questions

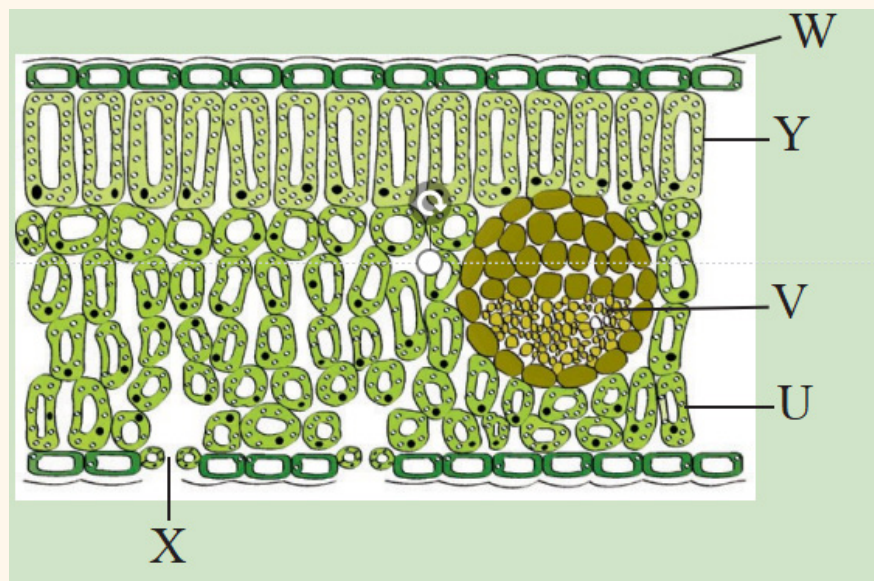
1. Which parts of the respiratory system must be included in your model?
2. What locally available materials can best represent the lungs, diaphragm, and air passages?
3. How will your model demonstrate inhalation and exhalation?
4. How does gas exchange occur in the alveoli?
5. What are the strengths and limitations of your model in representing the real respiratory system?
6. How can understanding gas exchange help people maintain good respiratory health?

8.5.2. Assessment questions

A. Knowledge and understanding

1. One of the following is not a respiratory surface. Which one is it?
 - a) Skin
 - b) Nostrils
 - c) Gills
 - d) Mouth cavity
2. Terrestrial animals obtain respiratory gases from
 - a) The surrounding water
 - b) The atmosphere
 - c) A and B
 - d) None of the above
3. Explain why stomata are usually located on the underside of leaves.
 - a) To maximize light absorption
 - b) To minimize water loss through transpiration
 - c) To facilitate pollination
 - d) To protect against herbivores
4. Suggest why organisms develop a gas exchange system.
5. Briefly explain the sequence of events that occurs during breathing.
6. Do you think the Rwanda government should ban cigarettes sales?
7. Give two characteristics of leaves that support the exchange gases.

8. (a) Name the parts labelled on the internal structure of the leaf below.



(b) What is the importance of parts W and X?

9. Explain how you would help a person who has just had an asthma attack.

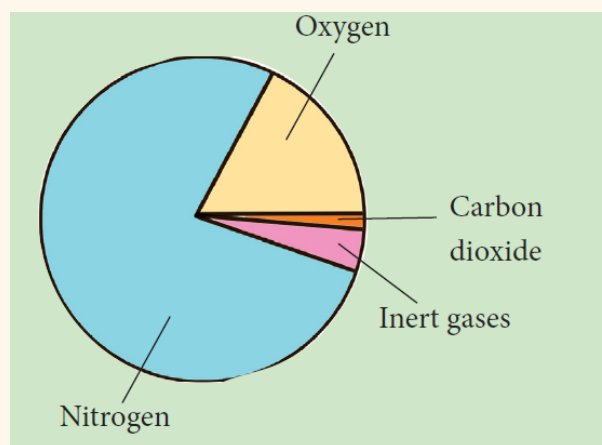
10. How does exercise affect the breathing rate?

B. Skills

11. Two leaves were immersed in a beaker containing warm water. Bubbles of a gas were observed to form and escape from the leaf surfaces. In leaf A, bubbles formed on both surfaces but more on the lower surface than on the upper surface. In leaf B, very many bubbles formed on the upper surface but none on the lower surface.

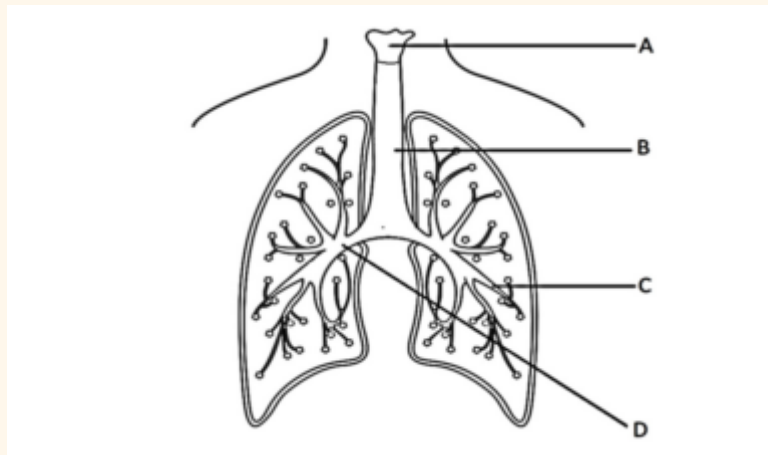
- What does the production of bubbles depict about the types of plants?
- What does the production of bubbles indicate about the internal structure of the leaf?
- Suggest the possible habitats of the leaves A and B.

12. The atmospheric air contains 79% nitrogen and 20% oxygen as shown below.

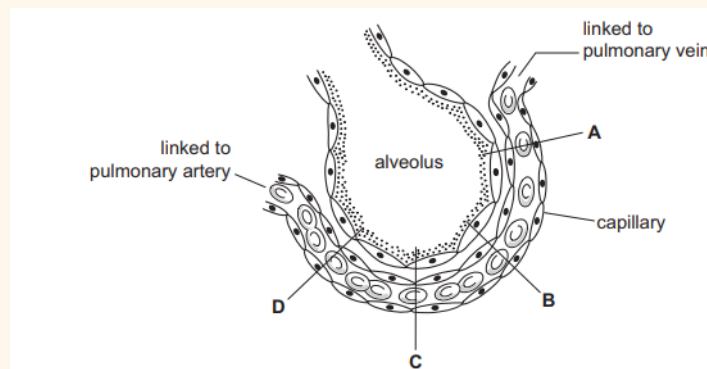


Explain why we breathe oxygen and not nitrogen, which is more plentiful.

13. The diagram below represents a human respiratory system



- Identify the structures labelled A, B C and D in the diagram.
 - What prevents structure B from collapsing?
 - Name a muscle that contracts during inhalation.
 - Describe the change in air pressure inside the chest during inhalation.
 - Inhalation is an active process, and exhalation is a passive process.
 - Explain the terms active and passive.
14. The diagram below shows an alveolus and an associated blood capillary.



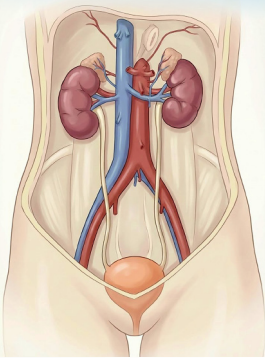
- In which organ is the alveolus located in the human body?
- Observe the diagram and identify a letter where the highest rate of diffusion of carbon dioxide will occur. Why?

C. Attitudes and values

- Why is tree planting considered a responsible practice for the environment?
- Why should people avoid air pollution?
- Why is regular physical exercise encouraged in connection with gas exchange?

UNIT 9

EXCRETION IN HUMANS



Key Unit Competence:

Relate the structures of urinary system to their functions.

Learning objectives

By the end of this unit you should be able to:

Knowledge and understanding

1

- Explain excretion.
- Outline the importance of excretion.
- Match excretory organs to respective excretory products in humans.
- Describe the structure of the kidney, limited to the cortex, medulla, and ureter.
- Describe the nephron and functioning.
- Explain the process of urine formation limited to ultra-filtration and selective re-absorption.

Skills

2

- Analyze the anatomy and physiology of human urinary system.
- Relate habitual practices to urine concentration.
- Draw a human urinary system elaborating the kidneys, ureters, bladder, urethra, and other parts.
- Model a well-labelled structured nephron.
- Relate the effect of water intake, temperature, and exercise on the volume and concentration of urine produced.

Attitudes and values

3

- Develop good habits to maintain a healthy urinary system.

9.0. Introduction

Excretion is an essential life process through which living organisms remove metabolic waste products and toxic substances from their bodies. These wastes are produced during normal cellular activities such as respiration, digestion, and protein metabolism. If allowed to accumulate, they can be harmful and interfere with the normal functioning of the body. In humans and other animals, specialized excretory organs help maintain a stable internal environment by removing waste materials and regulating the balance of water and salts.

The kidneys are the main excretory organs in humans. They contain millions of microscopic structures called nephrons, which are the structural and functional units of the kidney. Nephrons work together to filter blood, remove waste products, and produce urine. Understanding excretion helps students appreciate the importance of waste removal, homeostasis, and the role of the excretory system in maintaining health.

Key Concepts: excretion, metabolic waste, excretory system, kidney, nephron, urine formation, filtration, reabsorption, secretion, homeostasis, urea, water balance, salts, osmoregulation.

Introductory activity



1.



A



B

Two healthy people have drunk the same amount of water (0.5L). One went to watch movies for one hour while the second one went to do sport for one hour. But the urine of the first was abundant and colourless while that of the second was scanty and deeply coloured.

- What may be the color of the urine which is deeply coloured?
- Why is the urine of the second one less and more deeply coloured?

In this unit, you will learn how urine is formed and how waste products are excreted from our body.

9.1. Need for excretion

Just as vehicles remove substances that can damage their engine if left to accumulate, human beings also remove toxic waste products from their bodies. This happens through a process called **excretion**. Therefore, we can define excretion as the process by which organisms remove waste products of metabolism from the body. Through excretion, organisms control osmotic pressure. It also enables them to promote homeostasis; that is, the balance of the organism's internal body environment.

Learning activity 9.1



Discussion on necessity of excretion

1. Discuss the following with your classmates
 - a) The importance of excretion.
 - b) The health implications of not emptying the bladder and bowels promptly.
2. Research on human excretory organs. Use your answers to fill the table below

Table 9.1: Excretory organs and their excretory products and incidental loss

Excretory organ	Main excretory products

3. Compare your findings with those of your classmates.
4. Share your work with the rest of the class.

Excretion is one of the main characteristics of living things. Every living cell undergoes life processes involving chemical reactions. In some of these reactions, complex materials are chemically broken into simpler materials and in others new compounds are formed from simple materials. All of these reactions are collectively called metabolism. During **metabolism**, waste products are formed. When waste products accumulate in or around cells, they can reach very high concentrations, affect the normal cellular functions and potentially kill the cell. For example:

1. Carbon dioxide is a by-product of respiration. When it accumulates in a cell it changes the pH of the cell. It then interferes with the way some enzymes work. If this situation is not corrected, the affected cells eventually die.
2. The body does not have a mechanism to store excess amino acids that it gets from proteins digestion. The excess amino acids are then taken up into the liver, where they are broken down into urea and glycogen in a process called **deamination**. Insignificant amounts of uric acid are also produced by protein breakdown. Urea and uric acid are nitrogenous waste products that are excreted or removed from the body by the urine.

Many other cellular reactions also produce by-products that can be toxic. It is therefore necessary to remove these waste products from the cells. Waste products, if not deposited of, may accumulate to toxic levels, leading to deterioration of health and eventually death.

Unlike plants, animals have difficulty getting rid of waste substances for several reasons:

- Animals are more active than plants. Therefore, their metabolic processes occur at a higher rate and produce a great amount of waste products.
- Animals do not convert most of their waste products to other uses as occurs in plants.
- Animals ingest certain substances in their diet beyond their needs. These extra substances, for example proteins, are broken down with the formation of toxic substances such as ammonia.

For these reasons, animals have a more complex excretory system than plants.

Table 9.2: Summarizes the excretory organs and their waste products in humans.

Excretory organ	Excretory product
Skin	Urea, lactic acid, excess salts and excess water in form of sweat.
Kidney	Excess salt, excess water and nitrogenous waste in form of urine.
Lungs	Carbon dioxide and excess water in form of water vapour
Liver	Bile pigments
	Nitrogenous compounds such as ammonia, urea, uric acid and trimethylamine oxide.

9.1.1. The role of the liver in excretion

The liver is the second largest organ in the body after the skin. It plays a very important role in the body. It is described as the metabolic centre. The liver has many functions including maintenance of a constant internal environment (**homeostasis**) and excretion. The excretory functions of the liver are described below.

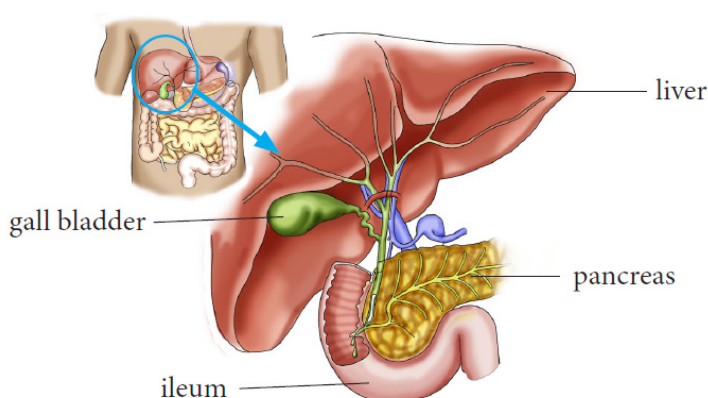


Fig 9.1: Location of liver in the body

a. Deamination

The body cannot store proteins or amino acids. This is because their breakdown forms toxic substances in the body. Amino acids are the byproducts of protein digestion. Surplus amino acids are broken down in the liver in a process called **deamination**.

From each amino acid, the amino group (NH_2) is converted to ammonia (NH_3). Ammonia should not be allowed to accumulate in the body because it is highly toxic. The rest of the amino acid molecule is converted into glycogen or triglyceride for storage.

The ammonia produced from the amino group is quickly converted to a less toxic substance called **urea**. During this transformation, ammonia is initially combined with carbon dioxide in a series of enzymatic catalysed reactions. The resulting urea is taken up by the kidney in blood and is eliminated from the body in the urine.

b. Detoxification

The liver removes harmful substances such as drugs and hormones from the blood. Within the liver, these substances are converted into inactive or less dangerous forms, for instance, hydrogen peroxide, a highly toxic by-product of certain metabolic processes, is rapidly split into water and oxygen by enzyme catalase in the liver.



Thus, the liver purifies or detoxifies blood. The inactive substances formed in the liver are returned to the bloodstream and are finally excreted from the body by the kidneys.

c. Elimination of Haemoglobin

Haemoglobin from old worn-out red blood cells is broken down by the liver cells into pigments. These pigments are further broken down and eliminated in the bile, giving urine its characteristic yellow colour.

d. Elimination of sex hormones and cholesterol

After sex hormones have performed their functions, some are modified chemically by the liver cells. Others are sent to the kidney for renal excretion while others are expelled in bile.

Excess **cholesterol** is also excreted in bile. If there is a considerable excess amount of cholesterol in the blood, some may be deposited in the walls of blood arteries obstructing flow.

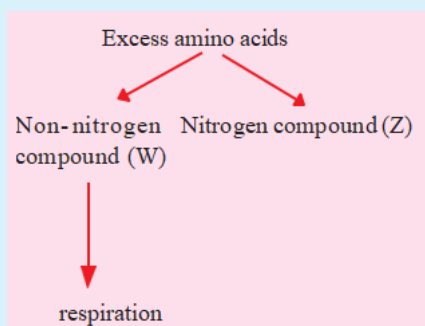
Application activity 9.1



1. Undigested and indigestible waste products are not considered as excretory wastes, explain.
2. Copy and fill the blank spaces in the passage that follows.

Excretion involves the removal of products from the body. Urea is produced by the during while carbon dioxide is a waste product of Nitrogenous waste products are excreted by the while carbon dioxide is excreted by the

3. Proteins are very important in our bodies. However, excess amino acids from protein digestion cannot be stored. Study the diagram below and answer the questions that follow.



- (i) Name the process illustrated in the diagram.
(ii) In which organ does the process you have named in (i) above occur?
(iii) Name compound Z.
(iv) Compound W is used as source of energy, name compound W.

4. Other than urea, name two other waste products excreted by the liver.
5. Describe how urea is transported in blood to the kidneys.

9.2. Structure of the human urinary system

Learning activity 9.2



(a) Identifying parts of the human urinary system.

Requirements

- Manila papers and marker pens
- Charts, blackboard diagrams, and photographs of the human urinary system.
- A drawing of the internal structure of the mammalian kidney

Procedure

1. Study the chart provided and use them to draw the human urinary system on Manila paper and label the kidneys, ureters, bladder and urethra.
2. Draw a labelled diagram of the internal structure of the mammalian kidney.
3. In your groups, discuss the positions of the cortex, medulla, pyramid and pelvis.
4. Share your findings with other class.

(b) The examination of the external and internal structure of a mammalian kidney

Requirements

- Fresh kidney of sheep, goat, or cow
- Sharp razor, knife, or scalpel
- Small dissecting board
- Hand lens or compound microscope

Procedure

1. Examine the whole kidney. Note the various tubes attached to it.
 - What is the outer colour of the kidney?
2. Draw and label the external structure of the kidney.
3. Place the kidney on the dissecting board.
4. Use the scalpel, razor, or knife to cut the kidney along its length at the middle.
5. Use a hand lens to identify the following parts: cortex, medulla, pelvis, renal artery, renal vein and urethra.
6. Draw and label the internal structure of the kidney.

The human urinary system is made up of two kidneys, urinary bladder, two ureters and a single urethra. Kidneys participate in filtering blood and separating waste metabolic substances from it. The main purpose of the urinary system is to remove urine from the body. The draining of urine from the bladder, through the urethra and out of the body is known as **urination**.

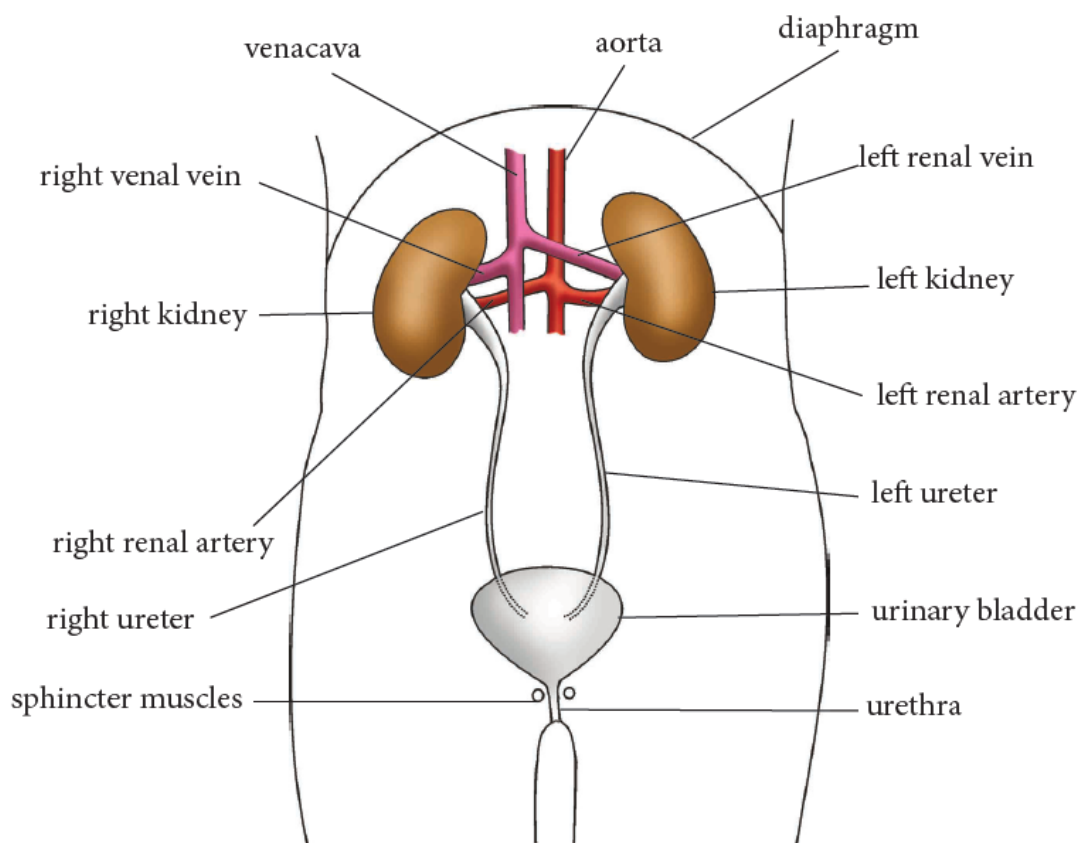


Fig 9.1: Human urinary system

Function of parts of the urinary system

The urinary system keeps everything in balance by removing waste, such as urea, extra salt, extra water and other things the body does not need. Urea is produced when protein, found in meats, is broken down in the body.

❖ The kidneys

There are a pair of kidneys that are red-brown and are located below the ribs in the middle of the back. Their function is to:

- Remove waste from the blood in the form of urine.
- Make erythropoietin, a hormone which helps make red blood cells.
- Make vitamin D active.
- Regulate blood pressure.

❖ Ureter

Each kidney has a narrow tube called the ureter, which carries urine from the kidney to the bladder. Muscles in the ureter walls tighten and relax forcing urine down this tube, away from the kidneys.

❖ Bladder

The bladder is a triangle-shaped, hollow organ located in the lower abdomen. It is held in place by ligaments attached to the pelvic bones. The bladder's walls relax and expand to store urine, and contract and flatten to empty urine through the urethra.

❖ Sphincter muscle

Circular muscles that help keep urine from leaking by closing tightly like a rubber band around the opening of the bladder.

❖ Urethra

The urethra is the tube that allows urine to pass outside the body. The brain signals the bladder muscles tighten, which squeezes urine out of the bladder. At the same time, the brain signals the sphincter muscles to relax to let urine exit the bladder through the urethra. When all the signals occur in the correct order, normal urination occurs.

Kidneys are bean-shaped and are red-brown in colour. They lie near the back of the abdominal cavity about the level of the waistline. Each kidney weighs approximately 142.5 g. It is about the size of a clenched fist. The right kidney is slightly lower than the left. The kidney is surrounded by a layer of fat which helps to cushion it from mechanical or physical injury.

The kidney is supplied with blood from the general circulatory system via the **renal artery** which branches off from the aorta. Blood from the kidney goes back to the general circulation through the **renal vein** which joins the vena cava. A tube called the ureter connects each kidney to the bladder located in the lower abdomen.

The bladder is stretchy to hold large quantities of urine. It stores urine temporarily. From the bladder, another tube called the urethra opens to the exterior of the organism. Two rings of sphincter muscles encircle the urethra. They control the emptying of the bladder.

Internal structure of the kidney

A frontal section through the kidney reveals three main regions. The outer part is called cortex, inner part called medulla and the pelvis.

1. Cortex

This is the outer part which is dark in colour. It contains a dense network of blood capillaries that form the **glomeruli of nephrons**. Nephron is the functional unit of the kidney.

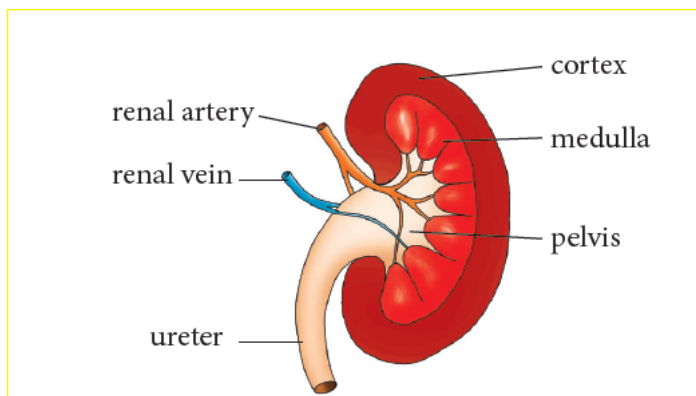


Fig. 9.2: Cross-section of kidney

2. Medulla

This part is pale red in colour and lies between the cortex and the pelvis. It contains several cone-like extensions called **pyramids**.

3. Pelvis

This part is white in colour. It narrows to form the ureter. Pelvis is a collecting space leading to the ureter, which takes urine to the bladder.

The nephron

The most important function of the kidney as an excretory organ is to filter wastes from the blood. This takes place in tiny units called nephrons or renal tubules. A nephron is therefore referred to as the functional unit of the kidney. Each kidney has about 1.25 million nephrons. One part of the nephron is in the cortex and the other part in the medulla.

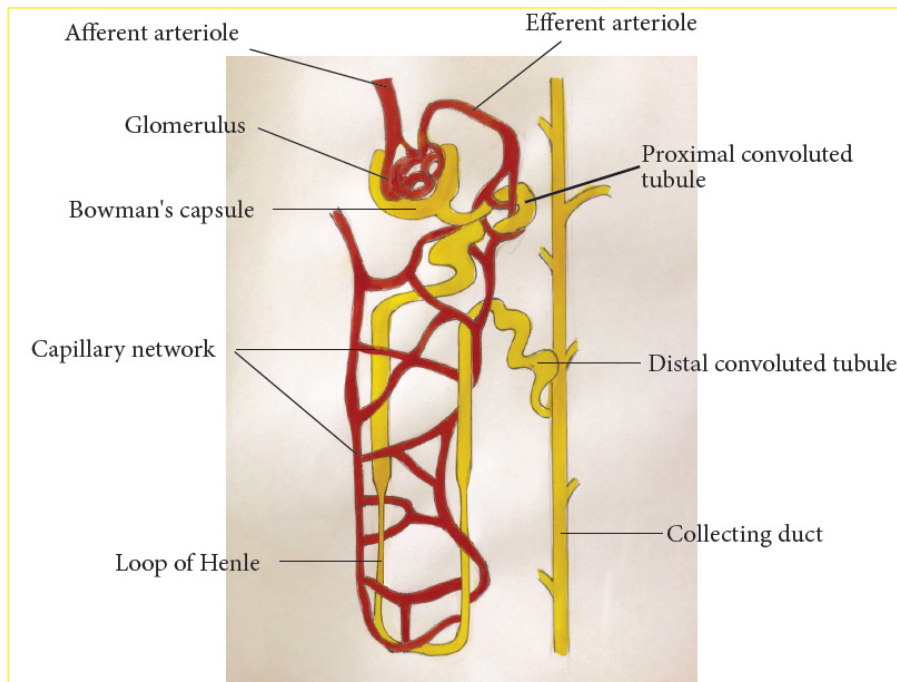


Fig 9.3: Structure of the nephron

The nephron has three distinct coiled parts:

- The proximal convoluted tubule
- A U-shaped loop of Henle
- A distal convoluted tubule

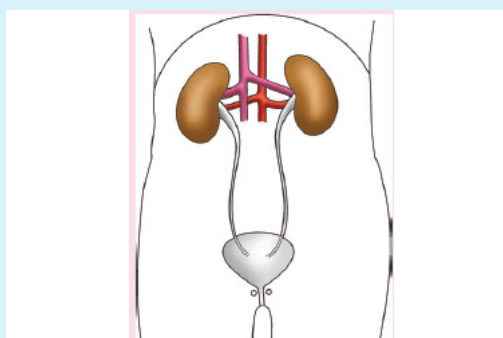
Both the proximal convoluted tubule and the distal convoluted tubule are located in the cortex. The loop of Henle is in the medulla. One end of the nephron is modified to form a cup shaped structure called the **Bowman's capsule**.

The nephron is supplied with an extensive network of blood capillaries. In the Bowman's capsule, the capillaries form a knot called the glomerulus (plural glomeruli) which branches from an afferent arteriole that originates from the renal artery. The Glomeruli capillaries reunite to form an efferent arteriole which channels blood away from the glomerulus.

Application activity 9.2



1. The diagram below shows the mammalian urinary system. On it, indicate the renal artery, urethra, ureter and left kidney.



2. Given the following parts of the nephron:

- | | |
|--------------------------------|--------------------------------|
| (i) Bowman's capsule | (iv) Collecting duct |
| (ii) Loop of Henle | (v) Proximal convoluted tubule |
| (iii) Distal convoluted tubule | |

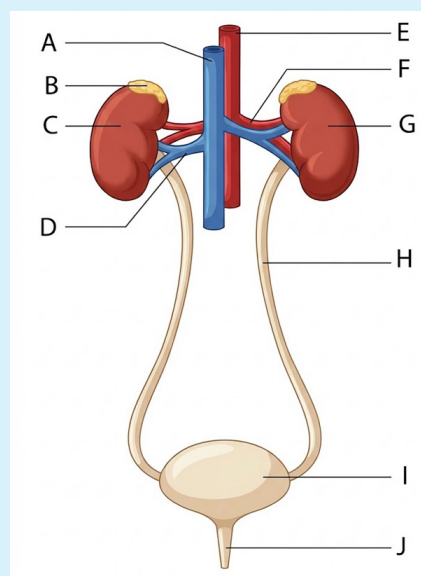
Arrange the parts in order beginning with the region of the nephron from where the filtrate begins to flow.

3. The figure below shows the human urinary system.

- (a) Name the parts represented by the letters A to J.

- (b) Name the parts with the functions below:

- (i) Stores urine
- (ii) produces urine.
- (iii) Carries urine to the outside.



9.3. Urine formation

Learning activity 9.3



Research on the process of the urine formation

Search for the following topics:

1. The nephron
2. The steps involved in urine formation
3. The place where each step starts in the kidney

Study questions:

1. What are nephrons?
2. What are the main steps involved in urine formation?
3. Where do each of the steps stated above take place?

Excretion in the nephron is conducted in three stages: **ultrafiltration**, **selective reabsorption** and **tubular secretion**. Blood coming into the kidney from the artery contains both waste substances and useful substances. Both substances must enter the nephron, where separation takes place by ultrafiltration. The body must not lose the useful substances. Therefore, useful substances must be taken back into the blood so that they are not lost. This process is known as **selective reabsorption**.

a. Ultrafiltration

Ultrafiltration takes place in the glomerulus. The afferent arteriole that takes blood to the glomerular capillaries has a wider lumen than the efferent arteriole that takes blood away from it.

Due to the difference in afferent and efferent arteriole size, a high pressure of blood is created in the glomerulus. This pressure forces **water, mineral ions** and small molecules like **glucose, amino acids** and urea out of the glomerulus. These pass through the tiny pores in the walls of the glomerular capillaries into Bowman's capsule. The liquid collected in the Bowman's capsule is called **glomerular filtrate**.

The larger molecules in the blood, like blood proteins, white **blood cells**, **red blood cells** and **platelets** cannot pass through the capillary walls of the glomerulus. These remain in the blood and continue to flow to the efferent arteriole. The glomerular filtrate flows down the nephron where re-absorption will take place as it flows along.

b. Selective reabsorption

As the glomerular filtrate passes along the nephron, some substances that are useful to the body are selectively taken back or reabsorbed into the blood capillaries network surrounding the nephron.

- All **amino acids** and **glucose** are reabsorbed by active transport in the proximal convoluted tubule.

Some **salt** and **water** are reabsorbed depending on how much of them the body still needs. Water is reabsorbed by osmosis and salts by active transport. Salts are mainly **reabsorbed in the proximal convoluted tubule**. Water is reabsorbed in both the proximal and distal convoluted tubules. However, most of the water is reabsorbed in the region of the collecting duct.

- No **urea** is reabsorbed into the blood.

c. Tubular secretion

d. Tubular secretion is the process of actively transporting other substances from the blood (the peritubular capillaries) to the tubule of the nephron. It helps to eliminate waste and maintain the internal balance of the body.

By the time the filtration from the glomerulus completes its movement down the nephron, it has a high concentration of urea, some salts and water. The liquid is now called urine. Several nephrons empty into one collecting duct, and all the collecting ducts of a kidney empty into the ureter. The process of urine formation is a continuous one, and the ureter continuously receives small amounts of urine. The bladder stores the urine until it is full, then one begins to experience an uncomfortable feeling. The sphincter muscles must then be relaxed in order to empty the bladder.

Factors that affect urine production

The volume, colour, odour of urine and frequency of urination is affected by many factors. They include:

1. Amount of fluids taken

Large intake of fluids lowers the osmotic pressure of blood. This leads to reduced reabsorption of water in the kidney tubules resulting in the production of large amounts of dilute urine.

2. Amount of salt taken

Intake of a salty meal raises the osmotic pressure of blood. This leads to increased reabsorption of water in the kidney tubules resulting in the production of coloured, little and smelly urine.

3. Weather

In hot and dry weather conditions, the body tends to lose a lot of water through sweating, thereby raising the osmotic potential of blood. In this case a lot of water is reabsorbed resulting in coloured, little, and smelly urine.

During cold weather the frequency of urination increases because sweating is so minimal.

4. Physical activity

During exercises like running, jumping and playing, we sweat a lot. The kidney reabsorbs more water resulting in little, coloured and smelly urine.

5. Diseases

Certain diseases that affect the secretion of hormones controlling the reabsorption of water in the kidney tubules can lead to the production of large amounts of urine. An example is diabetes insipidus.

Application activity 9.3



1. What are the units of kidney function that produce urine?
2. What are the main steps involved in urine formation?
3. Where does each of the above-mentioned actions take place?
4. Name the main parts of the nephron.

9.4. Practices that maintain healthy urinary system

Learning activity 9.4



Discussion on the ways of maintaining the health of the urinary system

1. Discuss good habits to promote a healthy urinary system.
2. Present your work to the rest of the class.

Study questions:

1. Which measures can be taken to improve urinary health?
2. Why is it important to drink enough water?
3. Why is it not advisable to keep urine in the bladder for so long?

The best way to prevent urinary tract infection is to keep the kidneys and the entire urinary system healthy. This can be done through proper diet and nutrition. Some methods that keep the urinary system and kidneys functioning normally include:

- (i) Drinking a lot of water, at least 10 glasses of water a day to flush out toxins in the body.
- (ii) Exercising regularly to keep fit. Maintain a healthy weight according to your age to avoid putting excess strain on all bodily systems.



Fig. 9.4: Exercising through running

- (iii) Avoid taking too many drugs especially pain killers. Stick to prescription drugs from a qualified Medical Physician.
- (iv) Visit a doctor (urologist) regularly to check the health. of the urinary system.
- (v) Eat healthy by avoiding junk food. Eat more fresh fruits and green vegetables. Choose foods low in sodium, sugar and fats but high in fiber content.
- (vi) Be informed about the causes and prevention methods of kidney diseases and urinal tract infection causes.
- (vii) Avoid smoking and alcohol intake.

Application activity 9.4



Write True (T) or False (F).

1. Drinking truly little water improves kidney function.
2. Urinary tract infections can affect kidneys if not treated early.
3. Excessive use of painkillers may damage kidneys.
4. Sweating reduces the need for the kidneys to work.
5. A balanced diet helps maintain a healthy urinary system.

9.5. End of Unit Assessment



9.5.1. Assessment questions

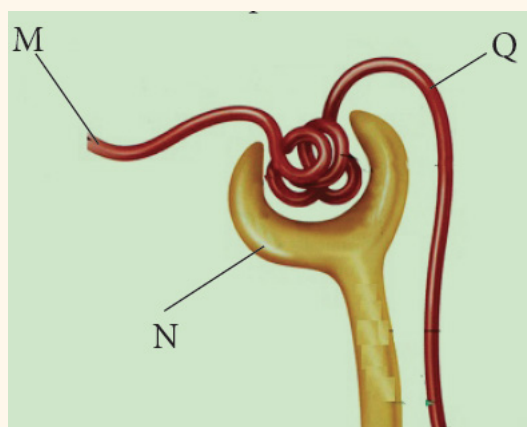
A. Knowledge and understanding

1. In which organ are excess amino acids broken down?
 - A. Lungs
 - B. Heart
 - C. Kidneys
 - D. Liver
2. The active uptake of substances from the nephron into the blood is called
 - A. Tubular secretion
 - B. Selective re-absorption
 - C. Ultra filtration
 - D. Micro nutrition

3. The kidney function is achieved by
 - A. Ultra filtration
 - B. Secretion
 - C. Selective re-absorption
 - D. Both A and B
 - E. All of the above.
4. Iradukunda eats a full bag of salty chips. What effect does this have on the quantity and concentration of urine?
5. Differentiate between excretion and egestion.

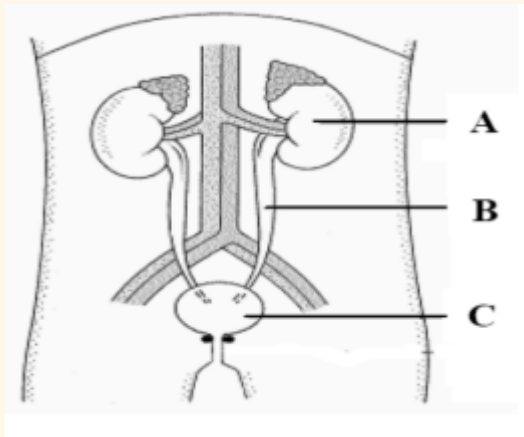
B. Skills

6. Given that the volume of blood filtered by the kidneys is $1.18 \text{ dm}^3 \text{ min}^{-1}$;
 - (i) Calculate the total volume of blood filtered in 24 hours. Show your workings.
 - (ii) If 1.7 dm^3 of urine is produced in 24 hours, calculate the percentage volume of the filtered blood is excreted as urine in 24 hours.
7. Explain why the concentration of urea in the urine increases by the time the filtrate reaches the collecting duct.
8. The figure below represents a mammalian nephron.



- a) State and give the importance of the structural difference between parts labelled **M** and **Q**.
 - b) Name the process that occurs at the labelled **N**.
 - c) In a healthy person, all glucose and amino acids are completely reabsorbed from the filtrate. Some water is also reabsorbed. Name the physiological processes by which these substances are reabsorbed.
9. Two filtrates were obtained from various parts of the nephron. Filtrate A contained glucose, amino acids, water and urea while filtrate B contained water and urea.
 - (i) From which part of the nephron was filtrate A and B extracted? Give reasons for your answer.
 - (ii) Account for the absence of proteins and blood cells in either filtrate.
 10. To avoid losing too much water, a desert kangaroo excretes very concentrated urine. Explain how its nephron is adapted to enable this.

11. The diagram shows the human urinary system.



- Name the parts labelled A and B.
- What role is played by the part labelled C.
- Suggest two substances excreted by part A
- Identify the region of part A in which each of the following take place:
(i) Filtration. (ii) Reabsorption.
- Give a function of part A other than excretion.

12. (a) Define the term excretion.

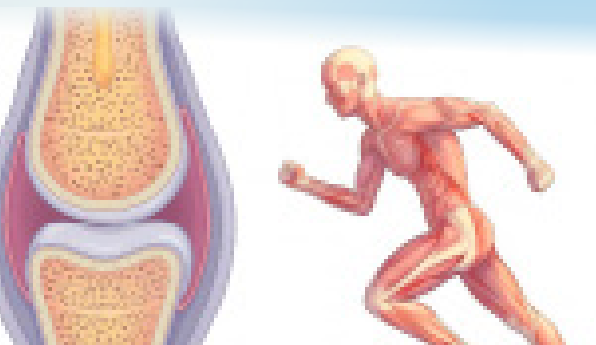
(b) Identify four human excretory organs and for each organ, mention the metabolic waste product produced.

C. Attitudes and values

- Why is it important to avoid sharing towels or underwear?
- Why is avoiding excessive alcohol and drugs important for urinary health?
- Why is regular urination important?

UNIT 10

JOINTS AND MOVEMENT



Key Unit Competence:

Relate main types of joints to their functions.

Learning objectives

By the end of this unit you should be able to:

Knowledge and understanding

1

- Explain the function of bones, ligaments, tendons, cartilage, nerves and synovial fluid in the joint.

Skills

2

- Analyse the structure of immovable and movable joints.
- Apply acquired knowledge to classify joints in the human body as immovable or movable.
- Differentiate between hinge, ball and socket joints.
- Draw and label a typical synovial joint.
- Demonstrate how antagonistic muscles bring about movement at a hinge joint.

Attitudes and values

3

- Appreciate the importance of linking skeletal muscles and bones in movement.
- Develop good habits that maintain safety of the body joints.

10.0. Introduction

Movement is an important characteristic of living organisms and enables animals and humans to perform daily activities such as walking, running, grasping objects, and maintaining posture. Movement occurs through the coordinated action of bones,

muscles, and joints. Joints are places where two or more bones meet and allow different types and degrees of movement.

In this unit, students will explore the structure and functions of joints, the different types of joints found in the human body, and how muscles work together to produce movement. Students will also investigate the importance of maintaining healthy joints and muscles for efficient body movement and overall well-being.

Key concepts: movement, joint, skeleton, muscle, ligament, tendon, cartilage, fixed joint, slightly movable joint, freely movable joint, flexion, extension, posture, locomotion.

Introductory activity



In previous levels, you might have learnt about the skeletal systems of organisms.



A



B

- What did we say about the importance of the skeleton in our body?
- Can you give examples of moving and fixed joints?
- What is the role played by the joints in the above-mentioned A and B image?

10.1. Types of joints

Learning activity 10.1



Determining the position of hinge joint

- When you open and close a door, how does it move?
- Ask a friend to open and close a door as you watch.
 - What do you notice?

You may have realized that the door can move only towards you or away from you. Notice too that the door is held in place by hinges which allow this movement.

3. Is there any other possible way the door can move? Try it and find it out.

The movement of the door can be described as a back-and-forth movement only. When something held at a point, moves in such a way, it is said to move along one plane only.

4. Now, try and move different parts of your arm or leg. At which joints do you think there is movement in one plane only?

5. Name the position of these joints.

6. Use an arm model with a rubber band to demonstrate movement of joints.

We are able to move different parts of our body because bones are connected by joints, and muscle act on these joints to produce movement. But, there are some joints that aren't moving at all. In the following lessons, you will learn about the different types of joints, how they work and how to keep your joints healthy.

A joint is a point where a bone meets another bone. Except for pelvic, sacral, skull and sternal bones, joints allow movement and provide mechanical support. Muscles pull on bones to make joints move. Therefore, without joints, movement would not be possible. Most joints only allow slight movement while other joints allow no movement at all.

Joints develop between adjacent bones. But the movement is important to determine the type of joint that develops. Restricted movement even in a freely movable joint can transform it into an immovable joint.

- **Bone:** is a hard, tough connective tissue composed of mineral salts such as calcium and phosphate. It is abundant in all animal skeletons.
- **Cartilage:** this is a skeletal connective tissue which is softer than a bone. It supports the trachea, nose, oesophagus and pinna of the ear.
- **Ligament:** this is a fibrous tissue which joins one bone to another. They are elastic to allow movement at a joint.
- **Tendon:** a tough connective tissue which attaches a muscle to a bone. They are inelastic.
- **Muscle :** this is contractile tissue, specialized for contraction and relaxation. They cover the skeleton. Muscles are responsible for locomotion and other types of movement in animals.

Joints are classified structurally depending on how the bones articulate or functionally depending on the degree of movement between the articulating bones.

i) Structural classification

- a) **Cartilaginous joint :** the bones are connected by cartilage.
- b) **Fibrous :** the bones are connected by dense fibrous tissue rich in collagen.
- c) **Synovial :** there is a space between the bones called synovial cavity that is filled with a fluid known as synovial fluid.

ii) Functional classification

a) Movable joints : allow some degree of movement. There are two categories.

- Those which allow slight mobility.
- Those which are freely movable.

b) Fixed or immovable joints: allow very little or no mobility.

Fixed or immovable joints

These are joints that do not allow any movement, for example, joints in the cranium commonly known as sutures. These joints have seams between the bones of the skull. They are not smooth but rather have interlocking finger-like processes that increase stability. Between the bones are dense fibrous tissues.

Fixed joints are also found in the pelvic girdle.

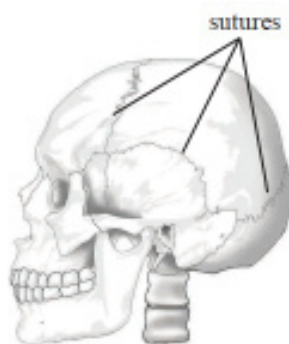


Figure 10.1: The skull

Movable joints or synovial joints

These are joints that allow movement of body parts to take place. Some movable joints allow only a small degree of movement while others allow a wide range of movement. There are several movable joints. These joints are classified depending on the shape of the adjoining articular surface. They include:

i) Hinge joints

Hinge joints allow movement in one plane. The joints at the elbow, knee and finger joints show this back-and-forth movement as the only type of movement.

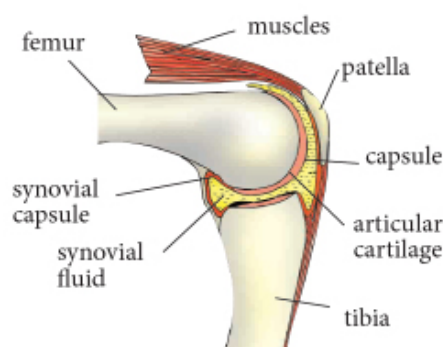


Figure 10.2: Hinge joint of the knee

ii) Pivot joints

It consists of a cylindrical bony process of one bone that rotates within a ring partially composed of bone and cartilage. Examples include articulation of the axis with the atlas and the articulation between the head of radius and the proximal end of the ulna.

iii) Ball and socket joint

Move your arm at the elbow. Then move your arm at the shoulder in as many ways as possible. What was the difference in the type of movements at the elbow joint and at the shoulder joint?

These types of joints allow the greatest flexibility of movement of all joints. It consists of a ball-shaped end of one bone that fits into the cavity (socket) of an adjacent bone. This type of joint allows wide range of movement in almost any direction. The shoulder joint is an example of a ball and socket joint. The other is the hip joint. Bones at hinge joint and ball and-socket joint are held in place by ligaments which join them together.

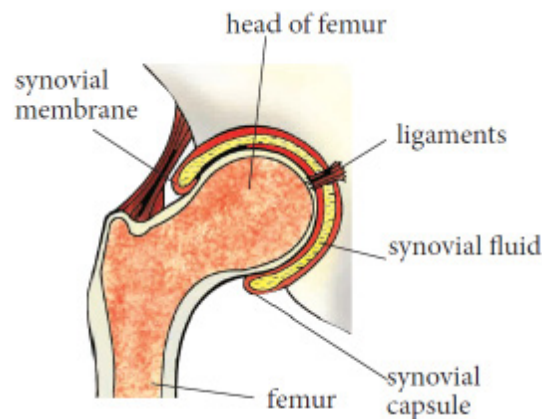


Figure 10.3: Ball and socket joint at the hip

Structure of a synovial joint

The articular surfaces of bones at synovial joint are covered with articular cartilage that provides a smooth surface where the bones meet. The adjoining bones are enclosed within a synovial cavity that is surrounded by a **fibrous capsule** that holds the bones together. Some portions of the capsule may thicken to form ligaments.

A **synovial membrane** lines the synovial cavity. The membrane produces **synovial fluid** that is a complex mixture of cells and nutrients such as proteins, polysaccharides and fats. The synovial fluid covers the surfaces of the joint and provides a lubricating film. Examples include hip joint, shoulder joint and knee joint.

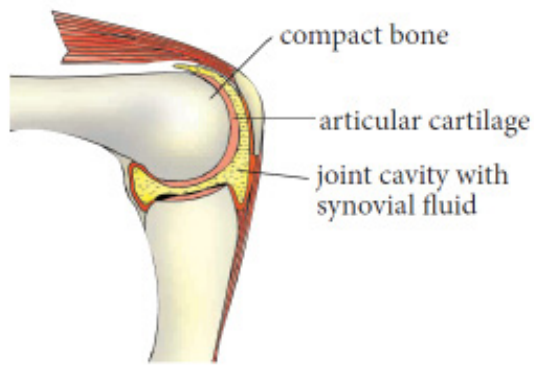


Figure 10.4: A synovial joint

Application activity 10.1



1. Assume that a sharp object penetrated a synovial joint. From the list of structures write the order in which they would most likely be penetrated.
 - 1 Muscle or tendon
 - 2 Ligament
 - 3 Articular cartilage
 - 4 Fibrous capsule
 - 5 Skin
 - 6 Synovial membrane

a) 5,1,2,4,3,6 b) 5,1,2,4,6,3 c) 5,1,2,6,4,3 d) 5,2,1,4,3,6
2. Which two joints are moving at the same time?
 - a) Knee joint and shoulder joint
 - b) Knee joint and elbow joint
 - c) Neck joint and knee joint
 - d) Neck joint and ankle joint.
3. Which of the following is used to classify joints in the body?
 - a) The number of bones that articulate.
 - b) The embryonic tissues formed at the joint.
 - c) The connective tissues that bind the bones together.
 - d) The degree of movement at the joint.
 - e) Both C and D.

10.2. Action of antagonistic muscles when moving the hinge joint

Learning activity 10.2



Demonstration of antagonistic muscle action

1. What is a hinge joint?
2. Bend your arm. Which muscle is contracting when the arm is bent?
3. Extend your arm. Which muscle is contracting when the arm is extended?
4. What are antagonistic muscles? Give examples.
5. What are the main components of a synovial joint?

Roll up your sleeves. Make a fist with your hand tightly. Bend your arm at the elbow. Observe the inner part of the upper arm. You may have noted that the inner muscles of your upper arm bulge. What do you think causes the bulge?

The biceps and triceps are referred to as **antagonistic muscles**. They are found in the upper part of the forearm. To lift the arm, the biceps contract while the triceps relax. Joints provide a fulcrum point for muscles to act. The joint action in the upward movement of the arm is **flexion** resulting in the decreasing of the joint angle. Biceps muscles are therefore called **flexor** muscles. To straighten the arm, triceps muscles contract while biceps relax; the joint action is **extension** that results an increase ease in the joint angle.

Antagonistic muscles occur in pairs and oppose a specific movement such that when one muscle contracts, the other relaxes. This means that they never contract or relax at the same time. When the lower arm is straightened at the elbow, we say that the arm is extended. When the arm is bent at the elbow, we say the arm is flexed. When the arm is extended, the triceps muscles are in a contracted state and so, they are tight and short in length. The bicep muscles on the other hand are relaxed and therefore stretched.

Basically, the movement of the arm is purely brought about by the opposing actions of the biceps and triceps.

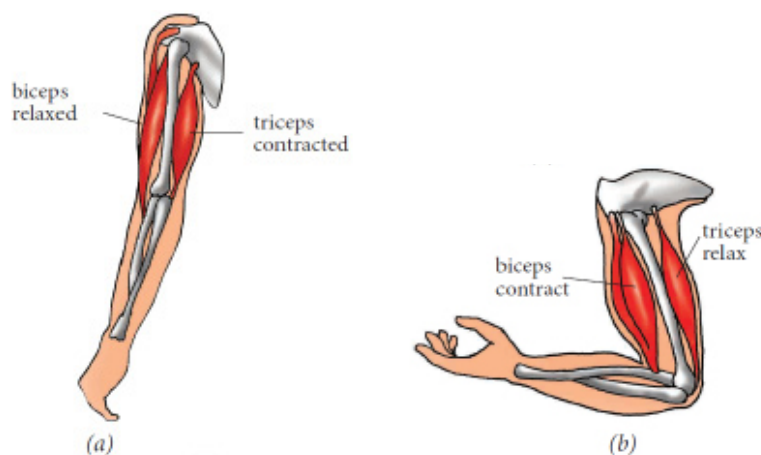


Figure 10.5: (a) Extended arm and (b) bent arm at the elbow

Application activity 10.2



1. Suppose the bicep muscles are paralysed, which arm movement will be impossible?
2. Why do muscles have to work in antagonistic way at the joints?

10.3. Discussion on good practices to promote healthy bones and joints

Learning activity 10.3



Discuss the following topics:

1. What practices can cause bone fractures?
2. Why should we eat foods rich in calcium, zinc, and magnesium ions to maintain healthy bones and promote quick recovery from fractures?
3. Explain how sunlight enhances strong bones.
4. Explain why elderly people develop a bent back.
5. Present your work to the rest of the class.

Keeping your bones and joints healthy is more important as you get older. Serious conditions such as **osteoporosis** and **arthritis** can make it hard to move around. They may lead to even more medical problems. There are simple things that you can do to reduce your likelihood of developing these conditions or at least prevent them from becoming worse.

Simple lifestyle changes can help to protect bones and joints. They include:

1. Eating healthy food appropriate for bone health, for example:
 - a) Incorporating more calcium rich foods into the diet. Getting enough calcium is essential in keeping the bones healthy and strong. Calcium rich foods include:
 - Low-fat dairy products such as milk.
 - Green leafy vegetables, such as Dodo or cabbages.
 - Foods fortified with calcium, such as cereals and bread.
 - b) Eating foods that contain Vitamin D. Foods that contain vitamin D include egg yolk and beef liver. Vitamin D is also synthesized when our skin gets exposed to the sun.
 - c) Getting enough Vitamin C, which is necessary for repairing tissues, including the cartilage in joints. Foods that are rich in vitamin C include: fruits, such as oranges, lemons, watermelon, sweet potatoes and tomatoes.
2. Avoiding alcohol consumption. Taking alcohol interferes with the body's ability to absorb vitamins and minerals. It also results in an increase in hormones that deplete bone density.

3. Weight bearing exercises help the body to build more bone mass and maintain the bone density. Some examples of weight bearing exercises include:

- Walking
- Running
- Dancing
- Playing soccer
- Playing basketball
- Playing tennis

4. Avoiding smoking. Smoking has been shown to cause bone mass depletion. Smoking also causes many other serious health problems as well.



Figure 10.6: No smoking symbol

5. Maintaining a healthy body weight and posture. It is important to maintain a healthy body weight to keep your bones and joints healthy. If you are underweight, you are at a higher risk of bone loss. If you are overweight, you may be causing extra stress on your joints. This may put you at risk of developing osteoarthritis.

Application activity 10.3



1. Explain why it is important to maintain a healthy weight.
2. Give reasons why we should avoid alcohol and smoking.
3. Indicate the benefits of wearing the correct shoe size and properly fitting clothes. Give the benefits of putting on the right size of shoes or non-fitting clothes.

10.4 End of unit assessment

End of unit assessment



A. Knowledge and understanding

1. The strong tissues that hold bones together at a joint and protect them from dislocation are.....
a) Cartilages b) Ligaments c) Tendons d) Muscles

2. Flexion of the forearm is caused by.....
 - a) Relaxation of the biceps and contraction of the triceps.
 - b) Relaxation of both biceps and triceps.
 - c) Contraction of both biceps and triceps.
 - d) Contraction of the biceps and relaxation of the biceps.
3. Bones are held together by.....
 - a) Ligaments b) Cartilages c) Tendons d) Muscles
4. The skull below shows a type of joint.



- a) What specific name is given to that type of joint?
- b) Classify the joint as movable, slightly movable or immovable.
- c) Of what benefit is the joint(s) to the body?

B. Skills

5. Draw the synovial joint and describe its function.
6. Explain why we should always eat a balanced diet with regard to bones and joints health.
7. Animals move from one place to another, whereas plants don't. Do you think it's a wastage of time and energy for animals to move around?
8. Ngaboyisonga is growing old. Design a diet suitable for his bones.
9. Which two joints in our skeleton move in the same way?
 - a) Knee joints and neck joints
 - b) Knee joints and elbow joints
 - c) Shoulder joints and elbow joints
 - d) Ankle joints and neck joints
10. What tasks would be difficult to complete if your knee joints were not functioning properly?
11. What food can keep our joints healthy?

C. Attitudes and values

12. How does warm-up before exercise demonstrate good behaviour?
13. How can participation in team sports or physical activity promote positive values?
14. Why is understanding the function of joints important for daily life?

UNIT 11

INFECTIOUS DISEASES, IMMUNITY AND VACCINATION



Key Unit Competence:

Assess natural and artificial methods that fight against infection.

Learning objectives

By the end of this unit you should be able to:

Knowledge and understanding

1

- Explain the role of pathogens in the transmission diseases
- Describe how vaccinations enhanced the body's defense system.

Skills

2

- Assess the effect of autoimmune system type 1 diabetes.
- Model the antigen- antibody binding site.
- Differentiate modes of transmission of infectious diseases
- Show complementarity of pathogens and antigens to passive and active immunity.

Attitudes and values

3

- Advocate for vaccination and breast feeding as sustainable disease prevention.
- Be aware of vaccination programs in their communities.

11.0. Introduction

Infectious diseases are illnesses caused by pathogens such as bacteria, viruses, fungi, and parasites that can spread from one person to another, through contaminated food and water, or by vectors. The human body has natural defense mechanisms that help protect against these disease-causing organisms. Immunity enables the body to recognize and fight infections, while vaccination strengthens immunity by preparing the body to resist specific diseases.

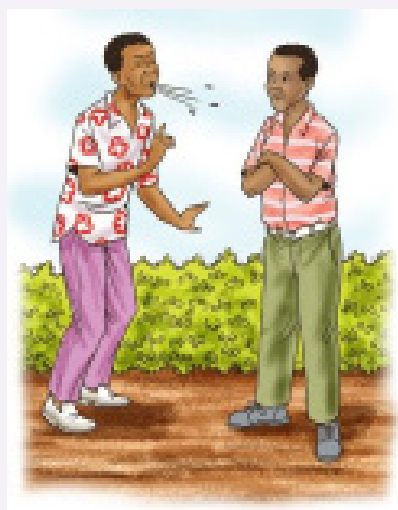
In this unit, students will explore the causes, transmission, prevention, and control of infectious diseases. They will also learn about the body's immune system, the role of immunity in disease prevention, and the importance of vaccination in protecting individuals and communities from infectious diseases.

Key concepts: infectious disease, pathogen, bacteria, virus, fungus, parasite, transmission, infection, immunity, immune system, antibody, antigen, vaccination, vaccine, prevention, control, vector.

Introductory activity



Infectious diseases are also known as **communicable diseases**. They are diseases that are transmitted from one person to another. These diseases are usually caused by harmful micro-organisms. Examples of micro-organisms include viruses, fungi, protozoa and bacteria. Look at the picture and then answer the questions below.



- Can you tell what happens to the other person who doesn't cough?
- Can the action of the person in the image lead to the transmission of a disease or not?
- If so, could all the diseases we know be transmitted this way? Why is that?
- What does that say about the diseases that you're going to learn about in this unit?

11.1. Transmission of infectious diseases

Learning activity 11.1



a) Discussion on transmission of infectious diseases

- In groups, use student books, handouts and the internet, learn and discuss the following topics:

• Pathogen • Epidemic • Pandemic • Host • Infectious disease • Vector

Study questions:

2. How does the body defend itself against diseases?

3. Write your findings in your notebook.

4. Share your findings with the class.

b) An interactive talk with a healthcare officer

1. Prepare a questionnaire by formulating a series of questions for the healthcare worker on transmission of infectious diseases.

You may include the following questions.

- How is Ebola spread from one person to another?
- Who are the most vulnerable people to be infected with malaria?
- Why is tuberculosis a very dangerous disease?
- Is smoking related to tuberculosis?
- What has the government done to control HIV and AIDS in the Rwandan population?
- What is the role of pathogens in spreading diseases?

2. Submit your research to the teacher for assessment.

Infectious diseases can be transmitted from one person to another either directly or indirectly. They are caused by pathogens, which are microorganisms that cause disease. Examples of pathogens include bacteria, protozoa, and viruses; these are also commonly known as germs.

When pathogens reach the body, they may cause diseases. However, the body has natural mechanisms to recognize and eliminate such foreign materials. The ability of the body to defend itself against foreign bodies and organisms that cause disease is called **immunity**. The system that is responsible for protecting the body against diseases is known as the **immune system**.

- **Transmissible disease:** These are diseases that can be passed from person-to-person such as common cold.
- **Host:** an organism that harbours and supports the survival or development of pathogenic microorganisms or parasites. for example, freshwater snails act as intermediate hosts for schistosomes that cause bilharzia.
- **Vector:** This is a living organism that can transmit infectious diseases between human beings or from animals to human beings. Female anopheles' mosquito is a vector to the plasmodium parasite malaria.
- **Carrier:** This is a person who has been infected but develops no signs or symptoms. This person can transmit the disease to other people.
- **Signs:** These are visible expressions brought about by a disease, for example, red spots on the body.
- **Symptoms:** A physical or mental feature that is regarded as indicating a condition

of disease, particularly such a feature that is felt by the patient, for example, nausea, dizziness and headache.

- **Epidemiology:** The study of all factors that contribute to the appearance of a particular disease.

Main types of pathogens

Bacteria: single-celled organisms (e.g. *Vibrio cholerae* causing cholera)

Viruses: tiny particles that reproduce only inside living cells (e.g. measles virus, HIV)

Fungi: include yeasts and molds (e.g. ringworm fungi)

Protozoa: single-celled eukaryotes (e.g. *Plasmodium* causing malaria)

Note: *Not all microorganisms are harmful; only the ones that cause disease are referred to as pathogens. Pathogens can enter the body via air, food, water, cut or wounds, sexual contact, or vectors such as mosquitoes. In short, pathogens are agents that cause infectious diseases.*

Infections can be spread in several different ways. Some of them are listed below:

1. Diseases that are transmitted through the air are referred to as **air-borne** diseases. Tuberculosis and common cold are examples.
2. Some diseases are transmitted through the consumption of contaminated water and food. They are referred to as **waterborne** or **foodborne** diseases respectively. Examples are cholera and typhoid fever.
3. Through blood transfusion or contact with contaminated blood. Examples of diseases that can be spread through contaminated blood are HIV and AIDS and malaria.

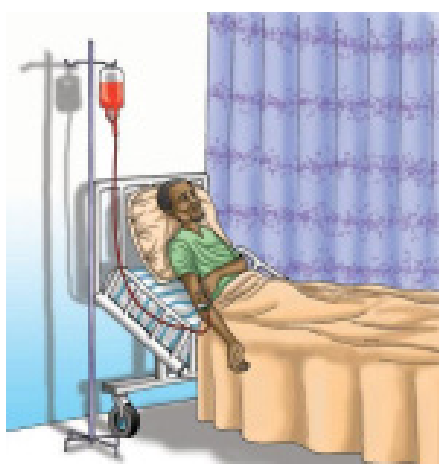


Figure 11.1: Contaminated blood used during blood transfusion can lead to transmission of diseases

4. Physical contact with an infected person can transmit a disease. These infectious diseases are termed as contagious. They include ringworms, scabies, Ebola and COVID 19.
5. Communicable diseases are also spread by vectors. They are therefore known as **vector-borne** diseases, for example, sleeping sickness is spread by tsetse flies, malaria is spread by female *Anopheles* mosquito. Yellow fever on the other hand is spread by black flies and some species of mosquitoes.

6. Some infections can also spread through unprotected sexual intercourse with an infected person. These diseases are known as **sexually transmitted infections** (STIs), for example, HIV and AIDS.
7. Germs can also spread from a mother to her unborn child, usually through blood or other body fluids. They can also pass through contact with skin or mucous membranes during delivery.

Application activity 11.1



1. Which of the following is the vector of sleeping sickness?
a) Tsetse fly b) Mosquito c) Black fly d) Trypanosoma
2. If you have come in contact with a person who has been coughing, what diseases are you likely to have contracted.
3. Name two diseases transmitted through contaminated blood transfusions.

11.2. Symptoms, prevention, control and treatment of common infectious diseases

Learning activity 11.2



Visit to health centre to learn about infectious diseases

During your field visit to the medical centre, learn about these diseases and complete the worksheet below.

Table 11.1: Common infectious diseases

Infectious disease	Causes	Symptoms	Prevention and control
1. Malaria			
2. Cholera			
3. HIV/AIDS			
4. Tuberculosis			
5. Ebola			
6. Salmonellosis			
7. Typhoid			

a. Malaria

Malaria is a protozoan disease caused by a parasite called *Plasmodium*. The parasites are spread through the bites of infected female *Anopheles* mosquitoes, which are the malaria vectors. There are several species of *Plasmodium* parasite that cause malaria in human beings. They include *P. falciparum* and *P. vivax*. Malaria poses the greatest threat to human health.

Malaria affects some groups in the population more than others. These groups are prone to developing a severe form of the disease, than others. They include infants, children under 5 years of age, pregnant women and patients suffering from HIV and AIDS.

Symptoms of malaria

- i. High fever and periodic chills that may be accompanied with sweating.
- ii. Frontal headache.
- iii. Pain in the joints.
- iv. General body weakness.
- v. Lack of appetite and nausea that may be accompanied by vomiting.

If malaria is not treated within 24 hours it can progress to severe illness, often leading to death. Children with severe malaria frequently develop severe anaemia and cerebral malaria. In adults, it may involve malfunctioning of several organs.

Prevention and treatment of malaria

Mosquito control is the main way to prevent and reduce malaria transmission. The following are some of the ways mosquitoes can be controlled.

- i. Clearing bushes around homes.
- ii. Draining marshes, pools and stagnant water around homes. This destroys breeding places for mosquitoes.
- iii. Spray a thin layer of used oil on stagnant water that cannot be drained. Oil sprayed on stagnant water kills mosquito larvae by suffocating them.
- iv. Spraying the inside walls of our houses with insecticides.
- v. Closing doors and windows during the evening time to prevent mosquitoes from entering. Windows can also be screened with window proof wire mesh.
- vi. Removing old tins, broken pots, polythene bags and other things that can hold water from the compound. This prevents mosquitoes from breeding.
- vii. Fish that eat mosquitoes can be introduced into ponds and pools to feed on mosquito larvae and pupae.
- viii. Preventative medicines can be taken regularly but upon prescription from a qualified medical doctor.
- ix. Sleeping under a treated mosquito net.



Figure 11.2: Sleeping under a mosquito net

b. Cholera

Cholera is an acute diarrheal infection caused by ingestion of food or water contaminated with the bacterium *Vibrio cholerae*. It has a short incubation period of 2 hours to 5 days, a factor that triggers its pattern of outbreaks. Cholera is endemic in poor countries and strikes areas with poor sanitary conditions where drinking water is not well treated. Cholera affects both children and adults and can cause death within hours.

Most people infected with cholera do not develop any symptoms. The bacterium is present in their faeces for 1-10 days after infection and is shed back into the environment, endangering the health of other people.

Most people who develop symptoms have mild or moderate symptoms, while others develop acute watery diarrhoea with severe dehydration (lack of water in the body). This can lead to death if left untreated.

Symptoms of cholera

- i. Severe watery diarrhea (rice water stool) and vomiting.
- ii. General body weakness.
- iii. Low blood pressure.
- iv. The patient loses weight rapidly.
- v. The patient develops wrinkled skin and sunken eyes because of dehydration.

Control and treatment of cholera

- i. Drinking water should be treated or boiled and stored properly.
- ii. Proper disposal of human waste. A well-managed sewage disposal system in towns and use of latrines in the rural areas.
- iii. Covering cooked food to avoid contamination by house flies.
- iv. Vaccination during epidemics or when one is visiting an infested area.
- v. Cholera is treated using antibiotics. Prevention of dehydration also helps in the cure.
- vi. Washing hands before and after eating and after visiting the toilet.



Figure 11.3: *Sleeping under a mosquito net*

c. HIV and AIDS

The Human Immunodeficiency Virus (HIV) attacks the immune system. This weakens a person's defense mechanism against infections and some types of cancer. As the virus destroys and impairs the function of immune cells, infected individuals gradually become immune deficient. Immune function is typically measured by number of white blood cells in the body (**CD4 cell count**).

Immunodeficiency leads to increased susceptibility to a wide range of infections. These infections (**opportunistic infections**) are diseases that can be fought by people with a healthy immune system. This is when it is said that the patient has AIDS. HIV and AIDS is transmitted by different means from person to person. They include:

- i. By birth, from an infected mother to her unborn baby.
- ii. By having unprotected sexual intercourse with an infected person.
- iii. By transfusion of infected blood.
- iv. By sharing unsterilized surgical and skin piercing instruments with infected people.

Symptoms of HIV and AIDS

The symptoms depend on the stage of infection. People living with HIV and AIDS tend to be most infectious in the first few months. The first few weeks after initial infection, individuals may experience no symptoms or an influenza-like illness including fever, headache, rash or sore throat.

As the infection progresses, the immune system weakens. An individual can develop other signs and symptoms, such as swollen lymph nodes, weight loss, fever, diarrhea and cough. Without treatment, they could also develop severe illnesses such as tuberculosis, meningitis, and cancers such as lymphomas and Kaposi's sarcoma, among others.

Prevention and control

Preventing the spread of AIDS requires health and sexual education to be the most effective. Everyone must be aware of the risk of contracting the virus through an inappropriate lifestyle.

HIV and AIDS patients should be given anti-**retroviral** (ARVs) medicines to extend their life. In order to prevent AIDS, it is also necessary to avoid the means by which it

is transmitted. The following are ways for a person to prevent contracting the disease.

- i. Abstaining from sex till marriage.
- ii. Being faithful to one partner.
- iii. Using condoms during sex.
- iv. Mothers should give birth in hospitals and go for regular medical check-up.
- v. Blood should be screened before transfusion.
- vi. Avoid sharing piercing and cutting instruments like razorblades and safety pins.

d. Tuberculosis

Tuberculosis (TB) is caused by a bacterium called *Mycobacterium tuberculosis* that most often affects the lungs. TB spreads through the air from person to person. When someone with pulmonary TB coughs, sneezes, they spread the bacteria around in the air. If a healthy person nearby inhales the bacteria he or she may be infected.

People with a low immune system, such as people living with HIV and AIDS, people who are malnourished, people who have diabetes, or who smoke, are at much greater risk of contracting the disease.

Symptoms of pulmonary TB

Common symptoms of active pulmonary TB are:

- i. Cough with sputum and blood at times
- ii. Chest pains
- iii. General weakness
- iv. Weight loss
- v. Fever
- vi. Night sweats

When a person develops active tuberculosis, the symptoms; cough, fever, night sweats and weight loss may be mild for many months. This can lead to delays in seeking medical care, resulting in transmission of the bacteria to other people.

Prevention and treatment of tuberculosis

- i. Isolate patients to prevent transmission of the disease to other people.
- ii. Immunize children with tuberculosis vaccine known as BCG. This may prevent the spread of tuberculosis. But the vaccine does not necessarily protect against pulmonary tuberculosis.

Avoid overcrowded places. Opening windows in overcrowded rooms to allow free circulation of air can prevent contracting the disease.

- iii. Drinking boiled milk. Some types of tuberculosis is transmitted from bovine.
- iv. Once infected, tuberculosis are treated with antibiotics. Standard anti-TB drugs

have been used for some time, and resistance to the medicines is widespread.

e. Ebola

The Ebola virus causes an acute, serious illness which usually leads to death if untreated. Ebola virus is introduced into the human body through contact with body fluids of infected animals such as chimpanzees, gorillas, fruit bats, monkeys, forest antelope and porcupines found ill or dead in the forest.

Ebola is then spread from person to person through direct contact (through broken skin or mucous membranes) with the blood, secretions, organs or other body fluids of infected persons and with surfaces and materials such as bedding and clothing contaminated by these fluids.

Medical personnel also get infected when treating Ebola patients. Burial ceremonies in which mourners are in direct contact with the body of the dead patient may also be conducive to. People stay contagious as long as there's a virus in their blood.



Figure 11.4: Wear protective gear when handling Ebola patients

Symptoms of Ebola virus disease

The incubation period: the time interval from infection with the virus to onset of symptoms is 2 to 21 days. Human beings are not infectious until they develop symptoms.

- i. First symptoms are the sudden onset of fever, fatigue, muscle pain, headache and sore throat.

This is followed by vomiting, diarrhoea, rash, symptoms of impaired kidney and liver function.

- ii. Internal and external bleeding with blood oozing from the gums, ears and eyes.

Prevention of Ebola

- i. Avoid direct contact with wild animals like bats and chimpanzees.
- ii. Avoid direct contact with people showing signs of Ebola virus.
- iii. Use gloves and other protective clothing when handling Ebola patients.
- iv. Wash your hands regularly with soap and water.
- v. Couples who have just recovered from the virus should abstain from sexual intercourse or use condoms.

vi. Burial of the dead patients should be done by health workers.

f. Salmonellosis

This is an infection caused by *Salmonella* bacteria. The elderly, infants, and those with impaired immune systems are more likely to develop severe forms of illness.

Salmonella infection usually results from ingestion of the bacteria from contaminated food or water. Eggs, milk, meat or poultry are particularly high-risk foods. Fruit and vegetables may also be contaminated, especially if manure has been used as fertiliser. People may become infected if they transfer animal faeces containing *Salmonella* bacteria from their hands to their mouths, for example, eating after touching animals and failing to wash hands.

Person-to-person spread may occur when contaminated hands, objects or food are handled. In most cases, the illness lasts for 4 to 7 days. Most people recover without any kind of treatment. In some cases, diarrhea may be severe enough to cause severe dehydration and require hospitalisation.

Symptoms of salmonellosis

Most people infected with salmonella experience diarrhoea, chills with fever, nausea with or without vomiting, headache, blood in the stool and abdominal cramps 12 to 72 hours after infection.

Prevention and treatment of salmonellosis

- i. People with salmonella infections should not be allowed to handle food or take care of children.
- ii. Food should be well cooked.
- iii. Do not purchase dirty or cracked eggs.
- iv. Boil milk thoroughly before drinking.
- v. Always wash your hands with soap and clean water.
- vi. Wash fruits (including melons) and vegetables before eating.
- vii. Swimming pools should be disinfected regularly.
- viii. Recovery from *Salmonella* infection usually occurs within a week and antibiotic treatment is not normally required.

g. Typhoid

Typhoid fever is a bacterial infection. It is caused by a bacterium called *Salmonella typhi*, which only lives in humans. It is related to the bacteria that cause salmonella food poisoning. It can spread throughout the body, affecting many organs. Without prompt treatment, it can cause serious complications and can be fatal.

Typhoid fever is highly contagious. A small number of people (carriers) recover from typhoid fever but continue to have bacteria. Both ill people and carriers shed bacteria in their faeces and urine. The bacteria are spread by eating food or drinking water contaminated by faeces or urine of patients and carriers.

Typhoid is common in areas where hygiene standards are poor, and water is likely to be contaminated by sewage waste.



Figure 11.5: Water contaminated with sewage waste

Symptoms of typhoid

- i. Sustained high fever.
- ii. Continuous headache.
- iii. Malaise (feeling unwell).
- iv. Decreased appetite.
- v. Enlarged spleen may cause abdominal discomfort.
- vi. Dry cough in the early stage of the illness.
- vii. A flat rose-coloured rash which may be visible on the trunk.
- viii. Constipation or diarrhea : constipation occurs more often than diarrhea in adults.
- ix. Illness varies from mild with low-grade fever, to severe with multiple complications.

Prevention and treatment of typhoid

- i. Isolation of the patients to avoid spread of the disease by contact and sterilizing clothes used by patients.
- ii. Food handlers in institutions, such as hospitals, schools and restaurants should undergo regular medical examinations and treatment if they are found to be infected.
- iii. Proper disposal of faeces in toilet and pit latrines.
- iv. Water treatment and purification to kill the bacteria.
- v. Boiling drinking water and cooking of food properly to kill the bacteria.
- vi. Practice good hygienic methods like washing hands properly after visiting the toilet.
- vii. Washing fruits and vegetables with clean water before eating.
- viii. Vaccinations in the case of outbreaks of the disease in epidemics.
- ix. In case of infection, seek appropriate treatment from qualified medical practitioners. Antibiotic treatment helps shorten the duration of infection.

Application activity 11.2



1. Which of the following disease is the most contagious?
 - a) Malaria
 - b) Ebola
 - c) HIV and AIDS
 - d) Typhoid
2. What is the easiest thing you can do to prevent getting sick?
3. State what ways can you use to sensitize the community to the dangers of infectious diseases?

11.3. Defense against infection

Learning activity 11.3



Make research and discuss defense against infection

1. Discuss the ways your body fights infections.
2. Share your findings with other class members.

Immunity is the ability of the body to resist disease infection. This means a person can be exposed to factors that can cause a disease, yet they do not become sick. The body's defense mechanism protects an organism from infections.

If pathogens do not encounter resistance from the body's defense mechanism, almost all diseases would be fatal. Organisms must find a way of defending themselves against harmful micro-organisms like viruses, bacteria and fungi.

The body has many ways of defending itself from invasion by harmful micro-organisms. In most cases the body's defense mechanism prevents this from happening. Some of these defense mechanisms are made in a way to block pathogens from entering the body.

Other mechanisms eliminate pathogens that are already in the body. There are also defense mechanisms that attack micro-organisms when they persist inside the body. The ability of the body to prevent itself from diseases using its lines of defense is called **resistance**.

Body defense mechanisms can either be first line or second line.

a. Body's first line of defense

This line of defence prevents harmful micro-organisms from entering the body. It is the external defence system of the body. It includes both physical and chemical defence lines. The body does this using various barriers that intercept micro-organisms hence prevent them from entering the body.

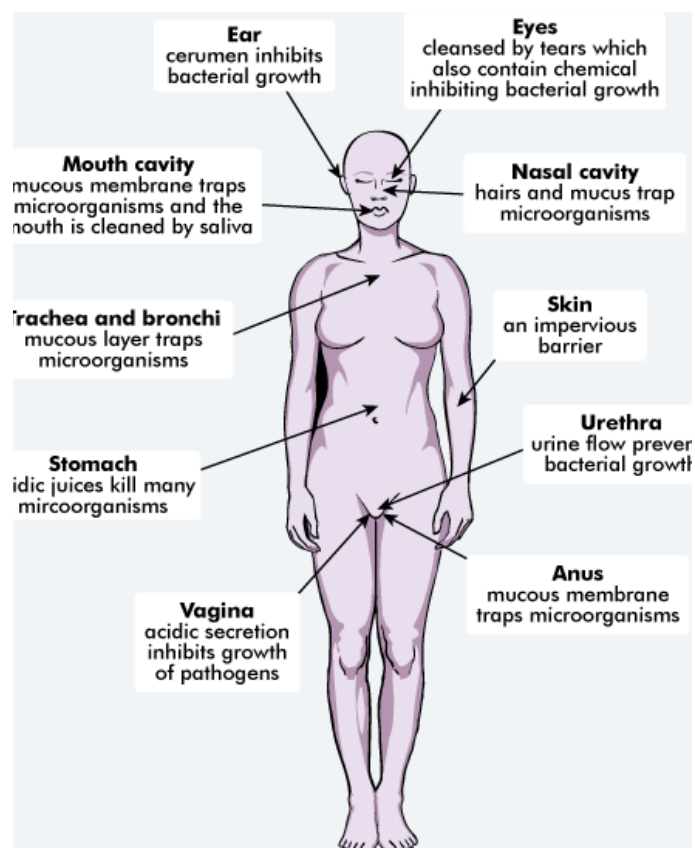


Figure 11.6: The body's first line of defence mechanisms

Example of the body's defense mechanism include:

- The **skin** creates a physical barrier that protects the cells inside the body against the entrance of pathogens. Healthy skin is rarely penetrated by pathogens. Certain chemical secretions produced by the skin help to stop the growth of bacteria and fungi.
- The nose and passages leading to the lungs are lined with cells that produce sticky fluid called **mucus** that traps invading microbes and dust. Tiny hairs called **cilia** line the trachea. They can move back and forth in a wave-like motion. Cilia trap microbes and dust particles and prevent them from entering the lungs. These particles then combine with mucus and are either coughed, sneezed out or swallowed and then passed out of the body in faeces.
- The stomach produces hydrochloric acid which destroys many of the microbes that enter the body in food and drinks we take.
- Tears act as a barrier to pathogens. Tears are a watery secretion produced by tear glands located in the outer eye. They contain a powerful enzyme that can digest and break down harmless substances. Therefore, harmful microorganisms cannot enter the body through the opening of the eye.
- Clotting of blood occurs when an open cut or wound exposes blood to air. Such a cut causes a break in the skin, exposing the body to harmful micro-organisms. When a clot forms, it seals the opening and enables the wound to heal.

b. Body's second line of defense

When pathogens can get past the first line of defense, such as, a cut on the skin,

disease can develop. This prompts the second line of defense to become active. This happens through a sequence of steps called the **immune response**. The immune system attacks the pathogens. The body has a range of defense mechanisms that operate to prevent or destroy pathogens. These responses only react to the presence of any pathogen or foreign material.

White blood cells known as **phagocytes** are adapted to fight these disease-causing micro-organisms in the following ways:

- They move by amoeboid action. They send out cell extensions called **pseudopodia** which allow them to surround invading microbes and engulf them.
- Phagocytes release digestive enzymes which break down the trapped microbes before they can do any harm. This process is called **phagocytosis**.
- They also have the ability to change their shape. This enables them to fit in the narrow capillaries hence reach the pathogens

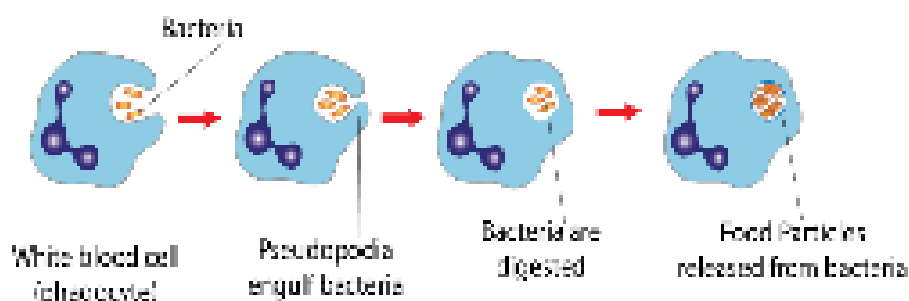


Figure 11.7: Phagocytosis in white blood cells

Immunity

This is the ability of an organism to resist a particular infection or toxin by the action of specific antibodies or some white blood cells.

An antibody is a protein produced by the body's immune system when it detects harmful substances, called **antigens**. Examples of antigens include micro-organisms such as bacteria, fungi, parasites, viruses, and chemicals. Each type of antibody is unique and defends the body against one specific type of antigen.

Antibodies detect and neutralize the antigen by binding to it. This signals the production of other cells of the immune system to get rid of the invading microbes before it spreads or attacks the body.

Immunity may be active or passive. In active immunity, the body produces its own antibodies against the pathogen while in passive immunity the body receive antibodies from other people or an animal. Either type of immunity may occur **naturally** or be induced **artificially**, for treatment and prevention purposes. Thus, we can recognize four classes of immunity under this scheme.

- **Natural active immunity:** This is the production of antibodies by the body as a result of infection or other natural exposure to the antigen.
- **Artificial active immunity:** This is the production of antibodies by the body, as a result of vaccination against diseases such as smallpox, tetanus or influenza.

- **Natural passive immunity:** this is temporary immunity that resulting from the production of antibodies by another individual. The only natural way this happens is for the fetus to get antibodies from the mother via the placenta before it is born or for the baby to get them from the colostrum or breast milk after it is born.
- **Artificial passive immunity:** This is temporary immunity which is obtained by injecting an immune serum obtained from another person or from animals (e.g. horses) that has produced antibodies against a pathogen. The immunological serum is used as an emergency treatment for snake bites, botulism, rabies, and other diseases.

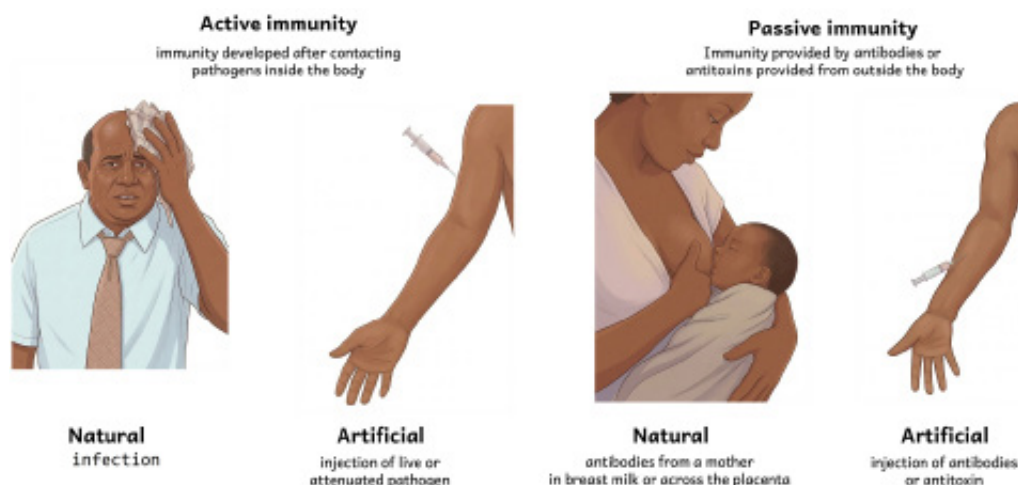


Figure 11.8: Types of immunity

Application activity 11.3



1. Fill in the following table.

Terms	Definition
Immunity	
Defense	
	Harmful microorganisms.
	An animal or plant on or in which a parasite lives.

2. Differentiate between the first and second line of defence system in the human body.

11.4. Antibodies and antigens

Learning activity 11.4



Investigating antibodies and antigens

Observe the diagrams on the chart showing an antigen and antibody, then identify the following:

- Shape of the antigen and that of the antigen-binding site on the antibody.
- The reaction centers of antibodies.

Task

- Why are antibodies specific to only one type of antigen?
- Draw an antibody.
- Describe the antibody-antigen complex.

An antibody, also known as an **immunoglobulin** is a large Y-shaped protein. It is produced mainly by plasma cells. It is used by the immune system to identify and neutralize pathogens such as bacteria and viruses. The antibody recognizes a unique molecule of the harmful agent, called an **antigen**. The antibody then binds to this antigen. Antibodies are specific to their antigens.

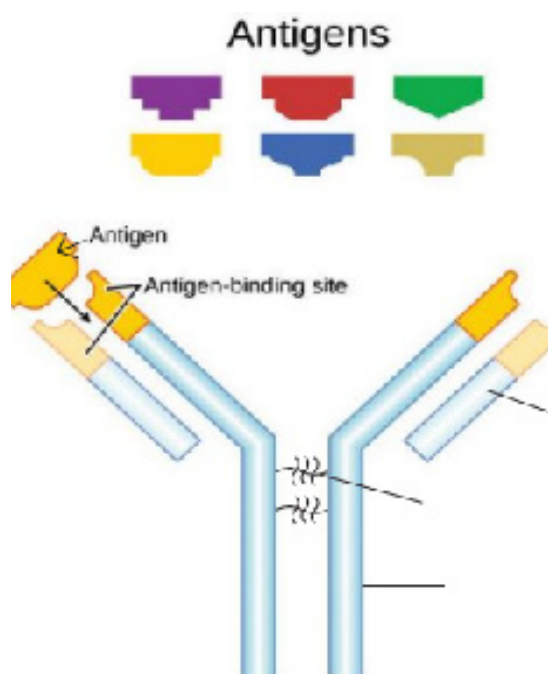


Figure 11.9: Structure of antibody and antigens

Antigen

This is any substance that can stimulate an immune response, specifically activating the body to produce antibodies. In general, two main divisions of antigens are recognized: foreign antigens and self-antigens.

- Foreign antigens come from outside the body. All pathogens are examples of foreign antigens. Viruses and micro-organisms such as bacteria and protozoa, and certain proteins in foods all lie under this group.
- Self-antigens, on the other hand, come from within the body. Normally, the body can distinguish self from non-self. However, in individuals with autoimmune disorders, normal body substances provoke an immune response, leading to the generation of autoantibodies.

Antigen–antibody reaction

An antigen and antibody reaction works like a lock and key mechanism. Antibodies themselves do not destroy antigens. They inactivate and tag antigens for destruction by phagocytes.

All antibodies form an antigen-antibody (immune) complex When they combine with antigens.

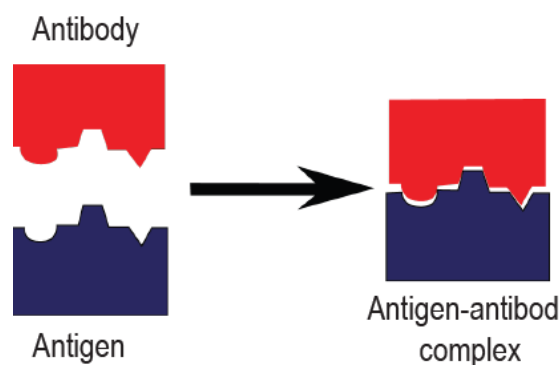


Figure 11.10: Formation of an antigen-antibody complex

Each antigen stimulates the production of a specific antibody. All antibodies are Y-shaped. They, however, differ at the antigen binding site. Each antibody has a site complementary to a certain antigen. This means that each antibody can only bind to a specific antigen.

In response to the antigen, the antibody wraps its two sites like a 'lock' around the 'key' of the antigen sites to destroy it. An antibody's mode of action varies with different types of antigens. With its two-armed Y-shaped structure, the antibody can attack two antigens at the same time with each arm.

Application activity 11.4



1. Draw a key and lock and relate it to antibody- antigen reaction.
2. How does the body defend itself from infections?

11.5. Immune response

Learning activity 11.5



Observing immune response

1. Carefully observe the simulations of the immune response of organisms
2. Differentiate immune responses of organisms. Share your findings with the class.

Immune response refers to the process by which the body recognizes and defends itself against bacteria, viruses and substances that appear foreign and harmful. The

immune system is a collection of cells, tissues and molecules that protect the body from numerous pathogens and toxins in our environment.

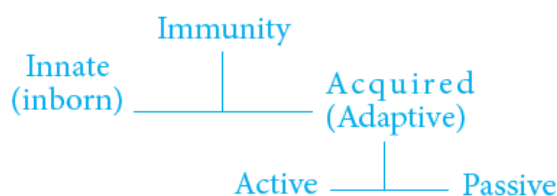


Figure 11.11: The immune system

This defense against microbes has been divided into two general types of reaction:

- Reactions of innate immunity
- Reactions of adaptive immunity.

a) Innate immunity

Innate immunity, also called nonspecific immunity, is the body defense system that one is born with. It protects against all antigens. Innate immunity involves barriers that keep harmful materials from entering your body. It consists of cells and proteins that are always present and ready to fight microbes at the site of infection.

The main components of the innate immune system are:

- i. Physical epithelial barriers
- ii. Phagocytic leukocytes
- iii. Dendritic cells
- iv. A special type of white blood cells called a natural killer (NK) cell
- v. Circulating plasma proteins

Innate immunity includes the components of the body's first line of defense. These are enzymes in tears and skin oils, mucus, skin and acid in the stomach. If an antigen gets past these barriers, it is attacked and destroyed by other parts of the immune system.

b) Adaptive immunity

The adaptive immune system, also known as acquired immunity is immunity that develops after exposure to various antigens. The immune system builds a defense against that specific antigen. It is called into action against pathogens that can evade or overcome innate immune defenses. Components of the adaptive immune system are normally silent; however, when activated, these components "adapt" to the presence of infectious agents by creating mechanisms for neutralizing or eliminating the microbes.

The immune system includes certain types of white blood cells, for example, **lymphocytes**. There are B and T types of lymphocytes.

- **B** lymphocytes are cells that produce antibodies. Antibodies attach to a specific antigen and present it to other immune cells for destruction.

- **T** lymphocytes are cells that attack antigens directly specifically targeting infected cells or cancer cells in a process called cell-mediated immunity and help control the immune response.

During growth and development of white blood cells, they always learn to differentiate between individuals' own body tissues and foreign substances. Once B cells and T cells are formed, a few of those cells will multiply and provide "**memory**" for the immune system. This allows the immune system to respond faster and more efficiently the next time an individual is exposed to the same antigen. In many cases it will prevent you from getting sick, for example, a person who has had measles or has been immunised against measles is immune from getting it again.

11.5.1. Immunisation and autoimmunity

Immunization is the process by which an immune response is triggered by the administration of a vaccine towards an infectious disease. Small amounts of an antigen, such as dead viruses, are given to activate immune system "memory". Memory allows your body to react quickly and efficiently to future exposure. Such small doses of antigens are referred to as vaccines.

Immunisation is an important method of disease prevention. By receiving a vaccination, usually in the form of an injection, a person can be 'immunised' against a disease and reduce their likelihood of developing the illness. Immunisation can also reduce the spread of the disease in the population.

Autoimmunity is a condition whereby the immune system attacks and kills its own cells. Any disease that results from such an immune response is termed an **autoimmune disease**.

An efficient immune response protects against many diseases and disorders. Wrong immune response causes immune system disorders. An overactive immune response can lead to the development of autoimmune diseases, in which antibodies form against the body's own tissues.

Complications from altered immune responses include:

- i. Allergy or hypersensitivity
- ii. Autoimmune disorders
- iii. Immunodeficiency disorders
- iv. Serum sickness
- v. Transplant rejection

11.5.2 Type 1 diabetes and the immune system

Type 1 diabetes was initially known as **juvenile** diabetes or insulin-dependent diabetes mellitus. It is a chronic condition in which the pancreas produces little or no **insulin**. Insulin is a hormone needed to enable sugar (glucose) to enter cells to produce energy. The pancreas is the organ responsible for controlling the amount of sugar (glucose) in the body. Specific cells in the pancreas known as **beta** cells are responsible for manufacturing insulin. Insulin released into the blood regulates the amount of sugar in the body keeping it at normal levels.

Type 1 diabetes is caused by **destruction of insulin-producing beta cells** in the pancreas, leading to little or no insulin production.

Main causes

Autoimmune reaction: the immune system mistakenly attacks pancreatic beta cells

Genetic predisposition: inherited susceptibility

Environmental triggers: often viral infections (e.g. enteroviruses) that trigger the autoimmune response.

Note: Type 2 diabetes mellitus occurs when the body cells become resistant to insulin or when less insulin is produced by the body.

Application activity 11.5



1. Differentiate between active and passive immunity.
2. What is the immune system made up of?
3. Outline the importance of breastfeeding.

11.6. End of Unit Assessment



11.6.1. Project activity: Investigating the Spread and Prevention of Infectious Diseases in the Community

1. Context/Background

Infectious diseases such as malaria, tuberculosis, cholera, influenza, and COVID-19 affect millions of people worldwide. These diseases can spread through contaminated food and water, air, vectors, or direct contact. Understanding how infectious diseases spread and how they can be prevented helps individuals and communities maintain good health.

2. Challenge/Problem Statement

Many infectious diseases continue to spread in communities due to poor hygiene, inadequate sanitation, and lack of awareness about disease prevention.

Key Question

How can knowledge of infectious disease transmission help communities reduce the spread of diseases?

3. Purpose of the Project

The purpose of this project is to help students investigate common infectious diseases in their community, understand their modes of transmission, and propose practical prevention measures.

4. Instructions

- Identify at least two infectious diseases common in your community.
- Investigate their causes, modes of transmission, symptoms, and prevention measures.
- Collect information through interviews, observations, textbooks, health facilities, or reliable online sources.
- Present findings using charts, posters, reports, or presentations.

5. Your Tasks

- Identify common infectious diseases in the community.
- Investigate how these diseases are transmitted.
- Explain their effects on individuals and communities.
- Suggest practical measures to prevent their spread.
- Develop an awareness message promoting disease prevention.

6. Expected Student Outputs

- A report on selected infectious diseases.
- Charts or tables showing modes of transmission and prevention measures.
- A health awareness poster or brochure.
- Presentation of findings and recommendations.

7. Constraints and Limitations

Scope: Focus on diseases found in the local community.

Materials and Methodology: Use interviews, observations, textbooks, and reliable health information sources.

Time Frame: 2 weeks.

8. Guiding Questions

- i. What infectious diseases are common in your community?
- ii. How are these diseases transmitted?
- iii. What factors contribute to their spread?
- iv. What prevention measures can reduce infections?
- v. How can communities promote healthy practices?

11.6.2. Assessment Questions

A. Knowledge and understanding

1. Write true or false for each of the following:

- a) Another name for antibody is antigen.
 - b) Antigens are proteins in nature.
 - c) Diabetes Type I is an example of an autoimmune disease.
 - d) Immunisation is an example of artificial passive immunity.
2. The following are methods used by antibodies in fighting antigens. Which one is not?
- a) Precipitation b) Agglutination c) Autoimmunity d) Neutralization
3. All the following statements are true. Which one is not?
- a) An antibody is a protein that is produced by the body as a result of presence of an antigen in the body.
 - b) Immunisation involves the introduction of alternated antigens into the body
 - c) An antibody cannot destroy its own cells.
 - d) Breastfeeding is a source of immunity to a new-born baby.
4. Fill in the blank spaces.
- a) Antigens bind at the..... site forming complexes.
 - b)immunity is a type of immunity that children acquire before they are born.
 - c) cells in the pancreas are responsible for the production of a hormone called that regulates blood glucose.
5. Explain the effects of malaria to a developing country.
6. What causes multi-drug resistance in the treatment of TB.
7. Sub-Saharan Africa has the highest incidences of HIV and AIDS. Suggest why the rate is so high.
8. Explain why HIV virus has a devastating effect on our body's ability to fight diseases.
9. Mukantagara was involved in an accident and lost a lot of blood. His family and friends offered to donate blood. As a doctor, what factors should you consider before blood transfusion is done?
10. Dukuzumuremyi's parents' rear fish which is the family's source of income. Her younger brother recently complained of headaches, chills and fever, general body weakness and pain in the joints.
- i. Dukuzumuremyi's brother is to be suffering from
 - ii. Dukuzumuremyi's parents are away on vacation. What should she do?
 - a) Give her brother herbal medicine.
 - b) Go to the local chemistry and buy drugs.

- c) Wait for their parents to come back.
- d) Take her brother to the hospital.

iii. What should she do to prevent the same in future?.

11. Saliva, mucus and tears are all first line defence mechanisms in the body. Explain
12. Write a letter to your District Mayor advising him/her on how to prevent infectious diseases in your district.

B. Skills

13. Do you think your family or society can live without infectious diseases?
14. People are aware of the dangers of malaria and yet it has persisted. Why do you think this is so?
15. Role-play measures to prevent the spread of covid-19.
16. How is breast milk a form of immunity for the baby?
17. Explain the key and lock mechanism in antibody-antigen reaction?
18. a) Explain why you were vaccinated when you were young.
b) What would have happened if you were not vaccinated

C. Attitudes and values

19. What should be the attitude of people towards vaccination?
20. Why should people carry out a full vaccination course?
21. Why should people avoid spreading rumors about infectious diseases?
22. Why is it important to respect isolation and quarantine guidelines in an epidemic?
23. Mukantagara has refused to take his baby for vaccination against polio. His friend Hakizimana told him that polio vaccines are laced with a virus to control population.
a) Were his concerns true?
b) What would you advise him to do?
c) Have you heard such rumours in your village? What was it about?

UNIT 12

SEXUAL BEHAVIOUR AND SEXUAL RESPONSE



Key Unit Competence:

Practice positive sexual behaviours, living free from side effects of irresponsible sexuality..

Learning objectives

By the end of this unit you should be able to:

Knowledge and understanding

1

- Explain different ways couples can show love and affection.
- Describe male and female sexual responses
- Discuss responses to sexual attraction and excitement at puberty by boys and girls

Skills

2

- Demonstrate responsible sexual behaviour and its impact on health and relationship
- Recognize the physical and emotional changes associated with sexual response.
- Apply knowledge of safe practices to prevent unintended pregnancies and STIs
- Assess how condoms and other contraceptive methods reduce the risk of sexual behaviour consequences
- Summarize the concept of transactional sexual activity.

Attitudes and values

3

- Recognize that sexual relationships require emotional and physical maturity.
- Develop self-confidence and control towards sexual thoughts and feelings.
- Adopt assertiveness and negotiation skills to help one to resist unwanted sexual pressure or reinforce the intention to practice abstinence.

12.0. Introduction

Sexual behaviour refers to actions, attitudes, feelings, and decisions related to sexuality. It is influenced by biological, psychological, social, cultural, religious, and personal factors. Sexual behaviour varies among individuals and societies and may include forming relationships, expressing affection, and making decisions about sexual activity. Understanding sexual behaviour helps individuals make responsible and informed choices, avoid sexually transmitted infections (STIs) and unintended pregnancies, develop healthy and respectful relationships, and appreciate the importance of consent, self-discipline, and personal values.

Sexual response refers to the physical and emotional changes that occur in the body when a person becomes sexually aroused. These responses are controlled by the nervous and hormonal systems and are a normal part of human growth and development.

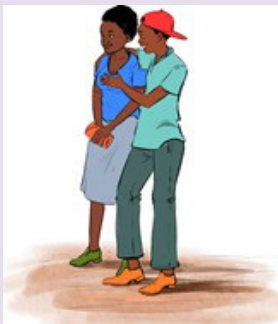
Sexual behaviour and sexual responses are natural aspects of human life. Learning about them equips individuals with the knowledge and skills needed to make healthy, responsible, and respectful decisions, contributing to personal well-being, positive relationships, and social health.

Key concepts in this unit: sexual behaviour, sexual response, sexuality, consent, relationships, decision-making, self-discipline, values, sexual health, and responsible behaviour.

Introductory activity



You may have read about adolescence in primary school.



- What does adolescence mean?
- What are some of the risks we encounter during adolescence?
- What should we do to avoid them?
- What is going on in the picture? Is it good or bad?
- Suggest more constructive activities the two people could do instead of what they are currently doing.

12.1. Male and female sexual responses

Learning activity 12.1



Discussion on male and female sexual responses

- Discuss the following topics:
 - Sexual responses in males and females.
 - Sexual excitement for physical response.

2. Note down and share your findings with the rest of the class.
 - i. Study questions: What do you observe happening in the picture?
 - ii. What type of relationship do you think exists between the two people?
 - iii. Is the behavior of the boy appropriate? Explain your answer.
 - iv. What risks can arise from such behavior among adolescents?
 - v. What advice would you give to both individuals in the picture?
 - vi. What lessons can young people learn from this situation?

Sexual response refers to the physical and emotional changes that occur in the body when a person experiences sexual arousal. These changes are normal biological processes that help prepare the body for sexual activity.

The human sexual response cycle is commonly divided into four stages:

i) Excitement phase

The excitement phase is the first stage of the body's response to sexual feelings. It prepares the body for sexual activity and may be triggered by seeing someone attractive, kissing, hugging, or gentle touching. During this stage, the heart rate and breathing rate increase, blood flow to the reproductive organs rises, and a person may experience feelings of attraction and excitement. This phase can last from a few minutes to several hours.

In boys: Increased blood flow to the penis causes it to become firm, a process called an erection.

In girls: The body produces natural moisture, known as lubrication, and the tissues around the genital area may swell slightly due to increased blood flow.

ii) Plateau phase

The plateau phase follows the excitement phase. During this stage, the body's physical responses become stronger and more noticeable. Heart rate, breathing rate, and blood pressure continue to increase as the body prepares for the next stage of the sexual response cycle. Feelings of excitement and tension also become more intense. This phase lasts until the climax (orgasmic phase) is reached.

iii) Orgasmic phase

The orgasmic phase is the peak or climax of the sexual response cycle. It is usually the shortest phase and lasts only a few seconds. During this stage, the heart rate, breathing rate, and blood pressure reach their highest levels. The body experiences intense feelings of pleasure, accompanied by involuntary muscle contractions. In males, this phase is usually associated with the release of semen (ejaculation). After orgasm, the body begins to return to its normal resting state.

iv) Resolution phase

The resolution phase is the final stage of the sexual response cycle. During this stage, the body gradually returns to its normal resting state. Heart rate, breathing rate, and blood pressure decrease to their usual levels. Feelings of excitement reduce, and the body becomes relaxed. Many people experience a sense of calmness, comfort, and well-being during this phase, and some may feel tired or sleepy.

Sexual stimulation

Sexual stimulation refers to any physical, emotional, or mental factor that can cause **sexual arousal** (feelings of sexual excitement). It may come from different sources, such as touch, sight, sound, thoughts, or emotions.

Sexual stimulation can include:

- **Physical stimulation**, such as touching certain parts of the body.
- **Visual stimulation**, such as seeing an attractive person.
- **Auditory stimulation**, such as hearing a pleasant voice or certain sounds.
- **Mental or emotional stimulation**, such as thoughts, memories, feelings of affection, or imagination.

When sexual stimulation is strong enough, it may lead to further stages of the sexual response cycle, including orgasm. While sexual arousal can sometimes occur without direct physical contact, physical stimulation is often important for reaching orgasm.

Key Points

- Sexual stimulation is anything that triggers sexual arousal.
- It can be physical, emotional, mental, visual, or auditory.
- Sexual stimulation causes changes in the body, such as increased heart rate and blood flow to the reproductive organs.
- The type and level of stimulation that causes arousal varies from person to person.

Sexual feelings and responses are natural biological processes that develop during and after puberty. Understanding them helps individuals learn about their bodies and reproductive health.

Application activity 12.1



1. Which of the following best describes sexual response?
 - a) Growth of reproductive organs
 - b) Physical and emotional changes that occur during sexual arousal
 - c) Fertilization of the ovum
 - d) Puberty changes
2. Which change occurs during sexual arousal in males?

- a) Menstruation b) Erection of the penis c) Ovulation d) Lactation
3. Which hormone is mainly involved in female sexual response?
a) Testosterone b) Insulin c) Estrogen d) Thyroxine
4. Which change occurs during sexual arousal in females?
a) Erection of the penis b) Ejaculation c) Vaginal lubrication
d) Voice deepening
5. The release of semen from the penis is called:
a) Ovulation b) Ejaculation c) Menstruation d) Fertilization

12.2. Puberty in boys and girls

Learning activity 12.2



Investigating puberty

1. Share your experiences on emotional attraction related to sexuality.
2. Watch a film showing physical and emotional changes during puberty.
3. Participate in a role-play of a boy–girl relationship.
4. Share your findings and experiences with the rest of the class.

Puberty is a period during which adolescents reach sexual maturity and become capable of reproduction. Numerous changes take place both in males and females at this stage. Puberty comes as a result of hormones that originate from the gonads: testicles in men and ovaries in females.

Teenagers normally face challenges during this period of growth and development. The greatest challenge is increase in sexual desire in both females and males. These challenges include unwanted pregnancies and sexually transmitted diseases. Normally there is no big physical difference between males and females before puberty apart from the sexual organs.

Puberty in boys

Puberty in males takes place between the ages of 11 and 13 on average. When a boy reaches puberty at an early stage, he may have benefits. The benefits include popularity with his fellow peers because of being tall and therefore a leader. Late onset of puberty can lead to low self-esteem in males, and such may also face problems of anxiety and fear. Sexual desire is at its peak. Boys start dating girls at this stage.

This is a critical time where boys need guidance and counseling because most of them face problems; associated with increased body activity and sexual desire. The problems faced at this stage include:

- Abuse of drugs and alcohol
- Lack of respect for authority
- Fighting among peer groups
- Parental hatred
- Poor hygiene and general body cleanliness.

Puberty in girls

On average, girls begin puberty at an early age of about 10-11 years. Puberty in girls occurs faster than in boys. Menstruation in girls is a characteristic that shows the onset of puberty. This is brought about by the growth and development of the ovaries.

Early onset of puberty in girls is detrimental, causing low self-esteem and stress. This is due to breast enlargement and a lot of fat deposition on the body that causes different body shapes between the girl and her peers. Early onset is caused by several factors which include diet and environmental exposure.

Puberty in girls is more challenging to parents. Girls tend to become more private and sensitive emotionally. Parents should therefore be more careful. Social behaviours of girls also tend to change. Girls become more attracted to males than females. The main challenge that girls face during puberty is unwanted pregnancies. Other challenges include rejection from peers, clothes to wear and physical appearance.



Figure 12.1: Peer pressure may lead to bad behaviour like this

Application activity 12.2



1. Describe the emotional changes that occur during puberty in boys and girls..
2. Draw a diagram showing sexual response in men and women.
3. Is there any way to avoid unnecessary sexual stimulation?

12.3. People and sexual relationships

Learning activity 12.3



People and sexual relationships

1. Discuss the following questions with your classmates.
 - What is the meaning of love?
 - What are some things we do to show love?
 - Does love go together with sex?
 - How can one show love to a neighbour?
 - What is sexuality and life skills education?
 - Why do we need this education now?
2. Note down your findings.
3. Share your findings with the rest of the class members.

People develop different types of relationships ranging from acquaintances to friendships to romantic relationships. Most people form romantic relationships. This involves falling in love and making a commitment. Intimate relationships bring a sense of stability and security. Partners in a healthy relationship should have characteristics below:

- Treat each other with love and respect.
- Be honest with each other.
- Spend time with each other.
- Take special interest in activities that each person likes. Respect each other's emotional, physical and sexual limits. Speak honestly about their feelings.



Figure 12.2: Teenagers engaging in fun activities

It is important to note that sexual relationships between close members of a family like father and daughter are unacceptable. It is considered a taboo in society and is also punishable by law. This form of sexual relationship is called incest. If it happens,

children born normally suffer from genetic disorders and lack of vigor. The following are forms of sexual relationships:

- a) **Exogamy:** Since incest is a taboo in the Rwandan society, individuals are advised to find their mates outside their family circles. This is known as exogamy. Endogamy is the opposite of exogamy where individuals are allowed to marry their distant cousins.
- b) **Monogamy:** This is a long-term relationship between a man and a woman in which none has other sexual partners. In the Rwandan society, monogamy is the legal form of sexual relationship.
- c) **Polygamy:** This is a long-term sexual relationship of having more than one wife or husband at the same time. An example is a man having more than one wife or a woman with more than one man. The Rwandan government is not in support of polygamous sexual relationships. These relationships are a source of sexually transmitted diseases and contribute to poor family planning that hinders development in families.
- d) **Casual relationships:** This is an act of having sex with someone you do not have an attachment to. This is the case when one engages with a sex worker. Casual relationships are dangerous since they can be sources of sexually transmitted diseases including HIV and AIDS and other forms of sexual and physical abuse.
- e) **Dating relationships:** This is when two people (a boy and a girl) go out together. It leads to engagement and later marriage. Dating should be done carefully. You should avoid sexual intercourse until an appropriate time e.g. after marriage. Sexual intercourse can lead to unexpected pregnancies or sexually transmitted diseases such as HIV and AIDS.



Figure 12.3: A teenage girl refusing advances from a boy

Sexual problems and disappointment

A sexual problems or sexual dysfunction refers to a problem experienced during any phase of the sexual response cycle. Such a problem prevents the individual or couple from experiencing satisfaction from the sexual activity. The following are some of the sexual disappointments.

In males

- a) **Premature ejaculation:** This is a condition whereby a man cannot delay ejaculation long enough to satisfy the woman.

- b) **Impotence:** This is the inability to produce or maintain an erection. Impotence occurs because of fear, anxiety, disease such as diabetes and drug abuse.
- c) **Inflammation:** Blockage of the urethra occurs as a result of an infection or due to swelling of the prostate gland.

In females

- a) **Vaginal discharges:** This is any abnormal discharge from the vagina. It can be a white discharge with irritating smell especially after sexual intercourse.
- b) **Functional disorders:** This is the inability of a woman to experience sexual pleasure and satisfaction. This results from emotional problems, feeling of guilt and trauma.

Sexuality, age and culture

Sexuality refers to anything to do with being male or female. Human beings have engaged in sexual relationships for long. Each society, however, interprets sexuality and sexual activity in different ways. Human sexuality is part of social life.

Society dictates the norms; what is considered acceptable or unacceptable. Sexual behaviors are based on the morals and values of society. The following sexual behaviors are unacceptable in the Rwandan culture:

- i. **Incest:** This is the act of having sex with a close relative.
- ii. **Homosexuality:** This is when two people of the same sex are engaged in a sexual relationship.
- iii. **Masturbation:** The act of self- stimulation that leads to sexual satisfaction.
- iv. **Anal sexual intercourse:** This is a type of sexual activity in which the penis is inserted into the anus.
- v. **Rape:** The act of forcing someone into sexual intercourse without consent.
- vi. **Adultery:** Having sex outside wedlock.
- vii. **Fornication:** Engaging in sex before marriage.
- viii. **Sex slavery:** The act of taking advantage of vulnerable individuals in a forced commercial sex.
- ix. **Prostitution:** Engaging in sexual relationships for financial gains.

Ageing is process of growing old. The ageing process does not reduce the need or desire for sex. For many older adults, the concept of sexuality includes practices such as kissing, hugging and fondling. Such activity may replace actual intercourse and become increasingly important for those who no longer desire or are incapable of sexual intercourse.

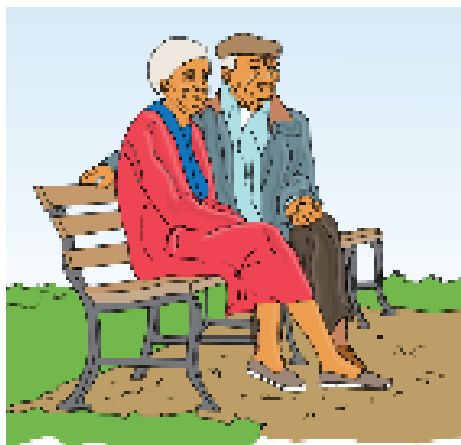


Figure 12.4: An old couple seated in a park talking about their past

Ageing in men

In an ageing male, the most common sexual dysfunction is erectile dysfunction. This can be due to hormonal changes as part of normal ageing or due to underlying conditions such as diseases. Testosterone hormones in men decline with age. This hormone is produced by the testes in man. It increases sexual urge and libido.

Ageing in women

The most common sexual dysfunction in women, is lack of desire and arousal. This is due to decrease in hormones, which begins at menopause and continue as women age. This decrease in hormones may also produce reduced muscles, loss of bone mass and declining energy. Sexually, it may result in vaginal dryness, decreased libido, and decreased ability to achieve orgasm.

Violation of human rights

Look at the picture below. What is happening? In your opinion, should children be treated this way? Suggest better methods of discipline.



Figure 12.5: Child abuse is not allowed in the modern society

Human rights are rights that are considered to be legitimate for all people. The rights should be enjoyed by everyone, regardless of their position in society. Human rights violations include sexual harassment or abuse, coercion and human trafficking.

Child sexual exploitation

Child sexual exploitation is a type of sexual abuse in which children are sexually exploited for money, power or status.

Children or young people are tricked into sexual relationships by being offered money. They might be invited to parties and given drugs and alcohol. They may also be recruited online. Some children and young people are illegally taken into foreign countries for the purpose of sexual exploitation. This is part of human trafficking.

Sexual assault

Sexual assault is any forced or involuntary sexual contact or behaviour which occurs without consent. Sexual assault includes rape and attempted rape, child abuse, and sexual harassment or threat.

Application activity 12.3



1. What are the effects of sex outside marriage?
2. Suggest factors that propagate sexual abuse of children.
3. How does culture affect sexuality?

12.4 End of unit assessment



12.4.1. Project activity: Promoting Positive Sexual Behaviour and Responsible Decision-Making Among Adolescents

1. Context/Background

Adolescence is a period of physical, emotional, and social development during which young people face various situations that require responsible decision-making. Peer pressure, media influence, cultural beliefs, and personal values can affect behaviour and choices. Developing positive sexual behaviour and decision-making skills helps adolescents maintain healthy relationships, avoid risky behaviours, and promote their overall well-being.

2. Challenge/Problem Statement

Many adolescents face challenges in making responsible decisions related to relationships and personal behaviour due to peer pressure, misinformation, and limited awareness of the consequences of risky choices.

Key Question

How can adolescents promote positive sexual behaviour and make responsible decisions that support their health and well-being?

3. Purpose of the Project

The purpose of this project is to help students understand the importance of positive sexual behaviour, healthy relationships, self-discipline, and responsible decision-making. It also aims to empower students to make informed choices that promote personal and social well-being.

4. Instructions

- Investigate factors that influence adolescent behaviour and decision-making.
- Identify behaviours that promote healthy and respectful relationships.
- Collect information from textbooks, school guidance programs, health education materials, and reliable sources.
- Develop awareness messages that encourage responsible decision-making.

5. Your Tasks

- Identify factors that influence adolescent behaviour and choices.
- Explain the characteristics of positive sexual behaviour.
- Discuss the importance of self-discipline, respect, and personal values in decision-making.
- Identify consequences of risky behaviours.
- Design a poster, brochure, role-play, or awareness campaign promoting responsible decision-making.
- Present your findings to the class.

6. Expected Student Outputs

- A report on positive sexual behaviour and responsible decision-making.
- A poster, brochure, or awareness message promoting healthy choices.
- A role-play or presentation demonstrating decision-making skills.
- Recommendations for promoting healthy relationships among adolescents.

7. Constraints and Limitations

Scope: The project should focus on positive sexual behaviour, healthy relationships, personal values, self-discipline, and responsible decision-making among adolescents.

Materials and Methodology: Use textbooks, school guidance materials, interviews with teachers or counsellors, and reliable educational resources. Information should be presented through reports, posters, presentations, or role-plays.

Time Frame: 2 weeks.

8. Guiding Questions

- i. What factors influence adolescent behaviour and decision-making?
- ii. What are the characteristics of positive sexual behaviour?
- iii. Why are self-discipline, respect, and personal values important in decision-making?
- iv. What are the consequences of risky behaviours?
- v. How can adolescents resist peer pressure and make responsible choices?
- vi. What strategies can school and communities use to promote healthy relationships and responsible behaviour?

12.4.2. Assessment Questions

A. Knowledge and understanding

- Sexual behaviour mainly refers to:
 - The structure of reproductive organs
 - How individuals express sexuality and relate to others?
 - Hormone production in the body
 - Cell division during reproduction
- Which of the following is a responsible sexual behaviour?
 - Having multiple sexual partners
 - Ignoring consent
 - Practising abstinence
 - Early unprotected sex
- Sexual response is best described as:
 - Emotional attraction only
 - Physical and psychological changes during sexual activity
 - Hormonal imbalance
 - Fertilization of the ovum
- The first stage of the human sexual response cycle is:
 - Orgasm
 - Resolution
 - Arousal (excitement)
 - Plateau
- Ejaculation in males occurs during:
 - Arousal
 - Plateau
 - Orgasm
 - Resolution
- How can couples show love and affection?
- Teenagers are at a higher risk of contracting STIs. Explain
- What do you understand by the term sexual maturity? How does it help in making decisions? Why is understanding the function of joints important for daily life?

B. Skills

- How can you assist in preventing violation of human rights in your community?
- What are the implications of reckless sexual behavior?
- Sebahire and Sentwali recently celebrated 40 years in marriage. Suggest reasons why their relationship is still strong.
- During adolescence, teenagers are exposed to challenges.
 - What are these challenges?
 - How can teenagers overcome these challenges?
- Is sexual orientation a choice?
- In your opinion, how would you deal with sexual assault?

C. Attitudes and values

15. Suggest ways of dealing with indiscipline in children.
16. Why is mutual respect important in sexual relationships?

UNIT 13

PREGNANCY PREVENTION



Key Unit Competence:

Apply knowledge of pregnancy prevention in sexual and reproductive decisions.

Learning objectives

By the end of this unit you should be able to:

Knowledge and understanding

1

- Identify effective ways of preventing unintended pregnancy.
- Describe forms of contraceptive methods.

Skills

2

- Demonstrate confidence in discussing different contraceptive methods.
- Design a model that shows efficient use of male and female condoms.
- Classify benefits and possible side effects of various contraceptive methods.

Attitudes and values

3

- Appreciate the importance of making informed choices about reproduction and family size.
- Value the benefits of child spacing.

13.0. Introduction

Pregnancy prevention refers to the methods and practices used to avoid unintended pregnancies. It involves making informed and responsible decisions about reproductive health and understanding the different ways pregnancy can be prevented. Knowledge of pregnancy prevention helps individuals protect their health, achieve their educational and life goals, and make responsible choices about their future.

Various methods can be used to prevent pregnancy, including abstinence, natural family planning methods, and modern contraceptive methods. Each method has its advantages, limitations, and level of effectiveness. Understanding these methods

enables individuals to make informed decisions and seek appropriate guidance when necessary.

This unit explores the concept of pregnancy prevention, the different methods used to prevent pregnancy, and the importance of responsible decision-making in reproductive health. It also highlights the role of individuals, families, communities, and health services in promoting reproductive health and preventing unintended pregnancies.

Key concepts: pregnancy prevention, reproductive health, abstinence, contraception, family planning, fertility, responsible decision-making, unintended pregnancy, and sexual and reproductive health.

Introductory activity



From what you learned in Unit 12, identify some examples of positive sexual behaviour. Suppose you have a friend who engages in risky sexual behaviours. What health, social, or emotional risks might he or she face? What advice would you give to help him or her avoid these risks?

Observe the pictures below and answer the following questions:



1. What materials or items are shown in the pictures?
2. What are these materials used for?
3. How can they help prevent unintended pregnancies?
4. What are the advantages and possible limitations of using them?
5. Why is it important to make informed and responsible decisions about pregnancy prevention?

13.1. Contraceptives

Learning activity 13.1



Research on contraceptive methods

1. Search about the following topics:
 - a) What contraceptive is
 - b) Contraceptive methods
 - c) Common myths about contraceptives
 - d) Importance of use of contraceptives
2. Share your findings with the rest of the class.

What is a Contraceptive?

A contraceptive is any method, device, or substance used to prevent pregnancy. Contraceptives help individuals and couples plan if and when to have children by preventing fertilization or implantation.

Contraceptive Methods

Contraceptive methods are used to prevent pregnancy. Common methods include:

- **Abstinence** – choosing not to engage in sexual activity.
- **Barrier methods** – such as male and female condoms, which prevent sperm from reaching the egg.
- **Hormonal methods** – such as pills, injections, and implants, which prevent ovulation.
- **Intrauterine devices (IUDs)** – devices placed in the uterus to prevent pregnancy.
- **Natural family planning methods** – based on monitoring fertility signs and avoiding pregnancy during fertile periods.
- **Permanent methods** – surgical procedures that permanently prevent pregnancy.

Common Myths About Contraceptives

The following are common myths or misconceptions about contraceptives:

1. Birth control pills always cause significant weight gain and cancer.
2. The birth control pill must be taken at exactly the same time every day to work.
3. Mothers who are breastfeeding do not need any form of contraception.
4. Using birth control pills for a long time makes it difficult to become pregnant later.
5. Newer contraceptive methods are less safe than older methods.
6. Intrauterine devices (IUDs) can only be used by women who already have children.
7. It is unhealthy to use certain contraceptive methods that may reduce or stop menstrual periods.
8. A person cannot become pregnant the first time they have sexual intercourse.
9. Natural family planning methods are always effective without careful and consistent use.
10. Contraceptives always cause infertility.
11. All contraceptive methods provide complete protection against sexually transmitted infections (STIs).

These beliefs are myths and should be replaced with accurate information from qualified health professionals and reliable sources.

Facts

- Most contraceptive methods are safe when used correctly and under appropriate guidance.

- Fertility usually returns after discontinuing most contraceptive methods.
- Pregnancy can occur the first time a person has sexual intercourse.
- Natural family planning methods require careful monitoring and correct use to be effective.
- Most contraceptives do not protect against STIs; condoms are among the few methods that can help reduce the risk of STI transmission.
- Breastfeeding may reduce the chance of pregnancy under specific conditions, but it is not always a reliable method of pregnancy prevention.

Importance of Using Contraceptives and Family Planning

Family planning involves the use of contraceptive methods to help individuals and couples decide the number of children they want and when to have them. It also helps families plan and space births according to their needs and circumstances.

The use of contraceptives helps prevent unintended pregnancies and promotes responsible parenthood. It enables individuals and couples to achieve their desired family size while improving their quality of life.

Contraceptives contribute to maternal and child health by reducing health risks associated with closely spaced pregnancies. Proper birth spacing helps women maintain good health during pregnancy, childbirth, and after delivery, while also improving the health and well-being of children.

Family planning can also support educational and career goals by allowing individuals to make informed decisions about their future. In addition, it helps reduce the number of unsafe abortions and their associated health risks.

Some contraceptive methods, such as condoms, provide additional protection by helping to reduce the risk of sexually transmitted infections (STIs). Therefore, contraceptives play an important role in promoting reproductive health and overall well-being.

Application activity 13.1



1. What are contraceptives?
2. What is the importance of contraceptives?
3. State 5 common myths about contraceptives.

13.2. Natural contraceptives

Learning activity 13.2



Discussion on natural contraceptives

Search on the following topics

1. Types of natural contraceptive methods
2. Advantages and disadvantages of natural contraception

Natural family planning methods relies on knowledge of the menstrual cycle of the woman. This knowledge enables couples to avoid having sex when the woman is fertile. Natural birth control methods teach women to determine the fertile phase of their menstrual cycle. To avoid pregnancy, women avoid intercourse on their fertile days. To effectively use the method, a woman needs to accept responsibility for charting and interpreting her fertility on a day-to-day basis.

Advantages of natural contraception

- a) Effective method of birth control.
- b) Have no negative health side effects.
- c) An alternative for women who cannot or do not want to use hormonal methods.
- d) Promotes positive body awareness.
- e) Consistent with many religious beliefs and lifestyles.
- f) Alerts women to reproductive health and fertility concerns.
- g) Fosters communication between partners and encourages male involvement.

Disadvantages of natural contraception

- a) Provides no protection from sexually transmitted infections.
- b) Often it is difficult to find trained instructors.
- c) Requires time to learn (usually 3 to 6 cycles).
- d) Requires discipline and commitment to chart fertility signs and follow the rules to avoid pregnancy.
- e) Times of abstinence from intercourse may be a challenge for some couples.

Types of natural contraception

a. Abstinence

Abstinence is the conscious and voluntary decision to refrain from sexual activity.

In the context of health education, it is most commonly understood as abstaining from sexual intercourse. Because it involves the total avoidance of the sexual contact that leads to reproduction or the transmission of infections, it is categorized as the most effective way to prevent unintended pregnancy and sexually transmitted infections (STIs).

Key characteristics of abstinence include:

Behavioral Choice: It is a lifestyle decision that is entirely under an individual's control and does not require medical intervention or supplies.

Complete Protection: When practiced consistently, it is the only method that offers 100% effectiveness against both pregnancy and STIs.

Values-Based: It is often a reflection of an individual's personal, cultural, or religious values regarding timing, maturity, and emotional readiness for sexual activity.

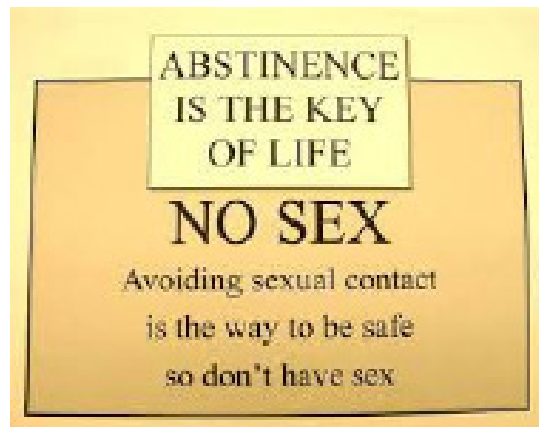


Figure 13.1: A poster *advocating for abstinence*

Advantages of abstinence

- a) Minimal risk of misuse.
- b) Prevents transmission of STIs including HIV and AIDS.
- c) No physical side effects.
- d) No need to visit a health care provider.
- e) No cost involved.

b. Lactation Amenorrhea Method (LAM)

The lactation Amenorrhea Method (LAM) is used by a mother who has just given birth and is exclusively breastfeeding. This method is highly effective for the first six months after childbirth. The mother must breastfeed the baby at least every four hours during the day and every six hours through the night. She also must be aware of her menstrual period. After six months fertility may return at any time.

c. Temperature method

The temperature method is also known as the basal body temperature method. It is a type of natural family planning where a woman measures her basal body temperature: It is believed that ovulation may cause a slight increase in basal body temperature. Elevated waking temperature for three days in a row is considered confirmation of the post-ovulatory less fertile phase.

d. Calendar method

The calendar method is also known as Rhythm method or cycle beads. It is a much less effective natural birth control method. It predicts a woman's fertile days using calculations based on the length of past cycles and not daily observations of fertility signs.

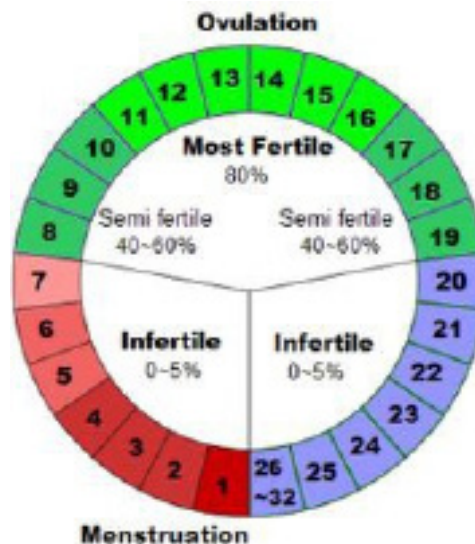


Figure 13.2: Calendar chart

e. The Billings Ovulation Method

This is a natural method of fertility management. It enables a woman to recognize her body's natural signal of fertility through production of cervical mucus. The cervix produces different types of mucus in response to changing hormone levels. Once a woman is familiar with these changes, she can identify their patterns of fertility and infertility in their cycle.

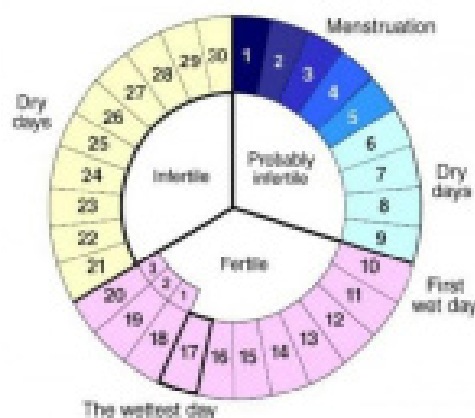


Figure 13.3: Billing chart

Advantages

It allows a woman to understand her fertile and infertile days and helps to safeguard the woman's reproductive health.

Disadvantages

It is not an effective method since the body system can fail.

f. Withdrawal method

The withdrawal method (also known as *coitus interruptus*) is a form of natural birth control where the male partner removes his penis from the vagina before ejaculation occurs. The goal is to ensure that sperm does not enter the vagina, thereby preventing fertilization.

Effectiveness and Limitations

In the context of reproductive health education, it is critical to present the withdrawal method with caution, as it is generally considered less reliable than modern contraceptive methods.

Advantages

- **No Cost or Medical Intervention:** It requires no hormones, devices, or medical supplies.
- **Readily Available:** It can be used in situations where other forms of contraception are not accessible.
- **No Physical Side Effects:** It does not introduce chemicals or medical risks to the body.

Disadvantages and Risks

- **High Failure Rate:** It has a high "typical use" failure rate. Many individuals struggle to withdraw in time or may not realize that pre-ejaculate fluid (the clear fluid that appears before ejaculation) can contain sperm, which may be sufficient to cause pregnancy.
- **Requires High Self-Control:** It relies entirely on the male partner's awareness and self-control. During the intensity of sexual activity, this can be extremely difficult to maintain consistently.
- **No Protection Against STIs:** Crucially, the withdrawal method offers zero protection against sexually transmitted infections (STIs) or HIV, as it involves direct skin-to-skin contact and the exchange of bodily fluids.
- **Psychological Impact:** The need to focus on the timing of withdrawal can create anxiety or reduce the quality of sexual intimacy for both partners.

Application activity 13.2



1. is the avoidance of vaginal intercourse. It is the most effective family planning method.
2. Which of the following is a natural contraceptive method?
 - a) Oral pills
 - b) Condom
 - c) Abstinence
 - d) IUD
3. The rhythm (safe days) method is based on:
 - a) Use of hormones
 - b) Surgical procedures
 - c) Knowing the fertile period of the menstrual cycle
 - d) Use of condoms

13.3. Artificial contraceptives

Learning activity 13.3



Observing artificial contraceptives

1. Observe the models of artificial contraceptives, then answer the questions below.
 - Can you identify the models?
 - How are the models used in relation to real life situation?
2. Share your findings with the rest of the class.

Artificial birth control is any product, procedure or practice that uses artificial or unnatural means to prevent pregnancy. Barrier methods such as condoms and diaphragms, hormonal methods such as the pill and IUDs, and surgical sterilization procedures such as vasectomy or hysterectomy are all artificial birth control methods. Most types of artificial contraceptives work by:

- Preventing an egg from being released every month (hormones).
- Preventing sperms from reaching the egg (barrier and some IUD methods).
- Blocking the reproductive function in men or women (sterilization).
- Preventing a fertilized egg from implanting in the uterus (hormones).

Barrier methods

a. The male condom

This line of defence prevents harmful micro-organisms from entering the body. It is the external defence system of the body. It includes both physical and chemical defence lines. The body does this using various barriers that intercept micro-organisms hence prevent them from entering the body.



Figure 13.4: (a) Male condom (b) How to put on a male condom

Advantages

- i. It is easy to use.
- ii. Affordable
- iii. It offers protection against sexually transmitted infections.
- iv. It can be bought over the counter without prescription from a doctor.

Disadvantages

- A new condom is required every time one engages in sex.
- If used improperly it can break, this could lead to unwanted pregnancy or transmission of STIs.

b. The female condom

This is a loose rubber sheath with two flexible plastic rings at either end. It is inserted into the vagina before sexual intercourse.

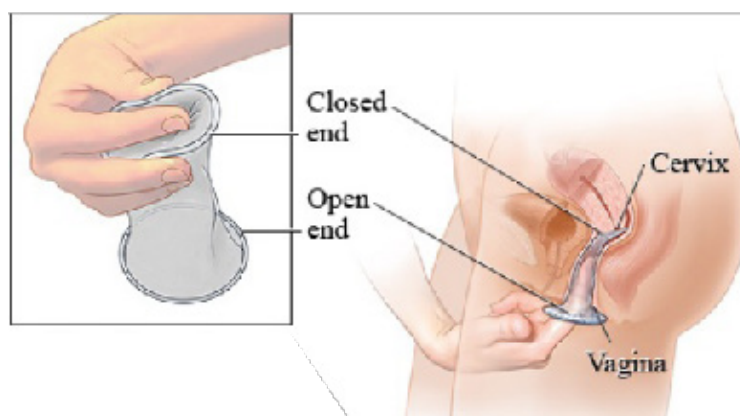


Figure 13.5: (a) Female condom (b) How to put on a female condom

It offers 95% effective protection against pregnancy, as well as some protection against STIs. Female condoms are generally more expensive than the male ones, but they are less likely to burst. They can be inserted up to eight hours before sex.

c. Spermicide

Spermicide is a birth control method that contains chemicals that stop sperms from fertilising the ova. Spermicides are available in different forms, including creams, film, foams and gels. Spermicide can be used alone, or it can be used with other birth control methods such as a diaphragm, to make them more effective.

Spermicide does not always offer the best protection against pregnancy. It has very few associated side-effects, but it does not protect against STIs.

d. The Diaphragm

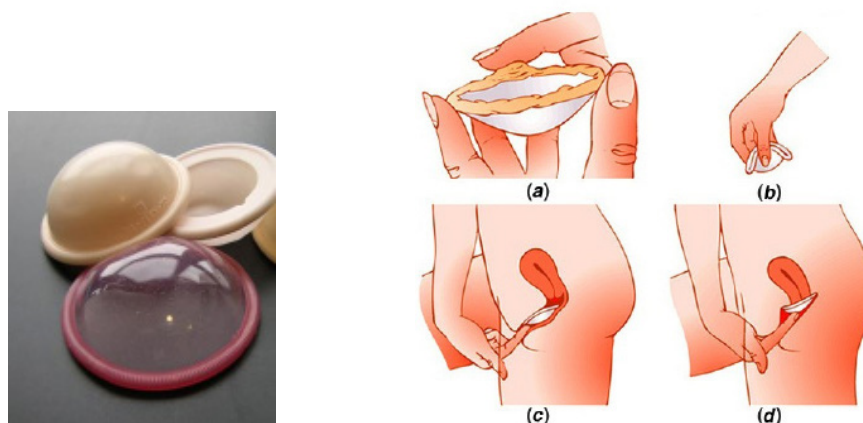


Figure 13.6: (a) Diaphragms (b) How to put on a diaphragm

This is a rubber cap inserted into the vagina to cover the cervix. It prevents sperms from entering the uterus. Diaphragm should be used together with spermicides which kill the sperms. It is inserted at least six hours before sex, and it needs to be removed after 24 hours for cleaning. Depending on the material and type of diaphragm, it can be reused many times.

The diaphragm does not protect against STIs, and a doctor is needed to provide guidance on how it should be used.

Hormonal based contraceptives

This is the use of a mixture of artificial hormones resembling oestrogen and progesterone. They are administered in three ways:

- Orally (oral contraceptives)
- Through injections
- Administration of implants

a. The Pill

It is the most popular form of female contraception. It is taken orally in the form of a tablet, every day at almost the same time.

The contraceptive pill will prevent 95% pregnancy. It comes close to providing 99% protection if one pill is taken every day as prescribed.

The pill does not provide any protection against STIs, and a doctor's prescription is required to buy it.

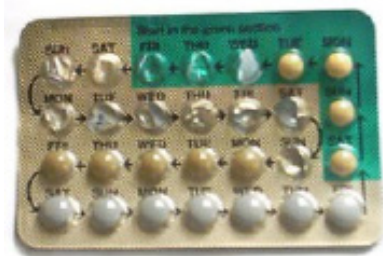


Figure 13.7: The pill

b. Contraceptive Injections

This is an injection that contains artificial progesterone (progestin). One shot of hormones lasts in the body for 8 to 12 weeks (3 months) and has the same effect as the pill.

Injections are about 99% effective. It fails if the user forgets to renew the contraceptive shot in time. However, once the shot is given, it cannot be reversed. One becomes effectively infertile for the next three months. Just like the pill, contraceptive injections do not protect against STIs.

c. The contraceptive implant

The implant offers long term protection. It lasts for about three years on average. The contraceptive implant contains progestin (progesterone), the same hormone as the

contraceptive pill. The hormone is released into the body at a steady, slow pace for three years, producing the same effects as the pill.

The implant is inserted in the arm by a doctor and must be removed after three years. Since the risk of human mistake is ruled out, the implant has a much higher effectiveness rate than the pill around 99.99%. The implant does not protect against STIs.

d. The vaginal ring

The vaginal contraceptive ring is a small, transparent plastic ring that is inserted in the vagina and kept for three weeks. It is removed during monthly periods and replaced with a new one after that.

The ring contains the same hormones as the contraceptive pill (progesterone and oestrogen), therefore provides the same kind of effective protection. A doctor's prescription is required and just like other hormonal contraception, the vaginal ring does not protect from STIs.

e. The contraceptive patch

The contraceptive patch is the same thing as the contraceptive pill but in the form of a patch. It provides the same effective protection against pregnancy. It does not protect from STIs. There is however a risk of skin irritation, and a (rare) chance that the patch accidentally comes off.

f. Emergency contraception

Emergency contraception assists to stop pregnancy in cases of eventualities such as rape or unprotected sex. This method is for one-off occasions and is not recommended for daily use. After 72 hours (3 days) the effectiveness drops to below 50%.

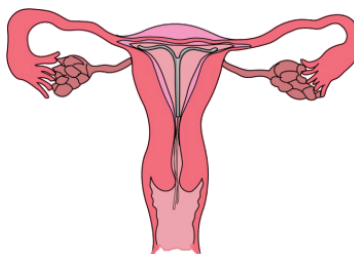
Non-hormonal based contraception methods

a. The Intrauterine Device (IUD)

An intrauterine device (IUD or coil) is a small contraceptive device, often 'T'-shaped. It often contains either copper or hormones. It is inserted into the uterus. It offers long-acting reversible contraception. Therefore, it is the most effective type of reversible birth control. It is kept inside the vagina for up to five or ten years.



Figure 13.8: (a): IUD



(b) Properly inserted IUD

The effectiveness rate for IUDs is above 99%. However, they provide no protection against STIs. It requires a doctor to properly insert it and to make periodic follow-up visits.

b. Permanent contraception methods

These are for people who do not want children in the future (Vasectomy for men and tubal ligation for women)

Sterilisation is a procedure that closes or blocks fallopian tubes in women and sperm duct in men. This prevents movement of the ova and sperm respectively. Sterilisation is an option available to both men and women.

i) Vasectomy

This involves tying off and cutting the tubes that carry sperms, the vas deferens. It provides no protection against STIs, and the effects are permanent. In very rare cases (less than 1%), the tubes can grow back, making pregnancy a risk.

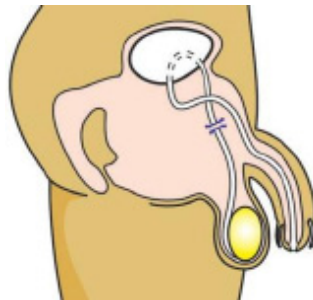


Figure 13.9: Vasectomy

ii) Tubal ligation

- Surgical sterilization: This involves cutting and tying the fallopian tubes so that they cannot link the ovaries with the uterus anymore. The effects are permanent.



Figure 13.10: Tubal ligation

- Non-surgical sterilization: This involves placing a coil in each fallopian tube through the vagina and uterus. Scars appear and eventually block each tube completely. The scars may take up to 3 months to completely block the tubes, so you need to use another method of contraception in the meantime.

Both options also offer more than 99% protection against pregnancy. In rare cases the blocked tubes can grow back and reconnect. However, this method offers no protection against STIs.

Abortion

Abortion is the deliberate process of prematurely terminating a pregnancy before the embryo or foetus is capable of independent life. Depending on the number of weeks it has, a pregnancy can be terminated either by taking medication or through a surgical procedure.

The following are some of the reasons why a woman might decide to have an abortion:

- Personal circumstances include risk to the well-being of existing children.
- Health risk to the mother.
- A high chance that the baby will have a serious genetic or physical abnormality.

Menopause

This is a term used to describe that state when a woman stops having monthly periods and is no longer able to get pregnant naturally. Monthly periods become less frequent over a few months or years before they stop altogether. In some women they stop abruptly.

Most women will experience menopausal symptoms. Some of these can be quite severe and have a significant impact on a woman's everyday activities. Common symptoms include:

- a) Hot flushes
- b) Night sweats
- c) Vaginal dryness and discomfort during sex
- d) Difficulty in sleeping
- e) Low mood or anxiety
- f) Reduced sex drive (libido).
- g) Problems with memory and concentration

Menopausal symptoms can begin months or even years before the periods stop. They can last around four years after the last period, although some women experience them for much longer.

Application activity 13.3



1. Should abortion be allowed for unwanted pregnancies in teenagers?
2. Why are spermicides not good for preventing pregnancy?
3. The following are artificial contraceptives.
 - Male and female condoms, diaphragm, implants, the pill and spermicides
 - Identify the ones that require a doctor's prescription.



13.4.1. Project: Promoting Informed Decision-Making on Pregnancy Prevention

1. Context/Background

Adolescents face many situations that require responsible decision-making. Lack of accurate information, peer pressure, and misconceptions about pregnancy prevention can lead to unintended pregnancies, which may affect health, education, and future opportunities. Making informed decisions helps young people protect their health and achieve their goals.

2. Challenge/Problem Statement

Many adolescents lack adequate information and skills to make informed decisions about pregnancy prevention.

Key Question

How can adolescents make informed decisions that help prevent unintended pregnancies?

3. Purpose of the Project

The purpose of this project is to help students understand the importance of informed decision-making in pregnancy prevention and promote responsible behaviour among adolescents.

4. Instructions

- Investigate factors that influence decision-making among adolescents.
- Research pregnancy prevention methods and their importance.
- Identify common myths and misconceptions about pregnancy prevention.
- Develop awareness materials that promote informed decision-making.

5. Your Tasks

- Explain the meaning of informed decision-making.
- Identify factors that influence adolescent decisions.
- Describe ways of preventing unintended pregnancies.
- Identify common myths about pregnancy prevention and explain the facts.
- Design a poster, brochure, drama, song, poem, or digital presentation promoting informed decision-making.

6. Expected Student Outputs

- A report on informed decision-making and pregnancy prevention.
- A poster, brochure, or awareness campaign.

- A presentation, role-play, poem, or drama.
- Recommendations for promoting responsible behaviour among adolescents.

7. Constraints and Limitations

Scope: Focus on informed decision-making and pregnancy prevention among adolescents.

Materials and Methodology: Use textbooks, health education materials, interviews with teachers or health workers, and reliable online resources.

Time Frame: 2 weeks.

8. Guiding Questions

- What is informed decision-making?
- What factors influence adolescent decisions?
- Why is pregnancy prevention important?
- What myths exist about pregnancy prevention?
- How can adolescents make responsible choices?

13.4.2. Assessment Questions

A. Knowledge and understanding

- The male sterilisation operation known as vasectomy is carried out by
 - Blocking the ureter
 - Removing the testis
 - Cutting the scrotum
 - Tying the vas deferens
- Which hormones are present in implants?
 - Testosterone
 - Progesterone
 - LH
 - FSH
- Do you agree that circumcision reduces HIV infection?
- Family planning should be compulsory. Discuss
 - In your opinion, what family planning method could you use?
- Compare natural methods of family planning and artificial methods.

Natural methods of family planning	Artificial methods of family planning

- Do you think some of the common myths about contraceptives are true?

B. Skills

7. Musabyimana has been using contraceptives to prevent unwanted pregnancy. She was shocked recently to discover she was pregnant. What could have gone wrong?
8. Describe the general limitations of natural contraceptive methods.
9. Why do you think abortion is dangerous?
10. a) What factors contribute to the rise in teenage pregnancies?
b) What options do you think young girls facing unwanted pregnancy have?
c) What should be the partner's responsibility during a teenage pregnancy?

C. Attitudes and values

11. What is the position of young people in relation to early or unplanned pregnancies?
12. Why is abstinence considered a reliable method of pregnancy prevention?
13. How does self-control help in pregnancy prevention?
14. Why is mutual decision-making important in the use of contraceptives?

UNIT 14

REDUCING RISK OF STIS AND HIV, CARE AND SUPPORT



Key Unit Competence:

Develop preventive measures against STIs and HIV transmission.

Learning objectives

By the end of this unit you should be able to:

Knowledge and understanding

1

- Explain how STIs and HIV are transmitted, treated and prevented.

Skills

2

- Recognize symptoms and complications of STIs and HIV.
- Demonstrate communication skills to negotiating sex abstinence and refusing unsafe sexual practices.
- Demonstrate awareness to prevent STIs such as chlamydia, gonorrhea, syphilis, HIV and HPV (genital human papilloma virus).
- Clarify the correct use of condoms is an efficient way of reducing the risk of HIV transmission and other STIs.
- Analyze why abstinence is the most effective protection against HIV and other STIs.
- Examine how age disparity/intergenerational relationships can increase the risks of acquiring HIV.
- Assess the role of sexual health services, help people limit the risks of STIs and AIDS.
- Justify how culture, gender and peer norms affect decision making about sexual behaviour related to STIs and AIDS.

Attitudes and values

3

- Appreciate behaviours that reduce the risk of STIs and HIV transmission.
- Acknowledge the importance of regular testing and treatment of STIs and HIV.

- Show empathy toward PLHIV.
- Acknowledge the effect of gender and culture on personal decision-making regarding sexual relationships.

14.0. Introduction

Sexually transmitted infections (STIs), including Human Immunodeficiency Virus (HIV), are infections that can be transmitted from one person to another through sexual contact. They can affect people of all ages and may have serious health, social, and economic consequences if not prevented or treated appropriately. Understanding how STIs and HIV are transmitted is important for making informed decisions that promote health and well-being.

Many STIs and HIV can be prevented through responsible behaviour, accurate knowledge, and access to appropriate health services. Learning about risk factors, prevention methods, testing, and treatment helps individuals protect themselves and others from infection. Early diagnosis and proper management can reduce complications and improve quality of life.

People living with HIV and those affected by STIs require care, support, and respect. Communities, families, schools, and health services all play an important role in reducing stigma and discrimination while promoting healthy lifestyles and access to treatment and support services.

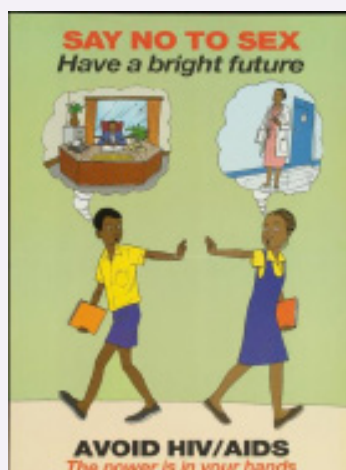
This unit explores the causes, transmission, prevention, and management of STIs and HIV. It also examines the importance of care, support, responsible decision-making, and community involvement in reducing the impact of these infections.

Key concepts: sexually transmitted infections (STIs), HIV, AIDS, transmission, prevention, risk reduction, testing, treatment, care, support, stigma, discrimination, healthy relationships, and responsible decision-making.

Introductory activity



Perhaps you have been sick before and had to visit a health centre for treatment. Did you enjoy being there? What could you do to avoid such a situation in the future? Now, look at the picture alongside and answer the questions below:



- What message is being conveyed in this image?
- What other ways can be used to reduce the spread of HIV and AIDS?
- What measures can be taken to prevent other sexually transmitted infections (STIs)? any prevention?

14. 1. Transmission of STIs

Learning activity 14.1



a) Discussion on the transmission of chlamydia, gonorrhea, syphilis and HIV infection

- Discuss only one disease under the following headings:
 - Causative agent (include scientific name)
 - Incubation period
 - Mode of transmission
- Write your findings on a manila paper.
- Present your work to the rest of the class.

b) Interactive talk about HIV transmission

- Listen attentively to a medical professional's presentation or a video presentation on HIV transmission .
- Take short notes during the presentation.
- Share your findings with the rest of the class.

Safe sex practices can help us prevent contracting sexually transmitted diseases. On the other hand, irresponsible sexual behaviour increases chances of individuals contracting sexually transmitted infections.

Most sexually transmitted diseases can be treated and cured. This becomes easy when diagnosis is done early, and treatments start during the early stages of the disease. It is therefore important for sexually active people to visit health Centres regularly for checkups. Though there is still no cure for HIV, early detection of the disease makes its management easy. This enables patients to live a normal life.

Sexually transmitted infections (STI) are infections that are passed on by infected people to healthy people during sexual intercourse. Examples of these diseases are chlamydia, gonorrhoea, syphilis, HIV and AIDS. They are discussed below:

1. Chlamydia

Chlamydia is caused by the bacterium *Chlamydia trachomatis*. The disease is spread by oral, vaginal or anal sex, and also through touch, for example, touching the eyes with a contaminated hand, may lead to conjunctivitis. Chlamydia can also be passed

to the infant during birth. It causes inflammation of the cervix in women, urethra and rectum in both men and women. Occasionally, other parts of the body like eyelids and throat may be affected. Other complications like pelvic inflammatory disease (PID) or infertility may occur from this infection.

Any sexually active person is at risk of contracting the disease. However, it is more common among young people. The disease is known as a 'silent' infection because it is mainly asymptomatic, thus the symptoms can be mild or confused with gonorrhea.

Signs and symptoms of chlamydia

In males

- Pain when passing out urine.
- White discharge from the penis.
- The testicles may be painful or swollen.
- Swelling of skin around the anus.

In females

- Painful and frequent urination.
- Smelly yellowish and abnormal vaginal discharge.
- Pain in the lower abdomen.
- Swollen skin in the vagina or around the anus.

Treatment

Chlamydia can be treated and cured with antibiotics prescribed by a healthcare professional. Early treatment helps prevent complications and reduces the spread of the infection. Individuals should complete the prescribed medication and ensure that their sexual partners are tested and treated if necessary.

2. Gonorrhoea

Gonorrhoea is transmitted through sexual contact with the penis, vagina, mouth or anus of an infected partner.

Gonorrhoea can also be spread from mother to baby during childbirth.

Gonorrhoea is caused by a bacterium called *Neisseria gonorrhoeae*. The bacteria attach to the epithelial cells of the vagina or male urethra. This results in inflammation and discharge of pus. If left untreated, the infection spreads to the other reproductive parts and may eventually block the passages resulting in infertility.

Signs and symptoms of gonorrhoea

Some men with gonorrhea may have no symptoms at all. However, men who do have symptoms may have:

- A burning sensation when urinating.
- A white, yellow, or green discharge from the penis.
- Painful or swollen testicles (although this is less common).

Most women with gonorrhea do not have any symptoms. Even when a woman has symptoms, they are often mild and can be mistaken for a bladder or vaginal infection. Women with gonorrhea are at risk of developing serious complications from the infection, even if they do not have any symptoms. Symptoms in women can include:

- Painful or burning sensation when urinating.
- Increased vaginal discharge.
- Vaginal bleeding between periods.

Treatment

Gonorrhoea is a sexually transmitted bacterial infection (STI) that can be treated and cured with appropriate antibiotics prescribed by a qualified health professional. It is important for infected individuals to seek medical care as soon as symptoms appear or if they suspect they have been exposed to the infection. Early treatment helps prevent complications such as infertility and reduces the spread of the disease to others.

People receiving treatment should follow the healthcare provider's instructions carefully and complete the full course of medication. They should also avoid sexual contact until treatment is completed and the infection has cleared up. In addition, sexual partners should be informed, tested, and treated if necessary to prevent reinfection and further transmission.

3. Syphilis

Syphilis is a sexually transmitted infection (STI) caused by a bacterium called *Treponema pallidum*. It is transmitted from one person to another through direct contact with a syphilitic sore, known as a chancre (painless sore). These sores are highly infectious and commonly appear on the external genital organs, vagina, anus, rectum, lips, or inside the mouth. Transmission usually occurs during unprotected vaginal, anal, or oral sexual contact with an infected person. Syphilis develops in stages if it is not treated early:

Primary stage:

A small, hard, painless sore (chancre) appears at the site where the bacteria entered the body, such as the genitals or mouth. The sore may heal on its own, but the infection remains in the body.

Secondary stage:

Occurs a few weeks after the chancre heals. It is characterized by skin rashes (often on the palms and soles), mild fever, fatigue, sore throat, and swollen lymph nodes. These symptoms may disappear without treatment, but the infection continues to progress.

Latent stage:

There are no visible signs or symptoms, but the bacteria remain in the body. This stage may last for years.

Tertiary stage:

If untreated, the infection can cause severe damage to vital organs such as the brain,

heart, nerves, eyes, and bones. This may result in paralysis, mental illness, blindness, heart disease, and can eventually lead to death.

Treatment

Syphilis can be effectively treated using antibiotics, especially penicillin, which is the most recommended drug. Alternatives such as erythromycin or tetracycline may be used for people allergic to penicillin. Early diagnosis and treatment are very important because they prevent complications and stop the spread of the infection.

4. Transmission of HIV (Human immunodeficiency virus) infection

HIV is the virus that causes AIDS (Acquired immune deficiency syndrome). It is mainly transmitted through:

- a) Certain body fluids: blood, semen, pre-seminal fluid, rectal fluids, vaginal fluids and breast milk from a person who has HIV. These fluids must meet a mucous membrane or damaged tissue or be directly injected into the bloodstream (from a needle or syringe) for transmission to occur. Mucous membranes are found inside the rectum, vagina, penis and mouth.
- b) Having anal or vaginal sex with someone who is HIV-positive, without using a condom or taking antiretroviral medicines for prevention or treatment of HIV. The riskiest sexual behaviour is anal sex. Vaginal sex is the second riskiest sexual behaviour.
- c) Sharing needles, syringes or other equipment used for preparing medicines for injections with someone who is infected with HIV.



Figure 14.1: Sharing syringes **such as this could lead to transmission of diseases**

Less commonly, HIV may be spread through:

- d) Mother to child during pregnancy, birth or breastfeeding. Although the risk can be high if a mother is living with HIV and not taking medicine.



Figure 14.2: Pregnant mother can transfer HIV to unborn child

- e) Being stuck with an HIV-contaminated needle or other sharp object. This is a risk mainly for health care workers.

Note: In some rare cases, HIV has been transmitted through:

- a) Receiving blood transfusions, blood products, organ or tissue transplants that are contaminated with HIV,
- b) Eating food that has been pre- chewed by a HIV-infected person. The only known cases are among infants.
- c) Being bitten by a person with HIV. There is no risk of transmission if the skin is not broken.
- d) Contact between broken skin, wounds or mucous membranes and HIV-infected blood or blood- contaminated body fluids.
- e) Deep, open mouth kissing if both partners have sores or bleeding gums. HIV is not spread through saliva.
- f) Oral sex.

Prevention of HIV infection (Sexual education focus)

- Practicing abstinence (delaying or avoiding sexual activity)
- Using condoms correctly and consistently during sexual intercourse
- Being in a faithful relationship with an uninfected partner
- Avoid sharing sharp instruments
- Ensuring safe blood transfusion through screened blood
- Getting tested regularly for HIV

Sexual education

Human Immunodeficiency Virus (HIV) is a virus that attacks the body's immune system, making it difficult for the body to fight infections and diseases. If not managed, HIV can progress to acquired immunodeficiency syndrome (AIDS).

Understanding how HIV is transmitted is an important part of sexual education because it helps individuals protect themselves and others. For example, such as being aware of:

- Sugar mummies and sugar daddies. They lure young people with money and promise a good life in exchange for sexual favours. These acts could result in the transmission of sexually transmitted infections and HIV and AIDS.
- Abstinence as most effective protection against STI and HIV and AIDS.

Sexual education helps students to:

- Understand how HIV is transmitted and prevented
- Make informed and responsible decisions
- Reduce risky sexual behaviours
- Protect themselves and their partners from infection.

Application activity 14.1



1. Which is the most effective protection against HIV and other STI?
2. Given the following STI: Gonorrhoea, syphilis, chlamydia and HIV. Choose the ones that can be prevented?
3. What is the common transmission mode for all the STI?

14.2. Ways of reducing STI and HIV infection

Learning activity 14.2



a) Discussion on the ways of reducing STI and HIV infection

1. Discuss the following topics:
 - Ways of reducing STIs and HIV infection.
 - Ways of prevention and treatment of STI and HIV infection.
 - Role of artificial contraceptives (barrier method- male and female condoms) and their correct use.
2. Present your work to the rest of the class.

b) Visit to a medical care centre

1. Listen attentively to a voluntary counselling and testing (VCT) medical personnel's presentation on the treatment of HIV.
2. Take notes during the presentation.
3. Write a report and present your work in class

Reducing sexually transmitted infections (STIs), including HIV infection, requires informed decisions, responsible behavior, and access to appropriate health services such as:

- a) Abstinence is the only sure way to prevent STIs.
- b) Being faithful to one trusted partner.
- c) Using condoms every time when engaging in sexual intercourse. Condoms are not 100% effective at preventing disease or pregnancy. However, they are extremely effective if they are used properly and correctly.
- d) Reduce the number of sexual partners.
- e) Avoid sharing towels or underclothing.
- f) Get a vaccination for hepatitis B.
- g) Get HIV tested.
- h) Avoiding alcohol consumption and abuse of drugs. Individuals who are drunk or on drugs often fail to have safe sex.

Treatment of sexually transmitted infections (STIs) depends on the type of infection and requires proper medical care. While many STIs are curable, HIV is manageable with lifelong treatment, and early intervention is key to better health outcomes.

1. Bacterial STI can be cured using antibiotics, if treatment begins early enough. Antibiotics, often in a single dose, can cure many sexually transmitted bacterial and parasitic infections, including gonorrhoea, syphilis, chlamydia and trichomoniasis. Treatment of gonorrhoea and chlamydia is always done at the same time because the two infections often appear together.
2. Viral STIs cannot be cured, but the symptoms can be controlled with medications. There is, for example a hepatitis B vaccine, but it won't help if you already infected.
3. HIV cannot be cured but can be managed. A person may choose to visit a voluntary counselling and testing (VCT) centre to undergo HIV and AIDS counselling. After counseling he or she can make an informed decision about whether to be tested for HIV or not.

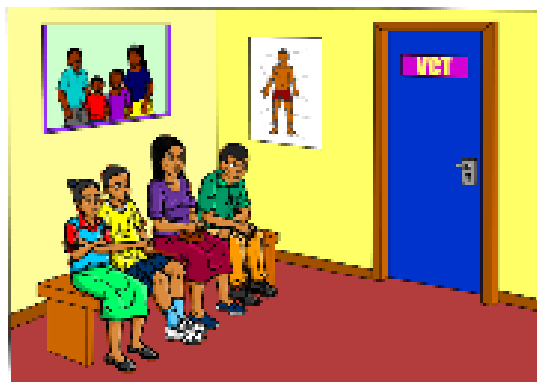


Figure 14.3: People waiting to enter a VCT centre

At the VCT centre pre-test and post-test counselling is offered to assist people to learn more about their status. Counselling involves getting information and learning how to live positively with the virus. This means learning to accept the fact that one is HIV-infected, seeking emotional support, eating a healthy diet, learning how to control the amount of stress in our life, making sure there is no re-infection and planning for the future.

Sex education

Sex education plays an important role in helping individuals understand how sexually transmitted infections (STIs) and HIV are transmitted and how they can be prevented. Accurate information enables people to make informed and responsible decisions that protect their health and the health of others. Understanding the Post-Exposure Prophylaxis (PEP) or short-term anti-retroviral therapy (ART) may reduce risk of HIV infection following potential exposure to HIV. Antiretroviral therapy (ART) is the use of medicines to treat HIV infection. People who are on ART take daily combinations of HIV and AIDS medicines. People with HIV infection and AIDS should start treatment with ART as soon as possible. ART does not cure HIV infection or AIDS but helps people infected to live longer and healthier lives. ART also reduces the risk of HIV and AIDS transmission. Potential risks of ART include unwanted side effects due to interactions

between drugs and HIV and AIDS. This includes not taking the medication every day as prescribed. This could lead to resistance to the drugs and to failure of treatment.

Application activity 14.2



1. What happens in a VCT centre?
2. What do you understand by the term ART?

14.3. Rights of people living with HIV (PLHIV)

Learning activity 14.3



Interactive talk on the rights of PLHIV

1. Listen attentively to a PLHIV community's presentation on rights of people living with HIV
2. Take notes during the presentation.
3. Write a report and present your work in class
4. Task:
 - a) What are the laws social and health policies in Rwanda related to the PLHIV?
 - b) How are the government and society supporting the PLHIV community?
 - c) What are the myths surrounding HIV transmission?

HIV related discrimination and stigma refer to negative attitude, prejudice and violence towards people with HIV. They may be deliberately avoided (**shunned**) by their families, peers and society and some may receive poor treatment in healthcare institutions and in the public sphere. The main reason for this stigma and discrimination is the myths surrounding HIV transmission.

These myths, disinformation ignorance and fear combined with other factors have led many people to falsely believe that:

- HIV and AIDS are always associated with death.
- HIV and AIDS are the result of irresponsible behaviour and deserve to be punished.
- HIV and AIDS are only transmitted through sexual intercourse, which is a taboo subject in many cultures.
- A person may be infected if he or she is associated with persons with HIV and AIDS.

The PLHIV face a variety of challenges associated with stigma and discrimination. These include loss of income and livelihoods, reputation, marriage and hope.

Rights of PLHIV

- i. No one living with HIV and AIDS should be discriminated...
- ii. PLHIV have the right to medical treatment and care from the health care institutions
- iii. No one can be dismissed from work because he or she is HIV-positive.
- iv. No one should be forced to take a HIV test in any situation, for example, before getting a job.
- v. Children living with HIV and AIDS should be allowed to attend any school.
- vi. Pregnant women living with HIV and AIDS have the right to decide about their pregnancy.
- vii. The test results may not be shown to anyone else without the consent of the person who took the test.

Application activity 14.3



1. Which of the following can protect a person from contracting HIV?
a) Morning after pills b) Spermicides c) Condoms d) Coil
2. HIV causes illness because
a) It reduces the number of red blood cells.
b) It attacks the immune system
c) It increases the body's core temperature.
d) It causes loss of weight.
3. Can a HIV-positive woman give birth to a HIV-negative baby?
4. How does HIV-related stigma and discrimination affect the fight against the disease?
5. Fill in table below by entering (T) for true and (F) for false in the appropriate columns

Statement about HIV and AIDS	True/ False
If you live in rural areas, you really don't need to be afraid of getting HIV and AIDS.	
HIV is the virus that causes AIDS.	
The cure to HIV/AIDS has been found.	
Infected mother can pass on HIV and AIDS to her unborn child.	
If you swim in a pool where someone with HIV and AIDS is swimming, you will be infected with HIV and AIDS	
Doctors and nurses who treat patients with HIV and AIDS get infected with the virus.	
A person with HIV and AIDS is bewitched.	

Statement about HIV and AIDS	True/ False
If you hug someone with HIV or AIDS, you may get HIV or AIDS.	
HIV and AIDS is transmitted through mosquito bites.	
HIV and AIDS is transmitted through sharing of public toilets.	
A person with HIV has AIDS.	

14.4. Sexual education and programs promoting positive living in the context of HIV

Learning activity 14.4



Learning about sexual education and HIV and AIDS prevention campaign

1. Identify organizations in Rwanda that carry out educational and outreach programs to reduce HIV prevalence and promote positive living.
2. Learn about the activities of the organizations, through pamphlets or online sources. Use the following questions as guidelines.
 - a) How should public education campaign programs be managed to increase their effectiveness?
 - b) Identify certain slogans that can be used in HIV prevention and awareness-raising campaigns.

Education programmes and awareness campaigns play a vital role in reducing the spread of HIV and AIDS. Many organizations implement prevention and education programmes using various communication channels such as television advertisements, billboards, pamphlets, social media, and radio broadcasts. These campaigns aim to increase public awareness and provide accurate information about HIV and AIDS to community members. While the messages target the general population, special attention is given to vulnerable groups who are at a higher risk of HIV infection. These vulnerable groups include:

- a) Sexually active youths.
- b) Sex workers.
- c) Men and women who are already HIV-positive and who spread the virus to others through unprotected sexual activity.
- d) Women who are in relationship with partners who are HIV- positive and who do not have the right to decide to use condoms.

Public sex education programmes aim to encourage people to:

- Undergo testing by visiting voluntary counselling and testing centres (VCT).
- Protect themselves from spreading the virus.
- Access to treatment and care if they are infected.
- Treat PLHIV with the dignity they deserve.

Most people who are HIV-positive are unaware of their status and therefore unwittingly transmit the virus at the early beginning of the infection. Education programmes should highlight the need for everyone to get tested to prevent the spread of the virus and to promote positive living.

Politicians, religious leaders, traditional leaders and teachers should be involved in promoting HIV positive campaigns.

Positive living is a lifestyle adopted by an HIV positive person to live life to the fullest and to slow down the progression of AIDS. The adoption of positive lifestyle practices has made a remarkable contribution to the quality of life of PLHIV patients. Important aspects of a positive life for PLHIV include positive choices to take care of their mental and physical health, positive outlook on life and avoidance of risky behaviours. Positive living include:

- i. Preventing HIV spread. Patients receiving antiretroviral (ART) therapy may still transmit HIV even if treatment is successful. There is also the possibility of re-infection with another strain, which may speed up the infection.
- ii. Taking medications as prescribed by a healthcare professional to prevent further infections.
- iii. Working as the energy allows and avoids stress.
- iv. Good nutrition is essential to improve health. They should eat a balanced diet consisting of regular meals (even if they are not hungry). The diet should contain protein, fat, carbohydrates and vitamins.
- v. Getting regular exercise and medical care.

Application activity 14.4



1. What issues should be supported in any campaign to prevent HIV infection?
2. Apart from preventing the spread of HIV, explain why it is important that people with HIV practise safe sex using condoms.

14.5. Support groups and mechanisms for PLHIV

Learning activity 14.5



Interactive talk with people living with HIV

1. Listen to PLHIV, visit your school or listen to the video of people living with HIV and AIDS.
2. Ask questions during the presentation. For example:
 - a) What are some of the consequences of stigma and insufficient support for people with HIV?
 - b) State some of the effects of PLHIV support.
3. Take notes during the presentation.

The stigma attached to HIV and AIDS makes life difficult for PLHIV and their families. Loneliness, anxiety, stress, confusion, bitterness and depression can make people more susceptible to disease unless they have support. HIV patients require a great deal of emotional, spiritual, psychological, social, physical and clinical support.

Different people and different institutions can offer support, but it is important that HIV patients come together and support each other. They would know better what their hopes, their joys, their fears and their needs are, and it is they who should define how they want to be understood and treated.

Some of the common needs of PLHIV are:

- Health and medical supplies and skilled medical services.
- Counselling to reduce isolation and promote acceptance.
- Community support groups to provide a safe space to share feelings and advice.
- Spiritual support such as prayer groups and home visits by religious leaders.
- Social acceptance helps them to feel welcome by visiting and treating them as friends.
- Physical care such as bathing, housekeeping when sick and any other thing they cannot do themselves.
- Nutrition assistance to help families improve their diet through low-cost and accessible food.
- Safe drinking water that has been boiled or treated with chlorine.
- Accurate information on HIV and AIDS.

Support groups may be set up for various reasons. Here are a few:

- Provide emotional support and coping mechanisms for those going through a difficult period in their lives who are willing to participate in support groups.
- Provide information to its members so that they can educate their family members, friends, colleagues and neighbours on the disease.
- Draw strength and exchange information from other members and experiences.
- Educate people on antiretroviral (ARV) medicines and encourage them to continue taking them.
- Ensure that they live a positive and productive life.
- To demystify the disease and to promote the de-stigmatization and acceptance of PLHIV and their families in society.

In Rwanda, several organizations are offering PLHIV support for such as:

- i. The AIDS Healthcare Foundation Rwanda (AHF) has been operating in Rwanda since 2006. It has started operations in the district hospitals of Shyira, Kibagabaga and Kanombe. Currently, AHF Rwanda is operating in 8 Districts (Kicukiro, Nyarugenge, Gasabo, Nyabihu, Musanze, Rubavu, Huye and Nyanza) and supports 20 Healthcare centres.

AHF activities include:

- The provision of medical treatment and advocacy irrespective of the ability to pay.

- Increasing access to care and treatment services for HIV and AIDS through a targeted model. This will ensure greater coverage of the population with regard to HIV counselling and Testing (HCT).
 - Access to antiretroviral therapy (ART)
- ii. INACOS (Initiative des Amis Combatant le SIDA) is the HIV and AIDS programme of Friends Church in Rwanda. It was founded in 2002 by Quakers Fidele Nsengiyumva and his wife Antoinette Runiga, both of whom are HIV-positive. She died in 2004, but he's still carrying on work that they started. INACOS is trying to be a community-changing agent by:
- Preventing new infections.
 - Reducing the impact on PLWHA (People Living with HIV and AIDS) or affected by it.

INACOS has targeted three groups in particular:

- Vulnerable people living with or affected by HIV and AIDS, including children.
- Vulnerable families are at risk of HIV infection, e.g. orphans and widows.
- Men and religious leaders who live with or affected by HIV and AIDS.

The main activities were sensitisation and awareness-raising, workshops, counselling, care and support, and developing income generating activities.

Application activity 14.5



1. Identify the support groups in your area and explain their activities
2. Are support groups needed to deal with the HIV and AIDS scourge?

14.6. Non-discrimination against PLHIV

Learning activity 14.6



Discussion on Non-discrimination Against PLHIV

Discuss ways to promote non-discriminatory behaviours and the inclusion of People Living with HIV (PLHIV) in society.

Stigma and discrimination against People Living with HIV (PLHIV) remain significant challenges in many communities. Stigma refers to negative attitudes, beliefs, and stereotypes about people with HIV, while discrimination involves unfair treatment based on a person's HIV status.

PLHIV may experience discrimination in various settings, including their homes, schools, workplaces, healthcare facilities, the media, and public institutions. Such discrimination can lead to social isolation, exclusion from educational opportunities, loss of employment, denial of healthcare services, emotional distress, and reduced quality of life.

Stigma and discrimination create a culture of fear, silence, shame, and secrecy. As a result, some people may avoid HIV testing, counselling, treatment, and support services because they fear being judged or rejected. This can hinder efforts to prevent the spread of HIV and reduce the effectiveness of treatment programmes, including the use of antiretroviral (ARV) medicines.

The negative perceptions and attitudes encountered by PLHIV can affect their self-esteem, mental health, and willingness to participate fully in community life. Therefore, promoting respect, acceptance, and equal opportunities for all people, regardless of their HIV status, is essential.

Ways to Promote Non-discrimination and Inclusion of PLHIV

- Providing accurate information and education about HIV and AIDS to reduce myths and misconceptions.
- Treating PLHIV with dignity, respect, and compassion.
- Encouraging inclusive schools, workplaces, and community environments.
- Protecting the rights and confidentiality of PLHIV.
- Challenging stigma, stereotypes, and discriminatory behaviours whenever they occur.
- Supporting PLHIV to access healthcare, education, employment, and social services.
- Encouraging community members to offer emotional and social support to PLHIV.

By promoting non-discrimination and inclusion, communities can create a supportive environment that enables PLHIV to live healthy, productive, and fulfilling lives.

Application activity 14.6



1. Why do you think that PLHIV are concerned about stigma?
2. How can you help to reduce the stigma surrounding PLHIV in your community?

14.7. End of Unit Assessment



A. Knowledge and understanding

1. The most effective way of preventing HIV and other STI is:
 - a) Using antibiotics
 - b) Abstinence from sexual intercourse
 - c) Taking painkillers
 - d) Washing after sex
2. Which of the following behaviours increases the risk of HIV infection?
 - a) Faithfulness to one uninfected partner
 - b) Correct, and consistent condom use
 - c) Sharing sharp objects
 - d) Abstinence

3. Which method helps reduce the risk of STI during sexual intercourse?
- a) Withdrawal method b) Use of condoms c) Oral contraceptive pills
d) Injectables
4. HIV can be transmitted through:
- a) Sharing food b) Mosquito bites c) Unprotected sexual intercourse
d) Handshakes
5. Supporting people living with HIV means:
- a) Avoiding them b) Denying them education or work
c) Treating them with dignity and respect d) Blaming them
6. Sexually transmitted infections are diseases that arise due to
7. For each of the named infections on table below. Name the causative agent and prevention measures.

Name of the disease	Causative agent	Preventive measures
Gonorrhoea		
Syphilis		
Chlamydia		

8. Which is the surest and best method of preventing contracting STI.
- a) Using a condom properly b) Having two faithful partners
c) Visiting a VCT centre d) Abstinence
9. If left untreated, which STI leads to deafness and death in later stages?
- a) Syphilis b) Chlamydia c) Gonorrhoea d) Yeast infection
10. Which of the following increases your risk of HIV exposure?
- a) Unprotected anal, oral or vaginal sex b) Sharing needles
c) Drug use d) Infection with another sexually transmitted disease
11. If I think that I have been exposed, the best thing to do is: .
- a) Not to worry - one exposure is not enough to get infected.
b) Do nothing and hope for the best.
c) Get tested at VCT centre.
d) It does not matter - if I am infected there is no hope for me anyway.
12. What are the social economic effects of HIV and AIDS?
13. HIV and AIDS are incurable but treatable. Explain
14. What is being done at VCT Centres?
15. Why are young people vulnerable to STI?

B. Skills

16. Discuss briefly the following
- Does having another STI place a person at greater risk of infection if they are exposed to HIV?
 - Does using a condom only some of the time offer any protection from STIs, including HIV?
 - Will washing the penis or vagina after sex lower the risk of becoming infected with an STI?
 - How well do condoms help protect against HIV infection?
 - Is there any truth to rumors that condoms are coated with HIV?
17. Why do you think guidance and counselling among the youth is important?
18. a) What is the difference between HIV and AIDS?
b) What is the impact of HIV and AIDS in society?
19. are the most effective method in protecting against transmission of HIV.
20. a) Suggest a diet plan for PLHIV.
b) Why is information important when dealing with PLHIV?
c) Explain the precautions you will take when caring for PLHIV.
21. a) In what ways can PLHIV live longer?
b) The rights of PLHIV are the rights of everyone. Explain.
c) What do you think of marriage relationships PLHIV?
22. Propose ways in which sexual education should be delivered to reach the majority.
23. Explain the truthfulness of the following statements.
- Children should not be allowed to go to school if they are HIV- positive.
 - I can get HIV from touching infected people or being in the same room with them.
24. Fill the following table with missing information.

Rights of PLHIV	How the issue should be addressed
Confidentiality and privacy	
Health and support services, public benefits, medical schemes and insurance	
Education on HIV and AIDS	
The responsibilities of the media	

Rights of PLHIV	How the issue should be addressed
The right to safer sex	
The rights of prisoners	
Liberty, autonomy, security of the person and freedom of movement	

C. Attitudes and values

25. Why is responsibility important for the prevention of STI and HIV?
26. What approach should young people take when it comes to HIV testing and advice?
27. Why is abstinence considered a responsible decision to reduce the risk of STI and HIV?
28. Why should peer pressure be resisted in sexual behaviour matters?